JUTZ-SCHARKUS

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MATERIALS

NUMERICAL QUANTITIES

FORMS

WESTERMANN

TABLES

REVISED TO INDIAN STANDARDS

FOR THE METAL TRADE

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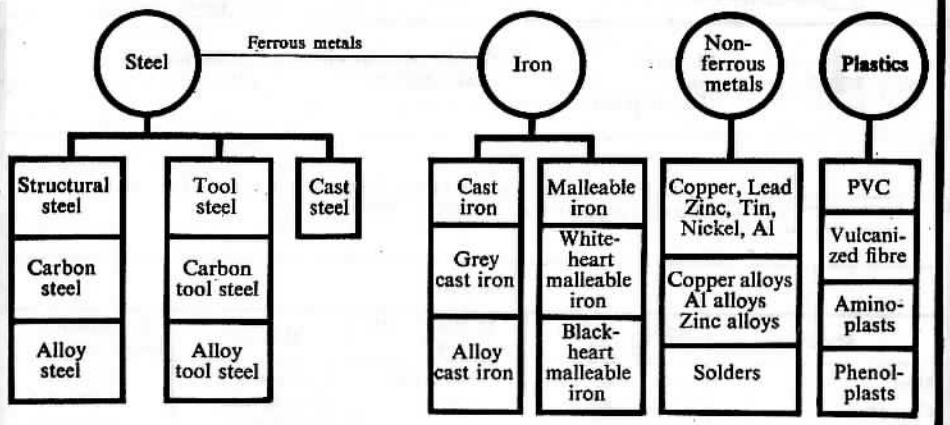
Materials

Classification and categories



General properties of materials

Chemical elements

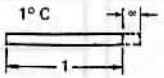
Specific weight—Melting points—Coefficient of linear (thermal) expansion

Symbol	Element	Specific weight gf/cm³	Melting or solidification point °C	Coefficient of linear (thermal) expansion α	Symbol	Element	Specific weight gf/cm³	Melting or solidification point °C	Coefficient of linear (thermal) expansion α
Ag	Silver	10.5	961	0.000 020	Ni	Nickel	8.9	1453	0.000 013
Al	Aluminium	2.7	660	0.000 024	P	Phosphorus	1.82	44	0.000 124
Au	Gold	19.3	1063	0.000 014	Pb	Lead	11.35	327	0.000 029
Ba	Barium	3.74	704		Pt	Platinum	21.45	1769	0.000 009
Be	Beryllium	1.85	1283	0.000 012	Ra	Radium	5.00	700	
Bi	Bismuth	9.75	271	0.000 013	S	Sulphur	2.06	113	0.000 064
C	Carbon				Sb	Antimony	6.69	630	0.000 011
	Graphite	2.25	3550	0.000 008	Se	Selenium	4.5	217	0.000 037
	Diamond	3.52	3600	0.000 001	Si	Silicon	2.4	1410	0.000 008
Ca	Calcium	1.55	850		Sn	Tin	7.3	232	0.000 023
Cd	Cadmium	8.64	321	0.000 029	Ta	Tantalum	16.6	3030	0.000 007
Ce	Cerium	6.9	775		Th	Thorium	11.2	1827	0.000 011
Co	Cobalt	8.8	1492	0.000 013	Ti	Titanium	4.52	1812	0.000 009
Cr	Chromium	7.1	1800	0.000 007	U	Uranium	18.7	1132	
Cu	Copper	8.9	1083	0.000 017	V	Vanadium	5.96	1730	
Fe	Iron	7.86	1535	0.000 012	W	Tangsten	19.27	3380	0.000 004
Ir	Iridium	22.42	2443	0.000 006	Zn	Zinc	7.13	420	0.000 026
K	Potassium	0.86	63	0.000 084	Zr	Zirconium	6.5	1852	0.000 005
La	Lanthanum	6.18	826		Hg	Mercury	13.5	— 39	
Li	Lithium	0.53	180	0.000 058	Cl	Chlorine		—101	
Mg	Magnesium	1.74	650	0.000 026	H	Hydrogen		—259	
Mn	Manganese	7.3	1244	0.000 023	He	Helium		— 272	
Mo	Molybdenum	10.21	2610	0.000 005	N	Nitrogen		—210	
Na	Sodium	0.97	98	0.000 071	Ne	Neon		—249	
Nb	Niobium	8.55	2415	0.000 007	O	Oxygen		—219	

Specific Weight—Melting Point—Coefficient of Thermal Expansion—Shrinkage

Specific weight = Weight per unit volume (gf/cm³ or kgf/dm³)

Melting point (Fusion point) = Temperature at which particular material starts melting



Coefficient of linear (thermal) expansion α = Increase in length of unit length of a solid for temperature rise of 1°C.

Materials

Material	Specific weight gf/cm³	Melting point °C	Material	Coefficient of linear expansion
Steel	7.85	1350...1450	Iron and steel	0.000 012
Cast steel	7.85		Chrome steel	0.000 010
Gray cast iron	7.2	1150...1250	Nickel steel	0.000 012
High-speed steel	9.0	≈2000	Tungsten carbide	0.000 006
Tungsten carbide	14.75	≈2000	Invar	0.000 001 5
Constantan	8.89	≈1600	Chromium	0.000 007
Invar (36 % Ni)	8.7	1450	Constantan	0.000 015
Brass	8.5	≈ 900	Electron	0.000 024
Al bronze	8.4		Aluminium	0.000 023
Al cast bronze	7.6		Magnesium	0.000 026
Tin bronze	8.6	≈ 900	Gold	0.000 014
Lead bronze	9.5		Silver	0.000 019
Al-alloy (Al, Cu, Mg)	2.8	≈ 650	Zinc	0.000 030
Mg-alloy	1.8	≈ 650	Tin	0.000 023
Babbitt metal	7.5...10.1	300...400	Lead	0.000 029
Plexiglass	1.2		Nickel	0.000 013
Alcohol at 18°C	0.79	−110	Platinum	0.000 009
Petrol at 15°C	0.72	−150	Brass	0.000 018
Copper sulphate	1.11		Brouce	0.000 017
Water at 4°C	1.0	0	Plexiglass	0.000 010
Acetylene at 0°C	1.17 kg/m³	−84	Glass	0.000 008
Carbon dioxide at 0°C	1.90 kg/m³	−78	Porcelain	0.000 003
Air at 0°C	1.29 kg/m³	−194		
Propane at 0°C	2.00 kg/m³	−43		

Shrinkage = difference in volume of the mould compared with the volume of the casting after cooling, in percent

Material	Shrinkage	Material	Shrinkage
Grey cast iron	1 %	Brass	1.5%
Cast steel	2 %	Copper	1%
Malleable iron	1.6 %	Tin, lead	1%
Brouce	1.5 %	Zinc alloys	1.5%
Gun metal	1.5 %	Al, Mg alloys	1.25%

System of Designation of Iron and Steel

Steel

Plain carbon steels — < 0.5 % Silicon, < 0.8 % Manganese, < 0.1 % Al or Ti, < 0.25 % or Copper — **Alloy steels** — > 0.5 % Silicon, > 0.8 % Manganese, > 0.1 % Al or Ti, > 0.25 % or Copper

Steels not required to receive heat treatment	Steels required to receive heat treatment	Carbon tool steels	Low alloy steels < 5% special alloying element	High alloy steels > 5% special alloying element

The System of Designation is as follows

1. Letter St 2. Minimum tensile strength in kgf/mm²	1. Letter C for carbon 2. Index number for carbon following letter C, denoting average Carbon content in hundredths of a percent	Letter T for Tool Steels Index Number for Carbon following letter T, denoting average Carbon content in hundredths of a percent	1. Average C content in hundredths of a percent without prefix C and with prefix T for Alloy Tool Steels 2. Chemical symbols of the significant elements arranged in descending order of percentage contents 3. Alloy Index indicating the average percentage of each alloying element	
e.g. St. 42 Steel having a minimum tensile strength of 42 kgf/mm²	e.g. C 35 Carbon steel having an average of 0.35% Carbon	e.g. T 90 Tool steel having an average of 0.90% Carbon	e.g. 15 Cr 65 Chrome steel with average percentages of C=0.15 and Cr=0.65	e.g. 20 Cr 18 Ni 2 Chrome Nickel Steel with average percentages of C=0.20; Cr=18 and Ni=2.00
Applicable for steels which are standardized on the basis of their tensile strength without detailed chemical composition	Steels with special limits for maximum S & P, receive the suffix "K", e.g. C 35 K	To indicate the treatment given to the steel, symbols are used, e.g. T 90 a, "a" is used to indicate annealing (ref. page 4, Add. symbols)	Alloy index number is assigned as follows:	

	Nominal or average alloy content	Alloy index number
1.	Up to 1 percent.	Average alloy content up to 2 decimal places underlined by a bar
2.	1 percent and over.	Rounded to the nearest whole number. Up to 0.5 rounded down, 0.5 and over rounded up.

Castings

System of Designation of Plain Castings

1. Symbols indicating the type of castings
2. Symbol for mechanical properties

OR

1. Symbols indicating the type of castings
2. Symbol for chemical composition similar to the designation of steels

System of Designation of Alloy Castings

1. Symbols indicating the type of castings
2. Average carbon content in hundredths of a percent following the type symbols of castings
3. Chemical symbols for the significant elements arranged in descending order
4. Alloy index number for the average percentages of alloying elements

CS-Steel Castings	FG-Grey Iron Castings	SG-Spherical or Nodular Graphite Iron Castings	Malleable Iron Castings
CS 125—Unalloyed steel castings with minimum tensile strength 125 kgf/mm² CSM 35—Unalloyed special steel castings with minimum tensile strength 35 kgf/mm² GS 50 Cr 1 V 20—Alloy steel castings with average percentage of C=0.50; Cr=1.00; V=2.20	FG 15—Grey iron castings with minimum tensile strength 15 kgf/mm² FG 35 Si 15—Special grey iron castings with minimum total carbon percentage=3.5 and average Silicon percentage=1.50	SG 80/2—Spheroidal or Nodular graphite iron castings with minimum Tensile strength 80 kgf/mm² and minimum elongation 2% on gauge length equal to five times the diameter of test bar	BM 35—Black heart malleable iron castings with minimum tensile strength 35 kgf/mm² PM 70—Pearlitic malleable iron castings with minimum tensile strength 70 kgf/mm² WM 42—White heart malleable iron castings with minimum tensile strength 42 kgf/mm²
CSH—Heat resistant steel castings CSC—Corrosion resistant steel castings	AFG—Austenitic flake graphite iron castings	ASG—Austenitic spheroidal or nodular graphite iron castings	ABR—Abrasion resistant iron castings

Tensile strengths are on 30 mm Dia Test Bars as-cast.

Additional symbols denoting special properties

Steel quality		Treatment given	
A–Non-ageing quality E–Stabilized against stress corrosion L–Control cooled to ensure freedom from flakes D–Fully killed D_s–Semi killed	R–Rimming quality G–Grain size controlled H–Hardenability controlled I–Inclusion controlled M–Structural homogeneity guaranteed by Macro-etch test	a–Annealed or softened c–Case carburized d–Hard drawn, cold reduced h–Hot-rolled n–Normalized	o–Spherodized p–Patented q–Hardened and tempered s–Stress relieved t–Tempered

e.g., St 42 An–Non-ageing steel with 42 kgf/mm² minimum tensile strength-normalized

15 Cr 3c–Chromium steel with average percentages of C = 0.15, Cr = 3.0 and case carburized

E–Electric Furnace Steel; R–Open Hearth Steel; BO–Basic Oxygen

Grey iron castings

IS:210–1970

Code for designation	Grades	Tensile strength Min, kgf/mm²	Transverse test: Breaking load Min, kgf	Transverse test: Corresponding transverse rupture stress kgf/mm²	Transverse test: Deflection Min, mm	Typical applications
FG 15	15	15	800	34.0	4.0	Parts requiring no special grades for general structural purposes
FG 20	20	20	900	38.2	4.5	
FG 25	25	25	1000	42.4	5.0	
FG 30	30	30	1100	46.7	5.5	Parts subjected to severe strains such as cylinder parts, etc.
FG 35	35	35	1350	57.3	5.5	
FG 40	40	40	1500	63.7	5.5	For extraordinary use

Malleable iron castings

IS:2108–1962
IS:2640–1964
IS:2107–1962

Code for designation	Grades	Tensile strength, Min, kgf/mm²	0.5% Proof stress, Min, kgf/mm²	Elongation % (gauge length=3 dia of test bars) Min	Brinell hardness HB Max	Phosphorous contact % Max	Typical applications
BM 35	A	35	21	14	149	0.12	Thin walled castings; mass production parts wheels, keys
BM 30	C	30	—	6	163	0.20	
PM 70	A	70	55	2	241 to 285	0.12	
PM 45	E	45	28	7	149 to 201	0.12	
WM 42	A	42	26	4	217	0.15	Parts for locks and sewing machine parts
WM 35	B	35	—	3	217	0.15	

Steel castings

IS:1030–1962

Code for designation	Grades	Tensile strength Min, kgf/mm²	Elongation % on gauge length $5.65\sqrt{S_0}$ Min,	S% Max	P% Max	Typical applications
CS 55	1	55	12	0.060	0.060	Used for general engineering purposes instead of grey iron castings if greater strength and tenacity are to be met.
CS 47	2	47	17	0.060	0.060	
CS 41	3	41	18	0.060	0.060	
CS 65	1	65	17	0.050	0.050	High strength, good toughness and high abrasion resistance properties; used in transportation equipment and agricultural machinery parts.
CS 85	2	85	12	0.050	0.050	
CS 125	3	125	5	0.050	0.050	

Alloy steel castings for high temperature service

IS:3038–1965
IS:2856–1964

Grades	Tensile strength Min, kgf/mm²	Elongation % on 5.56 $\sqrt{S_0}$ gauge length, Min	Yield stress or 0.5% proof stress Min, kgf/mm²	C%	Si%	Mn%	S% Max	P% Max	Typical applications
1	55	17	35	0.20-0.25	0.15-4.40	1.25-1.45	0.050	0.050	Cast parts which preferably are to withstand temperatures between 300°C to 525°C
2	47	17	25	0.25 Max	0.20-0.50	0.50-1.00	0.050	0.050	
3	52	15	31	0.15 Max	0.40 Max	0.40-0.80	0.050	0.050	
4	49	17	28	0.20 Max	0.60 Max	0.50-0.80	0.050	0.050	
5	52	17	31	0.08-0.15	0.35 Max	0.30-0.70	0.050	0.050	
6	63	15	43	0.20 Max	0.75 Max	0.40-0.70	0.050	0.050	
7	63	15	43	0.20 Max	1.00 Max	0.30-0.70	0.050	0.050	
CS_w-C 20	42	20	21	0.25 Max	0.60 Max	0.70 Max	0.050	0.050	Parts which are to be fusion welded
CS_w-C 25	49	18	25	0.30 Max	0.60 Max	1.00 Max	0.050	0.050	

Specification on Structural and Heat-treatable Steels

General structural steels

IS:1977–1969; IS:2062–1969
IS: 226–1969; IS: 961–1962

Designation of steel	Tensile strength kgf/mm²	Yield strength for thicknesses up to 20 mm	Yield strength for thicknesses 20–40 mm	Elongation % on gauge length $5.65\sqrt{S_0}$ Min	C% Max	S% Max	P% Max	Typical applications
St 32–O	32–44	—	—	26	—	0.07	0.07	Intended for general engineering purposes.
St 42–O	42–54	26.0	—	23	—	0.07	0.07	
St 42–S	42–54	26.0	24.0	23	0.25	0.055	0.055	Intended for all types of structures weldable upon certain conditions.
St 42–W	42–54	26.0	24.0	23	0.20	0.055	0.055	Can be subjected to fusion welding.
St 58–HT	58 Min	36.0	35.0	20	0.27	0.055	0.055	Intended for use in structures where fabrication is done by methods other than welding.
St 55–HTW	55 Min	36.0	35.0	20	0.20	0.055	0.055	Intended for use in structures where welding is employed for fabrication and where guaranteed weldability is required.

Standard sizes of hot-rolled products made of general structural steels

IS Number	Product	Page	IS Number	Product	Page
808	Beam, channel and angle sections	21	1732	Round and square bars	19
			1863	Bulb plates	—
1173	Tee bars	22	1864	Unequal angles	21
1252	Bulb angles	—	3954	Channel sections for general engineering purposes	22
1730	Plates, sheet and strip	20			
1731	Flats	20			

Case hardening steels

IS:4432–1967

Designation of steel	Case hardened: Tensile strength Min kgf/mm²	Case hardened: Elongation % Min	Temperatures for °C: Carburizing	Softening	Case hardening	Annealing	Typical applications
C 10, C 14, 19 S 11	50	17	900–920	650–680	760–780	—	These steels are used for components requiring high wear resistant surfaces, coupled with tough cores to resist shock loads and strength to give longer service life.
14 Mn IS 14, 11 Mn 2	60	17	900–930	650–680	760–780	800–920	
15 Cr 65	60	13	900–930	650–680	770–800	870–900	
17 Mn 1 Cr 95	80	10	900–930	650–680	810–840	850–880	
20 Mn Cr 1	100	8	900–930	650–680	810–840	850–880	
16 Ni 80 Cr 60	70	15	880–920	650–680	780–820	850–880	
16 Ni 1 Cr 80	85	12	900–930	650–660	780–820	850–880	
13 Ni 3 Cr 80	85	12	900–930	620–650	760–780	860–880	
15 Ni 4 Cr 1	135	9	900–930	600–630	760–780	860–880	
20 Ni 2 Mo 25	85	12	880–920	650–660	760–780	—	
20 Ni 55 Cr 50 Mo 20	90	11	880–920	650–660	780–820	—	
15 Ni Cr 1 Mo 12	100	9	900–930	630–650	780–820	860–880	
15 Ni 2 Cr 1 Mo 15	110	9	900–930	630–650	780–820	860–880	
16 Ni Cr 2 Mo 20	135	9	900–930	630–650	800–820	850–880	

Flame and induction hardening steels

IS:3930–1966

Designation of steel	Properties in quenched and tempered conditions: Tensile range kgf/mm²	Properties in quenched and tempered conditions: 0.2% proof stress, Min kgf/mm²	Izod impact Min. kgf.m	Surface hardness obtainable HRC	Hardening temperature: For oil quench	Hardening temperature: For water quench	Typical applications
C 30	60 to 75	36	5.5	45–50	860–890	860–890	These wrought unalloyed and alloyed steels for flame and induction hardening are used when high core strength and good impact properties are required.
C 45	70 to 85	44	3.5	55–61	830–860	820–850	
T 70	70 to 85	40	2.8	60–63	810–840	780–810	
37 Mn 2	60 to 75	40	4.8	53–59	850–870	840–860	
40 Mn 2 S 12	70 to 85	46	4.8	53–59	850–870	840–860	
35 Mn 2 Mo 45	80 to 95	56	5.5	53–59	840–860	830–850	
50 Cr 1	80 to 95	48	2.8	57–62	850–870	840–860	
50 Cr 1 V 23	80 to 95	48	2.8	57–62	850–870	840–860	
40 Ni 3	80 to 95	56	5.5	54–60	830–860	840–870	
40 Ni 2 Cr 1 Mo 28	90 to 105	66	5.5	54–60	830–840	810–830	
31 Ni 3 Cr 65 Mo 55	90 to 105	66	5.5	49–54	850–880	820–840	

Steels for hardening and tempering

IS:5517-1969

Designation of steel	Properties in hardened and tempered condition: Tensile strength kgf/mm²	Properties in hardened and tempered condition: Yield stress Min. kgf/mm²	Normalizing temperature °C	Hardening temperature °C	Quenching medium	Tempering temperature °C	Typical applications
C 30	60 to 75	40	860 to 890	860 to 890	Water or oil	550 to 660	These wrought unalloyed and alloyed steels in the form of billets and bars for general engineering purposes are intended to be used in the hardened and tempered condition
C 35 Mn 75	60 to 75	40	850 to 880	840 to 880	Water or oil	530 to 760	
C 40	60 to 75	38	830 to 860	830 to 860	Water or oil	550 to 660	
C 45	60 to 75	38	830 to 860	830 to 860	Water or oil	530 to 670	
C 50	80 to 95	54	810 to 840	810 to 840	Oil	550 to 660	
C 55 Mn 75	80 to 95	54	810 to 840	810 to 840	Oil	550 to 660	
40 S 18	70 to 85	48	830 to 860	830 to 860	Oil	550 to 660	
40 Mn 2 S 12	60 to 75	40	840 to 870	840 to 870	Oil	550 to 660	
20 Mn 2	60 to 75	44	860 to 900	860 to 900	Water or oil	550 to 660	
27 Mn 2	70 to 85	46	840 to 880	840 to 880	Water or oil	550 to 660	
35 Mn 2 Mo 45	100 to 115	80	—	840 to 860	Oil	550 to 660	
55 Cr 70	90 to 105	66	800 to 850	800 to 850	Oil	500 to 700	
40 Cr 1	80 to 95	60	850 to 880	850 to 880	Oil	550 to 700	
40 Cr 1 Mo 28	80 to 95	60	850 to 880	850 to 880	Oil	550 to 720	
40 Cr Al 1 Mo 18	90 to 105	70	—	850 to 900	Oil	550 to 700	
40 Ni 3	90 to 105	70	830 to 860	850 to 860	Oil	550 to 650	
35 Ni 1 Cr 60	90 to 105	70	—	820 to 850	Water or oil	550 to 660	
30 Ni 4 Cr 1	120 to 135	130	—	810 to 830	Air or oil	>250	
40 Ni 2 Cr 1 Mo 28	120 to 135	130	—	830 to 850	Oil	550 to 660	
31 Ni 3 Cr 65 Mo 55	120 to 135	10	—	830 to 850	Oil	upto 660	
40 Ni 3 Cr 65 Mo 55	120 to 135	0 130	830 to 850	830 to 850	Oil	upto 660	

Cold rolled carbon steel sheets

IS:513-1963

Types	Tensile Strength (for design purpose only) kgf/mm²	C % Max	Mn % Max	S % Max	P % Max	For all types: Delivery condition	For all types: Surface finish	Typical applications
O:Ordinary	28	0.15	—	0.060	0.060	(1) Scale-free	Coarse or rough	Course or rough for enamelling and lacquering
D:Drawing	28	0.12	0.50	0.050	0.050			
DD:Deep drawing	28	0.10	0.50	0.040	0.040	(2) Improved surface	Medium or dull	Medium or dull for general purposes (not suitable for plating)
EDD:Extra deep drawing	28	0.10	0.50	0.035	0.035	(3) Best surface	Fine or bright	Fine or bright for electroplating

Note: Sheet conforming to this standard are of weldable quality and are suitable both for fusion and spot welding.

Hot rolled carbon steel sheet and strip

IS:1079-1968

Grade	Tensile strength kgf/mm²	Yield stress kgf/mm²	Elongation % Min	C % Max	Mn % Max	S % Max	P % Max	Delivery condition	Typical applications
O-1079	—	—	—	—	—	0.060	0.060	Hot-rolled Annealed Normalized and Descaled	Used for cold formed structural members and for other general engineering purposes
D-1079	—	—	—	0.12	0.50	0.050	0.050		
DD-1079	27–40	—	23	0.10	0.50	0.040	0.040		
EDD-1079	27–39	—	25	0.10	0.50	0.035	0.035		
St 34-1079	34–42	21.0	25	0.15	—	0.050	0.050		
St 42-1079	42–50	24.0	22	0.25	—	0.050	0.050		
St 50-1079	50–60	30.0	20	0.30	—	0.050	0.050		
St 52-1079	52–62	36.0	20	0.22	—	0.050	0.050		

Springsteel

Hot-rolled spring steel

IS:3431–1965

Designation of steel	Grade	C %	Mn %	Si %	S % max	P % max	Cr %	V %	Typical applications
50 Cr IV 23	1	0.45–0.55	0.50–0.80	0.10–0.35	0.050	0.050	0.90–1.20	0.1–0.30	Steels in the form of Barm and flats for manufacture of volute, helical and laminated springs for automative suspension.
55 Si 2 Mn 90	2	0.50–0.60	0.80–1.00	1.50–2.00	0.050	0.050	—	—	

Cold-rolled steel strip for springs

IS:2507–1965

Designation of steel	Grade	Tensile strength kgf/mm²: Hardened and tempered	Tensile strength kgf/mm²: Annealed max	C %	Si %	Hardened in oil at °C	Annealed at °C	Typical applications
C 45	1	120–145	60	0.40–0.50	0.10–0.35	830–860	600–650	Cold rolled steel strip for the manufacture of springs for various purposes.
C 65	3	120–145	60	0.60–0.70	0.10–0.35	810–840	600–650	
C 75	5	120–160	65	0.70–0.80	0.10–0.35	780–810	600–650	
C 98	8	160–180	70	0.90–1.05	0.10–0.35	770–800	620–660	
55 Si 2 Mn 90	9	160–200	80	0.50–0.60	1.50–2.00	830–860	640–680	
50 Cr 1	10	170–230	80	0.45–0.55	0.10–0.35	830–860	640–80	
50 Cr IV 23	11	190–240	80	0.450–.55	0.10–0.35	830–860	600–680	

Spring steels for use under elevated temperatures

IS:4454–1967

Grades	Classification	Tensile strength (for wire dia up to 7 mm) min	C %	Si %	Cr %	Va %	Typical applications
1S	S denotes static stressed springs; D denotes dynamic stressed springs	150	0.45–0.55	0.15–0.35	0.90–1.20	0.15–0.30	Used for manufacturing cold formed helical springs, volute springs, etc. working under elevated temperatures.
1D		145	0.45–0.55	0.15–0.35	0.90–1.20	0.15–0.00	
2S		175	0.50–0.60	1.20–1.60	0.50–0.80	—	
2D		175	0.50–0.60	1.20–1.60	0.50–0.80	—	

Steels for Screws Manufacture

Carbon steel wire for the manufacture of machine screws

IS:1976–1960

Designation of steel	Grade	Tensile strength	C % max	Mn %	S % max	P % max	Typical applications
—	1	44–55 kgf/mm²	0.15	0.30–0.65	0.065	0.060	Used for the manufacture of machine screws by the cold heading process.
—	2	55–71 kgf/mm²	0.15	0.30–0.65	0.065	0.060	

Carbon steel wire for the manufacture of wood screws

IS:1673–1960

C 10	—	460 N/mm²	0.17	0.30–0.65	0.055	0.055	Used for the manufacture of wood screws by the cold heading process.
C 15	—	460 ,,	0.22	0.30–0.65	0.055	0.055	
10 S 11	—	460 ,,	0.17	0.60–0.95	0.08–0.15	0.055	

Boiler Steel Plates

IS:2002–1962

Grades	Tensile strength kgf/mm² min	Elongation % min	C% max	Si%	S% max	P% max	Typical applications
1	37–45	26	0.18	0.10–0.35	0.040	0.040	Plates which are required to be either welded, flanged or flame cut plates of non-flanging quality (low tensile)
2A	42–50	25	0.20	0.10–0.35	0.050	0.050	Plates of non-flanging quality (high tensile)
2B	52–62	20	0.22	0.10–0.35	0.050	0.050	

Seamless Steel Pipes

For high-temperature service

IS:2002–1962

Designation of steel	Tensile strength (normalised and tempered) N/mm² min	Elongation % min	C%	Si%	S% max	P% max	Typical applications
16 Mo 30	440–590	22	0.12–0.20	0.12–0.35	0.040	0.040	Used when the wall of pipes reach temperatures up to 580°C and are exposed to high pressure; can be fused and are welded; can be bent or folded in cold state.
15 Cr 90 Mo 55	440–590	22	0.10–0.20	0.10–0.35	0.040	0.040	
10 Cr 5 Mo 55	490–640	16	0.15 max	0.55 max	0.030	0.030	
14 Cr 45 Mo 60 V 27	460–610	15	0.10–0.35	0.10–0.35	0.040	0.040	

Seamless Steel Pipes

For high test line pipes

IS:1979–1971

Designation of steel	Tensile strength min kgf/mm²	Yield strength min kgf/mm²	C% max	C% max	S% max	P% max	Typical applications
Y St 30	42.2	29.5	0.29	1.25	0.04	0.05	Cover pipes intended for use in oil industry.
Y St 32	44.3	32.3	0.31	1.35	0.04	0.05	
Y St 37	46.4	36.6	0.29	1.25	0.04	0.05	

For dimensional requirements IS: 4431; 2507; 2591; 2002; 6630; 1979 may be referred

Cold Rolled Steel Strips for general engineering purposes — IS: 4030—1967

Temper of strips	Rockwell hardness (B Scale) min	max	C % max	Mn % max	S % max	P % max	Surface finish	Typical applications
No. 1—Hard	90	—	0.25	0.60	0.050	0.040	(a) Coarse or rough	Coarse or rough for enamelling and lacquering
No. 2—Half Hard	70	90	0.25	0.60	0.050	0.040		
No. 3—Quarter Hard	60	75	0.25	0.60	0.050	0.040	(b) Medium or dull	Medium or dull for general purpose
No. 4—Skin Rolled	—	65	0.15	0.60	0.050	0.040	(c) Fine or bright	Fine or bright for electroplating
No. 5—Dead Soft	—	55	0.15	0.60	0.050	0.040		

Steels for Rivet Bars — IS: 1148—1973, IS: 1149—1973

Designation of steel	Tensile strength kgf/mm²	Elongation % min	C % max	S % max	P % max	Typical applications
St 42 R	42 to 54	23	0.23	0.055	0.055	For manufacture of hot forged rivets for structural purposes
St 47 R	47 min	22	0.23	0.055	0.055	High tensile steel rivet bars for structural purposes

Free Cutting Steels — IS: 4431—1967

Designation of steel	Tensile strength kgf/mm²	Elongation % min	C %	Si %	Mn %	S %	P % max	Typical applications
10 S 11	37–49	24	0.15 max	0.05–0.30	0.60 to 0.90	0.08 to 0.13	0.060	Suitable also for case hardening
14 Mn 1S 14	44–54	22	0.10–0.18	0.05–0.30	1.20 to 1.50	0.10 to 0.18	0.060	
25 Mn 1S 14	50–60	20	0.20–0.30	0.25 max	1.00 to 1.50	0.10 to 0.18	0.060	These have good machinability and satisfactory chip-break
40 S 18	55–65	17	0.35–0.45	0.25 max	0.80 to 1.20	0.14 to 0.22	0.060	
13 S 25	37–49	22	0.08–0.18	0.10 max	0.80 to 1.20	0.22 to 0.30	0.060	(Rapid machining steel for repetition work)
40 Mn 2 S 12	60–70	15	0.35–0.45	0.25 max	1.30 to 1.70	0.08 to 0.15	0.060	

Black Bars for production of machined parts — IS: 2073—1970

Designation of steel	Tensile strength kgf/mm²	Elongation % min	C %	Si %	Mn %	S % max	P % max	Typical applications
C 14	37–45	26	0.10–0.18	–	0.40–0.70	0.055	0.055	These types are carbon steel black bars for production of machined parts for general engineering purposes
C 20	44–52	24	0.15–0.25	0.05–0.35	0.60–0.90	0.055	0.055	
C 30	50–60	21	0.25–0.35	0.05–0.35	0.60–0.90	0.055	0.055	
C 40	58–68	18	0.35–0.45	0.05–0.35	0.60–0.90	0.055	0.055	
C 45	63–71	15	0.40–0.50	0.05–0.35	0.60–0.90	0.055	0.055	
C 55 Mn 75	72 min	13	0.50–0.60	0.05–0.35	0.60–0.90	0.055	0.055	
C 65	75 min	10	0.60–0.70	0.05–0.35	0.50–0.80	0.055	0.055	

Symbolic Designation
of essential properties of materials
(iron and steel)

Examples and Explanations

IS No.	Title	See page	Designation (example)	Explanations
1977	Structural steels	5	St 32—0	St = Steel; 32 kgf/mm² minimum tensile strength O = Ordinary quality
1977	—do—	5	St 42—0	42 kgf/mm² minimum tensiles trength
226	—do—	5	St 42—S	S = Standard quality
226	—do—	5	St 42—Sc	c = Copper bearing quality
226	—do—	5	St 42—Kw	K = Special limits for max P and S w = Weldable
2062	—do—	5	St 42—W	W = Fusion welding quality
961	—do—	5	St 55—HTw	HT = High tensile steel w = Fusion weldable
1148	Rivet steels	8	St 42—R	R = Rivet bars
2002	Boiler plates	7	Grade 1	Plates required to be welded, flanged or flame-cut
2002	—do—	7	Grade 2A	Non-flanging quality (low tensile)
2002	—do—	7	Grade 2B	—do— (high tensile)
5517	Heat-treatable steels	6	C 30	C = Carbon 30 = Average C contents 0.30%
5517	—do—	6	T 50a	T = Tool steel; a = annealed
5517	—do—	6	C 35 Mn 75	C35 = Average carbon content 0.35 % Mn75 = Average manganese of 0.75%, represented without decimal point, underlined by a bar. (Applicable for alloying element upto 1 %)
4432	Case-hardening steels	5	C 10c	C = Carbon; c = case carburized
4432	—do—	5	11 Mn 2	Carbon average 0.11 %; Manganese average 1.5 %. (Average alloy content more than 1 % is rounded to the nearest whole number, upto 0.5 rounded down; 0.5 and over rounded up.
3431	Hot Rolled steels for springs	7	55 Si 2Mn 90h	h = Hot rolled
2507	Cold rolled steels strips for springs	7	C 45q	q = Hardened and tempered
4454	High temperature steels for springs	7	1S; 1D	S = Static stressed springs; D = Dynamic stressed
1079	Hot rolled carbon steel sheet and strip	6	0; D; DD; EDD	O = Ordinary; D = Drawn; DD = Deep drawn EDD = Extra deep drawn
513	Cold rolled carbon steel sheets	6	J; J2 J3; J4	J = Bright drawn or bright rolled; J2 = Precision ground; J3 = descalled; J4 = shot blast
513	—do—	6	F; F2 F3; F7	F = Black sheet; F3 = Pickled surface; F7 = Cold finished; F2 = Black sheet for enamelling and galvanizing
1030	Steel castings	4	CS 125	CS = Cast steel-unalloyed; 125 = Minimum tensile strength 125 kgf/mm²
210	Grey iron castings	4	FG 15	FG = Grey iron castings; 15 = Minimum tensile strength 15 kgf/mm²
2108	Malleable iron castings	4	BM 35	BM = Black heart malleable iron castings
2640	—do—	4	PM 70	PM = Pearlitic malleable iron castings
2107	—do—	4	WM 42	WM = White heart malleable iron casting For castings tensile strengths are on 30 mm dia test bars as cast

Tool and die steels

Tool and die steels for hot work

IS: 3748–1966

Designation of steel	C %	Si %	Mn %	Cr %	Mo %	V %	W %	Brinell hardness (annealed) HB, max	Typical applications
T33W9Cr3V38	0.25-0.40	0.10-0.35	0.20-0.40	2.80-3.30	—	0.25-0.50	8.00-10.0	241	Used for extrusion dies, hot swaging dies, forging die inserts, brass forging dies, hot shear blades, trimmer dies, die-casting dies for copper etc.
T35Cr5Mo1V30	0.30-0.40	0.80-1.20	0.25-0.50	4.75-5.25	1.20-1.60	0.20-0.40	—	229	
T35Cr5MoV1	0.30-0.40	0.80-1.20	0.25-0.50	4.75-5.25	1.20-1.60	1.00-12.0	—	229	
T35Cr5MoW1V30	0.30-0.40	0.80-1.20	0.25-0.50	4.75-5.25	1.20-1.60	0.20-0.40	1.20-1.60	229	
T55W14Cr3V45	0.50-0.60	0.10-0.35	0.20-0.40	2.80-3.30	—	0.30-0.40	13.0-15.0	248	

Tool and die steels for cold work

IS: 3749–1966

T50	0.45-0.55	0.10-0.35	0.60-0.90	—	—	—	—	240	Covers the requirements for plain carbon and alloy tool and die steels in the form of bars, blanks, rings, and other shapes for cold work, capable of being hardened and tempered. These are used for the making tools and dies for blanking, trimming, shaping and shearing.
T60	0.50-0.60	0.10-0.35	0.60-0.90	—	—	—	—	240	
T70Mn65	0.65-0.75	0.10-0.35	0.50-0.80	—	—	—	—	240	
T80Mn65	0.75-0.85	0.10-0.35	0.50-0.80	—	—	—	—	240	
T90	0.85-0.95	0.10-0.30	0.20-0.35	—	—	—	—	200	
T103	0.95-1.10	0.10-0.30	0.20-0.35	—	—	—	—	200	
T133	1.25-1.40	0.10-0.30	0.20-0.35	—	—	—	—	210	
T90V23	0.85-0.95	0.10-0.30	0.20-0.35	—	—	0.15-0.30	—	200	
T118Cr45	1.10-1.25	0.10-0.30	0.20-0.35	0.30-0.60	—	0.30 max	—	200	
T105Cr1Mn60	0.90-1.20	0.10-0.35	0.40-0.80	1.00-1.60	—	—	—	230	
T140W4Cr50	1.30-1.50	0.10-0.35	0.25-0.50	0.30-0.70	—	—	3.50-4.20	250	
T55Ni2Cr65Mo30	0.50-0.60	0.10-0.35	0.50-0.80	0.50-0.80	0.25-0.35	—	—	255	
T105W2Cr60V25	0.90-1.20	0.10-0.35	0.25-0.50	0.40-0.80	0.25 max	0.20-0.30	1.25-1.75	230	
T110W2Cr1	1.00-1.20	0.10-0.35	0.25-0.50	0.90-1.30	—	—	1.25-1.75	230	
T90Mn2W50Cr45	0.85-0.95	0.10-0.35	1.25-1.75	0.30-0.60	—	0.25 max	0.40-0.60	230	
T215Cr12	2.00-2.30	0.10-0.35	0.25-0.50	11.0-13.0	0.80 max	0.80 max	—	260	
T45Cr1Si95	0.40-0.50	0.80-1.10	0.55-0.75	1.20-1.60	—	—	—	230	
T55Cr70V15	0.50-0.60	0.10-0.35	0.60-0.80	0.60-0.80	—	0.10-0.20	—	230	
T55Si2Mn90Mo33	0.50-0.60	1.50-2.00	0.80-1.00	—	0.25-0.40	0.12-0.20	—	230	
T40W2Cr1V18	0.35-0.45	0.50-1.00	0.20-0.40	1.00-1.50	—	0.10-0.25	1.75-2.25	230	
T50W2Cr1V18	0.45-0.55	0.50-1.00	0.20-0.40	1.00-1.50	—	0.10-0.25	1.75-2.25	230	

Steels for die blocks for drop forgings

Designation of steel	C %	Si %	Mn %	Ni %	Cr %	Mo %	Brinell hardness HB Annealed max	Brinell hardness HB Hardened and tempered	Typical applications
T60	0.55-0.65	0.15-0.35	0.50-0.80	—	—	—	209	212-269	Steel for die blocks in square, rectangular and sections for drop forgings.
T60Ni1	0.55-0.65	0.15-0.35	0.50-0.80	1.0-1.4	—	—	209	212-269	
T55NiCr65	0.50-0.60	0.15-0.35	0.50-0.80	1.25-1.65	0.50-0.80	—	230	235-302	
T50NiCr35	0.48-0.53	0.15-0.35	0.45-0.65	0.80-1.00	0.80-1.00	0.30-0.40	255	269-477	

SPK

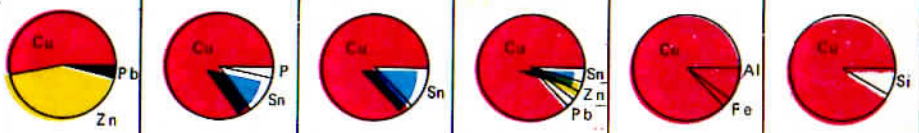
Classification of carbide tips according to their range of application

(IS: 2428–1964)

Designation	Increasing direction of the characteristic of		Range of application	
Identification colour	Carbide tip	Cutting	Material to be machined	Machining conditions
P01	Resistance to wear ↓ Toughness ↑	Cutting speed ↓ Feed ↓	Steel, steel casting	Precision turning and fine boring Cutting speed: high. Feed: Low
P10			Steel, steel casting	Turning, threading and milling Cutting speed: high. Feed: low or medium
P20			Steel, steel casting, malleable cast iron forming long chips	Turning, milling. Cutting speed and feed: medium Planing: with low feed rate
P30			Steel, steel casting, malleable cast iron forming long chips	Turning, planing, milling. Cutting speed: medium to low. Feed: medium to high even if operating conditions are unfavourable
P40			Steel, steel castings with sand inclusions or shrinkage cavities	Turning, planing, shaping. Cutting speed: low. Feed: high. Rake angle: high, for machining under unfavourable conditions and work on automatic machines
P50			Steel, steel castings of medium or low tensile strength with sand inclusions or shrinkage cavities	Turning, planing, shaping. Cutting speed: low. Feed: high. Rake angle large for machining under unfavourable conditions and work on automatic machines
M10	Resistance to wear ↓ Toughness ↑	Cutting speed ↓ Feed ↓	Steel, steel castings, manganese steel, grey cast iron, alloyed cast iron	Turning. Cutting speed: medium to high. Feed: low to medium
M20			Steel, steel casting, austenitic steel, manganese steel, grey cast iron, sphrodised cast iron and malleable cast iron	Turning, milling. Cutting speed: medium. Feed: medium
M30			Steel, steel casting, austenitic steel, grey cast iron, heat resisting alloys	Turning, milling, planing. Cutting speed: medium Feed: medium or high
M40			Free cutting steel, low tensile strength steel, brass and light alloy	Turning, profile turning, parting off especially in automatic machines
K01	Resistance to wear ↓ Toughness ↑	Cutting speed ↓ Feed ↓	Very hard grey cast iron, chilled castings of hardness up to 60 HRC. Alluminium alloys with high silicon content, hardened steel, plastics of abrasive type, hard board and cermics	Turning, precision turning and boring, milling, scraping
K10			Grey cast iron of hardness more than 220 HB, malleable cast iron formihg short chips, tempered steel, aluminium alloys containing silicon, copper alloys plastics, glass, hard rubber, hard cardboard, porcelain, stone	Turning, milling, boring, reaming, broaching, scraping
K20			Grey cast iron of hardness up to 220 HB, non-ferrous metals, such as copper, brass, aluminium, laminated wood of abrasive type	Turning, milling, planing, reaming, broaching
K30			Soft grey cast iron, low tensile strength steel, laminated wood	Turning, planing, shaping, milling. Rake angle: large, even under unfavourable conditions
K40			Soft or hard natural wood, nonferrous, metals	Turning, milling, planing, shaping. Rake angle: large, even under unfavourable machining conditions

Nomenclature	Grade	Minimum contents	Physical properties	Typical application
Copper				IS: 191-1967
Electrolytic tough pitch copper	ETP	99.9% Cu	Soft	For electrical parts
Fire refined high conductivity copper	FRHC	99.9% Cu	High conductivity	For conductors
Fire refined tough pitch copper	FRTP-1 FRTP-2	99.8% Cu 99.5% Cu	easy to cast	For castings
Tough pitch arsenical copper	ATP	99.2% Cu	Good bearing property	For bearings
Oxygen free high conductivity copper	OF	99.95% Cu	High conductivity	For conductors
Lead				IS: 27-1965
Pig lead Pig lead	Pb 99.99 Pb 98.94	99.99% Pb 99.94% Pb	Soft, can be cast, soldered and welded	Plates in storage batt For alloying
Zinc				IS: 4699-1968
Refined secondary zinc Refined secondary zinc	SZn 99.5 SZn 98.5	99.5% Zn 98.5% Zn	Can be cast, resistant to corrosion	For alloying For galvanizing
Tin				IS: 4280-1967
Refined secondary tin Refined secondary tin	Sn 99 Sn 96	99% Sn 96% Sn	Soft, can be cast, rolled to foils	For plating, casting For alloying
Aluminium				IS: 734-1967
Aluminium	F 1A	99.8% Al	Tensile strength 5.5kgf/mm²	May be cast, weldable
Aluminium	F 1B	99.5% Al	More resistant to corrosion	Available in the form of sheets plates, tubes, wires, forgings Used for cladding, on stronger alloys, food and chemical plants, electrical conductors and reflectors
Aluminium	F 1C	99.0% Al	Very ductile, resistant to corrosion, good conductor.	Available in the form of sheets, plates, tubes, wires, rods and forgings Used for panelling and moulding, lightly stressed and decorative assemblies, equipment for food, chemical and brewing industries, packing and cooking utensils
Aluminium (commercial quality)	AO	99% Al	Excellent, electrical, conductivity, resistant to corrosion.	For induction motor, rotors, power transmission cable accessories, vessels and fittings for food and chemical industries

Copper alloys



Brass	Phosphor bronze	Tin bronze	Leaded tin bronze	Aluminium bronze	Silicon bronze
Cu, Zn	Cu, Sn, P	Cu, Sn	Cu, Sn, Zn, Pb	Cu, Al, Fe	Cu Si

System of designation of copper alloys IS: 2378–1963

1. Chemical symbol of copper
2. This is followed by the chemical symbol of the next most significant element.
3. Other significant elements follow in the order of decreasing percentage or when equal in alphabetical order.
4. Nominal or average percentage of each alloying element are indicated by an index number as follows and these follow the respective chemical symbol.

System of alloy index

Nominal or average alloy content	Index number
Up to 1 percent	Only the alloy symbols are used in the descending order of percentage content. For differentiating an alloy, the average alloy content are expressed upto one decimal place and the decimal digit is underlined by a bar.
1 percent and over	Average or nominal alloying content is rounded to the nearest whole number, for example, values terminating upto 0.5 are rounded down and values terminating in 0.5 and over are rounded up.

Following symbols for the methods of casting precede the designation:
G: Sand cast; GK: Chill cast; GD: Die cast; GZ: Continuous cast
Note: Absence of these symbols indicate that the material is in wrought form.

Symbols for special requirements

K=symbol for rigid limits for purity

Surface finish and condition:
J=Bright rolled or drawn
J7=Bright pickled
J8=Bright annealed
M=Machined

Treatment:
a=Annealed; d=Cold worked (rolled or drawn)
c=Extruded; f=Forged; h=Hot rolled
1/4 H=Quarter hard; 1/2 H=Half hard
3/4 H=Three fourths hard; H=Full hard

Example: G Cu Sn 7 Zn 5 Pb 2—Leaded tin bronze sand castings with average percentages of Sn=7; Zn=5 and Pb=2

System of colour code for copper alloys—three colour bands system IS: 2376–1963

First band	Second band	Third band
First band signifies the alloy group to which the product belongs.	Second band in combination with the first and second bands identifies the particular product.	Third band in combination with first and second bands identifies the particular product.

Copper forging alloys

Nomenclature	Designation	Composition %	Residual constituent	Typical applications
Brass [IS: 3488–1966]	Cu Zn 42 Pb 2	Cu 56.5–60.0 Pb 1.00–2.50	Zn	Brass bars, rods and sections for forgings.
Naval brass [IS: 291–1961]	Grade I Grade II	Cu 61.0–64.0 Cu 59.0–62.0	Zn Zn	Brass having good mechanical properties combined with good resistance to corrosion by sea water.
Tin bronze [IS: 306–1968]	Cu Sn 8 Zn 4	Sn 7.5–8.5 Zn 3.5–4.5	Cu	Springs, sliding elements, membrane in chemical industries.
Aluminium bronze [IS: 305–1961]	Grade I Grade II Grade III	Al 8.5–10.5 Al 8.5–10.5 Al 9–11	Cu Cu Cu	Pipings, valve seats, etc. used in chemical industries.

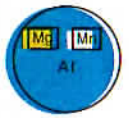
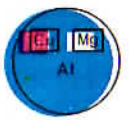
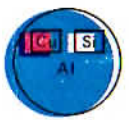
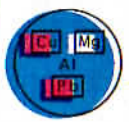
Copper casting alloys

Tin bronze casting [IS: 306–1968]	Cu Sn 8 Zn 4	Sn 7.5–8.5 Zn 3.5–4.5 Pb 1.5 max	Cu	Anti-friction bearings, worm wheels, gears, surfaces exposed to severe strains.
Brass castings [IS: 292–1961]	Grade I Grade II Grade III	Cu 71.0–81.0 Cu 67.0–74.0 Cu 64.0–71.0	Zn Zn Zn	Sand castings, fittings and chilled castings.
[IS: 1264–1965]	Cu Zn 40	Cu 61.0 min	Zn	Brass gravity die castings.
[IS: 304–1961]	Grade I	Cu 51.0 min	Zn	High tensile brass castings used for bushes, pinions, etc.
Aluminium bronze castings [IS: 305–1961]	Grade I Grade II Grade III	Al 8.5–10.5 Al 8.5–10.5 Al 9–11	Cu Cu Cu	Used in chemical industry, possess resistivity to wear, oxidation and corrosion by sea water.
Leaded tin bronze castings [IS: 318–1962]	Grade I	Sn 6.0–8.0; Pb 1.0–3.0	Cu	Meant for general castings used in low pressure valve fittings and pump castings.
	Grade II	Sn 4.0–6.0; Pb 4.0–6.0	Cu	
	Grade III	Sn 6.0–8.0; Pb 9.0–11.0	Cu	Suitable for high speed and heavy pressure bearings.
	Grade IV	Sn 6.0–8.0; Pb 14.0–16.0	Cu	General purpose bearing metal for moderate pressure.
	Grade V	Sn 4.0–6.0; Pb 22.0–26.0	Cu	For bearings under light loads and high speeds.
Phosphor bronze castings [IS: 28–1969]	P Cu Sn 7 P Cu Sn 10	P 0.30–0.50; Sn 6.0–8.0 P 0.50–0.70; Sn 9.0–11.0	Cu Cu	Used in slow speeds, heavy loads. Used for heavy duty bearings and bushes.
Nickel copper alloy castings [IS: 4131–1967]	Ni Cu 31 Mn 1 Si 1 Fe Ni Cu 29 Mn 1 Si 3 Fe	Ni 63.0–68.0 Ni 63.0–68.0	Cu Cu	Used for parts of pumps and valves.
Silicon bronze castings [IS: 1028–1970]	—	Cu + named elements 99.5 min. Si 1.0–5.0 Zn 5.0 max.	—	Used in sewage disposal equipment, chemical process equipment, marine hardware and anti-corrosive pipe, fittings such as valves and pumps.

Physical properties of brass

Requirements		Cu Zn 20	Cu Zn 30	Cu Zn 40
Tensile strength kgf/mm², min.	As mfd.	32.0	35.0	35 0
	Annealed	25.0	28.0	28.0
Elongation % min.	As mfd.	25.0	25.0	25.0
	Annealed	50.0	50 0	50.0
Supplied as	Sheets [IS: 422–1967]		O	
	Tubes [IS: 407–1966]		●	
	Rods [IS: 4170–1967]	●		●

Aluminium alloys

Binary alloys		Ternary alloys			Quarternary alloys
					
Al Mn Al Mg	Al Mg Mn	Al Mg Si	Al Cu Mg	Al Cu Si	Al Cu Mg Pb Al Mg Si Pb Al Zn Mg Cu

System of designation of aluminium alloys — IS: 6051–1970

Element—Number—Designation

Element	Group number	Element	Group number
Unalloyed aluminium (wrought and cast form)	1	Magnesium silicide (Mg_2Si)	6
Copper	2	Zinc	7
Manganese	3	Other elements (such as Ni, Ti,	
Silicon	4	Cr, Pb, Bi, etc.)	8
Magnesium	5	Unassigned	9

Wrought aluminium alloys (five digit system)

First digit	Second digit	Third, fourth and fifth digits
Identifies the major alloying element	Indicates rounded off mean value of the percentage of the major alloying element except for Group 4 containing silicon when the digit refers to the mean percentage halved and rounded off; and for Group 6 containing Mg_2Si the digit refers to five times the mean Magnesium percentage rounded off	Identify the minor alloying elements in the descending of their percentages and in the case of same alloy percentage in the serial order. For balanced compositions, the third digit will be zero. In case of high purity aluminium base alloy, the fifth digit will be 1.

Example: Alloying elements and their mean percentages
Designation 22450 — Cu = 1.5; Si = 1.0; Mg = 0.85

Casting alloys (four digit system)

First digit	Second digit	Third and fourth digits
Identifies the major alloying element	Indicates the mean value percentage of the major alloying element halved and rounded off	Identify the minor alloying element in the order of their decreasing percentage or in the case of same alloy percentage in the serial order

Example: Allowing elements and their mean percentages
Designation 2585 — Cu=9.8; Fe=1.0; Mg=0.25

Aluminium alloys—castings — IS: 617–1959

Designation of material	Characteristics	Tensile strength, min kgf/mm² Sand cast	Chill cast	Typical applications
A-1-M	Non-heat treatable alloy with excellent machining qualities	12.6	15.7	Used for medium-stressed castings such as automotive timing gears, spirit levels, moulding flasks, and drying plates.
A-4-M	Suitable as sand castings and gravity and pressure die castings	14.2	15.7	General purpose alloy used for a wide variety of castings.
A-5-M	Takes a high polish, has good anodizing characteristics	14.2	17.3	Mainly used as sand castings and gravity die castings. Suitable for architectural, ornamental and marine applications.
A-8-M	Has excellent castability and resistance to chemical attack	12.6	16.5	Air and water cooled cylinder blocks, cylinder heads, valve bodies. Truck body castings, storage tank fittings, pneumatic tools, etc.
A-11-W	Possesses high strength and good ductility. Its foundry characteristics are good and machining property excellent	22.0	26.8	Used for production of stressed parts such as flywheel housing and propellers, type writer key levers, power transmission cable accessories and wrenches.
A-11-WP		28.3	31.5	
A-14-WP	Noted for its stregnth at elevated temperatures and good machining property	22.0	28.3	Used mainly for pistons and cylinder heads.
A-16-W	Can be readily cast into intricate shapes; possesses good pressure tightness. Retains strength well at elevated temperatures.	17.3	20.5	Used for stressed parts in aircraft and automotive applications, fittings for fire fighting equipments.
A-16-WP		23.6	26.8	
A-18-M	Good castability and resistance to corrosion	11.8	14.2	Used for manufacture of electrical household appliances such as steam irons, vacuum cleaners, cast utensils.

Aluminium alloys—forgings

IS: 734–1967

Designation of material	Characteristics	Tensile strength in "O" condition, kgf/mm², min.	Elongation on $4\sqrt{S_o}$, %, min.	Available forms	Typical applications
NF 3	Has good workability, weldability and corrosion resistance	5.5	30	Sheet, plate extrusion, wire, rolled rod forgings	Building sheet, vehicle panelling, and sheet metal work, tanks cooking utensils.
NF 4	Work hardens rapidly becoming extremely tough; has high corrosion resistance	17.5	18	–do–	Panelling and structures exposed to marine atmospheric pressings, certain aircraft parts.
		(Fully heat treated condition)			
HF 9	Suitable for intricate extruded sections of medium strength	19.0	12	Extrusion tube, wire, rolled rod, forgings	Architectural members such as window frames, hand rails, medium strength.
HF 30	Medium strength alloy with good mechanical properties corrosion resistance and weldability	30.0	10	Sheet, plate, wire, extrusion tube, rolled rod forgings	For structural applications such as road and rail transport vehicles, bridges, cranes, roof trusses, rivets, etc.
HF 11	Good machining alloy which may be extruded in fairly complex sections	34.5	8	Extrusion, tube, forgings.	Aircraft components, repetition machined part.
HF 15	Combines high strength with fair ductility in the solution treated condition	44.0	8	Sheet, plate, extrusion, tube, wire, forgings	Stressed components of all types in aircraft and general engineering.

Antifriction bearing alloys

IS: 25–1966

Nomenclature	Grade	% Composition ≈	Pouring temp., °C	Typical applications
White metal	90	Sn 90 Sb 7 Cu 3 Pb	340-390	For lining of petrol and diesel engine bearings, cross-heads in steam engines and other bearings used at high speeds.
White metal	84	Sn 84 Sb 10 Cu 5 Pb	430-480	
White metal	60	Sn 60 Sb 11 Cu 3 Pb	370-400	For lining of bearings used for medium speeds, such as centrifugal pumps, electric motors, etc.
White metal	20	Sn 20 Sb 15 Cu 1.5 Pb	370-410	For low speed bearings such as pulp crushers, concrete mixers and rope conveyors.
White metal	5	Sn 5 Sb 15 Cu 0.5 Pb	350-390	For mill shaftings, railway carriage and wagon bearings.

Zinc alloy for die casting

IS: 742–1966

		Tensile strength kg/cm²	
Alloy 1	Al 4 Mg 0.05	27	For die casting.
Alloy 2	Al 4 Cu 1 Mg 0.05	30	

Soft solders

IS:193–1966

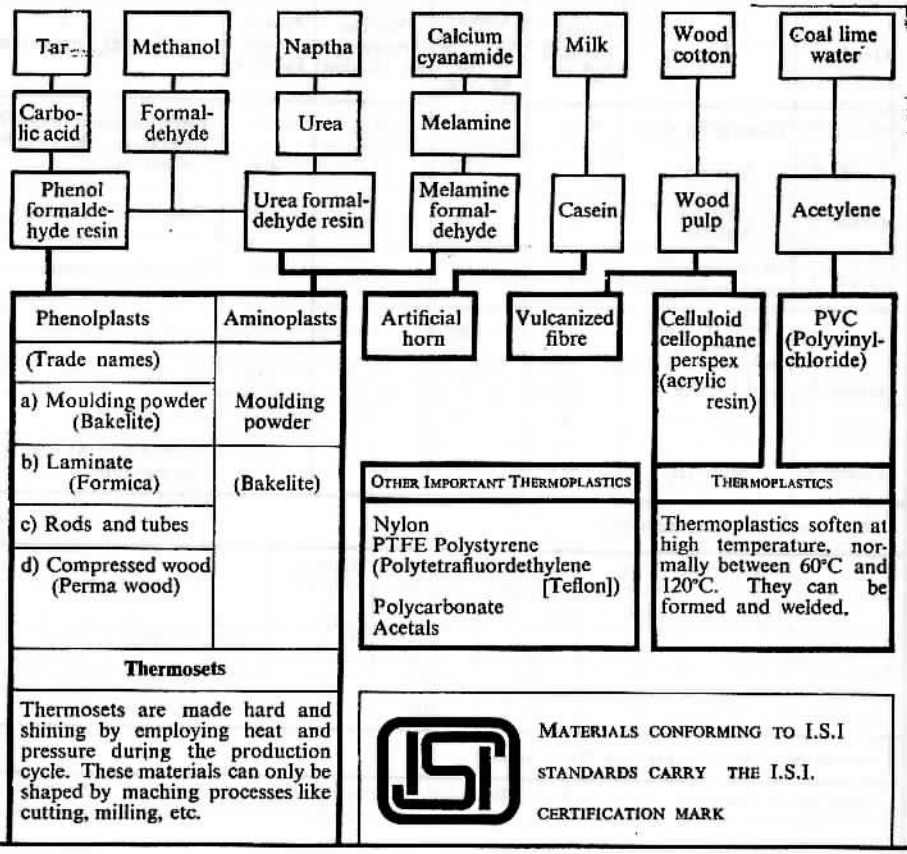
		Melting point °C	
Sn 60	Sn 60 Pb	188	For electrical and radio instrument assemblies.
Sn 50 Sb	Sn 50 Sb 2.5 Pb	204	Copper smith's and tinsmith's bit soldering and general machine soldering, e.g., can end-seams.
Sn 45 Sb	Sn 45 Sb 2.5 Pb	215	
Sn 40 Sb	Sn 40 Sb 2 Pb	227	For general soldering work.
Sn 35	Sn 35 Pb	247	For blow pipe soldering of side seams on high-speed body forming machines.
Sn 18 Sb	Sn 18 Sb 1 Pb	275	For lamp manufacture (dipping solder).
Sn 5 Ag15	Sn 5 Ag 1.5 Pb	301	For high temperature services.

Brazing alloys

IS: 2927–1964

Aluminium-silicon brazing alloys for Aluminium & Aluminium alloys	BA-AlSi 1	Si 5 Cu0.1 Fe0.3 Mn0.3 Al	630	General purpose brazing alloy for torch, dip and furnace brazing of aluminium and its alloys.
	BA-AlSi 2	Si 7 Cu0.2 Fe0.3 Mn0.3 Al	613	Suitable for dip and furnace brazing of Aluminium and its alloys.
	BA-AlSi 3	Si 12 Cu0.1 Fe0.3 Mn0.3 Al	595	General purpose brazing alloy having corrosion resistance properties. Used for torch brazing.
	BA-AlSi 4	Si 12 Cu3.5 Fe0.3 Mn0.3 Al	595	General purpose brazing alloy for torch, dip and furnace brazing. Not used for brazing food containers and utensils.
Spelter White brass	BA-CuZn 5	Cu 54 Pb 0.5 Zn	880	For general use.
	BA-CuZn 6	Cu 54 Sn 1 Pb 0.5 Zn	870	For general purpose torch brazing.
	BA-CuZn 7	Cu 50 Pb 0.5 Zn	870	For hearth brazing.
	BA-CuZn 8	Cu 48 Ni 10 Zn	980	For brazing steels, nickel and nickel alloys.
	BA-CuZn 9	Cu 43 Sn 0.5 Pb 0.7 Zn	845	Low melting point. Used where high brazing temperature has to be avoided.
	BA-CuAg 12	Ag50 Cu15 Zn15 Cd16 Ni3	688	For brazing carbide tips to shanks, and stainless steels.
	BA-CuAg 17	Ag40 Cu 36 Zn 24	617	For general use.

Engineering Plastics

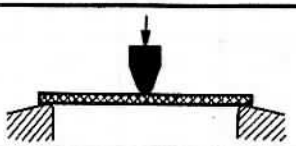


Materials conforming to I.S.I standards carry the I.S.I. certification mark

Methods of test for plastics

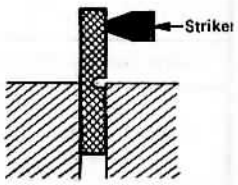
Cross Breaking Strength $[kgf/cm^2] = \frac{1.5\,WL}{BD^2}$

Where W = Load at fracture in kgf
B = Breadth of testpiece in cm
L = Distance between support in cm
D = Thickness of test piece in cm



Impact strength, edgewise [kgf-cm] $= \frac{\text{Energy absorbed} \times 12.7\text{ mm}}{\text{Width of notched force in mm}}$

Insulation resistance after immersion in water: The specimen is immersed in distilled water for 24 hours. After wiping dry the insulation resistance between pairs of electrodes is measured at a potential difference of 500 V DC. Temperature and humidity must be controlled.



Melt flow index: Plastic is extruded through a dye by a loaded piston under controlled conditions of temperature. Number of grams extruded per 10 minutes is determined.

Vicat softening point is the temperature at which a penetrator of 1 mm circular cross section will penetrate a thermoplastic specimen to a depth of 1 mm under a specified load using a uniform rate of temperature rise.

Phenolic moulding materials IS: 1300–1966

Grades	Type	Impact strength kgf–cm	Cross-breaking strength kgf/cm²	Surface resistivity log 10 ohm	Tensile strength kgf/cm²	Typical applications
Grade 1	General purpose Material	1.7	630	8.2	420	Electrical components for general use like switches, sockets, etc.
Grade 2		1.7	670	8.7	460	
Grade 3	Improved G. P. material	2	770	9.2	530	Telephone mouldings
Grade 4	Medium shock-resistant material	4	770	9.2	500	Toggle interruptor
Grade 5	Heat resistant material	1	570	10	320	Spools of resistors
Grade 6	Electrical Low loss	1.7	670	12.5	350	Spools
Grade 7		2.4	800	12	500	Fuse mountings
Grade 8	Material	2	770	11.5	530	Parts requiring high tensile strength

Polystyrene moulding materials IS: 2267–1972

Types	Melt flow index g/10 min	Vicat softening point, °C	Impact strength kgf–cm	Tensile strength kgf/cm²	Elongation %	Typical processing
Type 1	3.5	93	1.0	365	1	Injection moulding, extrusion
Type 2	3	87	2.0	240	5	Injection moulding, forming, extrusion
Type 3	2.5	82	4.0	200	15	Injection moulding, extrusion, forming

Laminated plastics (paper and fabric based)

Paper based laminates (Phenolic Resin Bounded, Thermo setting) IS: 2036–1962

Grade	Tensile strength kgf/cm²	Cross-breaking strength kgf/cm²	Insulation resistance Meg ohms	Properties/applications
P 1	840	1350	1	Mechanical parts; electrical properties of secondary importance
P 2	700	1350	50	Good mechanical and medium electrical properties
P 3	560	850	1,000	Very good electrical properties, good machining and hot punching properties
P 4	500	700	10,000	Exeptionally good electrical properties, exceilent hot punching properties
P 5	500	700	50,000	As P 4, but superior insulating resistance

Fabric based laminates (Phenolic Resin Bonded, Thermo setting) IS: 2038–1962

Grade	Tensile strength kgf/cm²	Cross-breaking strength kgf/cm²	Impact strength kgf–cm	Insulation resistance Meg ohms	Properties/Applications
F 1	700	1200	6.9	5	Made from fine-weave cotton fabric, for components requiring fine machining
F 2	600	1050	9.0	1	Made from medium-weave cotton fabric, suitable for small gears, etc.
F 3	625	980	11.7	0.5	For components requiring high impact strength

Rigid PVC sheets IS: 6307–1971

Types	Softening point °C, min	Tensile strength kgf/cm², min	Dimensional change at 120°C Extruded or calendered %	Calendered and laminated %	Applcations
Type 1	75	450	20	5	General purpose
Type 2	75	450	20	5	Parts requiring chemical resistance
Type 3	65	380	20	5	Parts requiring high impact strength
Type 4	50	380	—		For deep draw vacuum forming

Steel bars—dimensions and weight

Round, square and hexagon steel rods, hot rolled; specific weight: $\gamma = 7.85$ gf/cm³

Diameter d / Side width w	IS: 1723–1971	IS: 1732–1971	Width across flat w		Width across flat w	
5	0.154	0.196	5	0.170	20	2.72
6	0.222	0.283	6	0.245	22	3.29
8	0.395	0.502	7	0.333	24	3.96
10	0.617	0.785	8	0.435	25	4.25
12	0.888	1.13	9	0.551	27	4.96
14	1.21	—	10	0.680	30	6.12
16	1.58	2.01	11	0.824	32	6.96
18	2.00	—	12	0.979	36	8.81
20	2.47	3.14	13	1.149	40	10.88
22	2.98	—	14	1.33	41	11.4
25	3.85	4.91	15	1.53	46	14.4
28	4.83	—	16	1.74	50	17.0
32	6.31	8.04	17	1.96	55	20.6
36	7.99	—	18	2.20	60	24.5
40	9.85	12.6				
45	12.5	15.9				
50	15.4	19.6				
56	19.3	—				
63	24.5	31.2				
71	31.1	—				
80	39.5	50.2				
90	49.9	—				
100	61.7	78.5				

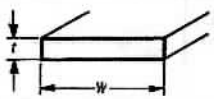
Examples for standard designation:

ISRO 20 : Round bar of 20 mm ⌀

ISSQ 20 : Square bar of side width 20 mm

Steel flats

IS: 1731–1971



Designation: Hot rolled steel flats confirming to this standard, shall be designated by their width in mm., followed by letters ISF and the thickness in mm.

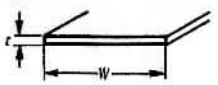
Example: A steel flat of width 10 mm. and thickness 3 mm. shall be designated as 10 ISF3.

Note: The weight per metre of flats is calculated on the basis that steel weighs 7.85 gf/cm³ and is rounded off to three significant figures.

Width mm	t – thickness, mm													
	4	5	6	8	10	12	16	18	20	25	32	45	50	60
	Weight in kgf/m													
10	0.236	0.393	0.471	—	—	—	—	—	—	—	—	—	—	—
20	0.471	0.785	0.942	1.26	1.57	1.88	2.51	2.83	—	—	—	—	—	—
25	0.589	0.981	1.18	1.57	1.96	2.36	3.14	3.53	—	—	—	—	—	—
30	0.707	1.18	1.41	1.88	2.36	2.83	3.77	4.24	4.71	—	—	—	—	—
35	0.824	1.37	1.65	2.20	2.75	3.30	4.40	4.95	5.5	6.87	8.79	—	—	—
40	0.942	1.57	1.88	2.51	3.14	3.77	5.02	5.65	6.28	7.85	10.0	—	—	—
45	—	1.77	2.12	2.83	3.53	4.24	5.65	6.36	7.07	8.83	11.3	—	—	—
50	1.18	1.96	2.36	3.14	3.93	4.71	6.28	7.06	7.85	9.81	12.6	—	—	—
55	—	2.16	2.59	3.45	4.32	5.18	6.91	7.77	8.64	10.8	13.8	—	—	—
60	1.41	2.36	2.83	3.77	4.71	5.65	7.54	8.48	9.42	11.8	15.1	21.2	—	—
65	—	2.55	3.06	4.08	5.10	6.12	8.16	9.18	10.2	12.8	16.3	23.0	—	—
70	—	2.75	3.30	4.40	5.50	6.59	8.79	9.89	11.0	13.7	17.6	24.7	27.5	—
75	—	2.94	3.53	4.71	5.89	7.07	9.42	10.6	11.8	14.7	18.8	26.5	29.4	—
80	—	3.14	3.77	5.02	6.28	7.54	10.0	11.3	12.6	15.7	20.1	28.3	31.4	—
90	—	3.53	4.24	5.65	7.07	8.48	11.3	12.7	14.1	17.7	22.6	31.8	35.3	42.4
100	—	3.93	4.71	6.28	7.85	9.42	12.6	14.1	15.7	19.6	25.1	35.3	39.2	47.1
120	—	4.71	5.65	7.45	9.42	11.3	15.1	17.0	18.8	23.6	30.1	42.4	47.1	56.5
140	—	—	—	8.79	11.0	13.2	17.6	19.8	22.0	27.5	35.2	49.5	55.0	65.9
160	—	—	—	10.0	12.6	15.1	20.1	22.6	25.1	31.4	40.2	56.5	—	—
180	—			11.3	14.1	17.0	22.6	25.4	28.3	35.3	45.2	63.6	—	—
200	—				15.7	18.8	25.1	28.3	31.4	39.2	50.2	70.6	—	—
300	—					28.3	37.7	42.4	47.1	58.9	75.4	10.6	—	—

Steel strip

IS: 1730–1961



Designation: Indian standard strip shall be designated as ISST followed by figures denoting width (mm)×thickness (mm).

Example: ISST 100×2.5

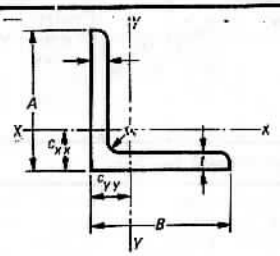
Note: The weight per metre of strip calculated on the basis that steel weighs 7.85 gf/cm³ and is rounded off to one decimal place.

Width in mm	t – thickness, mm													
	Weight per metre length, kg													
	t – thickness, mm													
	1.6	1.8	2.00	2.24	2.5	2.8	3.15	3.55	4.00	4.5	5.0	6.0	8.0	10.0
	Weight in kgf/m													
100	1.3	1.4	1.6	1.8	2.0	2.2	2.5	2.8	3.1	3.5	3.9	4.7	6.3	7.8
125	1.6	1.8	2.0	2.2	2.4	2.7	3.1	3.5	3.9	4.4	4.9	5.9	7.8	9.8
160	2.0	2.3	2.5	2.8	3.1	3.5	4.0	4.5	5.0	5.6	6.3	7.5	10.0	12.6
200	2.5	2.8	3.1	3.5	3.9	4.4	4.9	5.6	6.3	7.1	7.8	9.4	12.6	15.7
250	3.1	3.5	3.9	4.4	4.9	5.5	6.2	7.0	7.8	8.8	9.8	11.8	15.7	19.6
320	4.0	4.5	5.0	5.6	6.3	7.0	7.9	8.9	10.0	11.3	12.6	15.1	20.1	25.1
400	5.0	5.6	6.3	7.0	7.8	8.8	9.9	11.1	12.6	14.1	15.7	18.8	25.1	31.4
500	6.3	7.1	7.8	8.8	9.8	11.0	12.4	13.9	15.7	17.7	19.6	23.6	31.4	39.2
650	8.2	9.2	10.2	11.4	12.8	14.3	16.1	18.1	20.4	23.0	25.5	30.6	40.8	51.0
800	10.0	11.3	12.6	14.1	15.7	17.6	19.8	22.3	25.1	28.3	31.4	37.7	50.2	62.8
950	—	13.4	14.9	16.7	18.6	20.9	23.5	26.5	29.8	33.6	37.3	44.7	59.7	74.6
1050	—	—	16.5	18.5	20.6	23.1	26.0	29.3	33.0	37.1	41.2	49.5	65.9	82.4
1150	—	—	—	20.2	22.6	25.3	28.4	32.0	36.1	40.6	45.1	54.2	72.2	90.3
1250	—	—	—	—	24.5	27.5	30.9	34.8	39.2	44.2	49.1	58.9	78.5	98.1
1300	—	—	—	—	—	28.6	32.1	36.2	40.8	45.9	51.0	61.2	81.6	102.0
1450	—	—	—	—	—	—	35.8	40.4	45.5	51.2	56.9	68.3	91.1	113.8
1550	—	—	—	—	—	—	38.3	43.2	48.7	54.7	60.8	73.0	97.3	121.7

SPK

Indian standard equal angles

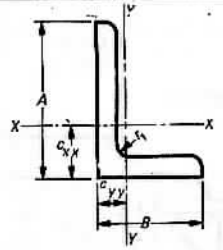
IS: 808–1964



Designation: Indian standard equal angle of size A=20 mm and B=20 mm may be designated as ISA 2020.

Indian standard unequal angles

IS: 808–1964



Designation: Indian standard unequal angle of size A=30 mm and B=20 mm may be designated as ISA 3020

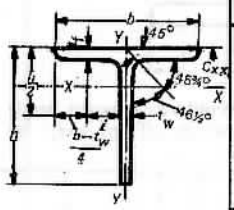
Size (A×B) mm×mm	Thickness (t) mm	Sectional area (a) cm²	Weight per metre (w) kgf/m	Radius at root (r_1) mm	Centre of gravity c_{xx} (c_{yy}) cm²	Moduli of section z_{xx} (z_{yy}) cm³
20×20	3.0	1.12	0.9	4.0	0.59	0.3
	4.0	1.45	1.1	4.0	0.63	0.4
25×25	3.0	1.41	1.1	4.5	0.71	0.4
	4.0	1.84	1.4	4.5	0.75	0.6
	5.0	2.25	1.8	4.5	0.79	0.7
30×30	3.0	1.73	1.4	5.0	0.83	0.6
	4.0	2.26	1.8	5.0	0.87	0.8
	5.0	2.77	2.2	5.0	0.92	1.0
35×35	3.0	2.03	1.6	5.0	0.95	0.9
	4.0	2.66	2.1	5.0	1.00	1.2
	5.0	3.27	2.6	5.0	1.04	1.4
	6.0	3.86	3.0	5.0	1.08	1.7
40×40	3.0	2.34	1.8	5.5	1.08	1.2
	4.0	3.07	2.4	5.5	1.12	1.6
	5.0	3.78	3.0	5.5	1.16	1.9
	6.0	4.47	3.5	5.5	1.20	2.3
45×45	3.0	2.64	2.1	5.5	1.20	1.5
	4.0	3.47	2.7	5.5	1.25	2.0
	5.0	4.28	3.4	5.5	1.29	2.5
	6.0	5.07	4.0	5.5	1.33	2.9
50×50	3.0	2.95	2.3	6.0	1.32	1.9
	4.0	3.88	3.0	6.0	1.37	2.5
	5.0	4.79	3.8	6.0	1.41	3.1
	6.0	5.68	4.5	6.0	1.45	3.6
55×55	5.0	5.27	4.1	6.5	1.53	3.7
	6.0	6.26	4.9	6.5	1.57	4.4
	8.0	8.18	6.4	6.5	1.65	5.7
	10.0	10.02	7.9	6.5	1.72	7.0
60×60	5.0	5.75	4.5	6.5	1.65	4.4
	6.0	6.84	5.4	6.5	1.69	5.2
	8.0	8.96	7.0	6.5	1.77	6.8
	10.0	11.00	8.6	6.5	1.85	8.4

Size (A×B) mm×mm	Thickness (t) mm	Sectional area (a) cm²	Weight per metre (w) kgf/m	Radius at root (r_1) mm	Centre of gravity (c_{xx}) (c_{yy}) cm cm	Moduli of section (z_{xx}) (z_{yy}) cm³ cm³
30×20	3.0	1.41	1.1	4.5	0.98 0.49	0.6 0.3
	4.0	1.84	1.4	4.5	1.02 0.53	0.8 0.4
	5.0	2.25	1.8	4.5	1.06 0.57	1.0 0.4
40×25	3.0	1.88	1.5	5.0	1.30 0.57	1.1 0.5
	4.0	2.46	1.9	5.0	1.35 0.62	1.4 0.6
	5.0	3.02	2.4	5.0	1.39 0.66	1.8 0.7
	6.0	3.56	2.8	5.0	1.43 0.69	2.1 0.9
45×30	3.0	2.18	1.7	5.0	1.42 0.69	1.4 0.7
	4.0	2.86	2.2	5.0	1.47 0.73	1.9 0.9
	5.0	3.52	2.8	5.0	1.51 0.77	2.3 1.1
	6.0	4.16	3.3	5.0	1.55 0.81	2.7 1.3
50×30	3.0	2.34	1.8	5.5	1.63 0.65	1.7 0.7
	4.0	3.07	2.4	5.5	1.68 0.70	2.3 0.9
	5.0	3.78	3.0	5.5	1.72 0.74	2.8 1.1
	6.0	4.47	3.5	5.5	1.76 0.78	3.4 1.3
60×40	5.0	4.76	3.7	6.0	1.95 0.96	4.2 2.0
	6.0	5.65	4.4	6.0	1.99 1.00	5.0 2.3
	8.0	7.37	5.8	6.0	2.07 1.08	6.5 3.0
65×45	5.0	5.26	4.1	6.0	2.07 1.08	5.0 2.5
	6.0	6.25	4.9	6.0	2.11 1.12	5.9 3.0
	8.0	8.17	6.4	6.0	2 19 1.20	7 7 3.9
70×45	5.0	5.52	4.3	6.5	2.27 1.04	5.7 2.5
	6.0	6.56	5.2	6.5	2.32 1.09	6 8 3.0
	8.0	8.58	6.7	6.5	2.40 1.16	8.9 3.9
	10.0	10.52	8.3	6.5	2.48 1.24	10.9 4.8
75×50	5.0	6.02	4.7	6.5	2.39 1.16	6.7 3.2
	6.0	7.16	5.6	6.5	2.44 1.20	8.0 3.8
	8.0	9.38	7.4	6.5	2.52 1.28	10.4 4.9
	10.0	11.52	9.0	6.5	2.60 1.36	12.7 6.0
80×50	5.0	6.27	4.9	7.0	2.60 1.12	7.5 3.2
	6.0	7.46	5.9	7.0	2.64 1.16	9.0 3.8
	8.0	9.78	7.7	7.0	2.73 1.24	11.7 4.9
	10.0	12.02	9.4	7.0	2.81 1.32	14.4 6.0

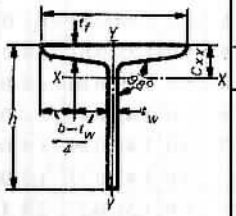
Tee Bars

IS: 1173–1967

Normal tee bar



Slit tee bars

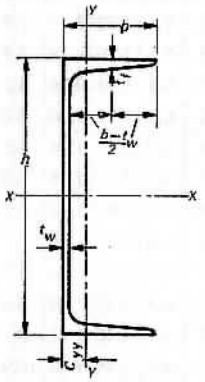


Designation	Dimensions in mm for h	b	t_w	t_f	Sectional area cm²	Wt. kgf/m	Centre of gravity position C_{xx}, cm	Moduli of section Z_{xx} cm³	Z_{vv} cm³
Indian standard normal tee bars									
ISNT 20	20	20	4.0	4.0	1.45	1.1	0.60	0.3	0.2
ISNT 30	30	30	4.0	4.0	2.26	1.8	0.82	0.8	0.5
ISNT 40	40	40	6.0	6.0	4.45	3.5	1.14	2.1	1.5
ISNT 50	50	50	6.0	6.0	5.66	4.4	1.35	3.4	2.3
ISNT 60	60	60	6.0	6.0	6.85	5.4	1.56	4.8	3.2
ISNT 75	75	75	9.0	9.0	12.69	10.0	2.04	11.4	7.8
ISNT 100	100	100	10.0	10.0	18.97	14.9	2.62	22.2	15.4
ISNT 150	150	150	10.0	10.0	28.88	22.7	3.61	47.5	33.4
Indian standard deep legged tee bars									
ISDT 100	100	50	5.8	10.0	10.37	8.1	3.03	14.2	3.8
ISDT 150	150	75	8.0	11.6	19.96	15.7	4.75	43.9	9.9
Indian standard slit light weight tee bars									
ISLT 200	200	165	8.0	12.5	36.22	28.4	4.78	83.3	43.4
ISLT 250	250	180	9.2	14.1	47.75	37.5	6.40	149.2	59.1
Indian standard slit medium weight tee bars									
ISMT 50	50	75	4.0	7.2	7.30	5.7	0.96	2.4	5.4
ISMT 62.5	62.5	75	4.4	7.6	8.30	6.5	1.30	4.3	5.8
ISMT 75	75	80	4.8	7.6	9.50	7.5	1.67	6.9	6.6
ISMT 87.5	87.5	90	5.5	8.6	12.31	9.7	1.98	10.7	9.4
ISMT 100	100	100	5.7	10.8	16.16	12.7	2.13	14.7	15.0

Channel Sections

(for structural use)

IS: 808–1964

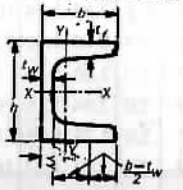


Designation	h	b	t_w	t_f	Sectional area cm²	Wt. kgf/m	C_{yy}	Z_{xx} cm³	Z_{yy} cm³
Indian standard junior channels									
ISJC 100	100	45	3.0	5.1	7.41	5.8	1.40	24.8	4.8
ISJC 125	125	50	3.0	6.6	10.08	7.9	1.64	43.2	7.6
ISJC 150	150	55	3.6	6.9	12.67	9.9	1.67	62.9	9.9
ISJC 175	175	60	3.6	6.9	14.24	11.2	1.75	82.3	11.9
ISJC 200	200	70	4.1	7.1	17.77	14.0	1.97	116.2	16.7
Indian standard light channels									
ISLC 75	75	40	3.7	6.0	7 26	5.7	1.35	17.6	4.3
ISLC 100	100	50	4.0	6.4	10.02	7.9	1.62	32.9	7.3
ISLC 125	125	65	4.4	6.6	13.67	10.7	2.04	57.1	12.8
ISLC 150	150	75	4.8	7.8	18.39	14.4	2.39	93.1	20.2
ISLC 175	175	75	5.1	9.5	22.42	17.6	2.40	131.4	24.8
ISLC 200	200	75	5.5	10.8	26.26	20.6	2.36	172.9	28.5
Indian standard medium weight channels									
ISMC 75	75	40	4.4	7.3	8.72	6.8	1.32	20.4	4.7
ISMC 100	100	50	4.7	7.5	11.75	9.2	1.54	37.5	7.5
ISMC 125	125	65	5.0	8.1	16.25	12.8	1.96	67.0	13.2
ISMC 150	150	75	5.4	9.0	20.94	16.4	2.23	104.3	19.4
ISMC 175	175	75	5.7	10.2	24.45	19.2	2.21	140.3	22.9
ISMC 200	200	75	6.1	11.4	28.28	22.2	2.18	182.5	26.4

Channel Sections

(for general use)

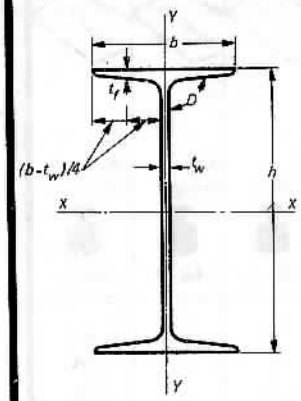
IS: 3954–1966



Designation	h	b	t_w	t_f	Sectional area cm²	Wt. kgf/m	C_{yy}	Z_{xx} cm³	Z_{yy} cm³
Indian standard channel sections for general use									
ISCG 16	16	10	3.2	3.2	0.971	0.76	0.38	0.37	0.12
ISCG 20	20	10	3.2	3.2	1.099	0.86	0.35	0.53	0.13
ISCG 40	40	32	5.0	7.5	6.144	4.82	1.25	6.79	2.95

Indian standard rolled steel beams

IS: 808–1964



Classification:

	Abbreviated reference symbol
(a) Indian Standard Junior Beams–ISJB	JB
(b) Indian Standard Light Weight Beams–ISLB	LB
(c) Indian Standard Medium Weight Beams–ISMB	MB
(d) Indian Standard Wide Flange Beams–ISWB	WB
(e) Indian Standard Column Sections-H-Beams–ISHB	HB
(f) Provisional Sizes—(P)	

Designation	Dimensions in mm h	b	t_f	t_w	Cross section cm²	Weight in kgf/m
ISJB 150	150	50	4.6	3.0	9.01	7.1
ISJB 175	175	50	4.8	3.2	10.28	8.1
ISJB 200	200	60	5.0	3.4	12.64	9.9
ISJB 225	225	80	5.0	3.7	16.28	12.8
ISLB 75	75	50	5.0	3.7	7.71	6.1
ISLB 100	100	50	6.4	4.0	10.21	8.0
ISLB(P) 100	100	50	7.0	4.3	10.97	8.6
ISLB 125	125	75	6.5	4.4	15.12	11.9
ISLB 150	150	80	6.8	4.8	18.08	14.2
ISLB 175	175	90	6.9	5.1	21.30	16.7
IBSL(P) 175	175	80	7.7	5.3	21.32	16.7
ISLB 200	200	100	7.3	5.4	25.27	19.8
ISLB(P) 200	200	100	8.0	5.6	26.85	21.1
ISLB 225	225	100	8.6	5.8	29.92	23.5
ISLB 250	250	125	8.2	6.1	35.53	27.9
ISLB 275	275	140	8.8	6.4	42.02	33.0
ISLB 300	300	150	9.4	6.7	48.08	37.7
ISLB(P) 300	300	140	11.6	7.0	52.89	41.5
ISLB 325	325	165	9.8	7.0	54.90	43.1
ISLB 350	350	165	11.4	7.4	63.01	49.5
ISLB 400	400	165	12.5	8.0	72.43	56.9
ISLB 450	450	170	13.4	8.6	83.14	65.3
ISLB 500	500	180	14.1	9.2	95.50	75.0
ISLB 550	550	190	15.0	9.9	109.97	86.3
ISLB 600	600	210	15.5	10.5	126.69	99.5
ISMB 100	100	75	7.2	4.0	14.60	11.5
ISMB(P)100	100	70	7.5	4.5	14.70	11.5
ISMB 125	125	75	7.6	4.4	16.60	13.0
ISMB(P)125	125	70	8.0	4.8	16.81	13.2
ISMB 150	150	80	7.6	4.8	19.00	14.9
ISMB(P)150	150	75	8.0	5.0	19.07	15.0
ISMB 175	175	90	8.6	5.5	24.62	19.3
ISMB(P)175	175	85	9.0	5.8	24.87	19.5
ISMB 200	200	100	10.8	5.7	32.33	25.4
ISMB 225	225	110	11.8	6.5	39.72	31.2
ISMB 250	250	125	12.5	6.9	47.55	37.3
ISMB 300	300	140	12.4	7.5	56.26	44.2
ISMB 350	350	140	14.2	8.1	66.71	52.4
ISMB 400	400	140	16.0	8.9	78.46	61.6
ISMB 450	450	150	17.4	9.4	92.27	72.4
ISMB 500	500	180	17.2	10 2	110.74	86.9
ISMB 550	550	190	19.3	11.2	132.11	103.7
ISMB 600	600	210	20.8	12.0	156 21	122.6
ISWB 150	150	100	7.0	5.4	21.67	17.0
ISWB 175	175	125	7.4	5.8	28.11	22.1
ISWB 200	200	140	9.0	6.1	36.71	28.8
ISWB 225	225	150	9.9	6.4	43.24	33.9
ISWB 250	250	200	9.0	6.7	52.05	40.9
ISWB 300	300	200	10.0	7.4	61.33	48.1
ISWB 350	350	200	11.4	8.0	72.5	56.9
ISWB 400	400	200	13.0	8.6	85.01	66.7
ISWB 450	450	200	15.4	9.2	101.15	79.4
ISWB 500	500	250	14.7	9.9	121.22	95.2
ISWB 550	550	250	17.6	10.5	143/34	112.5
ISWB 600	600	250	21.3	11.2	170.38	133.7
ISWB 600	600	250	23.6	11.8	184.86	145.1
ISHB 150	150	150	9.0	5.4	34.48	27.1
ISHB 150	150	150	9.0	8.4	38.98	30.6
ISHB 150	150	150	9.0	11.8	44.08	34.6
ISHB 200	200	200	9.0	6.1	47.54	37.3
ISHB 200	200	200	9.0	7.8	50.94	40.0
ISHB 225	225	225	9.1	6.5	54.94	43.1
ISHB 225	225	225	9.1	8.6	59.66	46.8
ISHB 250	250	250	9.7	6.9	64.96	51.0
ISHB 250	250	250	9.7	8.8	69.71	54.7
ISHB 300	300	250	10.6	7.6	74.85	58.8
ISHB 300	300	250	10.6	9.4	80.25	63.0
ISHB 350	350	250	11.6	8.3	85.91	67.4
ISHB 350	350	250	11.6	10.1	92.21	72.4
ISHB 400	400	250	12.7	9.1	98.66	77.4
ISHB 400	400	250	12.7	10.6	104.66	82.2
ISHB 450	450	250	13.7	9.8	111.14	87.2
ISHB 450	450	250	13.7	11.3	117.89	29.5

Sheet Metals

Plate thickness in mm	Sheet steel: Weight =kgf/m²	(pack rolled tin sheets) Plate thickness in mm	Sheet iron: Weight =kgf/m²	Plate thickness in mm	Sheet brass: Brass weight =kgf/m²	Sheet copper: Cu weight =kgf/m²	Sheet Al: Al weight =kgf/m²	Sheet Mg: Mg weight =kgf/m²	Sheet Zn: Zn weight =kgf/m²
0.18	1.44	0.15	1.27	0.10	0.85	0.89	—	—	—
0.2	1.6	0.19	1.54	0.15	1.28	1.34	—	—	1.08
0.22	1.76	0.22	1.76	0.20	1.70	1.78	0.54	—	1.44
0.24	1.92	0.24	1.92	0.25	2.13	2.23	0.675	—	1.80
0.28	2.24	0.27	2.16	0.30	2.55	2.67	0.810	0.546	2.15
0.32	2.56	0.28	2.24	0.35	2.98	3.12	0.945	—	2.51
0.38	3.04	0.31	2.48	0.40	3.40	3.56	1.08	0.728	2.87
0.44	3.52	0.32	2.56	0.45	3.83	4.01	—	—	3.23
0.5	4	0.36	2.88	0.50	4.25	4.45	1.35	0.910	3.59
0.56	4.48	0.37	2.96	0.55	4.68	4.90	—	—	3.95
0.63	5.04	0.41	3.28	0.60	5.10	5.34	1.62	1.09	4.31
0.75	6	0.43	3.39	0.65	5.53	5.79	—	—	4.67
0.88	7.04	0.46	3.63	0.70	5.95	6.23	—	—	5.03
1	8	0.50	3.90	0.75	6.38	6.68	—	—	5.38
1.13	9.04	0.52	4.11	0.80	6.80	7.12	2.16	1.46	5.74
1.25	10	0.57	4.50	0.85	7.23	7.52	—	—	—
1.38	11.04	0.58	4.58	0.90	7.65	8.01	—	—	6.46
1.5	12	0.64	4.99	1.00	8.50	8.90	2.70	1.82	7.18
1.75	14	0.66	5.15	1.10	9.35	9.79	—	—	—
2	16	0.70	5.46	1.20	10.20	10.70	3.24	2.18	8.62
2.25	18	0.75	5.85	1.30	11.10	11.60	—	—	—
2.5	20	0.80	6.24	1.40	11.90	12.50	—	—	10.1
2.75	22	0.90	7.02	1.50	12.80	13.40	4.05	2.73	10.8
3	24	1.00	7.8	1.60	13.60	14.20	—	—	—
3.5	28			1.70	14.50	15.10	—	—	—
4	32			1.80	15.30	16.00	4.86	3.28	12.9
4.5	36			2.00	17.00	17.80	5.40	3.64	14.4
4.75	38			2.50	21.30	22.30	6.75	4.55	18
5	40			2.80	23.80	24.90	7.56	—	20.1
6	48			3.00	25.50	26.70	8.10	5.46	21.5
7	56			3.20	27.20	28.50	—	—	—
8	64			3.50	29.80	31.20	9.45	6.37	25.1

Non-ferrous metals

Round, square and hexagonal bars

Diameter, d Side length, a Width across flat, w mm	Round: Copper kgf/m	Round: Brass kgf/m	Round: Aluminium alloy kgf/m	Round: Magnesium alloy kgf/m	Square: Brass kgf/m	Square: Aluminium alloy kgf/m	Square: Magnesium alloy kgf/m	Hexagonal: Brass kgf/m	Hexagonal: Aluminium alloy kgf/m	Hexagonal: Magnesium alloy kgf/m
3	0.063	0.060	0.0191	0.0129	0.0765	0.0243	—	0.0662	0.0210	—
3.5	0.086	0.082	0.0260	0.0175	0.104	0.0331	—	0.0902	0.0286	—
4	0.112	0.107	0.0339	0.0229	0.136	0.0432	0.0291	0.118	0.0374	0.0252
4.5	0.142	0.135	0.0429	0.0289	0.172	0.0547	0.0369	0.149	0.0473	0.0319
5	0.175	0.167	0.0530	0.0357	0.212	0.0675	0.0455	0.184	0.0585	0.0394
5.5	0.211	0.202	0.0641	0.0432	0.257	0.0817	0.0551	0.223	0.0707	0.0477
6	0.252	0.240	0.0763	0.0515	0.306	0.0972	0.0655	0.265	0.0842	0.0567
7	0.343	0.327	0.104	0.0700	0.416	0.132	0.0892	0.361	0.115	0.0772
8	0.447	0.427	0.136	0.0915	0.544	0.173	0.116	0.471	0.150	0.101
9	0.566	0.541	0.172	0.116	0.688	0.219	0.147	0.596	0.189	0.128
10	0.699	0.668	0.212	0.143	0.850	0.270	0.182	0.736	0.234	0.158
11	0.846	0.808	0.257	0.173	1.03	0.327	0.220	0.891	0.283	0.191
12	1.007	0.961	0.305	0.206	1.22	0.389	0.262	1.06	0.337	0.227
14	1.370	1.31	0.416	0.280	1.67	0.529	0.357	1.44	0.458	0.309
20	2.796	2.67	0.848	0.572	3.40	1.08	—	—	—	—
22	3.383	3.23	1.03	0.692	4.11	1.31	0.881	3.56	1.13	0.763
24	4.028	3.85	1.22	0.823	4.90	1.56	1.05	4.24	1.35	0.908
32	7.158	6.84	2.17	1.46	8.70	2.76	1.86	7.54	2.39	1.61
36	9.059	8.65	2.75	1.85	11.0	3.50	2.36	9.54	3.03	2.04
50	17.745	16.7	5.30	3.57	21.2	6.75	4.55	18.4	5.85	3.94

Flats

Nominal size w×t mm	Copper kgf/m	Brass kgf/m	Aluminium alloy kgf/m	Magnesium alloy kgf/m	Nominal size w×t mm	Copper kgf/m	Brass kgf/m	Aluminium alloy kgf/m	Magnesium alloy kgf/m
5×2	0.089	0.085	—	0.0182	30×5	1.34	1.28	0.405	0.273
5×3	0.13	0.12	—	0.0273	30×8	2.14	2.04	0.648	0.437
5×4	0.18	0.17	—	0.0364	30×10	2.67	2.55	0.810	0.546
10×2	0.18	0.17	0.054	0.0364	40×5	1.78	1.70	0.540	—
10×3	0.27	0.26	0.081	0.0546	40×8	2.85	2.72	—	—
10×4	0.36	0.34	0.108	0.0728	40×10	3.56	3.40	1.08	—
10×5	0.45	0.43	0.135	0.0910	40×15	5.34	5.10	1.62	1.09
10×8	0.71	0.68	0.216	0.146	40×20	7.12	6.80	2.16	1.46
15×2	0.27	0.26	0.081	0.0546	50×5	2.23	2.13	0.675	0.455
15×3	0.40	0.38	0.122	0.0819	50×8	3.56	3.40	1.08	0.728
15×5	0.67	0.64	0.202	0.137	50×10	4.45	4.25	1.35	0.910
15×8	1.07	1.02	0.324	0.218	50×15	6.68	6.38	2.02	1.37
15×10	1.34	1.28	0.405	0.273	50×20	8.90	8.50	2.70	1.82
20×5	0.89	0.85	0.270	0.182	60×5	2.67	—	0.810	—
20×8	1.42	1.36	0.432	0.291	60×8	4.27	—	1.30	—
20×10	1.78	1.70	0.540	0.364	60×10	5.34	5.10	1.62	—
20×15	2.67	2.55	0.810	0.546	60×15	8.01	—	2.43	—

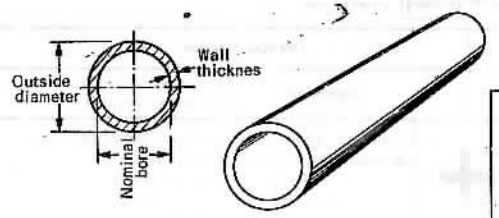
Light Metal Sections

The weights of Indian Standard Aluminium sections are based on density of 2.7 g/cm³. To get weights corresponding to density= 2.6 gf/cm³ and 2.8 gf/cm³, multiply the values by 0.963 and 1.037 respectively

Material: Aluminium sections as detailed below are to be extruded from aluminium alloys specified in IS: 733–1967

Details of the sections	Designation	Dimensions in mm for h	b	t	r	Sectional area cm²	Wt. per metre kgf/m	Centre of gravity C_x cm	Moduli of section Z_x cm³	Z_y cm³
T–Section IS: 6445–1971	ISALT 25×25–0.4	25	25	3.2	5.0	1.60	0.4	0.72	1.2	0.3
	„ 30×30–0.5	30	30	3.0	5.0	1.82	0.5	0.83	1.8	0.5
	„ 30×30–0.7	30	30	4.5	5.0	2.60	0.7	0.90	2.3	0.7
	„ 40×50–0.8	40	40	3.2	5.0	2.88	0.8	1.47	4.9	0.9
	„ 50×50–1.2	50	50	4.5	6.0	4.45	1.2	1.39	7.5	1.9
	„ 50×50–1.6	50	50	6.0	6.0	5.79	1.6	1.45	9.1	2.5
		A	B	t	r			C_x	C_y	
L–Section IS: 3909–1966	ISALA 20×10×1.5	20	10	1.5	4.0	0.46	0.12	0.69	0.23	0.14	0.04
	„ 20×10×2.0	20	10	2.0	4.0	0.59	0.16	0.72	0.25	0.18	0.05
	„ 20×15×1.5	20	15	1.5	4.0	0.54	0.14	0.60	0.37	0.15	0.09
	„ 20×15×2.0	20	15	2.0	4.0	0.69	0.19	0.63	0.39	0.19	0.11
	„ 20×15×3.0	20	15	3.0	4.0	0.99	0.27	0.67	0.43	0.28	0.16
	„ 30×15×2.0	30	15	2.0	5.0	0.91	0.25	1.03	0.33	0.42	0.12
	„ 30×15×3.0	30	15	3.0	5.0	1.31	0.35	1.09	0.37	0.61	0.17
	„ 30×20×2.0	30	20	2.0	5.0	1.01	0.27	0.94	0.47	0.44	0.21
	„ 30×20×3.0	30	20	3.0	5.0	1.46	0.40	0.99	0.51	0.64	0.31
	„ 30×20×4.0	30	20	4.0	5.0	1.89	0.51	1.03	0.55	0.83	0.39
		h	b	t_f	t_w	r					
I–Section IS: 3921–1966	ISALB 40×20–0.4	40	20	2.0	2.0	4.0	1.66	0.4	—	2.0	0.3
	„ 40×20–0.6	40	20	3.0	3.0	4.0	2.36	0.6	—	2.7	0.4
	„ 50×30–0.9	50	30	3.0	3.0	5.0	3.33	0.9	—	5.2	0.9
	„ 50×30–1.2	50	30	4.0	4.0	5.0	4.29	1.2	—	6.4	1.2
	„ 60×30–1.1	60	30	4.0	3.0	5.0	4.17	1.1	—	7.9	1.2
	„ 60×30–1.5	60	30	6.0	4.0	5.0	5.73	1.5	—	10.4	1.8
	„ 60×30–1.9	60	30	8.0	5.0	5.0	7.21	1.9	—	12.4	2.5
	„ 60×40–1.9	60	40	6.0	4.0	6.0	7.03	1.9	—	13.5	3.2
	„ 60×40–2.4	60	40	8.0	5.0	6.0	8.91	2.4	—	16.2	4.3
								C_y			
C–Section IS: 3921–1966	ISALC 40×20×2	40	20	2.0	2.0	5.0	1.63	0.44	0.56	1.99	0.40
	„ 40×20×3	40	20	3.0	3.0	5.0	2.33	0.63	0.60	2.69	0.58
	„ 50×30×3	50	30	3.0	3.0	6.0	3.27	0.88	0.91	5.10	1.33
	„ 50×30×4	50	30	4.0	4.0	6.0	4.23	1.14	0.95	6.32	1.72
	„ 60×30×3	60	30	4.0	3.0	7.0	4.17	1.13	0.94	7.87	1.75
	„ 60×30×4	60	30	6.0	4.0	7.0	5.73	1.55	1.03	10.37	2.48
	„ 60×30×5	60	30	8.0	5.0	7.0	7.21	1.95	1.09	12.39	3.17
	„ 60×40×4	60	40	6.0	4.0	7.0	6.93	1.87	1.46	13.29	4.35
	„ 60×40×5	60	40	8.0	5.0	7.0	8.81	2.38	1.53	16.01	5.57

Steel Tubes



IS: 1161–1968: Covers the requirements for hot finished welded (HFW), hot finished seamless (HFS) and electric resistance or induction butt welded (ERW) plain carbon steel tubes for structural purposes.

Nominal bore mm	Light: Outside diameter mm	Light: Wall thickness mm	Light: Weight kgf/m	Medium: Outside diameter mm	Medium: Wall thickness mm	Medium: Weight kgf/m	Heavy: Outside diameter mm	Heavy: Wall thickness mm	Heavy: Weight kgf/m
15	21.3	2.00	0.962	21.3	2.65	1.22	21.3	3.25	1.45
20	26.9	2.35	1.42	26.9	2.65	1.58	26.9	3.25	1.90
25	33.7	2.65	2.04	33.7	3.25	2.46	33.7	4.05	2.99
32	42.4	2.65	2.61	42.4	3.25	3.15	42.4	4.05	3.86
40	48.3	2.9	3.27	48.3	3.25	3.61	48.3	4.05	4.43
50	60.3	2.9	4.14	60.3	3.65	5.10	60.3	4.5	6.17
65	76.1	3.55	5.84	76.1	3.65	6.53	76.1	4.5	7.92
80	88.9	3.25	6.86	88.9	4.05	8.48	88.9	4.85	10.1
100	114.3	3.65	9.97	114.3	4.5	12.1	114.3	5.4	14.5
125	139.7	4.5	14.9	139.7	4.85	16.2	139.7	5.4	17.9
150	165.1	4.5	17.8	165.1	4.85	19.2	165.1	5.4	21.2

Aluminium tube [IS: 2678–1972]	Brass tube [IS: 407–1966]	Copper tube [IS: 5493–1969]
γ=2.7 kgf/dm³	γ=8.5 kgf/dm³	γ=8.9 kgf/dm³

Outside Ø mm	Weight in kgf/m, Wall thickness in mm: 0.5 Al	0.5 Brass	0.5 Cu	0.80 Al	0.75 Brass	0.75 Cu	1.0 Al	1.0 Brass	1.0 Cu	1.5 Al	1.5 Brass	1.5 Cu	2.0 Al	2.0 Brass	2.0 Cu
3	0.011			0.015											
4	0.015			0.022											
5	0.019	0.06		0.029	0.80			0.11	0.11		0.14				
6	0.023	0.07	0.077	0.035	0.10	0.11	0.042	0.13	0.14		0.18				
7	0.028	0.09	0.09	0.042	0.13	0.13	0.051	0.16	0.17		0.22	0.23			0.28
8	0.032	0.10	0.104	0.049	0.15	0.15	0.059	0.19	0.20	0.083	0.26	0.28			0.34
10	0.040	0.13	0.13	0.063	0.19	0.20	0.076	0.24	0.25	0.108	0.34	0.36	0.136		0.45
12	0.049	0.14		0.076	0.23		0.093	0.29	0.31	0.134	0.42	0.44	0.170	0.53	0.56
13	0.053			0.083			0.102	0.32	0.34	0.146	0.46	0.48	0.187	0.59	0.62
14	0.057			0.092			0.110		0.36	0.159		0.53	0.204		0.67
15	0.062			0.096			0.119	0.37	0.39	0.172	0.54	0.57	0.221	0.69	0.73
16	0.066			0.103		0.32	0.127	0.40	0.42	0.184	0.58	0.61	0.238		0.79
18	0.074		0.26	0.115			0.144	0.45	0.48	0.210	0.66	0.70	0.271	0.85	0.90
19	0.079			0.123			0.153	0.48		0.223	0.70	0.74	0.288		0.95
20	0.083			0.130			0.161	0.51	0.53	0.235	0.74	0.78	0.305	0.96	1.01
22	0.091			0.146			0.178	0.56	0.59	0.261	0.82	0.86	0.339	1.07	1.12
25	0.104			0.164			0.204	0.64	0.67	0.299	0.94	0.98	0.390	1.23	1.29
28	0.117			0.183			0.229	0.72	0.76	0.337	1.06	1.12	0.441	1.39	1.47
30	0.125			0.199			0.246	0.77	0.82	0.363	1.14	1.20	0.475	1.50	1.57
32				0.214			0.263	0.83	0.87	0.388	1.22	1.28	0.509	1.60	1.68
36				0.238			0.297		0.98	0.439		1.45	0.577		1.91

Number

Fundamental arithmetical operations

Vulgar fraction		Decimal fraction
Addition : Joining two or more numbers		**Subtraction** : Taking away a number from another number
Convert all fractions involved such that they have same denominator (L. C. M.), then proceed to add or subtract. $\frac{1}{2}+\frac{2}{3}=\frac{3}{6}+\frac{4}{6}=\frac{3+4}{6}=\frac{7}{6}=1\frac{1}{6}$ $\frac{3}{4}-\frac{2}{3}=\frac{9}{12}-\frac{8}{12}=\frac{9-8}{12}=\frac{1}{12}$	+ −	Treat as whole numbers, but remember that decimal point comes under decimal point. 14.370 + 2.035 = 16.405 14.370 − 2.035 = 12.335
Multiplication : The process of taking a number a given number of times		
Multiply numerator times numerator Multiply denominator times denominator $\frac{2}{3}\times\frac{3}{4}=\frac{2\times3}{3\times4}=\frac{6}{12}=\frac{1}{2}$	×	Treat as whole number. Shift the decimal point equal to number of decimals in both factors. 1.4×0.02 (1 2) → 3 decimal places $14\times2=28$ → 0.028
Division : The process of determining how many times one number is contained in another of same kind		
Multiply with the reciprocal of the second fraction. $\frac{3}{4}\div\frac{2}{3}=\frac{3}{4}\times\frac{3}{2}=\frac{9}{8}=1\frac{1}{8}$	÷	Shift the decimal point to the right as required to make full numbers. $1.5\div0.25$ $150\div25=6$ Decimal point 2 places to the right.

% Operations dealing with percentages

$\text{Percentage}=\frac{\text{Base}\times\text{Rate}}{100}$	Base : Rs. 750.00 Rate : 4½% $\text{Percentage}=\frac{750\times4\frac{1}{2}}{100}=33.75$ Rs.
$\text{Base}=\frac{\text{Percentage}\times100}{\text{Rate}}$	Percentage : Rs. 1920.00 Rate : 6% $\text{Base}=\frac{1920\times100}{6}=32000.00$ Rs.
$\text{Rate}=\frac{\text{Percentage}\times100}{\text{Base}}$	Percentage : Rs. 240.00 Base : Rs. 7200.00 $\text{Rate}=\frac{240\times100}{7200}=3\frac{1}{3}\%$

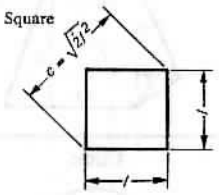
% Interest account

	Examples
$\text{Interest}=\frac{\text{Principal}\times\text{Rate of interest}\times\text{time}}{100}$	Principal : Rs. 5500.00 Rate of interest : 3% ∴ $\text{Interest}=\frac{5500\times3\times5}{100\times2}$ Period of time : 2½ years =Rs. 412.50
$\text{Principal}=\frac{\text{Interest}\times100}{\text{Period of time}\times\text{Rate of interest}}$	Rate of interest : 4% Period of time : 72 days ∴ $\text{Principal}=\frac{22.40\times360\times100}{72\times4}$ Interest : Rs. 22.40 =Rs. 2800.00
$\text{Rate of interest}=\frac{\text{Interest}\times100}{\text{Principal}\times\text{Period of time}}$	Principal : Rs. 2400.00 Interest : Rs. 30.00 ∴ $\text{Rate of interest}=\frac{30\times100\times12}{2400\times4}$ Period of time : 3 months =5%
$\text{Period of time}=\frac{\text{Interest}\times100}{\text{Principal}\times\text{Rate of Interest}}$	Principal : Rs. 2500.00 Interest : Rs. 300.00 $\text{time}=\frac{300\times100}{2500\times4}$ Rate of interest : 4% =3 years

Period of time : 1 year=360 days; 1 month=30 days.

Measurement of plane figures

Square — Area=Side × side

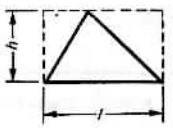


$A=l\times l$

$A=l^2$

$l=\sqrt{A}$

Triangle — $\text{Area}=\frac{\text{Base}\times\text{height}}{2}$

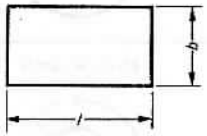


$A=\frac{l\times h}{2}$

$l=\frac{2A}{h}$

$h=\frac{2A}{l}$

Rectangle — Area=Length × Breadth

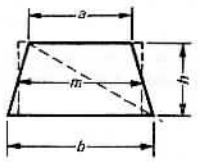


$A=l\times b$

$l=\frac{A}{b}$

$b=\frac{A}{l}$

Trapezoid — $\text{Area}=\frac{\text{Sum of two parallel sides}}{2}\times\text{height}$



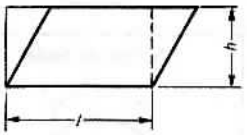
$A=\frac{a+b}{2}\times h$

$h=\frac{2A}{a+b}$

$a=\frac{2A}{h}-b$

$b=\frac{2A}{h}-a$

Parallelogram — Area=Length × Height

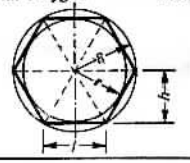


$A=l\times h$

$l=\frac{A}{h}$

$h=\frac{A}{l}$

Regular Polygon — Area=Area of each triangle × No. of sides

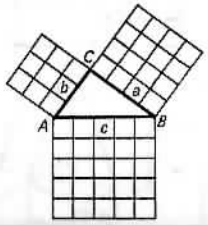


$A=\frac{l\times h}{2}\times n$

Calculation of regular polygons

The area can be calculated from l or R (radius of circumscribed circle) or r (radius of inscribed circle)
The length of a side l can be calculated from R or r
The radius R can be calculated from l or r; the radius r from R or l

n=no. of sides or =no. of corners	Area A=			Side l=		Radius R of circumscribed circle		Radius r of the inscribed circle	
	l^2 times	R^2 times	r^2 times	R times	r times	l times	r times	R times	l times
3 Sides	0.4330	1.2990	5.1962	1.7321	3.4641	0.5774	2.0000	0.5000	0.2887
4 ,,	1.0000	2.0000	4.0000	1.4142	2.0000	0.7071	1.4142	0.7071	0.5000
5 ,,	1.7205	2.3776	3.6327	1.1756	1.4531	0.8507	1.2361	0.8090	0.6882
6 ,,	2.5981	2.5981	3 4641	1.0000	1.1547	1.0000	1.1547	0.8660	0.8660
8 ,,	4.8284	2.8284	3.3137	0.7654	0.8284	1.3066	1.0824	0.9239	1.2071
10 ,,	7.6942	2.9389	3.2492	0.6180	0.6498	1.6180	1.0515	0.9511	1.5388
12 ,,	11.1960	3.0000	3.2154	0.5176	0.5359	1.9319	1.0353	0.9659	1.8660



The Pythagoras Theorem

The square of the hypotenuse is equal to the sum of the squares of the other two sides.

c= hypotenuse—opposite side to right angle

a, b=two sides forming right angle

$c^2=a^2+b^2$. $c=\sqrt{a^2+b^2}$ $a=\sqrt{c^2-b^2}$ $b=\sqrt{c^2-a^2}$

Example : $a=6$ cm $b=8$ cm $c=?$

$c=\sqrt{a^2+b^2}$ $=\sqrt{36+64}$ $=\sqrt{100}=10$ cm

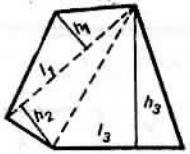
Irregular polygon

Reduce the total area of polygon into number of subareas, say triangles, rectangles, trapezoid etc.

A=Sum of all subareas

$A=A_1+A_2+A_3\ldots$

$$A=\frac{l_1\times h_1+l_1\times h_2+l_3\times h_3+\ldots}{2}$$



Circle

$A=\frac{\pi}{4}D^2$ Area $=\frac{\pi}{4}\times$ Diameter $\times$ Diameter

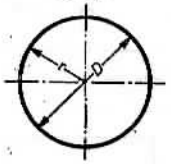
$=0.785\,D^2$

$A=\pi r^2$

$C=\pi D$ Circumference $=C=\pi\times$ Diameter

$=3.142\,D$

$=2\pi r$



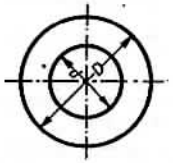
Hollow circle

A=Area of greater circle—Area of smaller circle

$=\frac{\pi}{4}D^2-\frac{\pi}{4}d^2$

$A=\frac{\pi}{4}(D^2-d^2)$

$=0.785\,(D^2-d^2)$



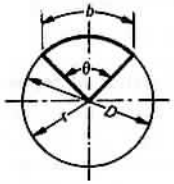
Sector of circle

$A=\frac{\text{Arc length}\times\text{Radius}}{2}$

$A=\frac{\pi\times r^2\times\theta}{360}$

$A=\frac{\pi}{4}\times\frac{D^2\times\theta}{360}$

$b=\frac{\pi\times D\times\theta}{360}$ $A=\frac{b\times r}{2}$



Segment of circle

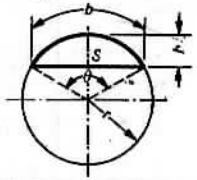
A=Area of sector of circle—Area of triangle

$A=\frac{\pi\times r^2\times\theta}{360}-\frac{S(r-h)}{2}$

$S=2r\sin\frac{\theta}{2}$

$h=\frac{S}{2}\tan\frac{\theta}{4}$

Approximate area of segment $A=\frac{2}{3}\times S\times h$



Ellipse

$A=\frac{\pi}{4}\times$ Major axis $\times$ Minor axis

$A=\frac{\pi}{4}\times D\times d$ $A=0.785\times D\times d$ $A=\pi\times R\times\gamma$

Circumference C depends on ratio $d:D$

Ratio $d:D$	$C=D$ times	$d:D$	$C=D$ times
0.9	2.9866	0.5	2.4221
0.8	2.8361	0.4	2.3013
0.7	2.6912	0.3	2.1930
0.6	2.5527	0.2	2.1010

Example

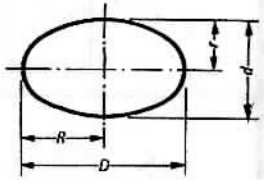
$D=150$ mm

$d=90$ mm

$d:D=90:150=0.6$

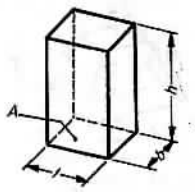
$C=150\times2.5527$

$C=382.9$ mm



Measurement of solid figures

Prism



Volume = Base $\times$ Height

$V=A\times h$

$V=A_\square\times h$

$V=A_\square\times h$

$V=l\times b\times h$

$V=A\times h$

Surface area for square base

$A_O=2A_\square+4A_\square$

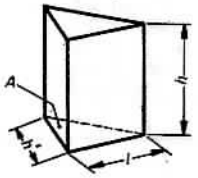
$=2\times$ Area of base $+$ $4\times$ Area of each side

$=2\times l\times b+4\times l\times h$

For Rectangle base

$A_O=2A_\square+2A_{\square1}+2A_{\square2}$

Prism



$V=A\times h$

$V=A_\triangle\times h$

$V=\frac{l\times h}{2}\times h$

$A_O=2A_\triangle+3A_{\square1}$

A = Equilateral triagle base

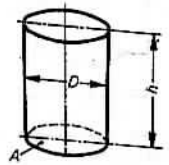
$A_O=2A_\triangle+A_{\square1}+A_{\square2}+A_{\square3}$

A = Scalene triangle base

$A_O=2A_\triangle+n\times A_\square$

Regular polygon of n sides

Cylinder



$V=A\times h$

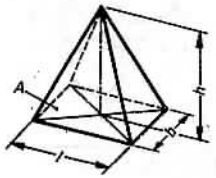
$V=A_O\times h$

$V=\frac{\pi}{4}\times D^2\times h$

Lateral area of the cylinder

$A_L=\pi\times D\times h$

Pyramid



$V=\frac{A\times h}{3}$

Volume $=\frac{\text{Base}\times\text{height}}{3}$

$V=\frac{A_\square\times h}{3}$

$V=\frac{l\times b\times h}{3}$

Surface

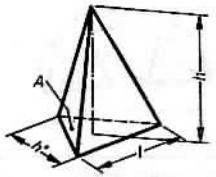
$A_O=A_\square+4A_{\triangle1}$

A = Square base

$A_O=A_\square+2A_{\triangle1}+2A_{\triangle2}$

A = Rectangle base

Pyramid



$V=\frac{A\times h}{3}$

$V=\frac{A_\triangle\times h}{3}$

$V=\frac{l\times h'\times h}{3}$

$A_O=A_\triangle+3A_{\triangle1}$

A = Equilateral triangle base

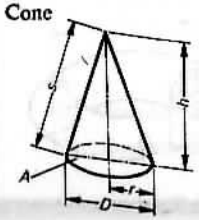
$A_O=A_\triangle+A_{\triangle1}+A_{\triangle2}+A_{\triangle3}$

A = Scalene triangle base

$A_O=A_\triangle+n\times A_{\triangle1}$

Regular polygon of n sides

Cone



$V=\frac{A\times h}{3}$

$V=A_O\times h$

$V=\frac{\pi}{4}D^2\times\frac{h}{3}$

$A_O=\frac{\pi}{4}D^2(D+2S)$

Lateral surface

$A_L=\frac{\pi\times D\times S}{2}$

$A_L=\pi\times r\times\sqrt{r^2+h^2}$

Surface area

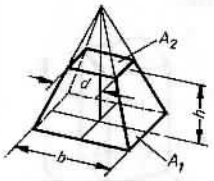
A_O = sum of individual areas

$V=\frac{h}{3}\times(a^2+a\times b+b^2)$ for A = square

$V\approx\left(\frac{a+b}{2}\right)^2\times h$; for A = square

$V=\frac{h}{3}(A_1+\sqrt{A_1\times A_2}+A_2)$ for A = n sided polygon

Frustum of a Pyramid



Lateral area of frustum

$A_L=\pi\times\frac{D+d}{2}\times S$

$S=\sqrt{h^2+(R-r)^2}$

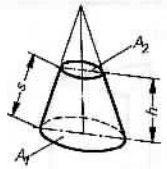
Surface

$A_O=A_1+A_2+A_L$

$V=\frac{\pi}{12}\times h\,(D^2+D\times d+d^2)$

$\frac{\pi}{12}=0.261$

Frustum of a cone



Surface

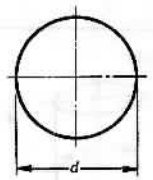
$A_O=\pi\times D^2$

$V=\frac{2}{3}\times\frac{\pi}{4}\times D^3$

$V=\frac{\pi D^3}{6}$

$V=0.5236D^3$

Sphere



Lateral area

$A_L=2\pi rh$

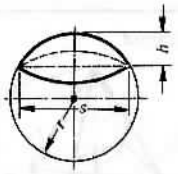
$A_L=\frac{\pi}{4}(S^2+4h^2)$

$V=\pi\times h^2\times\left(r-\frac{h}{3}\right)$

or

$V=\pi\times h\left(\frac{S^2}{8}+\frac{h^2}{6}\right)$

Spherical segment



Lateral area

$A_L=l\times\pi\times d_{S2}$

Surface

$A_O=C_A\times\pi\times d_{S1}$

C_A = circumference of cross-section

Pappus theorem:

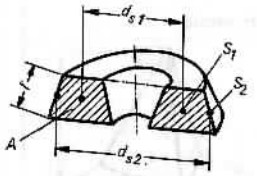
S_1 = centre of gravity of cross-section

A = cross-section

d_{S1} = diameter of circle of centres of gravity

$V=A\times\pi\times d_{S1}$

Solids of revolution



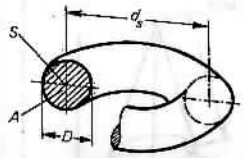
Lateral surface area

$A_O=A=C\times\pi\times D$

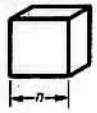
$A_O=\pi\times D\times\pi\times ds$

$V=\frac{\pi}{4}\times D^2\times\pi\times ds$

Ring of circular cross-section

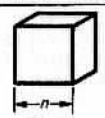


Example for the use of data tabulated: Values found under 376 can be used for following calculations

$C=\pi\times d$	$A=\frac{\pi}{4}\times d^2$	$A=n^2$	 $V=n^3$	Number d or n	$n=A$ $\sqrt{n}$	$n=V$ $\sqrt[3]{n}$
1181.2 Circumference of a circle if $d=376$ (mm, cm)	111 036 Area of a circle if $d=376$ (mm, cm)	141 376 Area of a square if $n=376$ (mm, cm)	53 157 376 Volume of a cube if $n=376$ (mm, cm)	376	19.3907 Side length of a square if $A=376$ (mm², cm²)	7.2177 Side length of a cube if $V=376$ (mm³, cm³)

Shifting the decimal point...

In case the basic number for a calculation is not 376 but 37.6 or 3.76 or 3760 the result can be found by appropriately shifting the decimal point of the value found in the table. If the decimal point of a number is 1 decimal off (left or right) the number which is tabulated, the decimal point of the values found on the table must be shifted by:

1 Decimal	2 Decimals	2 Decimals	3 Decimals	Number	
$C=\pi\times d$	$A=\frac{\pi}{4}\times d^2$	$A=n^2$	 $V=n^3$	d or n	Not applicable
1	2	2	3	1	Number of decimals to be shifted for $10\,n$ or $0.1\,n$
1181.2	111 036	141 376	53 157 376	376 Examples	As compared to the table value 376 is shifted by the decimal point
118.12	1110.36	1413.76	53157.376	37.6	1 Decimal (to the left)
11.812	11.1036	14.1376	53.157376	3.76	2 Decimals (to the left)
11812	11103600	14137600	53157376000	3760	1 Decimal (to the right)

Shifting the decimal point while extracting the root for values not tabulated

Example: $\sqrt{3.7}=?$ $\sqrt[3]{0.64}=?$

(1) Shift the decimal point of the basic number

(a) By 2 decimals when extracting square root (e.g., 3.7 changes to 370)

(b) By 3 decimals when extracting cube root (eg. 0.64 changes to 640) in order to arrive at a number which can be found in the table.

(2) The decimal point of the result found must be shifted back by 1 decimal to correct the change made.

Number d or n	$n=A$ $\sqrt{n}$	$n=V$ $\sqrt[3]{n}$
1	1/2	1/3
370	19.2354	
640		8.6177

$\sqrt{3.7}$ →

(1) 2 decimals to the right: 3.7 changes to 370

(2) 1 decimal to the left: 19.2354 becomes 1.92354

$\sqrt{3.7}=1.92354$

$\sqrt[3]{0.64}$ →

(1) 3 decimals to the right: 0.64 changes to 640

(2) 1 decimal to the left: 8.6177 becomes 0.86177

$\sqrt[3]{0.64}=0.86177$

Logarithms — Usage of numerical tables

Taking the logarithm is another inversion, so called the 2nd inversion of the process of raising a number to a power

Raising a number to a higher power	First inversion: Extracting the root of a number	Second inversion: Taking the logarithm of a number
$5^3=125$	$\sqrt[3]{125}=5$	$^{125}\log 125=3$

3 = the logarithm of the number 125 to the base 5

The column headed log in the table gives the Briggsian (or common) logarithms (log) to the base 10.
The logarithm of a number here is the index to the base 10
e.g. log 1000=3.0000, since $10^{3.0000}=1000$ 3.0000=the logarithm of 1000
Every logarithm of a number consists of the characteristic and the mantissa

Number — Logarithm
log 483 = 2.6839
antilogarithm — Mantissa / Characteristic

The mantissa can be taken from a logarithm table. The characteristic is to be added from case to case. It is obtained by merely counting the number of places of a given number, starting to the left from the decimal point onwards. Integral numbers from 1 to 1000, as listed in the column headed log have the characteristic added already.

Finding the characteristic

Let the number n has	the corresponding characteristic is then	Example number n	Example log n
4 places	3. ... one less than the number of digits to the left of the decimal point	4830	3.6839
3 places	2. ...	483	2.6839
2 places	1. ...	48.3	1.6839
1 place	0. ...	4.83	0.6839
0.... 1 naught	1. ... negative and equal to the number of naughts	0.483	1.6839
0.0... 2 naught	2. ...	0.0483	2.6839
0.00... 3 naught	3. ...	0.00483	3.6839

Significant figures having the same sequence here e.g. 483, 4.83, 0.0483 have the same mantissa, namely 6839.

Logarithms of these numbers differ from one another in that each number has its own characteristic in front.

Arithmatical operations involving logarithms in the computations

$a\times b$ multiplication is reduced to addition

$\log(a\times b)=\log a+\log b$

49.1×3.47

log 49.1 = 1.6911
+log 3.47 = 0.5403
2.2314

Mantissa, 2314 looked up in column headed log shows for

log	n
2305	170

2305 the antilog n = 170
Characteristic of 2 = 170
Interpolation* gives = 170.36

$49.1\times3.47=170.36$

$\frac{a}{b}$ division is reduced to subtraction

$\log\left(\frac{a}{b}\right)=\log a-\log b$

$3.47\div49.1$

log 3.47 = 0.5403
−log 49.1 = 1.6911
2.8492

Mantissa, 8492 looked up in column headed log shows for

log	n
8494	707

8494 the antilog n = 707
Characteristic of 2 = 0.0707
Interpolation* gives = 0.070666

$\frac{3.47}{49.1}=0.07067$

a^n raising a number to a higher power is reduced to multiplication

$\log(a^n)=n\times\log a$

16^3

log 16 = 1.2041
3 × log 16 = 3.6123
$16^3=4096$

Mantissa 6123 looked up in column headed log shows for

log	n
6117	409

6117 the antilog n = 409
Characteristic of 3 = 4090
Interpolation* gives = 4096

$\sqrt[n]{a}$ extracting the root of a number is reduced to division

$\log\sqrt[n]{a}=\frac{1}{n}\times\log a$

$\sqrt[3]{145}$

log 145 = 2.1614
$\frac{1}{3}$ log 145 = 0.72047
$\sqrt[3]{145}=5.2534$

Mantissa 72047 looked up in column headed log shows for

log	n
7202	525

7202 the antilog n = 525
Characteristic of 0 = 5.25
Interpolation* gives = 5.2534

*For Interpolation see page 35.

Interpolation

Finding the proportional part of the difference between the calculated values shown in the tables can be performed with sufficient accuracy assuming that the values listed are linear.

360.1 | $\pi\cdot360.1=?$

n	$\pi\times n$
360	1131.0
361	1134.1

3.1 difference

1.0 ≙ 3.1
0.1 ≙ 0.1 × 3.1 = 0.31
$\pi\times360.1=1131.0+0.31=1131.31$

200.25 | $\frac{\pi}{4}\times200.25=?$

n	$\frac{\pi}{4}\times n$
200	31415.9
201	31730.9

315 difference

1.00 ≙ 315.0
0.25 ≙ 0.25 × 315.0 = 78.75
$\frac{\pi}{4}\times200.25=31415.9+7875=31494.65$

stands for
corresponds

n	$\pi\times n$	difference 3.1
360	1131.0	
.1	1131.0+	0.1×3.1 = 1131.31
.2	1131.0+	0.2×3.1 = 1131.62
.3	1131.0+	0.3×3.1 = 1131.93
.4		
.5		
.6		
.65	1131.0+	0.65×3.1 = 1133.02
.7		
.8		
.9	1131.0+	0.9×3.1 = 1133.79
361	1134.1	

170.27 | $170.72^2=?$

n	n^2
170	28900
171	29241

341 difference

1.0 ≙ 341
0.27 ≙ 0.27 × 341 = 92.07
$170.27^2=28900+92.07=28992.07$

170.27 | $170.27^2=?$

n	n^2	difference
170	4913000	87211
171	5000211	

1.0 ≙ 87211
0.27 ≙ 0.27 × 87211 = 23547
$170.27^2=4913000+23547=4936547$

45.67 | $\sqrt{45.67}=?$ — $\sqrt{n}$

n	$\sqrt{n}$
45	6.7082
46	6.7823

0.0741 difference

1.0 ≙ 0.0741 Units of the fourth decimal
0.67 ≙ 0.67 × 741 = 496.47
$\sqrt{45.67}=6.7082+0.0496=6.7578$

687.63 | $\sqrt[3]{687.63}=?$ — $\sqrt[3]{n}$

n	$\sqrt[3]{n}$
687	8.8237
688	8.8280

0.0043 Units of the fourth decimal

1.0 ≙ 0.0043
$0.63\text{ ≙ }0.63\times43=\frac{4}{27.09}$
$\sqrt[3]{687.63}=8.8237+27.09=8.8264$

825.65 | log 825.65 = ? — log

n	log n
825	2.9165
826	2.9170

0.0005 difference

1 ≙ 5
0.65 ≙ 0.65 × 5 = 3.25
log 825.65 2.9165
+ 325
= 2.916825

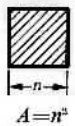
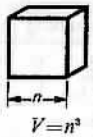
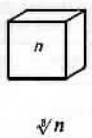
2.2314 | $n=?$ — log

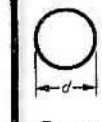
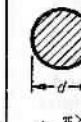
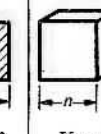
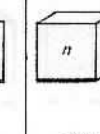
log n	n
2.2304	170
2.2330	171

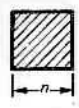
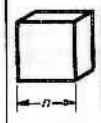
26 ≙ 1
1 ≙ 1.26 = 0.038
10 ≙ 0.38
2.2314 170
+ 0.38
= 170.38

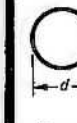
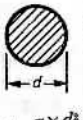
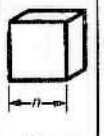
Inversion: It is commonly practised to start out from the column headed n to look for concerned values in the other columns, one can do likewise the opposite e.g.
$\sqrt{36480}=?$ 36481 read from column headed n^2
gives the result under column headed n: 191 $\sqrt{36480}$ 191

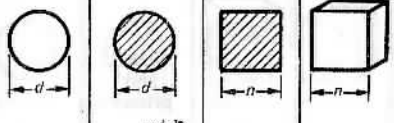
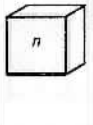
Let a number Z be known in column:	$\pi\times d$	$\frac{\pi}{4}\times d^2$	n^2	n^3	$\sqrt{n}$	$\sqrt[3]{n}$
column n gives the result:	d	d	$\sqrt{z}$	$\sqrt[3]{z}$	z^2	z^3

$C=\pi\times d$	$A=\frac{\pi\times d^2}{4}$	 $A=n^2$	 $V=n^3$	Number d or n	$\sqrt{n}$	 $\sqrt[3]{n}$	Factors and primes	log n
1	2	2	3		$\frac{1}{2}$	$\frac{1}{3}$	—	—
3.142	0.7854	1	1	1	1.0000	1.0000	—	0.0000
6.283	3.1416	4	8	2	1.4142	1.2599	—	0.3010
9.425	7.0686	9	27	3	1.7321	1.4422	—	0.4771
12.566	12.5664	16	64	4	2.0000	1.5874	2^2	0.6021
15.708	19.6350	25	125	5	2.2361	1.7100	—	0.6990
18.850	28.2743	36	216	6	2.4495	1.8171	2×3	0.7782
21.991	38.4845	49	343	7	2.6458	1.9129	—	0.8451
25.133	50.2655	64	512	8	2.8284	2.0000	2^3	0.9031
28.274	63.6173	81	729	9	3.0000	2.0801	3^2	0.9542
31.416	78.5398	100	1000	10	3.1623	2.1544	2×5	1.0000
34.558	95.0332	121	1331	11	3.3166	2.2240	—	1.0414
37.699	113.097	144	1728	12	3.4641	2.2894	$2^2\times3$	1.0792
40.841	132.732	169	2197	13	3.6056	2.3513	—	1.1139
43.982	153.938	196	2744	14	3.7417	2.4101	2×7	1.1461
47.124	176.715	225	3375	15	3.8730	2.4662	3×5	1.1761
50.265	201.062	256	4096	16	4.0000	2.5198	2^4	1.2041
53.407	226.980	289	4913	17	4.1231	2.5713	—	1.2305
56.549	254.469	324	5832	18	4.2426	2.6207	2×3^2	1.2553
59.690	283.529	361	6859	19	4.3589	2.6684	—	1.2788
62.832	314.159	400	8000	20	4.4721	2.7144	$2^2\times5$	1.3010
65.973	346.361	441	9261	21	4.5826	2.7589	3×7	1.3222
69.115	380.133	484	10648	22	4.6904	2.8020	2×11	1.3424
72.257	415.476	529	12167	23	4.7958	2.8439	—	1.3617
75.398	452.389	576	13824	24	4.8900	2.8845	$2^3\times3$	1.3802
78.540	490.874	625	15625	25	5.0000	2.9240	5^2	1.3980
81.681	530.929	676	17576	26	5.0990	2.9625	2×13	1.4150
84.823	572.555	729	19683	27	5.1962	3.0000	3^3	1.4314
87.965	615.752	784	21952	28	5.2915	3.0366	$2^2\times7$	1.4472
91.106	660.520	841	24389	29	5.3852	3.0723	—	1.4624
94.248	706.858	900	27000	30	5.4772	3.1072	$2\times3\times5$	1.4771
97.389	754.768	961	29791	31	5.5678	3.1414	—	1.4914
100.531	804.248	1024	32768	32	5.6569	3.1748	2^5	1.5052
103.673	855.299	1089	35937	33	5.7446	3.2075	3×11	1.5185
106.814	907.920	1156	39304	34	5.8310	3.2396	2×17	1.5315
109.956	962.113	1225	42875	35	5.9161	3.2711	5×7	1.5441
113.097	1017.88	1296	46656	36	6.0000	3.3019	$2^2\times3^2$	1.5563
116.239	1075.21	1369	50653	37	6.0828	3.3322	—	1.5682
119.381	1134.11	1444	54872	38	6.1644	3.3620	2×19	1.5798
122.522	1194.59	1521	59319	39	6.2450	3.3912	3×13	1.5911
125.66	1256.64	1600	64000	40	6.3246	3.4200	$2^3\times5$	1.6021
128.81	1320.25	1681	68921	41	6.4031	3.4482	—	1.6128
131.95	1385.44	1764	74088	42	6.4807	3.4760	$2\times3\times7$	1.6233
135.09	1452.20	1849	79507	43	6.5574	3.5034	—	1.6335
138.23	1520.53	1936	85184	44	6.6332	3.5303	$2^2\times11$	1.6435
141.37	1590.43	2025	91125	45	6.7082	3.5569	$3^2\times5$	1.6532
144.51	1661.90	2116	97336	46	6.7823	3.5830	2×23	1.6628
147.65	1734.94	2209	103823	47	6.8557	3.6088	—	1.6721
150.80	1809.56	2304	110592	48	6.9282	3.6342	$2^4\times3$	1.6812
153.94	1885.74	2401	117649	49	7.0000	3.6593	7^2	1.6902
157.08	1963.50	2500	125000	50	7.0711	3.6840	2×5^3	1.6990

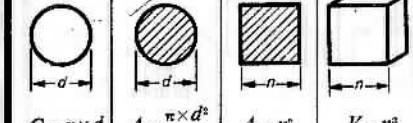
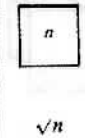
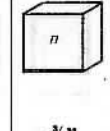
 $C=\pi\times d$	 $A=\frac{\pi\times d^2}{4}$	$A=n^2$	 $V=n^3$	Number d or n	$\sqrt{n}$	 $\sqrt[3]{n}$	Factors and primes	log n
1	2	2	3		$\frac{1}{2}$	$\frac{1}{3}$	—	—
160.22	2042.82	2601	132651	51	7.1414	3.7084	3×17	1.7076
163.36	2123.72	2704	140608	52	7.2111	3.7325	$2^2\times13$	1.7160
166.50	2206.18	2809	148877	53	7.2801	3.7563	—	1.7243
169.65	2290.22	2916	157464	54	7.3485	3.7798	2×3^3	1.7324
172.79	2375.83	3025	166375	55	7.4162	3.8030	5×11	1.7404
175.93	2463.01	3136	175616	56	7.4833	3.8259	$2^3\times7$	1.7482
179.07	2551.76	3249	185193	57	7.5498	3.8485	3×19	1.7559
182.21	2642.08	3364	195112	58	7.6158	3.8709	2×29	1.7634
185.35	2733.97	3481	205379	59	7.6811	3.8930	—	1.7709
188.50	2827.43	3600	216000	60	7.7460	3.9149	$2^3\times3\times5$	1.7782
191.64	2922.47	3721	226981	61	7.8102	3.9365	—	1.7853
194.78	3019.07	3844	238328	62	7.8740	3.9579	2×31	1.7924
197.92	3117.25	3969	250047	63	7.9373	3.9791	$3^2\times7$	1.7993
201.06	3216.99	4096	262144	64	8.0000	4.0000	2^6	1.8062
204.20	3318.31	4225	274625	65	8.0623	4.0207	5×13	1.8129
207.35	3421.19	4356	287496	66	8.1240	4.0412	$2\times3\times11$	1.8195
210.49	3525.65	4489	300763	67	8.1854	4.0615	—	1.8261
213.63	3631.68	4624	314432	68	8.2462	4.0817	$2^2\times17$	1.8325
216.77	3739.28	4761	328509	69	8.3066	4.1016	3×23	1.8389
219.91	3848.45	4900	343000	70	8.3666	4.1213	$2\times5\times7$	1.8451
223.05	3959.19	5041	357911	71	8.4261	4.1408	—	1.8513
226.19	4071.50	5184	373248	72	8.4853	4.1602	$2^3\times3^2$	1.8573
229.34	4185.39	5329	389017	73	8.5440	4.1793	—	1.8633
232.48	4300.84	5476	405224	74	8.6023	4.1983	2×37	1.8692
235.62	4417.86	5625	421875	75	8.6603	4.2172	3×5^3	1.8751
238.76	4536.44	5776	438976	76	8.7178	4.2358	$2^2\times19$	1.8808
241.90	4656.63	5929	456533	77	8.7750	4.2543	7×11	1.8865
245.04	4778.36	6084	474552	78	8.8318	4.2727	$2\times3\times13$	1.8921
248.19	4901.67	6241	493039	79	8.8882	4.2908	—	1.8976
251.33	5026.55	6400	512000	80	8.9443	4.3089	$2^4\times5$	1.9031
254.47	5153.00	6561	531441	81	9.0000	4.3267	3^4	1.9085
257.61	5281.02	6724	551363	82	9.0554	4.3445	2×41	1.9138
260.75	5410.61	6889	571787	83	9.1104	4.3621	—	1.9191
263.89	5541.77	7056	592704	84	9.1652	4.3795	$2^2\times3\times7$	1.9243
267.04	5674.50	7225	614125	85	9.2195	4.3968	5×17	1.9294
270.18	5808.80	7396	636056	86	9.2736	4.4140	2×43	1.9345
273.32	5944.68	7569	658503	87	9.3274	4.4310	3×29	1.9395
276.46	6082.12	7744	681472	88	9.3808	4.4480	$2^2\times11$	1.9445
279.60	6221.14	7921	704969	89	9.4340	4.4647	—	1.9494
282.74	6361.73	8100	729000	90	9.4868	4.4814	$2\times3^2\times5$	1.9542
285.88	6503.88	8281	753571	91	9.5394	4.4979	7×13	1.9590
289.03	6647.61	8464	778688	92	9.5917	4.5144	$2^2\times23$	1.9638
292.17	6792.91	8649	804357	93	9.6437	4.5307	3×31	1.9685
295.31	6939.78	8836	830584	94	9.6954	4.5468	2×47	1.9731
298.45	7088.22	9025	857375	95	9.7468	4.5629	5×19	1.9777
301.59	7238.23	9216	884736	96	9.7980	4.5789	$2^5\times3$	1.9823
304.73	7389.81	9409	912673	97	9.8489	4.5947	—	1.9868
307.88	7542.96	9604	941192	98	9.8995	4.6104	2×7^2	1.9912
311.02	7697.69	9801	970299	99	9.9499	4.6261	$3^2\times11$	1.9956
314.16	7853.98	10000	1000000	100	10.0000	4.6416	$2^2\times5^2$	2.0000

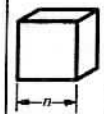
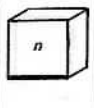
$C=\pi\times d$	$A=\frac{\pi\times d^2}{4}$	 $A=n^2$	 $V=n^3$	Number d or n	n $\sqrt{n}$	n $\sqrt[3]{n}$	Factors and primes	$\log n$
1	2	2	3		$\frac{1}{2}$	$\frac{1}{3}$	—	—
317.30	8011.85	10201	1030301	101	10.0499	4.6570	—	2.0043
320.44	8171.28	10404	1061208	102	10.0995	4.6723	$2\times3\times17$	2.0086
323.58	8332.29	10609	1092727	103	10.1489	4.6875	—	2.0128
326.73	8494.87	10816	1124864	104	10.1980	4.7027	$2^3\times13$	2.0170
329.87	8659.01	11025	1157625	105	10.2470	4.7177	$3\times5\times7$	2.0212
333.01	8824.73	11236	1191016	106	10.2956	4.7326	2×53	2.0253
336.15	8992.02	11449	1225043	107	10.3441	4.7475	—	2.0294
339.29	9160.88	11664	1259712	108	10.3923	4.7622	$2^3\times3^3$	2.0334
342.43	9331.32	11881	1295029	109	10.4403	4.7769	—	2.0374
345.58	9503.32	12100	1331000	110	10.4881	4.7914	$2\times5\times11$	2.0414
348.72	9676.89	12321	1367631	111	10.5357	4.8059	3×37	2.0453
351.86	9852.03	12544	1404928	112	10.5830	4.8203	$2^4\times7$	2.0492
355.00	10028.7	12769	1442897	113	10.6301	4.8346	—	2.0531
358.14	10207.0	12996	1481544	114	10.6771	4.8488	$2\times3\times19$	2.0569
361.28	10386.9	13225	1520875	115	10.7238	4.8629	5×23	2.0607
364.42	10568.3	13456	1560896	116	10.7703	4.8770	$2^2\times29$	2.0645
367.57	10751.3	13689	1601613	117	10.8167	4.8910	$3^2\times13$	2.0682
370.71	10935.9	13924	1643032	118	10.8628	4.9049	2×59	2.0719
373.85	11122.0	14161	1685159	119	10.9087	4.9187	7×17	2.0756
376.99	11309.7	14400	1728000	120	10.9545	4.9324	$2^5\times3\times5$	2.0792
380.13	11499.0	14641	1771561	121	11.0000	4.9461	11^2	2.0828
383.27	11689.9	14884	1815848	122	11.0454	4.9597	2×61	2.0864
386.42	11882.3	15129	1860867	123	11.0905	4.9732	3×41	2.0899
389.56	12076.3	15376	1906624	124	11.1355	4.9866	$2^2\times31$	2.0934
392.70	12271.8	15625	1953125	125	11.1803	5.0000	5^3	2.0969
395.84	12469.0	15876	2000376	126	11.2250	5.0133	$2\times3^2\times7$	2.1004
398.98	12667.7	16129	2048383	127	11.2694	5.0265	—	2.1038
402.12	12868.0	16384	2097152	128	11.3137	5.0397	2^7	2.1072
405.27	13069.8	16641	2146689	129	11.3578	5.0528	3×43	2.1106
408.41	13273.2	16900	2197000	130	11.4018	5.0658	$2\times5\times13$	2.1139
411.55	13478.2	17161	2248091	131	11.4455	5.0788	—	2.1173
414.69	13684.8	17424	2299968	132	11.4891	5.0916	$2^2\times3\times11$	2.1206
417.83	13892.9	17689	2352637	133	11.5326	5.1045	7×19	2.1239
420.97	14102.6	17956	2406104	134	11.5758	5.1172	2×67	2.1271
424.12	14313.9	18225	2460375	135	12.6190	5.1299	$3^3\times5$	2.1303
427.26	14526.7	18496	2515456	136	11.6619	5.1426	$2^3\times17$	2.1335
430.40	14741.1	18769	2571353	137	11.7047	5.1551	—	2.1367
433.54	14957.1	19044	2628072	138	11.7473	5.1676	$2\times3\times23$	2.1399
436.68	15174.7	19321	2685619	139	11.7898	5.1801	—	2.1430
439.82	15393.8	19600	2744000	140	11.8322	5.1925	$2^5\times5\times7$	2.1461
442.96	15614.5	19881	2803221	141	11.8743	5.2048	3×47	2.1492
446.11	15836.8	20164	2863288	142	11.9164	5.2171	2×71	2.1523
449.25	16060.6	20449	2924207	143	11.9583	5.2293	11×13	2.1553
452.39	16286.0	20736	2985984	144	12.0000	5.2415	$2^4\times3^2$	2.1584
455.53	16513.0	21025	3048625	145	12.0416	5.2536	5×29	2.1614
458.67	16741.5	21316	3112136	146	12.0830	5.2656	2×73	2.1644
461.81	16971.7	21609	3176523	147	12.1244	5.2776	3×7^2	2.1673
464.96	17203.4	21904	3241792	148	12.1655	5.2896	$2^2\times37$	2.1703
468.10	17436.6	22201	3307949	149	12.2066	5.3015	—	2.1732
471.24	17671.5	22500	3375000	150	12.2474	5.3133	$2\times3\times5^3$	2.1761

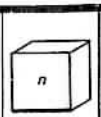
 $C=\pi\times d$	 $A=\frac{\pi\times d^2}{4}$	$A=n^2$	 $V=n^3$	Number d or n	n $\sqrt{n}$	n $\sqrt[3]{n}$	Factors and primes	$\log n$
1	2	2	3		$\frac{1}{2}$	$\frac{1}{3}$	—	—
474.38	17907.9	22801	3442951	151	12.2882	5.3251	—	2.1790
477.52	18145.8	23104	3511808	152	12.3288	5.3368	$2^3\times19$	2.1818
480.66	18385.4	23409	3581577	153	12.3693	5.3485	$3^3\times17$	2.1847
483.81	18626.5	23716	3652264	154	12.4097	5.3601	$2\times7\times11$	2.1875
486.95	18869.2	24025	3723875	155	12.4499	5.3717	5×31	2.1903
490.09	19113.4	24336	3796416	156	12.4900	5.3832	$2^2\times3\times13$	2.1931
493.23	19359.3	24649	3869893	157	12.5300	5.3947	—	2.1959
496.37	19606.7	24964	3944312	158	12.5698	5.4061	2×79	2.1987
499.51	19855.7	25281	4019679	159	12.6095	5.4175	3×53	2.2014
502.65	20106.2	25600	4096000	160	12.6491	5.4288	$2^5\times5$	2.2041
505.80	20358.3	25921	4173281	161	12.6886	5.4401	7×23	2.2068
508.94	20612.0	26244	4251528	162	12.7279	5.4514	2×3^4	2.2095
512.08	20867.2	26569	4330747	163	12.7671	5.4626	—	2.2122
515.22	21124.1	26896	4410944	164	12.8062	5.4737	$2^2\times41$	2.2148
518.36	21382.5	27225	4492125	165	12.8452	5.4848	$3\times5\times11$	2.2175
521.50	21642.4	27556	4574296	166	12.8841	5.4959	2×83	2.2201
524.65	21904.0	27889	4657463	167	12.9229	5.5069	—	2.2227
527.79	22167.1	28224	4741632	168	12.9615	5.5178	$2^3\times3\times7$	2.2253
530.93	22431.8	28561	4826809	169	13.0000	5.5288	13^2	2.2279
534.07	22698.0	28900	4913000	170	13.0384	5.5397	$2\times5\times17$	2.2305
537.21	22965.8	29241	5000211	171	13.0767	5.5505	$3^2\times19$	2.2330
540.35	23235.2	29584	5088448	172	13.1149	5.5613	$2^2\times43$	2.2355
543.50	23506.2	29929	5177717	173	13.1529	5.5721	—	2.2381
546.64	23778.7	30276	5268024	174	13.1909	5.5828	$2\times3\times29$	2.2406
549.78	24052.8	30625	5359375	175	13.2288	5.5934	$5^2\times7$	2.2430
552.92	24328.5	30976	5451776	176	13.2665	5.6041	$2^4\times11$	2.2455
556.06	24605.7	31329	5545233	177	13.3041	5.6147	3×59	2.2480
559.20	24884.6	31684	5639752	178	13.3417	5.6252	2×89	2.2504
562.35	25164.9	32041	5735339	179	13.3791	5.6357	—	2.2529
565.49	25446.9	32400	5832000	180	13.4164	5.6462	$2^5\times3^3\times5$	2.2553
568.63	25730.4	32761	5929741	181	13.4536	5.6567	—	2.2577
571.77	26015.5	33124	6028568	182	13.4907	5.6671	$2\times7\times13$	2.2601
574.91	26302.2	33489	6128487	183	13.5277	5.6774	3×61	2.2625
578.05	26590.4	33856	6229504	184	13.5647	5.6877	$2^3\times23$	2.2648
581.19	26880.3	34225	6331625	185	13.6015	5.6980	5×37	2.2672
584.34	27171.6	34596	6434856	186	13.6382	5.7083	$2\times3\times31$	2.2695
587.48	27464.6	34969	6539203	187	13.6748	5.7185	11×17	2.2718
590.62	27759.1	35344	6644672	188	13.7113	5.7287	$2^2\times47$	2.2742
593.76	28055.2	35721	6751269	189	13.7477	5.7388	$3^3\times7$	2.2765
596.90	28352.9	36100	6859000	190	13.7840	5.7489	$2\times5\times19$	2.2788
600.04	28652.1	36481	6967871	191	13.8203	5.7590	—	2.2810
603.19	28952.9	36864	7077888	192	13.8564	5.7690	$2^6\times3$	2.2833
606.33	29255.3	37249	7189057	193	13.8924	5.7790	—	2.2856
609.47	29559.2	37636	7301384	194	13.9284	5.7890	2×97	2.2878
612.61	29864.8	38025	7414875	195	13.9642	5.7989	$3\times5\times13$	2.2900
615.75	30171.9	38416	7529536	196	14.0000	5.8088	$2^2\times7^2$	2.2923
618.89	30480.5	38809	7645373	197	14.0357	5.8186	—	2.2945
622.04	30790.7	39204	7762392	198	14.0712	5.8285	$2\times3^3\times11$	2.2967
625.18	31102.6	39601	7880599	199	14.1067	5.8383	—	2.2989
628.32	31415.9	40000	8000000	200	14.1421	5.8480	$2^5\times5^2$	2.3010

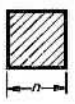
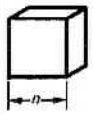
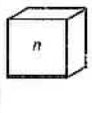
 $C=\pi\times d$	$A=\frac{\pi\times d^2}{4}$	$A=n^2$	$V=n^3$	Number d or n	$\sqrt{n}$	 $\sqrt[3]{n}$	Factors and primes	log n
1	2	2	3		$\frac{1}{2}$	$\frac{1}{3}$	—	—
631.46	31730.9	40401	8120601	201	14.1774	5.8578	3×67	2.3032
634.60	32047.4	40804	8242408	202	14.2127	5.8675	2×101	2.3054
637.74	32365.5	41209	8365427	203	14.2478	5.8771	7×29	2.3075
640.88	32685.1	41616	8489664	204	14.2829	5.8868	$2^2\times3\times17$	2.3096
644.03	33006.4	42025	8615125	205	14.3178	5.8964	5×41	2.3118
647.17	33329.2	42436	8741816	206	14.3527	5.9059	2×103	2.3139
650.31	33653.5	42849	8869743	207	14.3875	5.9155	$3^2\times23$	2.3160
653.45	33979.5	43264	8998912	208	14.4222	5.9250	$2^6\times13$	2.3181
656.59	34307.0	43681	9129329	209	14.4568	5.9345	11×19	2.3202
659.73	34636.1	44100	9261000	210	14.4914	5.9439	$2\times3\times5\times7$	2.3222
662.88	34966.7	44521	9393931	211	14.5258	5.9533	—	2.3243
666.02	35298.9	44944	9528128	212	14.5602	5.9627	$2^2\times53$	2.3263
669.16	35632.7	45369	9663597	213	14.5945	5.9721	3×71	2.3284
672.30	35968.1	45796	9800344	214	14.6287	5.9814	2×107	2.3304
675.44	36305.0	46225	9938375	215	14.6629	5.9907	5×43	2.3324
678.58	36643.5	46656	10077696	216	14.6969	6.0000	$2^6\times3^3$	2.3345
681.73	36983.6	47089	10218313	217	14.7309	6.0092	7×31	2.3365
684.87	37325.3	47524	10360232	218	14.7648	6.0185	2×109	2.3385
688.01	37668.5	47961	10503459	219	14.7986	6.0277	3×73	2.3404
691.15	38013.3	48400	10648000	220	14.8324	6.0368	$2^3\times5\times11$	2.3424
694.29	38359.6	48841	10793861	221	14.8661	6.0459	13×17	2.3444
697.43	38707.6	49284	10941048	222	14.8997	6.0550	$2\times3\times37$	2.3464
700.53	39057.1	49729	11089567	223	14.9332	6.0641	—	2.3483
703.72	39408.1	50176	11239424	224	14.9666	6.0732	$2^5\times7$	2.3503
706.86	39760.8	50625	11390625	225	15.0000	6.0822	$3^2\times5^2$	2.3522
710.00	40115.0	51076	11543176	226	15.0333	6.0912	2×113	2.3541
713.14	40470.8	51529	11697083	227	15.0665	6.1002	—	2.3560
716.28	40828.1	51984	11852352	228	15.0997	6.1091	$2^2\times3\times19$	2.3579
719.42	41187.1	52441	12008989	229	15.1327	6.1180	—	2.3598
722.57	41547.6	52900	12167000	230	15.1658	6.1269	$2\times5\times23$	2.3617
725.71	41909.6	53361	12326391	231	15.1987	6.1358	$3\times7\times11$	2.3636
728.85	42273.3	53824	12487168	232	15.2315	6.1446	$2^5\times29$	2.3655
731.99	42638.5	54289	12649337	233	15.2643	6.1534	—	2.3674
735.13	43005.3	54756	12812904	234	15.2971	6.1622	$2\times3^2\times13$	2.3692
738.27	43373.6	55225	12977875	235	15.3297	6.1710	5×47	2.3711
741.42	43743.5	55696	13144256	236	15.3623	6.1797	$2^2\times59$	2.3729
744.56	44115.0	56169	13312053	237	15.3948	6.1885	3×79	2.3748
747.70	44488.1	56644	13481272	238	15.4272	6.1972	$2\times7\times17$	2.3766
750.84	44862.7	57121	13651919	239	15.4596	6.2058	—	2.3784
753.98	45238.9	57600	13824000	240	15.4919	6.2145	$2^4\times3\times5$	2.3802
757.12	45616.7	58081	13997521	241	15.5242	6.2231	—	2.3820
760.27	45996.1	58564	14172488	242	15.5563	6.2317	2×11^2	2.3838
763.41	46377.0	59049	14348907	243	15.5885	6.2403	3^5	2.3856
766.55	46759.5	59536	14526784	244	15.6205	6.2488	$2^2\times61$	2.3874
769.69	47143.5	60025	14706125	245	15.6525	6.2573	5×7^2	2.3892
772.83	47529.2	60516	14886936	246	15.6844	6.2658	$2\times3\times41$	2.3909
775.97	47916.4	61009	15069223	247	15.7162	6.2743	13×19	2.3927
779.11	48305.1	61504	15252992	248	15.7480	6.2828	$2^3\times31$	2.3945
782.26	48695.5	62001	15438249	249	15.7779	6.2912	3×83	2.3962
785.40	49087.4	62500	15625000	250	15.8114	6.2996	2×5^3	2.3979

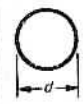
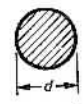
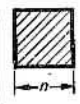
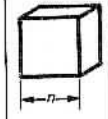
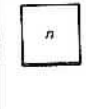
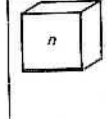
SPK

 $C=\pi\times d$	$A=\frac{\pi\times d^2}{4}$	$A=n^2$	$V=n^3$	Number d or n	 $\sqrt{n}$	 $\sqrt[3]{n}$	Factors and primes	log n
1	2	2	3		$\frac{1}{2}$	$\frac{1}{3}$	—	—
788.54	49480.9	63001	15813251	251	15.8430	6.3080	—	2.3997
791.68	49875.9	63504	16003008	252	15.8745	6.3164	$2^2\times3^2\times7$	2.4014
794.82	50272.6	64009	16194277	253	15.9060	6.3247	11×23	2.4031
797.96	50670.7	64516	16387064	254	15.9374	6.3330	2×127	2.4048
801.11	51070.5	65025	16581375	255	15.9687	6.3413	$3\times5\times17$	2.4065
804.25	51471.9	65536	16777216	256	16.0000	6.3496	2^8	2.4082
807.39	51874.8	66049	16974593	257	16.0312	6.3579	—	2.4099
810.53	52279.2	66564	17173512	258	16.0624	6.3661	$2\times3\times43$	2.4116
813.67	52685.3	67081	17373979	259	16.0935	6.3743	7×37	2.4133
816.81	53092.9	67600	17576000	260	16.1245	6.3825	$2^6\times5\times13$	2.4150
819.96	53502.1	68121	17779581	261	16.1555	6.3907	$3^2\times29$	2.4166
823.10	53912.9	68644	17984728	262	16.1864	6.3988	2×131	2.4183
826.24	54325.2	69169	18191447	263	16.2173	6.4070	—	2.4200
829.38	54739.1	69696	18399744	264	16.2481	6.4151	$2^3\times3\times11$	2.4216
832.52	55154.6	70225	18609625	265	16.2788	6.4232	5×53	2.4233
835.66	55571.6	70756	18821096	266	16.3095	6.4312	$2\times7\times19$	2.4249
838.81	55990.2	71289	19034163	267	16.3401	6.4393	3×89	2.4265
841.95	56410.4	71824	19248832	268	16.3707	6.4473	$2^2\times67$	2.4281
845.09	56832.2	72361	19465109	269	16.4012	6.4553	—	2.4298
848.23	57255.5	72900	19683000	270	16.4317	6.4633	$2\times3^3\times5$	2.4314
851.37	57680.4	73441	19902511	271	16.4621	6.4713	—	2.4330
854.51	58106.9	73984	20123648	272	16.4924	6.4792	$2^4\times17$	2.4346
857.65	58534.9	74529	20346417	273	16.5227	6.4872	$3\times7\times13$	2.4362
860.80	58964.6	75076	20570824	274	16.5529	6.4951	2×137	2.4378
863.94	59395.7	75625	20796875	275	16.5831	6.5030	$5^2\times11$	2.4393
867.08	59828.5	76176	21024576	276	16.6132	6.5108	$2^2\times3\times23$	2.4409
870.22	60262.8	76729	21253933	277	16.6433	6.5187	—	2.4425
873.36	60698.7	77284	21484952	278	16.6733	6.5265	2×129	2.4440
876.50	61136.2	77841	21717639	279	16.7033	6.5343	$3^3\times31$	2.4456
879.65	61575.2	78400	21952000	280	16.7332	6.5421	$2^5\times5\times7$	2.4472
882.79	62015.8	78961	22188041	281	16.7631	6.5499	—	2.4487
885.93	62458.0	79524	22425768	282	16.7929	6.5577	$2\times3\times47$	2.4503
889.07	62901.8	80089	22665187	283	16.8826	6.5654	—	2.4518
892.21	63347.1	80656	22906304	284	16.8523	6.5731	$2^2\times71$	2.4533
895.35	63794.0	81225	23149125	285	16.8819	6.5808	$3\times5\times19$	2.4548
898.50	64242.4	81796	23393656	286	16.9115	6.5885	$2\times11\times13$	2.4564
901.64	64692.5	82369	23639903	287	16.9411	6.5962	7×41	2.4579
904.78	65144.1	82944	23887872	288	16.9706	6.6039	$2^6\times3^2$	2.4594
907.92	65597.2	83521	24137569	289	17.0000	6.6115	17^2	2.4609
911.06	66052.0	84100	24389000	290	17.0294	6.6191	$2\times5\times29$	2.4624
914.20	66508.3	84681	24642171	291	17.0587	6.6267	3×97	2.4639
917.35	66966.2	85264	24897088	292	17.0880	6.6343	$2^2\times73$	2.4654
920.49	67425.6	85849	25153757	293	17.1172	6.6419	—	2.4669
923.63	67886.7	86436	25412184	294	17.1464	6.6494	$2\times3\times7^2$	2.4684
926.77	68349.3	87025	25672375	295	17.1756	6.6569	5×59	2.4698
936.19	69746.5	88804	26463592	296	17.2047	6.6644	$2^3\times37$	2.4713
929.91	68813.4	87616	25934336	297	17.2337	6.6719	$3^3\times11$	2.4728
933.05	69279.2	88209	26198073	298	17.2627	6.6794	2×149	2.4742
939.34	70215.4	89401	26730899	299	17.2916	6.6869	13×23	2.4757
942.48	70685.8	90000	27000000	300	17.3205	6.6943	$2^2\times3\times5^3$	2.4771

$C=\pi\times d$	$A=\frac{\pi\times d^2}{4}$	$A=n^2$	 $V=n^3$	Number d or n	$\sqrt{n}$	 $\sqrt[3]{n}$	Factors and primes	$\log n$
1	2	2	3		$\frac{1}{2}$	$\frac{1}{3}$	—	—
945.62	71157.9	90601	27270901	301	17.3494	6.7018	7×43	2.4786
948.76	71631.5	91204	27543608	302	17.3781	6.7092	2×151	2.4800
951.90	72106.6	91809	27818127	303	17.4069	6.7166	3×101	2.4814
955.04	72583.4	92416	28094464	304	17.4356	6.7240	$2^4\times19$	2.4829
958.19	73061.7	93025	28372625	305	17.4642	6.7313	5×61	2.4843
961.33	73541.5	93636	28652616	306	17.4929	6.7387	$2\times3^2\times17$	2.4857
964.47	74023.0	94249	28934443	307	17.5214	6.7460	—	2.4871
967.61	74506.0	94864	29218112	308	17.5499	6.7533	$2^2\times7\times11$	2.4886
970.75	74990.6	95481	29503629	309	17.5784	6.7606	3×103	2.4900
973.89	75476.8	96100	29791000	310	17.6068	6.7679	$2\times5\times31$	2.4914
977.04	75964.5	96721	30080231	311	17.6352	6.7752	—	2.4928
980.18	76453.8	97344	30371328	312	17.6635	6.7824	$2^3\times3\times13$	2.4942
983.32	76944.7	97969	30664297	313	17.6918	6.7897	—	2.4955
986.46	77437.1	98596	30959144	314	17.7200	6.7969	2×157	2.4969
989.60	77931.1	99225	31255875	315	17.7482	6.8041	$3^2\times5\times7$	2.4983
992.74	78426.7	99856	31554496	316	17.7764	6.8113	$2^3\times79$	2.4997
995.88	78923.9	100489	31855013	317	17.8045	6.8185	—	2.5011
999.03	79422.6	101124	32157432	318	17.8326	6.8256	$2\times3\times53$	2.5024
1002.2	79922.9	101761	32461759	319	17.8606	6.8328	11×29	2.5038
1005.3	80424.8	102400	32768000	320	17.8885	6.8399	$2^8\times5$	2.5052
1008.5	80928.2	103041	33076161	321	17.9165	6.8470	3×107	2.5065
1011.6	81433.2	103684	33386248	322	17.9444	6.8541	$2\times7\times23$	2.5079
1014.7	81939.8	104329	33698267	323	17.9722	6.8612	17×19	2.5092
1017.9	82448.0	104976	34012224	324	18.0000	6.8683	$2^2\times3^4$	2.5106
1021.0	82957.7	105625	34328125	325	18.0278	6.8753	$5^3\times13$	2.5119
1024.2	83469.0	106276	34645976	326	18.0555	6.8824	2×163	2.5132
1027.3	83981.8	106929	34965783	327	18.0831	6.8894	3×109	2.5146
1030.4	84496.3	107584	35287552	328	18.1108	6.8964	$2^3\times41$	2.5159
1033.6	85012.3	108241	35611289	329	18.1384	6.9034	7×47	2.5172
1036.7	85529.9	108900	35937000	330	18.1659	6.9104	$2\times3\times5\times11$	2.5185
1039.9	86049.0	109561	36264691	331	18.1934	6.9174	—	2.5198
1043.0	86569.7	110224	36594368	332	18.2209	6.9244	$2^2\times83$	2.5211
1046.2	87092.0	110889	36926037	333	18.2483	6.9313	$3^2\times37$	2.5224
1049.3	87615.9	111556	37259704	334	18.2757	6.9382	2×167	2.5238
1052.4	88141.3	112225	37595375	335	18.3030	6.9451	5×67	2.5250
1055.6	88668.3	112896	37933056	336	18.3303	6.9521	$2^4\times3\times7$	2.5263
1058.7	89196.9	113569	38272753	337	18.3576	6.9589	—	2.5276
1061.9	89727.0	114244	38614472	338	18.3848	6.9658	2×13^2	2.5289
1065.0	90258.7	114921	38958219	339	18.4120	6.9727	3×113	2.5302
1068.1	90792.0	115600	39304000	340	18.4391	6.9795	$2^3\times5\times17$	2.5315
1071.3	91326.9	116281	39651821	341	18.4662	6.9864	11×31	2.5328
1074.4	91863.3	116964	40001688	342	18.4932	6.9932	$2\times3^2\times19$	2.5340
1077.6	92401.3	117649	40353607	343	18.5203	7.0000	7^3	2.5353
1080.7	92940.9	118336	40707584	344	18.5472	7.0068	$2^3\times43$	2.5366
1083.8	93482.0	119025	41063625	345	18.5742	7.0136	$3\times5\times23$	2.5378
1087.0	94024.7	119716	41421736	346	18.6011	7.0203	2×173	2.5391
1090.1	94569.0	120409	41781923	347	18.6279	7.0271	—	2.5403
1093.3	95114.9	121104	42144192	348	18.6548	7.0338	$2^2\times3\times29$	2.5416
1096.4	95662.3	121801	42508549	349	18.6815	7.0406	—	2.5428
1099.6	96211.3	122500	42875000	350	18.7083	7.0473	$2\times5^3\times7$	2.5441

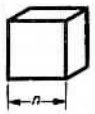
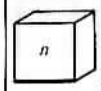
$C=\pi\times d$	$A=\frac{\pi\times d^2}{4}$	$A=n^2$	$V=n^3$	Number d or n	$\sqrt{n}$	 $\sqrt[3]{n}$	Factors and Primes	$\log n$
1	2	2	3		$\frac{1}{2}$	$\frac{1}{3}$	—	—
1102.7	96761.8	123201	43243551	351	18.7350	7.0540	$3^3\times13$	2.5453
1105.8	97314.0	123904	43614208	352	18.7617	7.0607	$2^5\times11$	2.5465
1109.0	97867.7	124609	43986977	353	18.7883	7.0674	—	2.5478
1112.1	98423.0	125316	44361864	354	18.8149	7.0740	$2\times3\times59$	2.5490
1115.3	98979.8	126025	44738875	355	18.8414	7.0807	5×71	2.5502
1118.4	99538.2	126736	45118016	356	18.8680	7.0873	$2^3\times89$	2.5515
1121.5	100098	127449	45499293	357	18.8944	7.0940	$3\times7\times17$	2.5527
1124.7	100660	128164	45882712	358	18.9209	7.1006	2×179	2.5539
1127.8	101223	128881	46268279	359	18.9473	7.1072	—	2.5551
1131.0	101788	129600	46656000	360	18.9737	7.1138	$2^6\times3^3\times5$	2.5563
1134.1	102354	130321	47045881	361	19.0000	7.1204	19^2	2.5575
1137.3	102922	131044	47437928	362	19.0263	7.1269	2×181	2.5587
1140.4	103491	131769	47832147	363	19.0526	7.1335	3×11^2	2.5599
1143.5	104062	132496	48228544	364	19.0788	7.1400	$2^2\times7\times13$	2.5611
1146.7	104635	133225	48627125	365	19.1050	7.1466	5×73	2.5623
1149.8	105209	133956	49027896	366	19.1311	7.1531	$2\times3\times61$	2.5635
1153.0	105785	134689	49430863	367	19.1572	7.1596	—	2.5647
1156.1	106362	135424	49836032	368	19.1833	7.1661	$2^4\times23$	2.5659
1159.2	106941	136161	50243409	369	19.2094	7.1726	$3^2\times41$	2.5670
1162.4	107521	136900	50653000	370	19.2354	7.1791	$2\times5\times37$	2.5682
1165.5	108103	137641	51064811	371	19.2614	7.1855	7×53	2.5694
1168.7	108687	138384	51478848	372	19.2873	7.1920	$2^2\times3\times31$	2.5705
1171.8	109272	139129	51895117	373	19.3132	7.1984	—	2.5717
1175.0	109858	139876	52313624	374	19.3391	7.2048	$2\times11\times17$	2.5729
1178.1	110447	140625	52734375	375	19.3649	7.2112	3×5^3	2.5740
1181.2	111036	141376	53157376	376	19.3907	7.2177	$2^9\times47$	2.5752
1184.4	111628	142129	53582633	377	19.4165	7.2240	13×29	2.5763
1187.5	112221	142884	54010152	378	19.4422	7.2304	$2\times3^3\times7$	2.5775
1190.7	112815	143641	54439939	379	19.4679	7.2368	—	2.5786
1193.8	113411	144400	54872000	380	19.4936	7.2432	$2^3\times5\times19$	2.5798
1196.9	114009	145161	55306341	381	19.5192	7.2495	3×127	2.5809
1200.1	114608	145924	55742968	382	19.5448	7.2558	2×191	2.5821
1203.2	115209	146689	56181887	383	19.5704	7.2622	—	2.5832
1206.4	115812	147456	56623104	384	19.5959	7.2685	$2^7\times3$	2.5843
1209.5	116416	148225	57066625	385	19.6214	7.2748	$5\times7\times11$	2.5855
1212.7	117021	148996	57512456	386	19.6469	7.2811	2×193	2.5866
1215.8	117628	149769	57960603	387	19.6723	7.2874	$2^8\times43$	2.5877
1218.9	118237	150544	58411072	388	19.6977	7.2936	$2^2\times97$	2.5888
1222.1	118847	151321	58863869	389	19.7231	7.2999	—	2.5900
1225.2	119459	152100	59319000	390	19.7484	7.3061	$2\times3\times5\times13$	2.5911
1228.4	120072	152881	59776471	391	19.7737	7.3124	17×23	2.5922
1231.5	120687	153664	60236288	392	19.7990	7.3186	$2^3\times7^2$	2.5933
1234.6	121304	154449	60698457	393	19.8242	7.3248	3×131	2.5944
1237.8	121922	155236	61162984	394	19.8494	7.3310	2×197	2.5955
1240.9	122542	156025	61629875	395	19.8746	7.3372	5×79	2.5966
1244.1	123163	156816	62099136	396	19.8997	7.3434	$2^2\times3^2\times11$	2.5977
1247.2	123786	157609	62570773	397	19.9249	7.3496	—	2.5988
1250.4	124410	158404	63044792	398	19.9499	7.3558	2×199	2.5999
1253.5	125036	159201	65521199	399	19.9750	7.3619	$3\times7\times19$	2.6010
1256.6	125664	160000	64000000	400	20.0000	7.3681	$2^4\times5^3$	2.6021

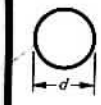
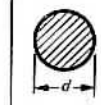
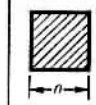
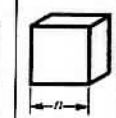
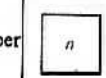
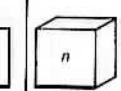
$C=\pi\times d$	$A=\frac{\pi\times d^2}{4}$	 $A=n^2$	 $V=n^3$	Number d or n	$\sqrt{n}$	 $\sqrt[3]{n}$	Factors and Primes	log n
1	2	2	3		$\frac{1}{2}$	$\frac{1}{3}$	—	—
1259.8	126293	160801	64481201	401	20.0250	7.3742	—	2.6031
1262.9	126923	161604	64964808	402	20.0499	7.3803	$2\times3\times67$	2.6042
1266.1	127556	162409	65450827	403	20.0749	7.3864	13×31	2.6053
1269.2	128190	163216	65939264	404	20.0998	7.3925	$2^6\times101$	2.6064
1272.3	128825	164025	66430125	405	20.1246	7.3986	$3^4\times5$	2.6075
1275.5	139462	164836	66923416	406	20.1494	7.4047	$2\times7\times29$	2.6085
1278.6	130100	165649	67419143	407	20.1742	7.4108	11×37	2.6096
1281.8	130741	166464	67917312	408	20.1990	7.4169	$2^3\times3\times17$	2.6107
1284.9	131382	167281	68417929	409	20.2237	7.4229	—	2.6117
1288.1	132025	168100	68921000	410	20.2485	7.4290	$2\times5\times41$	2.6128
1291.2	132670	168921	69426531	411	20.2731	7.4350	3×137	2.6138
1294.3	133317	169744	69934528	412	20.2978	7.4410	$2^2\times103$	2.6149
1997.5	133965	170569	70444997	413	20.3224	7.4470	7×59	2.6160
1300.6	134614	171396	70957944	414	20.3470	7.4530	$2\times3^3\times23$	2.6170
1303.8	135265	172225	71473375	415	20.3715	7.4590	5×83	2.6181
1306.9	135918	173056	71991296	416	20.3961	7.4650	$2^6\times13$	2.6191
1310.0	136572	173889	72511713	417	20.4206	7.4710	3×139	2.6201
1313.2	137228	174724	73034632	418	20.4450	7.4770	$2\times11\times19$	2.6212
1316.3	137885	175561	73560059	419	20.4695	7.4829	—	2.6222
1319.5	138544	176400	74088000	420	20.4939	7.4889	$2^4\times3\times5\times7$	2.6233
1322.6	139205	177241	74618461	421	20.5183	7.4948	—	2.6243
1325.8	139867	178084	75151448	422	20.5426	7.5007	2×211	2.6253
1328.9	140531	178929	75686967	423	20.5670	7.5067	$3^2\times47$	2.6263
1332.0	141196	179776	76225024	424	20.5913	7.5126	$2^3\times53$	2.6274
1335.2	141863	180625	76765625	425	20.6155	7.5185	$5^3\times17$	2.6284
1338.3	142531	181476	77308776	426	20.6398	7.5244	$2\times3\times71$	2.6294
1341.5	143201	182329	77854483	427	20.6640	7.5302	7×61	2.6304
1344.6	143872	183184	78402752	428	20.6882	7.5361	$2^2\times107$	2.6314
1347.7	144545	184041	78953589	429	20.7123	7.5420	$3\times11\times13$	2.6325
1350.9	145220	184900	79507000	430	20.7364	7.5478	$2\times5\times43$	2.6335
1354.0	145896	185761	80062991	431	20.7605	7.5537	—	2.6345
1357.2	146574	186624	80621568	432	20.7846	7.5595	$2^4\times3^3$	2.6355
1360.3	147254	187489	81182737	433	20.8087	7.5654	—	2.6365
1363.5	147934	188356	81746504	434	20.8327	7.5712	$2\times7\times31$	2.6375
1366.6	148617	189225	82312875	435	20.8567	7.5770	$3\times5\times29$	2.6385
1369.7	149301	190096	82881856	436	20.8806	7.5828	$2^3\times109$	2.6395
1372.9	149987	190969	83453453	437	20.9045	7.5886	19×23	2.6405
1376.0	150674	191844	84027672	438	20.9284	7.5944	$2\times3\times73$	2.6415
1379.2	151363	192721	84604519	439	20.9523	7.6001	—	2.6425
1382.3	152053	193600	85184000	440	20.9762	7.6059	$2^3\times5\times11$	2.6435
1385.4	152745	194481	85766121	441	21.0000	7.6117	$3^2\times7^2$	2.6444
1388.6	153439	195364	86350888	442	21.0238	7.6174	$2\times13\times17$	2.6454
1391.7	154134	196249	86938307	443	21.0476	7.6232	—	2.6464
1394.9	154830	197136	87528384	444	21.0713	7.6289	$2^2\times3\times37$	2.6474
1398.0	155528	198025	88121125	445	21.0950	7.6346	5×89	2.6484
1401.2	156228	198916	88716536	446	21.1187	7.6403	2×223	2.6493
1404.3	156930	199809	89314623	447	21.1424	7.6460	3×149	2.6503
1407.4	157633	200704	89915392	448	21.1660	7.6517	$2^8\times7$	2.6513
1410.6	158337	201601	90518849	449	21.1896	7.6574	—	2.6523
1413.7	159043	202500	91125000	450	21.2132	7.6631	$2\times3^4\times5^3$	2.6532

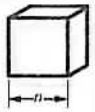
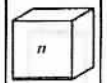
 $C=\pi\times d$	 $A=\frac{\pi\times d^2}{4}$	 $A=n^2$	 $V=n^3$	Number d or n	 $\sqrt{n}$	 $\sqrt[3]{n}$	Factors and primes	log n
1	2	2	3		$\frac{1}{2}$	$\frac{1}{3}$	—	—
1416.9	159751	203401	91733851	451	21.2368	7.6688	11×41	2.6542
1420.0	160460	204304	92345408	452	21.2603	7.6744	$2^2\times113$	2.6551
1423.1	161171	205209	92959677	453	21.2838	7.6801	3×151	2.6561
1426.3	161883	206116	93576664	454	21.3073	7.6857	2×227	2.6571
1429.4	162597	207025	94196375	455	21.3307	7.6914	$5\times7\times13$	2.6580
1432.6	163313	207936	94818816	456	21.3542	7.6970	$2^3\times3\times19$	2.6590
1435.7	164030	208849	95443993	457	21.3776	7.7026	—	2.6599
1438.8	164748	209764	96071912	458	21.4009	7.7082	2×229	2.6609
1442.0	165468	210681	96702579	459	21.4243	7.7138	$3^3\times17$	2.6618
1445.1	166190	211600	97336000	460	21.4476	7.7194	$2^4\times5\times23$	2.6628
1448.3	166914	212521	97972181	461	21.4709	7.7250	—	2.6637
1451.4	167639	213444	98611128	462	21.4942	7.7306	$2\times3\times7\times11$	2.6646
1454.6	168365	214369	99252847	463	21.5174	7.7362	—	2.6656
1457.7	169093	215296	99897344	464	21.5407	7.7418	$2^5\times29$	2.6665
1460.8	169823	216225	100544625	465	21.5639	7.7473	$3\times5\times31$	2.6675
1464.0	170554	217156	101194696	466	21.5870	7.7529	2×233	2.6684
1467.1	171287	218089	101847563	467	21.6102	7.7584	—	2.6693
1470.3	172021	219024	102503232	468	21.6333	7.7639	$2^2\times3^2\times13$	2.6703
1473.4	172757	219961	103161709	469	21.6564	7.7695	7×67	2.6712
1476.5	173494	220900	103823000	470	21.6795	7.7750	$2\times5\times47$	2.6721
1479.7	174234	221841	104487111	471	21.7025	7.7805	3×157	2.6730
1482.8	174974	222784	105154048	472	21.7256	7.7860	$2^3\times59$	2.6739
1486.0	175716	223729	105823817	473	21.7486	7.7915	11×43	2.6749
1489.1	176460	224676	106496424	474	21.7715	7.7970	$2\times3\times79$	2.6758
1492.3	177205	225625	107171875	475	21.7945	7.8025	$5^2\times19$	2.6767
1495.4	177952	226576	107850176	476	21.8174	7.8079	$2^2\times7\times17$	2.6776
1498.5	178701	227529	108531333	477	21.8403	7.8134	$3^2\times53$	2.6785
1501.7	179451	228484	109215352	478	21.8632	7.8188	2×239	2.6794
1504.8	180203	229441	109902239	479	21.8861	7.8243	—	2.6803
1508.0	180956	220400	110592000	480	21.9089	7.8297	$2^6\times3\times5$	2.6812
1511.1	181711	231361	111284641	481	21.9317	7.8352	13×37	2.6822
1514.2	182467	232324	111980168	482	21.9545	7.8406	2×241	2.6831
1517.4	183225	233289	112678587	483	21.9773	7.8460	$3\times7\times23$	2.6840
1520.5	183984	234256	113379904	484	22.0000	7.8514	$2^2\times11^2$	2.6849
1523.7	184745	235225	114084125	485	22.0227	7.8568	5×97	2.6857
1526.8	185508	236196	114791256	486	22.0454	7.8622	2×3^6	2.6866
1530.0	186272	237169	115501303	487	22.0681	7.8676	—	2.6875
1533.1	187038	238144	116214272	488	22.0907	7.8730	$2^3\times61$	2.6884
1536.2	187805	239121	116930169	489	22.1133	7.8784	3×163	2.6893
1539.4	188574	240100	117649000	490	22.1359	7.8837	$2\times5\times7^2$	2.6902
1542.5	189345	241081	118370771	491	22.1585	7.8891	—	2.6911
1545.7	190117	242064	119095488	492	22.1811	7.8944	$2^2\times3\times41$	2.6920
1548.8	190890	243049	119823157	493	22.2036	7.8998	17×29	2.6929
1551.9	191665	244036	120553784	494	22.2261	7.9051	$2\times13\times19$	2.6937
1555.1	192442	245025	121287375	495	22.2486	7.9105	$3^2\times5\times11$	2.6946
1558.2	193221	246016	122023936	496	22.2711	7.9158	$2^4\times31$	2.6955
1561.4	194000	247009	122763473	497	22.2935	7.9211	7×71	2.6964
1564.5	194782	248004	123505992	498	22.3159	7.9264	$2\times3\times83$	2.6972
1567.7	195565	249001	124251499	499	22.3383	7.9317	—	2.6981
1570.8	196350	250000	125000000	500	22.3607	7.9370	$2^3\times5^3$	2.6990

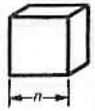
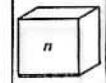
$C=\pi\times d$	$A=\frac{\pi\times d^2}{4}$	$A=n^2$	$V=n^3$	Number d or n	$\sqrt{n}$	$\sqrt[3]{n}$	Factors and primes	log n
1	2	2	3		$\frac{1}{2}$	$\frac{1}{3}$	—	—
1573.9	197136	251001	125751501	501	22.3830	7.9423	3×167	2.6998
1577.1	197923	252004	126506008	502	22.4054	7.9476	2×251	2.7007
1580.2	198713	253009	127263527	503	22.4277	7.9528	—	2.7016
1583.4	199504	254016	128024064	504	22.4499	7.9581	$2^3\times3^2\times7$	2.7024
1586.5	200296	255025	128787625	505	22.4722	7.9634	5×101	2.7033
1589.6	201090	256036	129554216	506	22.4944	7.9686	$2\times11\times23$	2.7042
1592.8	201886	257049	130323843	507	22.5167	7.9739	3×13^2	2.7050
1595.9	202683	258064	131096512	508	22.5389	7.9791	$2^2\times127$	2.7059
1599.1	303482	259081	131872229	509	22.5610	7.9843	—	2.7067
1602.2	204282	260100	132651000	510	22.5832	7.9896	$2\times3\times5\times17$	2.7076
1605.4	205084	261121	133432831	511	22.6053	7.9948	7×73	2.7084
1608.5	205887	262144	134217728	512	22.6274	8.0000	2^9	2.7093
1611.6	206692	263169	135005697	513	22.6495	8.0052	$3^3\times19$	2.7101
1614.8	207499	264196	135796744	514	22.6716	8.0104	2×257	2.7110
1617.9	208307	265225	136590875	515	22.6936	8.0156	5×103	2.7118
1621.1	209117	266256	137388096	516	22.7156	8.0208	$2^2\times3\times43$	2.7127
1624.2	209928	267289	138188413	517	22.7376	8.0260	11×47	2.7135
1627.3	210741	268324	138991832	518	22.7596	8.0311	$2\times7\times37$	2.7143
1630.5	211556	269361	139798359	519	22.7816	8.0363	3×173	2.7152
1633.6	212372	270400	140608000	520	22.8035	8.0415	$2^3\times5\times13$	2.7160
1636.8	213189	271441	141420761	521	22.8254	8.0466	—	2.7168
1639.9	214008	272484	142236648	522	22.8473	8.0517	$2\times3^2\times29$	2.7177
1643.1	214829	273529	143055667	523	22.8692	8.0569	—	2.7185
1646.2	215651	274576	143877824	524	22.8910	8.0620	$2^2\times131$	2.7193
1649.3	216475	275625	144703125	525	22.9129	8.0671	$3\times5^3\times7$	2.7202
1652.5	217301	276676	145531576	526	22.9347	8.0723	2×263	2.7210
1655.6	218128	277729	146363183	527	22.9565	8.0774	17×31	2.7218
1658.8	218956	278784	147197952	528	22.9783	8.0825	$2^4\times3\times11$	2.7226
1661.9	219787	279841	148035889	529	23.0000	8.0876	23^2	2.7235
1665.0	220618	280900	148877000	530	23.0217	8.0927	$2\times5\times53$	2.7243
1668.2	221452	281961	149721291	531	23.0434	8.0978	$3^2\times59$	2.7251
1671.3	222287	283024	150568768	532	23.0651	8.1028	$2^2\times7\times19$	2.7259
1674.5	223123	284089	151419437	533	23.0868	8.1079	13×41	2.7267
1677.6	223961	285156	152273304	534	23.1084	8.1130	$2\times3\times89$	2.7275
1680.8	224801	286225	153130375	535	23.1301	8.1180	5×107	2.7284
1683.9	225642	287296	153990656	536	23.1517	8.1231	$2^3\times67$	2.7292
1687.0	226484	288369	154854153	537	23.1733	8.1281	3×179	2.7300
1690.2	227329	289444	155720872	538	23.1948	8.1332	2×269	2.7308
1693.3	228175	290521	156590819	539	23.2164	3.1382	$7^2\times11$	2.7316
1696.5	229022	291600	157464000	540	23.2379	8.1433	$2^2\times3^3\times5$	2.7324
1699.6	229871	292681	158340421	541	23.2594	8.1483	—	2.7332
1702.7	230722	293764	159220088	542	23.2809	8.1533	2×271	2.7340
1705.9	231574	294849	160103007	543	23.3024	8.1583	3×181	2.7348
1709.0	232428	295936	160989184	544	23.3238	8.1633	$2^5\times17$	2.7356
1712.2	233283	297025	161878625	545	23.3452	8.1683	5×109	2.7364
1715.3	234140	298116	162771336	546	23.3666	8.1733	$2\times3\times7\times13$	2.7372
1718.5	234998	299209	163667323	547	23.3880	8.1783	—	2.7380
1721.6	235858	300304	164566592	548	23.4094	8.1833	$2^2\times117$	2.7388
1724.7	236720	301401	165469149	549	23.4307	8.1882	$3^3\times61$	2.7396
1727.9	237583	302500	166375000	550	23.4521	8.1932	$2\times5^3\times11$	2.7404

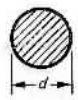
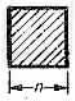
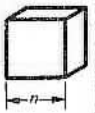
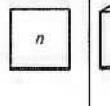
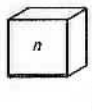
$C=\pi\times d$	$A=\frac{\pi\times d^2}{4}$	$A=n^2$	$V=n^3$	Number d or n	$\sqrt{n}$	$\sqrt[3]{n}$	Factors and primes	log n
1	2	2	3		$\frac{1}{2}$	$\frac{1}{3}$	—	—
1731.0	238448	303601	167284151	551	23.4734	8.1982	19×29	2.7412
1734.2	239314	304704	168196608	552	23.4947	8.2031	$2^3\times3\times23$	2.7419
1737.3	240182	305809	169112377	553	23.5160	8.2081	7×79	2.7427
1740.4	241051	306916	170031464	554	23.5372	8.2130	2×277	2.7435
1743.6	241922	308025	170953875	555	23.5584	8.2180	$3\times5\times37$	2.7443
1746.7	242795	309136	171879616	556	23.5797	8.2229	$2^3\times139$	2.7451
1747.9	243669	310249	172808693	557	23.6008	8.2278	—	2.7459
1753.0	244545	311364	173741112	558	23.6220	8.2327	$2\times3^2\times31$	2.7466
1756.2	245422	312481	174676879	559	23.6432	8.2377	13×43	2.7474
1759.3	246301	313600	175616000	560	23.6643	8.2426	$2^4\times5\times7$	2.7482
1762.4	247181	214721	176558481	561	23.6854	8.2475	$3\times11\times17$	2.7490
1765.6	248063	315844	177504328	562	23.7065	8.2524	2×281	2.7497
1768.7	248947	316969	178453547	563	23.7276	8.2573	—	2.7505
1771.9	249832	318096	179406144	564	23.7487	9.2621	$2^2\times3\times47$	2.7513
1775.0	250719	319225	180362125	565	23.7697	8.2670	5×113	2.7521
1778.1	251607	320356	181321496	566	23.7908	8.2719	2×283	2.7528
1781.3	252497	321489	182284263	567	23.8118	8.2768	$3^4\times7$	2.7536
1784.4	253388	322624	183250432	568	23.8328	8.2816	$2^3\times71$	2.7544
1787.6	254281	323761	184220009	569	23.8537	8.2865	—	2.7551
1790.7	255176	324900	185193000	570	23.8747	8.2913	$2\times3\times5\times19$	2.7559
1793.8	256072	326041	186169411	571	23.8956	8.2962	—	2.7566
1797.0	256970	327184	187149248	572	23.9165	8.3010	$2^2\times11\times13$	2.7574
1800.1	257869	328329	188132517	573	23.9374	8.3059	3×191	2.7582
1803.3	258770	329476	189119224	574	23.9583	8.3107	$2\times7\times41$	2.7589
1806.4	259672	330625	190109375	575	23.9792	8.3155	$5^2\times23$	2.7597
1809.6	260576	331776	191102976	576	24.0000	8.3203	$2^8\times3^2$	2.7604
1812.7	261482	332929	192100033	577	24.0208	8.3251	—	2.7612
1815.8	262389	334084	193100552	578	24.0416	8.3300	2×17^2	2.7619
1819.0	263298	335241	194104539	579	24.0624	8.3348	3×193	2.7627
1822.1	264208	336400	195112000	580	24.0832	3.3396	$2^3\times5\times29$	2.7634
1825.3	265120	337561	196122941	581	24.1039	8.3443	7×83	2.7642
1828.4	266033	338724	197137368	582	24.1247	8.3491	$2\times3\times97$	2.7649
1831.6	266948	339889	198155287	583	24.1454	8.3539	11×53	2.7657
1834.7	267865	341056	199176704	584	24.1661	8.3587	$2^3\times73$	2.7664
1837.8	268783	342225	200201625	585	24.1868	8.3634	$3^3\times5\times13$	2.7672
1841.0	269703	343396	201230056	586	24.2074	8.3682	2×293	2.7679
1844.1	270624	344569	202262003	587	24.2281	8.3730	—	2.7686
1847.3	271547	345744	203297472	588	24.2487	8.3777	$2^3\times3\times7^2$	2.7694
1850.4	272471	346921	204336469	589	24.2693	8.3825	19×31	2.7701
1853.5	273397	348100	205379000	590	24.2899	8.3872	$2\times5\times59$	2.7709
1856.7	274325	349281	206425071	591	24.3105	8.3919	3×197	2.7716
1859.8	275254	350464	207474688	592	24.3311	8.3967	$2^4\times37$	2.7723
1863.0	276184	351649	208527857	593	24.3516	8.4014	—	2.7731
1866.1	277117	352836	209584584	594	24.3721	8.4061	$2\times3^3\times11$	2.7738
1869.2	278051	354025	210644875	595	24.3926	8.4108	$5\times7\times17$	2.7745
1872.4	278986	355216	211708736	596	24.4131	8.4155	$2^2\times149$	2.7753
1875.5	279923	356409	212776173	597	24.4336	8.4202	3×199	2.7760
1878.7	280862	357604	213847192	598	24.4540	8.4249	$2\times13\times23$	2.7767
1881.8	281802	358801	214921799	599	24.4745	8.4296	—	2.7774
1885.0	282743	360000	216000000	600	24.4949	8.4343	$2^3\times3\times5^2$	2.7782

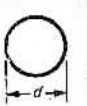
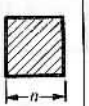
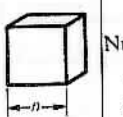
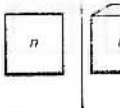
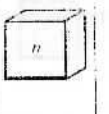
$C=\pi\times d$	$A=\frac{\pi\times d^2}{4}$	$A=n^2$	 $V=n^3$	Number d or n	$\sqrt{n}$	 $\sqrt[3]{n}$	Factors and primes	$\log n$
1	2	2	3		$\frac{1}{2}$	$\frac{1}{3}$	—	—
1888.1	283687	361201	217081801	601	24.5153	8.4390	—	2.7789
1891.2	284631	362404	218167208	602	24.5357	8.4437	$2\times7\times43$	2.7796
1894.4	285578	363609	219256227	603	24.5561	8.4484	$3^2\times67$	2.7803
1897.5	286526	364816	220348864	604	24.5764	8.4530	$2^2\times151$	2.7810
1900.7	287475	366025	221445125	605	24.5967	8.4577	5×11^2	2.7818
1903.8	288426	367236	222545016	606	24.6171	8.4623	$2\times3\times101$	2.7825
1906.9	289379	368449	223648543	607	24.6374	8.4670	—	2.7832
1910.1	290333	369664	224755712	608	24.6577	8.4716	$2^5\times19$	2.7839
1913.2	291289	370881	225866529	609	24.6779	8.4763	$3\times7\times29$	2.7846
1916.4	292247	372100	226981000	610	24.6982	8.4809	$2\times5\times61$	2.7853
1919.5	293206	373321	228099131	611	24.7184	8.4856	13×47	2.7860
1922.7	294166	374544	229220928	612	24.7386	8.4902	$2^2\times3^2\times17$	2.7868
1925.8	295128	375769	230346397	613	24.7588	8.4948	—	2.7875
1928.9	296092	376996	231475544	614	2417790	8.4994	2×307	2.7882
1932.1	297057	378225	232608375	615	24.7992	8.5040	$3\times5\times41$	2.7889
1935.2	298024	379456	233744896	616	24.8193	8.5086	$2^3\times7\times11$	2.7896
1938.4	298992	380689	234885113	617	24.8395	8.5132	—	2.7902
1941.5	299962	381924	236029032	618	24.8596	8.5178	$2\times3\times103$	2.7910
1944.6	300934	383161	237176659	619	24.8797	8.5224	—	2.7917
1947.8	301907	384400	238328000	620	24.8998	8.5270	$2^2\times5\times31$	2.7924
1950.9	302882	385641	239483061	621	24.9199	8.5316	$3^3\times23$	2.7931
1954.1	303858	386884	240641848	622	24.9399	8.5362	2×311	2.7938
1957.2	304836	388129	241804367	623	24.9600	8.5408	7×89	2.7945
1960.4	305815	389376	242970624	624	24.9800	8.5453	$2^4\times3\times13$	2.7952
1963.5	306796	390625	244140625	625	25.0000	8.5499	5^4	2.7959
1966.6	307779	391876	245314376	626	25.0200	8.5544	2×313	2.7966
1969.8	308763	393129	246491883	627	25.0400	8.5590	$3\times11\times19$	2.7973
1972.9	309748	394384	247673152	628	25.0599	8.5635	$2^2\times157$	2.7980
1976.1	310736	395641	248858189	629	25.0799	8.5681	17×37	2.7987
1979.2	311725	396900	250047000	630	25.0998	8.5726	$2\times3^2\times5\times7$	2.7993
1982.3	312715	398161	251239591	631	25.1197	8.5772	—	2.8000
1985.5	313707	399424	252435968	632	25.1396	8.5817	$2^3\times79$	2.8007
1988.6	314700	400689	253636137	633	25.1595	8.5862	3×211	2.8014
1991.8	315696	401956	254840104	634	25.1794	8.5907	2×317	2.8021
1994.9	316692	403225	256047875	635	25.1992	8.5952	5×127	2.8028
1998.1	317690	404496	257259456	636	25.2190	8.5997	$2^2\times3\times53$	2.8035
2001.2	318690	405769	258474853	637	25.2389	8.6043	$7^2\times13$	2.8041
2004.3	319692	407044	259694072	638	25.2587	8.6088	$2\times11\times29$	2.8048
2007.5	320695	408321	260917119	639	25.2784	8.6132	$3^3\times71$	2.8055
2010.6	321699	409600	262144000	640	25.2982	8.6177	$2^7\times5$	2.8062
2013.8	322705	410881	263374721	641	25.3180	8.6222	—	2.8069
2016.9	323713	412164	264609288	642	25.3377	8.6267	$2\times3\times107$	2.8075
2020.0	324722	413449	265847707	643	25.3574	8.6312	—	2.8082
2023.2	325733	414736	267089984	644	25.3772	8.6357	$2^2\times7\times23$	2.8089
2026.3	326745	416025	268336125	645	25.3969	8.6401	$3\times5\times43$	2.8096
2029.5	327759	417316	269586136	646	25.4165	8.6446	$2\times17\times19$	2.8102
2032.6	328775	418609	270840023	647	25.4362	8.6490	—	2.8109
2035.8	329792	419904	272097792	648	25.4558	8.6535	$2^3\times3^4$	2.8116
2038.9	330810	421201	273359449	649	25.4755	8.6579	11×59	2.8122
2042.0	331831	422500	274625000	650	25.4951	8.6624	$2\times5^2\times13$	2.8129

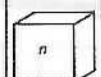
 $C=\pi\times d$	 $A=\frac{\pi\times d^2}{4}$	 $A=n^2$	 $V=n^3$	Number d or n	 $\sqrt{n}$	 $\sqrt[3]{n}$	Factors and primes	$\log n$
1	2	2	3		$\frac{1}{2}$	$\frac{1}{3}$	—	—
2045.2	332853	423801	275894451	651	25.5147	8.6668	$3\times7\times31$	2.8136
2048.3	333876	425104	277167808	652	25.5343	8.6713	$2^2\times163$	2.8143
2051.5	334901	426409	278445077	653	25.5539	8.6757	—	2.8149
2054.6	335927	427716	279726264	654	25.5734	8.6801	$2\times3\times109$	2.8156
2057.7	336955	429025	281011375	655	25.5930	8.6845	5×131	2.8162
2060.9	337985	430336	282300416	656	25.6125	8.6890	$2^4\times41$	2.8169
2064.0	339016	431649	283593393	657	25.6320	8.6934	$3^2\times73$	2.8176
2067.2	340049	432964	284890312	658	25.6515	8.6978	$2\times7\times47$	2.8182
2070.3	341083	434281	286191179	659	25.6710	8.7022	—	2.8189
2073.5	342119	435600	287496000	660	25.6905	8.7066	$2^3\times3\times5\times11$	2.8195
2076.6	343157	436921	288804781	661	25.7099	8.7110	—	2.8202
2079.7	344196	438244	290117528	662	25.7294	8.7154	2×331	2.8209
2082.9	345237	439569	291434247	663	25.7488	8.7198	$3\times13\times17$	2.8215
2086.0	346279	440896	292754944	664	25.7682	8.7241	$2^3\times83$	2.8222
2089.2	347323	442225	294079625	665	25.7876	8.7285	$5\times7\times19$	2.8228
2092.3	348368	443556	295408296	666	25.8070	8.7329	$2\times3^2\times37$	2.8235
2095.4	349415	444889	296740963	667	25.8263	8.7373	23×29	2.8241
2098.6	350464	446224	298077632	668	25.8457	8.7416	$2^3\times167$	2.8248
2101.7	351514	447561	299418309	669	25.8650	8.7460	3×223	2.8254
2104.9	352565	448900	300763000	670	25.8844	8.7503	$2\times5\times67$	2.8261
2108.0	353618	450241	302111711	671	25.9037	8.7547	11×61	2.8267
2111.2	354673	451584	303464448	672	25.9230	8.7590	$2^5\times3\times7$	2.8274
2114.3	355730	452929	304821217	673	25.9422	8.7634	—	2.8280
2117.4	356788	454276	306182024	674	25.9615	8.7677	2×337	2.8287
2120.6	357847	455625	307546875	675	25.9808	8.7721	$3^3\times5^2$	2.8293
2123.7	358908	456976	308915776	676	26.0000	8.7764	$2^2\times13^2$	2.8300
2126.9	359971	458329	310288733	677	26.0192	8.7807	—	2.8306
2130.0	361035	459684	311665752	678	26.0384	8.7850	$2\times3\times113$	2.8312
2133.1	362101	461041	313046839	679	26.0576	8.7893	7×97	2.8319
2136.3	363168	462400	314432000	680	26.0768	8.7937	$2^8\times5\times17$	2.8325
2139.4	364237	463761	315821241	681	26.0960	8.7980	3×227	2.8332
2142.6	365308	465124	317214568	682	26.1151	8.8023	$2\times11\times31$	2.8338
2145.7	366380	466489	318611987	683	26.1343	8.8066	—	2.8344
2148.8	367453	467856	320013504	684	26.1534	8.8109	$2^2\times3^2\times19$	2.8351
2152.0	368528	469225	321419125	685	26.1725	8.8152	5×137	2.8357
2155.1	369605	470596	322828856	686	26.1916	8.8194	2×7^3	2.8363
2158.3	370684	471969	324242703	687	26.2107	8.8237	3×229	2.8370
2161.4	371764	473344	325660672	688	26.2298	8.8280	$2^4\times43$	2.8376
2164.6	372845	474721	327082769	689	26.2488	8.8323	13×53	2.8382
2167.7	373928	476100	328509000	690	26.2679	8.8366	$2\times3\times5\times23$	2.8389
2170.8	375013	477481	329939371	691	26.2869	8.8408	—	2.8395
2174.0	376099	478864	331373888	692	26.3059	8.8451	$2^2\times173$	2.8401
2177.1	377187	480249	332812557	693	26.3249	8.8493	$3^2\times7\times11$	2.8407
2180.3	378276	481636	334255384	694	26.3439	8.8536	2×347	2.8414
2183.4	379367	483025	335702375	695	26.3629	8.8578	5×139	2.8420
2186.5	380459	484416	337153536	696	26.3818	8.8621	$2^3\times3\times29$	2.8426
2189.7	381553	485809	338608873	697	26.4008	8.8663	17×41	2.8432
2192.8	382649	487204	340068392	698	26.4197	8.8706	2×349	2.8439
2196.0	383746	488601	341582099	699	26.4386	8.8748	3×233	2.8445
2199.1	384845	490000	343000000	700	26.4575	8.8790	$2^2\times5^2\times7$	2.8451

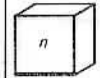
$C=\pi\times d$	$A=\frac{\pi\times d^2}{4}$	$A=n^2$	 $V=n^3$	Number d or n	$\sqrt{n}$	 $\sqrt[3]{n}$	Factors and primes	$\log n$
1	2	2	3		$\frac{1}{2}$	$\frac{1}{3}$	—	—
2202.3	385945	491401	344472101	701	26.4764	8.8833	—	2.8457
2205.4	387047	492804	345948408	702	26.4953	8.8875	$2\times3^3\times13$	2.8463
2208.5	388151	494209	347428927	703	26.5141	8.8917	19×37	2.8470
2211.7	389256	495616	348913664	704	26.5330	8.8959	$2^6\times11$	2.8476
2214.8	390363	497025	350402625	705	26.5518	8.9001	$3\times5\times47$	2.8482
2218.0	391471	498436	351895816	706	26.5707	8.9043	2×353	2.8488
2221.1	392580	499849	353393243	707	26.5895	8.9085	7×101	2.8494
2224.2	393692	501264	354894912	708	26.6083	8.9127	$2^2\times3\times59$	2.8500
2227.4	394805	502681	356400829	709	26.6271	8.9169	—	2.8507
2230.5	395919	504100	357911000	710	26.6458	8.9211	$2\times5\times71$	2.8513
2233.7	397035	505521	359425431	711	36.6646	8.9253	$3^2\times79$	2.8519
2236.8	398153	506944	360944128	712	26.6833	8.9295	$2^6\times89$	2.8525
2240.0	399272	508369	362467097	713	26.7021	8.9337	23×31	2.8531
2243.1	400393	509796	363994344	714	26.7208	8.9378	$2\times3\times7\times17$	2.8537
2246.2	401515	511225	365525875	715	26.7395	8.9420	$5\times11\times13$	2.8543
2249.4	402639	512656	367061696	716	26.7582	8.9462	$2^2\times179$	2.8549
2252.5	403765	514089	368601813	717	26.7769	8.9503	3×239	2.8555
2255.7	404892	515524	370146232	718	26.7955	8.9545	2×359	2.8561
2258.8	406020	516961	371694959	719	26.8142	8.9587	—	2.8567
2261.9	407150	518400	373248000	720	26.8328	8.9628	$2^6\times3^2\times5$	2.8573
2265.1	408282	519841	374805361	721	26.8514	8.9670	7×103	2.8579
2268.2	409415	521284	376367048	722	26.8701	8.9711	2×19^2	2.8585
2271.4	410550	522729	377933067	723	26.8887	8.9752	3×241	2.8591
2274.5	411687	524176	379503424	724	26.9072	8.9794	$2^2\times181$	2.8597
2277.7	412825	525625	381078125	725	26.9258	8.9835	$5^2\times29$	2.8603
2280.8	413965	527076	382657176	726	26.9444	8.9876	$2\times3\times11^2$	2.8609
2283.9	415106	528529	384240583	727	26.9629	8.9918	—	2.8615
2287.1	416248	529984	385828352	728	26.9815	8.9959	$2^3\times7\times13$	2.8621
2290.2	417393	531441	387420489	729	27.0000	9.0000	3^6	2.8627
2293.4	418539	532900	389017000	730	27.0185	9.0041	$2\times5\times73$	2.8633
2296.5	419686	534361	390617891	731	27.0370	9.0082	17×43	2.8639
2299.6	420835	535824	392223168	732	27.0555	9.0123	$2^6\times3\times61$	2.8645
2302.8	421986	537289	393832837	733	27.0740	9.0164	—	2.8651
2305.9	423138	538756	395446904	734	27.0924	9.0205	2×367	2.8657
2309.1	424293	540225	397065375	735	27.1109	9.0246	$3\times5\times7^2$	2.8663
2312.2	425447	541696	398688256	736	27.1293	9.0287	$2^3\times23$	2.8669
2315.4	426604	543169	400315553	737	27.1477	9.0328	11×67	2.8675
2318.5	427762	544644	401947272	738	27.1662	9.0369	$2\times3^2\times41$	2.8681
2321.6	428922	546121	403583419	739	27.1846	9.0410	—	2.8686
2324.8	430084	547600	405224000	740	27.2029	9.0450	$2^6\times5\times37$	2.8692
2327.9	431247	549081	406869021	741	27.2213	9.0491	$3\times13\times19$	2.8698
2331.1	432412	550564	408518488	742	27.2397	9.0532	$2\times7\times53$	2.8704
2334.2	433578	552049	410172407	743	27.2580	9.0572	—	2.8710
2337.3	434746	553536	411830784	744	27.2764	9.0613	$2^3\times3\times31$	2.8716
2340.5	435916	555025	413493625	745	27.2947	9.0654	5×149	2.8722
2343.6	437087	556516	415160936	746	27.3130	9.0694	2×373	2.8727
2346.8	438259	558009	416832723	747	27.3313	9.0735	$3^2\times83$	2.8733
2349.9	439433	559504	418508992	748	27.3496	9.0775	$2^2\times11\times17$	2.8739
2353 1	440609	561001	420189749	749	27.3679	9.0816	7×107	2.8745
2356.2	441786	562500	421875000	750	27.3861	9.0856	$2\times3\times5^3$	2.8751

$C=\pi\times d$	$A=\frac{\pi\times d^2}{4}$	$A=n^2$	 $V=n^3$	Number d or n	$\sqrt{n}$	 $\sqrt[3]{n}$	Factors and primes	$\log n$
1	2	2	3		$\frac{1}{2}$	$\frac{1}{3}$	—	—
2359.3	442965	564001	423564751	751	27.4044	9.0896	—	2.8756
2362.5	444146	565504	425259008	752	27.4226	9.0937	$2^6\times47$	2.8762
2365.6	445328	567009	426957777	753	27.4408	9.0977	3×251	2.8768
2368.8	446511	568516	428661064	754	27.4591	9.1017	$2\times13\times39$	2.8774
2371.9	447697	570025	430368875	755	27.4773	9.1057	5×151	2.8780
2375.0	448883	571536	432081216	756	27.4955	9.1098	$2^2\times3^3\times7$	2.8785
2378.2	450072	573049	433798093	757	27.5136	9.1138	—	2.8791
2381.3	451262	574564	435519512	758	27.5318	9.1178	2×379	2.8797
2384.5	452453	576081	437245479	759	27.5500	9.1218	$3\times11\times23$	2.8802
2387.6	453646	577600	438976000	760	27.5681	9.1258	$2^6\times5\times19$	2.8808
2390.8	454841	579121	440711081	761	27.5862	9.1298	—	2.8814
2393.9	456037	580644	442450728	762	27.6043	9.1338	$2\times3\times127$	2.8820
2397.0	457234	582169	444194947	763	27.6225	9.1378	7×109	2.8825
2400.2	458434	583696	445943744	764	27.6405	9.1418	$2^2\times191$	2.8831
2403.3	459635	585225	447697125	765	27.6586	9.1458	$3^2\times5\times17$	2.8837
2406.5	460837	586756	449455096	766	27.6767	9.1498	2×383	2.8842
2409.6	462041	588289	451217663	767	27.6948	9.1537	13×59	2.8848
2412.7	463247	589824	452984832	768	27.7128	9.1577	$2^8\times3$	2.8854
2415.9	464454	591361	454756609	769	27.7308	9.1617	—	2.8859
2419.0	465663	592900	456533000	770	27.7489	9.1657	$2\times5\times7\times11$	2.8865
2422.2	466873	594441	458314011	771	27.7669	9.1696	3×257	2.8871
2425.3	468085	595984	460099648	772	27.7849	9.1736	$2^2\times193$	2.8876
2428.5	469298	597529	461889917	773	27.8029	9.1775	—	2.8882
2431.6	470513	599076	463684824	774	27.8209	9.1815	$1\times3^2\times43$	2.8887
2434.7	471730	600625	465484375	775	27.8388	9.1855	$5^3\times31$	2.8893
2437.9	472948	602176	467288576	776	27.8568	9.1894	$2^3\times97$	2.8899
2441.0	474168	603729	469097433	777	27.8747	9.1933	$3\times7\times37$	2.8904
2444.2	475389	605284	470910952	778	27.8927	9.1973	2×389	2.8910
2447.3	476612	606841	472729139	779	27.9106	9.2012	19×41	2.8915
2450.4	477836	608400	474552000	780	27.9285	9.2052	$2^6\times3\times5\times13$	2.8921
2453.6	479062	609961	476379541	781	27.9464	9.2091	11×71	2.8927
2456.7	480290	611524	478211763	782	27.9643	9.2130	$2\times17\times23$	2.8932
2459.9	481519	613089	480048687	783	27.9821	9.2170	$3^3\times29$	2.8938
2463.0	482750	614656	481890304	784	28.0000	9.2209	$2^4\times7^2$	2.8943
2466.2	483982	616225	483736625	785	28.0179	9.2248	5×157	2.8949
2469.3	485216	617796	485587656	786	28.0357	9.2287	$2\times3\times131$	2.8954
2472.4	486451	619369	487443403	787	28.0535	9.2326	—	2.8960
2475.6	487688	620944	489303872	788	28.0713	9.2365	$2^2\times197$	2.8965
2478.7	488927	622521	491169069	789	28.0891	9.2404	3×263	2.8971
2481.9	490167	624100	493039000	790	28.1069	9.2443	$2\times5\times79$	2.8976
2485.0	491409	625681	494913671	791	28.1247	9.2482	7×113	2.8982
2488.1	492652	627264	496793088	792	28.1425	9.2521	$2^3\times3^2\times11$	2.8987
2491.3	493897	628849	498677257	793	28.1603	9.2560	13×61	2.8993
2494.4	495143	630436	500566184	794	28.1780	9.2599	2×397	2.8998
2497.6	496391	632025	502459875	795	28.1957	9.2638	$3\times5\times53$	2.9004
2500.7	497641	633616	504358336	796	28.2125	9.2677	$2^2\times199$	2.9009
2503.8	498892	635209	506261573	797	28.2312	9.2716	—	2.9015
2507.0	500145	636804	508169592	798	28.2489	9.2754	$2\times3\times7\times19$	2.9020
2510.1	501399	638401	510082399	799	28.2666	9.2793	17×47	2.9026
2513.3	502655	640000	512000000	800	28.2843	9.2832	$2^8\times5^3$	2.9031

$C=\pi\times d$	 $A=\frac{\pi\times d^2}{4}$	 $A=n^2$	 $V=n^3$	Number d or n	 $\sqrt{n}$	 $\sqrt[3]{n}$	Factors and primes	log n
1	2	2	3		$\frac{1}{2}$	$\frac{1}{3}$	—	—
2516.4	503912	641601	513922401	801	28.3019	9.2870	$3^2\times 89$	2.9036
2519.6	505171	643204	515849608	802	28.3196	9.2909	2×401	2.9042
2522.7	506432	644809	517781627	803	28.3373	9.2948	11×73	2.9047
2525.8	507694	646416	519718464	804	28.3549	9.2986	$2^2\times 3\times 67$	2.9053
2529.0	508958	648025	521660125	805	28.3725	9.3025	$5\times 7\times 23$	2.9058
2532.1	510223	649636	523606616	806	28.3901	9.3063	$2\times 13\times 31$	2.9063
2535.3	511490	651249	525557943	807	28.4077	9.3102	3×269	2.9069
2538.4	512758	652864	527514112	808	28.4253	9.3140	$2^3\times 101$	2.9074
2541.5	514028	654481	529475129	809	28.4429	9.3179	—	2.9080
2544.7	515300	656100	531441000	810	28.4605	9.3217	$2\times 3^4\times 5$	2.9085
2547.8	516573	657721	533411731	811	28.4741	9.3255	—	2.9090
2551.0	517848	659344	535387328	812	28.4956	9.3294	$2^2\times 7\times 29$	2.9096
2554.1	519124	660969	537367797	813	28.5132	9.3332	3×271	2.9101
2557.3	520402	662596	539353144	814	28.5307	9.3370	$2\times 11\times 37$	2.9106
2560.4	521681	664225	541343375	815	28.5482	9.3408	5×163	2.9112
2563.5	522962	665856	543338496	816	28.5657	9.3447	$2^4\times 3\times 17$	2.9117
2566.7	524245	667489	545338513	817	28.5832	9.3485	19×43	2.9122
2569.8	525529	669124	547343432	818	28.6007	9.3523	2×409	2.9128
2573.0	526814	670761	549353259	819	28.6182	9.3561	$3^2\times 7\times 13$	2.9133
2576.1	528102	672400	551368000	820	28.6356	9.3599	$2^3\times 5\times 41$	2.9138
2579.2	529391	674041	553387661	821	28.6531	9.3637	—	2.9143
2582.4	530681	675684	555412248	822	28.6705	9.3675	$2\times 3\times 137$	2.9149
2585.5	531973	677329	557441767	823	28.6880	9.3713	—	2.9154
2588.7	533267	678976	559476224	824	28.7054	9.3751	$2^3\times 103$	2.9159
2591.8	534562	680625	561515625	825	28.7228	9.3789	$3\times 5^3\times 11$	2.9165
2595.0	535858	682276	563559976	826	28.7402	9.3827	$2\times 7\times 59$	2.9170
2598.1	537157	683929	565609283	827	28.7576	9.3865	—	2.9175
2601.2	538456	685584	567663552	828	28.7750	9.3902	$2^2\times 3^3\times 23$	2.9180
2604.4	539758	687241	569722789	829	28.7924	9.3940	—	2.9186
2607.5	541061	688900	571787000	830	28.8097	9.3978	$2\times 5\times 83$	2.9191
2610.7	542365	690561	573856191	831	28.8271	9.4016	3×277	2.9196
2613.8	543671	692224	575930368	832	28.8444	9.4053	$2^6\times 13$	2.9201
2616.9	544979	693889	578009537	833	28.8617	9.4091	$7^3\times 17$	2.9207
2620.1	546288	695556	580093704	834	28.8791	9.4129	$2\times 3\times 139$	2.9212
2623.2	547599	697225	582182875	835	28.8964	9.4166	5×167	2.9217
2626.4	548912	698896	584277056	836	28.9137	9.4204	$2^2\times 11\times 17$	2.9222
2629.5	550226	700569	586376253	837	28.9310	9.4241	$3^3\times 31$	2.9227
2632.7	551541	702244	588480472	838	28.9482	9.4279	2×419	2.9232
2635.8	552858	703921	590589719	839	28.9655	9.4316	—	2.9239
2638.9	554177	705600	592704000	840	28.9828	9.4354	$2^5\times 3\times 5\times 7$	2.9243
2642.1	555497	707281	594823321	841	29.0000	9.4391	29^2	2.9248
2645.2	556819	708964	596947688	842	29.0172	9.4429	2×421	2.9253
2648.4	558142	710649	599077107	843	29.0345	9.4466	3×281	2.9258
2651.5	559467	712336	601211584	844	29.0517	9.4503	$2^2\times 211$	2.9263
2654.6	560794	714025	603351125	845	29.0689	9.4541	5×13^3	2.9269
2657.8	562122	715716	605495736	846	29.0861	9.4578	$2\times 3^2\times 47$	2.9274
2660.9	563452	717409	607645423	847	29.1033	9.4615	7×11^2	2.9279
2664.1	564783	719104	609800192	848	29.1204	9.4652	$2^4\times 53$	2.9284
2667.2	566116	720801	611960049	849	29.1376	9.4600	3×283	2.9289
2670.4	567450	722500	614125000	850	29.1548	9.4727	$2\times 5^3\times 12$	2.9294

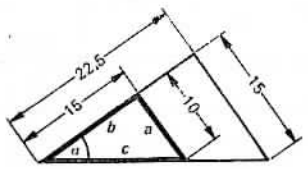
 $C=\pi\times d$	$A=\frac{\pi\times d^2}{4}$	 $A=n^2$	 $V=n^3$	Number d or n	 $\sqrt{n}$	 $\sqrt[3]{n}$	Factors and primes	log n
1	2	2	3		$\frac{1}{2}$	$\frac{1}{3}$	—	—
2673.5	568786	724201	616295051	851	29.1719	9.4764	23×37	2.9299
2676.6	570124	725904	618470208	852	29.1890	9.4801	$2^2\times 3\times 71$	2.9304
2679.8	571463	727609	620650477	853	29.2062	9.4838	—	2.9310
2682.9	572803	729316	622835864	854	29.2233	9.4875	$2\times 7\times 61$	2.9315
2686.1	574146	731025	625026375	855	29.2404	0.4912	$3^2\times 5\times 19$	2.9320
2689.2	575490	732736	627222016	856	29.2575	9.4949	$2^3\times 107$	2.9325
2692.3	576835	734449	629422793	857	29.2746	9.4986	—	2.9330
2695.5	578182	736164	631628712	858	29.2916	9.5023	$2\times 3\times 11\times 13$	2.9335
2698.6	579530	737881	633839779	859	29.3087	9.5060	—	2.9340
2701.8	580880	739600	636056000	860	29.3258	9.5097	$2^6\times 5\times 43$	2.9345
2704.9	582232	741321	638277381	861	29.3428	9.5134	$3\times 7\times 41$	2.9350
2708.1	583585	743044	640503928	862	29.3598	9.5171	2×431	2.9355
2711.2	584940	744769	642735647	863	29.3769	9.5207	—	2.9360
2714.3	586297	746496	644972544	864	29.3939	9.5244	$2^6\times 3^3$	2.9365
2717.5	587655	748225	647214625	865	29.4109	9.5281	5×173	2.9370
2720.6	589014	749956	649461896	866	29.4279	9.5317	2×433	2.9375
2723.8	590375	751689	651714363	867	39.4449	9.5354	3×17^2	2.9380
2726.9	591738	753424	653972032	868	29.4618	9.5391	$2^2\times 7\times 31$	2.9385
2730.0	593102	755161	656234909	869	29.4788	9.5427	11×79	2.9390
2733.2	594468	756900	658503000	870	29.4958	9.5464	$2\times 3\times 5\times 29$	2.9395
2736.3	595835	758641	660776311	871	29.5127	9.5501	13×67	2.9400
2739.5	597204	760384	663054848	872	29.5296	9.5537	$2^3\times 109$	2.9405
2742.6	598575	762129	665338617	873	29.5466	9.5574	$3^2\times 97$	2.9410
2745.8	599947	763876	667627624	874	29.5635	9.5610	$2\times 19\times 23$	2.9415
2748.9	601320	765625	669921875	875	29.5804	9.5647	$5^5\times 7$	2.9420
2752.0	602696	767376	672221376	876	29.5973	9.5683	$2^6\times 3\times 73$	2.9425
2755.2	604073	769129	674526133	877	29.6142	9.5719	—	2.9430
2758.3	605451	770884	676836152	878	29.6311	9.5756	2×439	2.9435
2761.5	606831	772641	679151439	879	29.6479	9.5792	3×293	2.9440
2764.6	608212	774400	681472000	880	29.6648	9.5828	$2^6\times 5\times 11$	2.9445
2767.7	609595	776161	683797841	881	29.6816	9.5865	—	2.9450
2770.9	610980	777924	686128968	882	29.6985	9.5901	$2\times 3^6\times 7^3$	2.9455
2774.0	612366	779689	688465387	883	29.7153	9.5937	—	2.9460
2777.2	613754	781456	690807104	884	29.7321	9.5973	$2^6\times 13\times 17$	2.9465
2780.3	615143	783225	693154125	885	29.7489	9.6010	$3\times 5\times 59$	2.9469
2783.5	616534	784996	695506456	886	39.7658	9.6046	2×443	2.9474
2786.6	617927	786769	697864103	887	29.7825	9.6082	—	2.9479
2789.7	619321	788544	700227072	888	29.7993	9.6118	$2^3\times 3\times 37$	2.9484
2792.9	620717	790321	702595369	889	29.8161	9.6154	7×127	2.9489
2796.0	622114	792100	704969000	890	39.8329	9.6190	$2\times 5\times 89$	2.9494
2799.2	623513	793881	707347971	891	29.8496	9.6226	$3^4\times 11$	2.9499
2802.3	624913	795664	709732288	892	29.8664	9.6262	$2^2\times 223$	2.9504
2805.4	626315	797449	712121957	893	29.8831	9.6298	19×47	2.9509
2808.6	627718	799236	714516984	894	39.8998	9.6334	$2\times 3\times 149$	2.9513
2811.7	629124	801025	716917375	895	29.9166	9.6370	5×179	2.9518
2814.9	630530	802816	719323136	896	29.9333	9.6406	$2^7\times 7$	2.9523
2818.0	631938	804609	721734273	897	29.9500	9.6442	$3\times 13\times 23$	2.9528
2821.2	633348	806404	724150792	898	29.9666	9.6477	2×449	2.9533
2824.3	634760	808201	726572699	899	29.9833	9.6513	29×31	2.9538
2827.4	636173	810000	729000000	900	30.0000	9.6549	$2^2\times 3^3\times 5$	2.9542

$C=\pi\times d$	$A=\frac{\pi\times d^2}{4}$	$A=n^2$	$V=n^3$	Number d or n	$\sqrt{n}$	 $\sqrt[3]{n}$	Factors and Primes	$\log n$
1	2	2	3		$\frac{1}{2}$	$\frac{1}{3}$	—	—
2830.6	637587	811801	731432701	901	30.0167	9.6585	17×53	2.9547
2833.7	639003	813604	733870808	902	30.0333	9.6620	$2\times 11\times 41$	2.9552
2836.9	640421	815409	736314327	903	30.9500	9.6656	$3\times 7\times 43$	2.9557
2840.0	641840	817216	738763264	904	30.0666	9.6692	$2^3\times 113$	2.9562
2843.1	643261	819025	741217625	905	30.0832	9.6727	5×181	2.9567
2846.3	644683	820836	743677416	906	30.0998	9.6763	$2\times 3\times 151$	2.9571
2849.4	646107	822649	746142643	907	30.1164	9.6799	—	2.9576
2852.6	647533	824464	748613312	908	30.1330	9.6834	$2^2\times 227$	2.9581
2855.7	648960	826281	751089429	909	30.1496	6.6870	$3^2\cdot 101$	2.9586
2858.8	650388	828100	753571000	910	30.1662	9.6905	$2\times 5\times 7\times 13$	2.9590
2862.0	651818	829921	756058031	911	30.1828	9.6941	—	2.9595
2865.1	653250	831744	758550528	912	30.1993	9.6976	$2^4\times 3\times 19$	2.9600
2868.3	654684	833569	761048497	913	30.2159	9.7012	11×83	2.9605
2871.4	656118	835396	763551944	914	30.2324	9.7047	2×457	2.9610
2874.6	657555	837225	766060875	915	30.2490	9.7082	$3\times 5\times 61$	2.9614
2877.7	658993	839056	768575296	916	20.2655	9.7118	$2^8\times 229$	2.9619
2880.8	660433	840889	771095213	917	30.2820	9.7153	7×131	2.9624
2884.0	661874	842724	773620632	918	30.2985	9.7188	$2\times 3^3\times 17$	2.9628
2887.1	663317	844561	776151559	919	30.3150	9.7224	—	2.9633
2890.3	664761	846400	778688000	920	30.3315	9.7259	$2^8\times 5\times 23$	2.9638
2893.4	666207	848241	781229961	921	30.3480	9.7294	3×307	2.9643
2896.5	667654	850084	783777448	922	30.3645	9.7329	2×461	2.9647
2899.7	669103	851929	786330467	923	30.3809	9.7364	13×71	2.9652
2902.8	670554	853776	788889024	924	30.3974	9.7400	$2^3\times 3\times 7\times 11$	2.9657
2906.0	672006	855625	791453125	925	30.4138	9.7435	$5^3\times 37$	2.9661
2909.1	673460	857476	794022776	926	30.4302	9.7470	2×463	2.9666
2912.3	674915	859329	796597983	927	30.4467	9.7505	$3^2\times 103$	2.9671
2915.4	676372	861184	799178752	928	30.4631	9.7540	$2^6\times 29$	2.9676
2918.5	677831	863041	801765089	929	30.4795	9.7575	—	2.9680
2921.7	679291	864900	804357000	930	30.4959	9.7610	$2\times 3\times 5\times 31$	2.9685
2924.8	680752	866761	806954491	931	30.5123	9.7645	$7^2\times 19$	2.9690
2928.0	672216	868624	809557568	932	30.5287	9.7680	$2^5\times 233$	2.9694
2931.1	683680	870489	812166237	933	30.5450	9.7715	3×311	2.9699
2934.2	685147	872356	814780504	934	30.5614	9.7750	2×467	2.9704
2937.4	686615	874225	817400375	935	30.5778	9.7785	$5\times 11\times 17$	2.9708
2940.5	688084	876096	820025856	936	30.5941	9.7819	$2^3\times 3^2\times 13$	2.9713
2943.7	689555	877969	822656953	937	30.6105	9.7854	—	2.9717
2946.8	691028	879844	825293672	938	30.6268	9.7889	$2\times 7\times 67$	2.9722
2950.0	692502	881721	827936019	939	30.6431	9.7924	3×313	2.9727
2953.1	693978	883600	830584000	940	30.5594	9.7959	$2^2\times 5\times 47$	2.9731
2956.2	695455	885481	833237621	941	30.6757	9.7993	—	2.9736
2959.4	696934	887364	835896888	942	30.6920	9.8028	$2\times 3\times 157$	2.9741
2962.5	698415	889249	838561807	943	30.7083	9.8063	23×41	2.9745
2965.7	699897	891136	841232384	944	30.7246	9.8097	$2^4\times 59$	2.9750
2968.8	701380	893025	843908625	945	30.7409	9.8132	$3^3\times 5\times 7$	2.9754
2971.9	702865	894916	846590536	946	30.7571	9.8167	$2\times 11\times 43$	2.9759
2975.1	704352	896809	849278123	947	30.7734	9.8201	—	2.9764
2978.2	705840	898704	851971392	948	30.7896	9.8236	$2^2\times 3\times 79$	2.9768
2981.4	707330	900601	854670349	949	30.8058	9.8270	13×73	2.9773
2984.5	708822	902500	857375000	950	30.8221	9.8305	$2\times 5^3\times 19$	2.9777

$C=\pi\times d$	$A=\frac{\pi\times d^2}{4}$	$A=n^2$	$V=n^3$	Number d or n	$\sqrt{n}$	 $\sqrt[3]{n}$	Factors and primes	$\log n$
1	2	2	3		$\frac{1}{2}$	$\frac{1}{3}$	—	—
2987.7	710315	904401	860085351	951	30.8383	9.8339	3×317	2.9782
2990.8	711809	906304	862801408	952	30.8545	9.8374	$2^3\times 7\times 17$	2.9786
2993.9	713306	908209	865523177	953	30.8707	9.8408	—	2.9791
2997.1	714803	910116	868250664	954	30.8869	9.8443	$2\times 3^3\times 53$	2.9796
3000.2	716303	912025	870983875	955	30.9031	9.8477	5×191	2.9800
3003.4	717804	913936	873722816	956	30.9192	9.8511	$2^2\times 239$	2.9805
3006.5	719306	915849	876467493	957	30.9354	9.8546	$3\times 11\times 29$	2.9809
3009.6	720810	917764	879217912	958	30.9516	9.8580	2×479	2.9814
3012.8	722316	919681	881974079	959	30.9677	9.8614	7×137	2.9818
3015.9	723823	921600	884736000	960	30.9839	9.8648	$2^6\times 3\times 5$	2.9823
3019.1	725332	923521	887503681	961	31.0000	9.8683	31^2	2.9827
3022.2	726842	925444	890277128	962	31.0161	9.8717	$2\times 13\times 37$	2.9832
3025.4	728354	927369	893056347	963	31.0322	9.8751	$3^2\times 107$	2.9836
3023.5	729867	929296	895841344	964	31.0483	9.8785	$2^2\times 241$	2.9841
3031.6	731382	931225	898632125	965	31.0644	9.8819	5×193	2.9845
3034.8	732899	933156	901428696	966	31.0805	9.8854	$2\times 3\times 7\times 23$	2.9850
3037.9	734417	935089	904231063	967	31.0966	9.8888	—	2.9854
3041.1	735937	937024	907039232	968	31.1127	9.8922	$2^3\times 11^2$	2.9859
3044.2	737458	938961	909853209	969	31.1288	9.8956	$3\times 17\times 19$	2.9863
3047.3	738981	940900	912673000	970	41.1448	9.8990	$2\times 5\times 97$	2.9868
3050.5	740506	942841	915498611	971	31.1609	9.9024	—	2.9872
3053.6	742032	944784	918330048	972	31.1769	9.9058	$2^2\times 3^5$	2.9877
3056.8	743559	946729	921167317	973	31.1929	9.9092	7×139	2.9881
3059.9	745088	948676	924010424	974	31.2090	9.9126	2×487	2.9886
3063.1	746619	950625	926859375	975	31.2250	9.9160	$3\times 5^3\times 13$	2.9890
3066.2	748151	952576	929714176	976	31.2410	9.9194	$2^4\times 61$	2.9895
3069.3	749685	954529	932574833	977	31.2570	9.9227	—	2.9899
3072.5	751221	956484	935441352	978	31.2730	9.9261	$2\times 3\times 163$	2.9903
3075.6	752758	958441	938313739	979	31.2890	9.9295	11×89	2.9908
3078.8	754296	960400	941192000	980	31.3050	9.9329	$2^7\times 5\times 7^3$	2.9912
3081.9	755837	962361	944076141	981	31.3209	9.9363	$3^3\times 109$	2.9917
3085.0	757378	964324	946966168	982	31.3369	9.9396	2×491	2.9921
3088.2	758922	966289	949862087	983	31.3528	9.9430	—	2.9926
3091.3	760466	968256	952763904	984	31.3688	9.9464	$2^9\times 3\times 41$	2.9930
3094.5	762013	970225	955671625	985	31.3847	9.9497	5×197	2.9934
3097.6	763561	972196	958585256	986	31.4006	9.9531	$2\times 17\times 29$	2.9939
3100.8	765111	974169	961504803	987	41.4166	9.9565	$3\times 7\times 47$	2.9943
3103.9	766662	976144	964430272	788	31.4325	9.9598	$2^2\times 13\times 19$	2.9948
3107.0	768214	978121	967361669	989	31.4484	9.9632	23×43	2.9952
3110.2	769769	980100	970299000	990	31.4643	9.9666	$2\times 3^2\times 5\times 11$	2.9956
3113.3	771325	982081	973242271	991	31.4802	9.9699	—	2.9961
3116.5	772882	984064	976191488	992	31.4960	9.9733	$2^5\times 31$	2.9965
3119.6	774441	986049	979146657	993	31.5119	9.9766	3×331	2.9970
3122.7	776002	988036	982107784	994	31.5278	9.9800	$2\times 7\times 71$	2.9974
3125.9	777564	990025	985074875	995	31.5436	9.9833	5×199	2.9978
3129.0	779128	992016	988047936	996	31.5595	9.9866	$2^2\times 3\times 83$	2.9983
3132.2	780693	994009	991026973	997	31.5753	9.9900	—	2.9987
3135.3	782260	996004	994011992	998	31.5911	9.9933	2×499	2.9991
3138.5	783828	998001	997002999	999	31.6070	9.9967	$3^3\times 37$	2.9996
3141.6	785398	1000000	1000000000	1000	31.6228	10.0000	$2^3\times 5^3$	3.0000

Trigonometric tables

Utilization of trigonometric tables

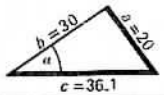


The ratio in right angled triangles that a bears to b remains constant, provided that the magnitude of the angle α remains unchanged.
e.g.

$$\alpha = 33°40' \quad \frac{a}{b} = \frac{10}{15} = 0.666 \quad \frac{a}{b} = \frac{15}{22.5} = 0.666$$

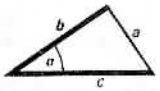
The same statement applies for the other ratios b/a, a/c, b/c. The ratios determine the magnitude of the angle as well as an angle does determine the ratios mentioned.

System of naming the elements in right angled triangles: The side c of a right angled triangle opposite the right angle is called the hypotenuse. The sides a, b including the right angle are called the adjacent sides (the catheti or the smaller sides of rectangular triangles).



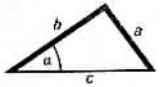
$\sin \alpha = \frac{a}{c}$

$\frac{a}{c} = \frac{20}{36.1} = 0.554$ corresponds to an angle $\alpha = 33°40'$



$\cos \alpha = \frac{b}{c}$

$\frac{b}{c} = \frac{30}{36.1} = 0.831$ corresponds to an angle $\alpha = 33°40'$



$\tan \alpha = \frac{a}{b}$

$\frac{a}{b} = \frac{20}{30} = 0.666$ corresponds to an angle $\alpha = 33°40'$

$\cot \alpha = \frac{b}{a}$

$\frac{b}{a} = \frac{30}{20} = 1.500$ corresponds to an angle $\alpha = 33°40'$

Reading angle magnitudes from tables – tables lists angles in 10′ incremental steps

Examples: $\alpha = 27°14'$ $\sin \alpha = ?$

$\sin 27°20' = 0.4592$ } values taken from tables
$\sin 27°10' = 0.4566$ }
Difference of $10' = 0.0026$
" " $1' = 0.00026$

$4' = 0.0010$
$\sin 27°10' = 0.4566$
$\sin 27°14' = 0.4576$

$0.5147 = \tan \alpha$ $\alpha = ?$

Difference 0.0015

$0.5169 = \tan 27°20'$ } values taken from tables
$0.5132 = \tan 27°10'$ }
Difference of $0.0037 = 10'$
" " $0.0001 = \frac{10'}{37}$

$0.0015 = \frac{15 \times 10}{37} = 4'$

$0.5132 = \tan 27°10'$
$0.5147 = \tan 27°14'$

Examples: $\alpha = 27°14'$ $\cos \alpha = ?$

$\cos 27°10' = 0.8897$ } values taken from tables
$\cos 27°20' = 0.8884$ }
Difference of $10' = 0.0013$
" " $1' = 0.00013$

$4' = 0.0005$
$\cos 27°10' = 0.8897$
$\cos 27°14' = 0.8892$

$1.9430 = \cot \alpha$ $\alpha = ?$

Difference 0.0056

$1.9486 = \cot 27°10'$ } values taken from tables
$1.9347 = \cot 27°20'$ }
Difference of $0.0139 = 10'$
" " $0.0001 = \frac{10'}{139}$

$0.0056 = \frac{56 \times 10}{139} = 4'$

$1.9486 = \cot 27°10'$
$1.9430 = \cot 27°14'$

Formulae for the oblique triangle

Theorem of the sines: $\frac{a}{\sin \alpha} = \frac{b}{\sin \beta} = \frac{c}{\sin \gamma}$

Theorem of the cosines:
$a^2 = b^2 + c^2 - 2bc \cos \alpha$
$b^2 = a^2 + c^2 - 2ac \cos \beta$
$c^2 = a^2 + b^2 - 2ab \cos \gamma$

Theorem of the tangent: $\frac{\tan \frac{\alpha - \beta}{2}}{\tan \frac{\alpha + \beta}{2}} = \frac{a-b}{a+b}$

Area A of a triangle: $A = \frac{1}{2} ab \sin \gamma = \frac{1}{2} bc \sin \alpha = \frac{1}{2} ac \sin \beta$

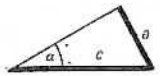
Sines 0...45°

$\sin \alpha = \frac{a}{c}$; $a = c \times \sin \alpha$; $c = \frac{a}{\sin \alpha}$

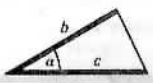
Degree	0′	10′	20′	30′	40′	50′	60′	
	Minutes							
0	0.0000	0.0029	0.0058	0.0087	0.0116	0.0145	0.0175	89
1	0.0175	0.0204	0.0233	0.0262	0.0291	0.0320	0.0349	88
2	0.0349	0.0378	0.0407	0.0436	0.0465	0.0494	0.0523	87
3	0.0523	0.0552	0.0581	0.0610	0.0640	0.0669	0.0698	86
4	0.0698	0.0727	0.0756	0.0785	0.0814	0.0843	0.0872	85
5	0.0872	0.0901	0.0929	0.0958	0.0987	0.1016	0.1045	84
6	0.1045	0.1074	0.1103	0.1132	0.1161	0.1190	0.1219	83
7	0.1219	0.1248	0.1276	0.1305	0.1334	0.1363	0.1392	82
8	0.1392	0.1421	0.1449	0.1478	0.1507	0.1536	0.1564	81
9	0.1564	0.1593	0.1622	0.1650	0.1679	0.1708	0.1736	80
10	0.1736	0.1765	0.1794	0.1822	0.1851	0.1880	0.1908	79
11	0.1908	0.1937	0.1965	0.1994	0.2022	0.2051	0.2079	78
12	0.2079	0.2108	0.2136	0.2164	0.2193	0.2221	0.2250	77
13	0.2250	0.2278	0.2306	0.2334	0.2363	0.2391	0.2419	76
14	0.2419	0.2447	0.2476	0.2504	0.2532	0.2560	0.2588	75
15	0.2588	0.2616	0.2644	0.2672	0.2700	0.2728	0.2756	74
16	0.2756	0.2784	0.2812	0.2840	0.2868	0.2896	0.2924	73
17	0.2924	0.2952	0.2979	0.3007	0.3035	0.3062	0.3090	72
18	0.3090	0.3118	0.3145	0.3173	0.3201	0.3228	0.3256	71
19	0.3256	0.3283	0.3311	0.3338	0.3365	0.3393	0.3420	70
20	0.3420	0.3448	0.3475	0.3502	0.3529	0.3557	0.3584	69
21	0.3584	0.3611	0.3638	0.3665	0.3692	0.3719	0.3756	68
22	0.3746	0.3773	0.3800	0.3827	0.3854	0.3881	0.3907	67
23	0.3907	0.3934	0.3961	0.3987	0.4014	0.4041	0.4067	66
24	0.4067	0.4094	0.4120	0.4147	0.4173	0.4200	0.4226	65
25	0.4226	0.4253	0.4279	0.4305	0.4331	0.4358	0.4384	64
26	0.4384	0.4410	0.4436	0.4462	0.4488	0.4514	0.4540	63
27	0.4540	0.4566	0.4592	0.4617	0.4643	0.4669	0.4695	62
28	0.4695	0.4720	0.4746	0.4772	0.4797	0.4823	0.4848	61
29	0.4848	0.4874	0.4899	0.4924	0.4950	0.4975	0.5000	60
30	0.5000	0.5025	0.5050	0.5075	0.5100	0.5125	0.5150	59
31	0.5150	0.5175	0.5200	0.5225	0.5250	0.5275	0.5299	58
32	0.5299	0.5324	0.5348	0.5373	0.5398	0.5422	0.5446	57
33	0.5446	0.5471	0.5495	0.5519	0.5544	0.5568	0.5592	56
34	0.5592	0.5616	0.5640	0.5664	0.5688	0.5712	0.5736	55
35	0.5736	0.5760	0.5783	0.5807	0.5831	0.5854	0.5878	54
36	0.5878	0.5901	0.5925	0.5948	0.5972	0.5995	0.6018	53
37	0.6018	0.6041	0.6065	0.6088	0.6111	0.6134	0.6157	52
38	0.6157	0.6180	0.6202	0.6225	0.6248	0.6271	0.6293	51
39	0.6293	0.6316	0.6338	0.6361	0.6383	0.6406	0.6428	50
40	0.6428	0.6450	0.6472	0.6494	0.6517	0.6539	0.6561	49
41	0.6561	0.6583	0.6604	0.6626	0.6648	0.6670	0.6691	48
42	0.6691	0.6713	0.6734	0.6756	0.6777	0.6799	0.6820	47
43	0.6820	0.6841	0.6862	0.6884	0.6905	0.6926	0.6947	46
44	0.6947	0.6997	0.6988	0.7009	0.7030	0.7050	0.7071	45
	60′	50′	40′	30′	20′	10′	0′	Degree
	Minutes							

$\cos \alpha = \frac{b}{c}$; $b = c \times \cos \alpha$; $c = \frac{b}{\cos \alpha}$ **Cosines 45...90°**

Sines 45...90°



Degree	Minutes 0′	10′	20′	30′	40′	50′	60′	
45	0.7071	0.7092	0.7112	0.7133	0.7153	0.7173	0.7193	44
46	0.7193	0.7214	0.7234	0.7254	0.7274	0.7294	0.7314	43
47	0.7314	0.7333	0.7353	0.7373	0.7392	0.7412	0.7431	42
48	0.7431	0.7451	0.7470	0.7490	0.7509	0.7528	0.7547	41
49	0.7547	0.7566	0.7585	0.7604	0.7623	0.7642	0.7660	40
50	0.7660	0.7679	0.7698	0.7716	0.7735	0.7753	0.7771	39
51	0.7771	0.7790	0.7808	0.7826	0.7844	0.7862	0.7880	38
52	0.7880	0.7898	0.7916	0.7934	0.7951	0.7969	0.7986	37
53	0.7986	0.8004	0.8021	0.8039	0.8056	0.8073	0.8090	36
54	0.8090	0.8107	0.8124	0.8141	0.8158	0.8175	0.8192	35
55	0.8192	0.8208	0.8225	0.8241	0.8258	0.8274	0.8290	34
56	0.8290	0.8307	0.8323	0.8339	0.8355	0.8371	0.8387	33
57	0.8387	0.8403	0.8418	0.8434	0.8450	0.8465	0.8480	32
58	0.8480	0.8496	0.8511	0.8526	0.8542	0.8557	0.8572	31
59	0.8572	0.8587	0.8601	0.8616	0.8631	0.8646	0.8660	30
60	0.8660	0.8675	0.8689	0.8704	0.8718	0.8732	0.8746	29
61	0.8746	0.8760	0.8774	0.8788	0.8802	0.8816	0.8829	28
62	0.8829	0.8843	0.8857	0.8870	0.8884	0.8897	0.8910	27
63	0.8910	0.8923	0.8936	0.8949	0.8962	0.8975	0.8988	26
64	0.8988	0.9001	0.9013	0.9026	0.9038	0.9051	0.9063	25
65	0.9063	0.9075	0.9088	0.9100	0.9112	0.9124	0.9135	24
66	0.9135	0.9147	0.9159	0.9171	0.9182	0.9194	0.9205	23
67	0.9205	0.9216	0.9228	0.9239	0.9250	0.9261	0.9272	22
68	0.9272	0.9283	0.9293	0.9304	0.9315	0.9325	0.9336	21
69	0.9336	0.9346	0.9356	0.9367	0.9377	0.9387	0.9397	20
70	0.9397	0.9407	0.9417	0.9426	0.9436	0.9446	0.9455	19
71	0.9455	0.9465	0.9474	0.9483	0.9492	0.9502	0.9511	18
72	0.9511	0.9520	0.9528	0.9537	0.9546	0.9555	0.9563	17
73	0.9563	0.9572	0.9580	0.9588	0.9596	0.9605	0.9613	16
74	0.9613	0.9621	0.9628	0.9636	0.9644	0.9652	0.9659	15
75	0.9659	0.9667	0.9674	0.9681	0.9689	0.9696	0.9703	14
76	0.9703	0.9710	0.9717	0.9724	0.9730	0.9737	0.9744	13
77	0.9744	0.9750	0.9757	0.9763	0.9769	0.9775	0.9781	12
78	0.9781	0.9787	0.9793	0.9799	0.9805	0.9811	0.9816	11
79	0.9816	0.9822	0.9827	0.9833	0.9838	0.9843	0.9848	10
80	0.9848	0.9853	0.9858	0.9863	0.9868	0.9872	0.9877	9
81	0.9877	0.9881	0.9886	0.9890	0.9894	0.9899	0.9903	8
82	0.9903	0.9907	0.9911	0.9914	0.9918	0.9922	0.9925	7
83	0.9925	0.9929	0.9932	0.9936	0.9939	0.9942	0.9945	6
84	0.9945	0.9948	0.9951	0.9954	0.9957	0.9959	0.9962	5
85	0.9962	0.9964	0.9967	0.9969	0.9971	0.9974	0.9976	4
86	0.9976	0.9978	0.9980	0.9981	0.9983	0.9985	0.9986	3
87	0.9986	0.9988	0.9989	0.9990	0.9992	0.9993	0.9994	2
88	0.9994	0.9995	0.9996	0.9997	0.9997	0.9998	0.99985	1
89	0.99985	0.99989	0.99993	0.99996	0.99998	0.99999	1.0000	0
	60′	50′	40′	30′	20′	10′	0′	Degree
	Minutes							



Cosines 0...45°

Tangent 0...45°

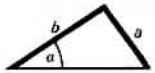
$\tan\alpha=\frac{a}{b}$; $a=b\times\tan\alpha$; $b=\frac{a}{\tan\alpha}$

Degree	Minutes 0′	10′	20′	30′	40′	50′	60′	
0	0.0000	0.0029	0.0058	0.0087	0.0116	0.0145	0.0175	89
1	0.0175	0.0204	0.0233	0.0262	0.0291	0.0320	0.0349	88
2	0.0349	0.0378	0.0407	0.0437	0.0466	0.0495	0.0524	87
3	0.0524	0.0553	0.0582	0.0612	0.0641	0.0670	0.0699	86
4	0.0699	0.0729	0.0758	0.0787	0.0816	0.0846	0.0875	85
5	0.0875	0.0904	0.0934	0.0963	0.0992	0.1022	0.1051	84
6	0.1051	0.1080	0.1110	0.1139	0.1169	0.1198	0.1228	83
7	0.1228	0.1257	0.1287	0.1317	0.1346	0.1376	0.1405	82
8	0.1405	0.1435	0.1465	0.1495	0.1524	0.1554	0.1584	81
9	0.1584	0.1614	0.1644	0.1673	0.1703	0.1733	0.1763	80
10	0.1763	0.1793	0.1823	0.1853	0.1883	0.1914	0.1944	79
11	0.1944	0.1974	0.2004	0.2035	0.2065	0.2095	0.2126	78
12	0.2126	0.2156	0.2186	0.2217	0.2247	0.2278	0.2309	77
13	0.2309	0.2339	0.2370	0.2401	0.2432	0.2462	0.2493	76
14	0.2493	0.2524	0.2555	0.2586	0.2617	0.2648	0.2679	75
15	0.2679	0.2711	0.2742	0.2773	0.2805	0.2836	0.2867	74
16	0.2867	0.2899	0.2931	0.2962	0.2994	0.3026	0.3057	73
17	0.3057	0.3089	0.3121	0.3153	0.3185	0.3217	0.3249	72
18	0.3249	0.3281	0.3314	0.3346	0.3378	0.3411	0.3443	71
19	0.3443	0.3476	0.3508	0.3541	0.3574	0.3607	0.3640	70
20	0.3640	0.3673	0.3706	0.3739	0.3772	0.3805	0.3839	69
21	0.3839	0.3872	0.3906	0.3939	0.3973	0.4006	0.4040	68
22	0.4040	0.4074	0.4108	0.4142	0.4176	0.4210	0.4245	67
23	0.4245	0.4279	0.4314	0.4348	0.4383	0.4417	0.4452	66
24	0.4452	0.4487	0.4522	0.4557	0.4592	0.4628	0.4663	65
25	0.4663	0.4699	0.4734	0.4770	0.4806	0.4841	0.4877	64
26	0.4877	0.4913	0.4950	0.4986	0.5022	0.5059	0.5095	63
27	0.5095	0.5132	0.5169	0.5206	0.5243	0.5280	0.5317	62
28	0.5317	0.5354	0.5392	0.5430	0.5467	0.5505	0.5543	61
29	0.5543	0.5581	0.5619	0.5658	0.5696	0.5735	0.5774	60
30	0.5774	0.5812	0.5851	0.5890	0.5930	0.5969	0.6009	59
31	0.6009	0.6048	0.6088	0.6128	0.6168	0.6208	0.6249	58
32	0.6249	0.6289	0.6330	0.6371	0.6412	0.6453	0.6494	57
33	0.6494	0.6536	0.6577	0.6619	0.6661	0.6703	0.6745	56
34	0.6745	0.6787	0.6830	0.6873	0.6916	0.6959	0.7002	55
35	0.7002	0.7046	0.7089	0.7133	0.7177	0.7221	0.7265	54
36	0.7265	0.7310	0.7355	0.7400	0.7445	0.7490	0.7536	53
37	0.7536	0.7581	0.7627	0.7673	0.7720	0.7766	0.7813	52
38	0.7813	0.7860	0.7907	0.7954	0.8002	0.8050	0.8098	51
39	0.8098	0.8146	0.8195	0.8243	0.8292	0.8342	0.8391	50
40	0.8391	0.8441	0.8491	0.8541	0.8591	0.8642	0.8693	49
41	0.8693	0.8744	0.8796	0.8847	0.8899	0.8952	0.9004	48
42	0.9004	0.9057	0.9110	0.9163	0.9217	0.9271	0.9325	47
43	0.9325	0.9380	0.9435	0.9490	0.9545	0.9601	0.9657	46
44	0.9657	0.9713	0.9770	0.9827	0.9884	0.9942	1.0000	45
	60′	50′	40′	30′	20′	10′	0′	Degree
	Minutes							

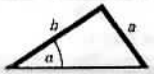
$\cot\alpha=\frac{b}{a}$; $b=a\times\cot\alpha$; $a=\frac{b}{\cot\alpha}$

Cotangent 45...90°

Tangent 45...90°



Degree	Minutes 0′	10′	20′	30′	40′	50′	60′	
45	1.0000	1.0058	1.0117	1.0176	1.0235	1.0295	1.0355	44
46	1.0355	1.0416	1.0477	1.0538	1.0599	1.0661	1.0724	43
47	1.0724	1.0786	1.0850	1.0913	1.0977	1.1041	1.1106	42
48	1.1106	1.1171	1.1237	1.1303	1.1369	1.1436	1.1504	41
49	1.1504	1.1571	1.1640	1.1708	1.1778	1.1847	1.1918	40
50	1.1918	1.1988	1.2059	1.2131	1.2203	1.2276	1.2349	39
51	1.2349	1.2423	1.2497	1.2572	1.2647	1.2723	1.2799	38
52	1.2799	1.2876	1.2954	1.3032	1.3111	1.3190	1.3270	37
53	1.3270	1.3351	1.3432	1.3514	1.3597	1.3680	1.3764	36
54	1.3764	1.3848	1.3934	1.4019	1.4106	1.4193	1.4281	35
55	1.4281	1.4370	1.4460	1.4550	1.4641	1.4733	1.4826	34
56	1.4826	1.4919	1.5013	1.5108	1.5204	1.5301	1.5399	33
57	1.5399	1.5497	1.5597	1.5697	1.5798	1.5900	1.6003	32
58	1.6003	1.6107	1.6213	1.6318	1.6426	1.6534	1.6643	31
59	1.6643	1.6753	1.6864	1.6977	1.7090	1.7205	1.7321	30
60	1.7321	1.7438	1.7556	1.7675	1.7796	1.7917	1.8041	29
61	1.8041	1.8165	1.8291	1.8418	1.8546	1.8676	1.8807	28
62	1.8807	1.8940	1.9074	1.9210	1.9347	1.9486	1.9626	27
63	1.9626	1.9768	1.9912	2.0057	2.0204	2.0353	2.0503	26
64	2.0503	2.0655	2.0809	2.0965	2.1123	2.1283	2.1445	25
65	2.1445	2.1609	2.1775	2.1943	2.2113	2.2286	2.2460	24
66	2.2460	2.2637	2.2817	2.2998	2.3183	2.3369	2.3558	23
67	2.3559	2.3750	2.3945	2.4142	2.4342	2.4545	2.4751	22
68	2.4751	2.4960	2.5172	2.5387	2.5605	2.5826	2.6051	21
69	2.6051	2.6279	2.6511	2.6746	2.6985	2.7228	2.7475	20
70	2.7475	2.7725	2.7980	2.8239	2.8502	2.8770	2.9042	19
71	2.9042	2.9319	2.9600	2.9887	3.0178	3.0475	3.0777	18
72	3.0777	3.1084	3.1397	3.1716	3.2041	3.2371	3.2709	17
73	3.2709	3.3052	3.3402	3.3759	3.4124	3.4495	3.4874	16
74	3.4874	3.5261	3.5656	3.6059	3.6470	3.6891	3.7321	15
75	3.7321	3.7760	3.8208	3.8667	3.9136	3.9617	4.0108	14
76	4.0108	4.0611	4.1126	4.1653	4.2193	4.2747	4.3315	13
77	4.3315	4.3897	4.4494	4.5107	4.5736	4.6383	4.7046	12
78	4.7046	4.7729	4.8430	4.9152	4.9894	5.0658	5.1446	11
79	5.1446	5.2257	5.3093	5.3955	5.4845	5.5764	5.6713	10
80	5.6713	5.7694	5.8708	5.9758	6.0844	6.1970	6.3138	9
81	6.3138	6.4348	6.5605	6.6912	6.8269	6.9682	7.1154	8
82	7.1154	7.2687	7.4287	7.5958	7.7704	7.9530	8.1444	7
83	8.1444	8.3450	8.5556	8.7769	9.0098	9.2553	9.5144	6
84	9.5144	9.7882	10.0780	10.3854	10.7019	11.0594	11.4301	5
85	11.4301	11.8262	12.2505	12.7062	13.1969	13.7267	14.3007	4
86	14.3007	14.9244	15.6048	16.3499	17.1693	18.0750	19.0811	3
87	19.0811	20.2056	21.4704	22.9038	24.5418	26.4316	28.6363	2
88	28.6363	31.2416	34.3678	38.1885	42.9641	49.1039	57.2900	1
89	57.2900	68.7501	85.9398	114.5887	171.885	343.774	∞	0
	60′	50′	40′	30′	20′	10′	0′	Degree
	Minutes							



Cotangent 0...45°

Inches and millimetres

(a) Inches to millimetres Basis: 1 inch = 25.4 millimetres

Inch	0	1/16	1/8	3/16	1/4	5/16	3/8	7/16
0		1.59	3.18	4.76	6.35	7.94	9.53	11.11
1	25.40	26.98	28.58	30.16	31.75	33.34	34.93	36.51
2	50.80	52.39	53.97	55.56	57.15	58.74	60.33	61.91
3	76.20	77.79	79.38	80.96	82.55	84.14	85.73	87.31
4	101.60	103.19	104.78	106.36	107.95	109.54	111.13	112.71
5	127.00	128.59	130.18	131.76	133.35	134.94	136.53	138.11
6	152.40	153.99	155.58	157.16	158.75	160.34	161.93	163.51
7	177.80	179.39	180.98	182.56	184.15	185.74	187.33	188.91
8	203.20	204.79	206.38	207.96	209.55	211.14	212.73	214.31
9	228.60	230.19	231.78	233.36	234.95	236.54	238.13	239.71
10	254.00	255.59	257.18	258.76	260.35	261.94	263.53	265.11
Inch	1/2	9/16	5/8	11/16	3/4	13/16	7/8	15/16
0	12.70	14.29	15.88	17.46	19.05	20.64	22.23	23.81
1	38.10	39.69	41.28	42.86	44.45	46.04	47.63	49.21
2	63.50	65.09	66.68	68.26	69.85	71.44	73.03	74.61
3	88.90	90.49	92.08	93.66	95.25	96.84	98.43	100.01
4	114.30	115.89	117.48	119.06	120.65	122.24	123.83	125.41
5	139.70	141.29	142.88	144.46	146.05	147.64	149.23	150.81
6	165.10	166.69	168.28	169.86	171.45	173.04	174.63	176.21
7	190.50	192.09	193.68	195.26	196.85	198.44	200.03	201.61
8	215.90	217.49	219.08	220.66	222.25	223.84	225.43	227.01
9	241.30	242.89	244.48	246.06	247.65	249.24	250.83	252.41
10	266.70	268.29	269.88	271.46	273.05	274.64	276.23	277.81

Example: $25^3/_4'' = \begin{cases} 20'' = (10 \times 2'' = 10 \times 50.8 =) & 508.00 \\ 5^3/_4'' = & 146.05 \end{cases} = 654.05$ mm

(b) Millimetres to inches Basis: 1 Millimetre = 0.039369 inch

mm	0	1	2	3	4	5	6	7	8	9
0		0.039	0.079	0.118	0.157	0.197	0.236	0.276	0.315	0.354
10	0.394	0.433	0.472	0.512	0.551	0.591	0.630	0.669	0.700	0.748
20	0.787	0.827	0.866	0.905	0.945	0.984	1.024	1.063	1.102	1.142
30	1.181	1.220	1.259	1.299	1.338	1.378	1.417	1.457	1.496	1.535
40	1.575	1.614	1.653	1.693	1.732	1.772	1.811	1.850	1.890	1.929
50	1.968	2.007	2.047	2.087	2.126	2.165	2.205	2.244	2.283	2.323
60	2.362	2.401	2.441	2.480	2.520	2.559	2.598	2.638	2.677	2.716
70	2.756	2.795	2.835	2.874	2.913	2.953	2.992	3.031	3.071	3.110
80	3.149	3.189	3.228	3.268	3.307	3.346	3.386	3.425	3.464	3.504
90	3.543	3.583	3.622	3.661	3.701	3.740	3.779	3.819	3.858	3.897
mm	0	100	200	300	400	500	600	700	800	900
0		3.94	7.87	11.81	15.75	19.68	23.62	27.56	31.49	35.43
1000	39.37	43.30	47.24	51.18	55.12	59.05	62.99	66.93	70.86	74.80
2000	78.74	82.67	86.61	90.55	94.48	98.42	102.36	106.30	110.23	114.17
3000	118.11	122.04	125.98	129.92	133.85	137.79	141.73	145.66	149.60	153.54
4000	157.47	161.41	165.35	169.20	173.22	177.16	181.10	185.03	188.97	192.91
5000	196.84	204.71	212.59	220.38	228.34	236.21	244.09	251.96	259.83	267.71

Example: 2256 mm = $\begin{cases} 2200 \text{ mm} = 86.61'' \\ 56 \text{ mm} = 2.205'' \end{cases} = 88.815''$

Greek alphabet

Α α	a alpha	Ζ ζ	z zeta	Λ λ	l lambda	Π π	p pi	Φ φ	ph phi
Β β	b beta	Η η	e eta	Μ μ	m mu	Ρ ρ	r rho	Χ χ	ch chi
Γ γ	g gamma	Θ θ	th theta	Ν ν	n nu	Σ σ	s sigma	Ψ ψ	ps psi
Δ δ	d delta	Ι ι	i iota	Ξ ξ	x xi	Τ τ	t tau	Ω ω	o omega
Ε ε	e epsilon	Κ κ	k kappa	Ο ο	o omicron	Υ υ	u upsilon		

Units and symbols of physical quantities

IS: 1890 (Part III)–1961 and IS: 3616–1961

	Symbol	Physical quantity	Conventional units	S.I. Units	Symbol for S.I. Unit
Geometrical Quantities	*l*	Length	m	Metre	m
	h	Height	m	Metre	m
	b	Width, breadth	m	Metre	m
	r	Radius	m	Metre	m
	d	Diameter	m	Metre	m
	d, *δ*	Wall thickness	m	Metre	m
	s	Length of path	m	Metre	m
	A (*S*)	Area	m^2	Square metre	m^2
	V (*v*)	Volume	m^3	Cubic metre	m^3
	α, β, γ etc.	Angle	°	Radian (1 rad = 57.3°)	rad
	λ	Wave length	km	Kilometre	km
	I, *Ia*	Second moment of area	cm^4	Metre to the fourth power	m^4
Mass	*m*	Mass	kg	Kilogram	kg
	e, ρ	Density	g/cm^3	Kilogram per cubic metre	$kg\ m^3$
	I, *J*	Moment of inertia (dynamic moment of inertia)	$kg.m^2$	Newton metre squared	$N.m^2$
Time	*T*	Time or time linterva	*s*	Second	*s*
	n	Rotational frequency	1/min	Reciprocal second	$1/s, s^{-1}$
	u, *v*, *w*, *c*	Velocity speed	m/min	Metre per second	m/s
	ω	Angular velocity	rad/s	Radian per second	rad/s
	g	Acceleration of freefall	m/s^2	Metre per second squared	m/s^2
	a	Acceleration	m/s^2	Metre per second squared	m/s^2
		Retardation	m/s^2	Metre per second squared	m/s^2
Force and pressure	*F*	Force	kgf	Newton (1kgf = 9.80665N)	N (kgm/s^2)
	G(*P*, *W*)	Weight,	kgf	Newton	N (kgm/s^2)
	γ	Specific weight	kgf/m^3	Newton per cubic metre	N/m^3
	M	Moment of force (force × distance)	kgf.m	Newton metre	N.m
	P	Pressure (force/area)	kgf/cm^2	Pascal, Newton per square metre	Pa, N/m^2
	σ	Normal stress	kgf/mm^2	bar (1 bar = 10 N/m)	
	τ	Shear stress	kgf/mm^2	bar	
	E	Modulus of elasticity	kgf/mm^2	Newton per square metre.	N/m^2
	G	Shear modulus	kfg/mm^2	Newton per square metre	N/m^2
	μ, (*f*)	Coefficient of friction			
Temperature	*t*	Temperature based on the melting point of ice	°C		
	t	Degree Kelvin	°K	Kelvin	K
		Coeff. of linear expansion	$°C^{-1}, K^{-1}$		
		Coeff. of cubial expansion			
Heat, work, energy	*A*, *W*	Work	kgfm	Joule (1 Joule = 1 N.m)	J (Nm)
	P	Power	kgfm/s	Watt	W (J/s)
	E, *W*	Energy	kgfm	Joule	J (Nm)
	η	Efficiency	—	—	—
	W, *A*, *E*, *Q*	Quantity of heat	kcal	Joule	J
		Specific heat	kcal/kgf°C	Joule per newton per degree Kelvin	J/N.°K
		Thermal conductivity	kcal/mh°C	Joule per metre per second per degree Kelvin	J/ms°K
Electrical quantities	*E*	Electric potential	V	Volt	V (W/A)
	E	Electromotive force	V	Volt	V (W/A)
	I	Electric current	A	Ampere	A
	R	Electric resistance	Ω	Ohm	Ω (V/A) Vm/A
	e	Specific resistance	Ω mm²/m		
		Conductance	Ω^{-1}	Siemens	S

Units of physical quantities

Units of length

Micron	1 μ	= 0.001 mm
Millimetre	1 mm	= 1000 μ
Centimetre	1 cm	= 10 mm
Decimetre	1 dm	= 10 cm
Metre	1 m	= 10 dm
Kilometre	1 km	= 1000 m
Inch	1″	= 25.4 mm
Foot	1′	= 0.305 m
Yard	1 Yd	= 0.914 m
Nautical mile	1	= 1852 m
Geographical mile	1	= 7420 m

Units of area

Square millimetre	1 mm^2	
,, centimetre	1 cm^2	= 100 mm^2
,, decimetre	1 dm^2	= 100 cm^2
,, metre	1 m^2	= 100 dm^2
Are	1 a	= 100 m^2
Hectare	1 ha	= 100 a
Square kilometre	1 km^2	= 100 ha
,, inch	1 sq. in	= 6.45 cm^2
,, foot	1 sq. ft	= 0.093 m^2
,, yard	1 sq. yd	= 0.84 m^2
Acre	1	= 40.5 a

Units of volume and capacity

Cubic millimetre	mm^3	
,, centimetre	1 cm^3	= 1000 mm^3
,, decimetre	1 dm^3	= 1000 cm^3
,, metre	1 m^3	= 1000 dm^3
Litre	1 *l*	= 1 dm^3
Hectolitre	1 hl	= 100 *l*
Cubic inch	1 cu. in	= 16.387 cm^3
,, foot	1 cu. ft	= 28317 cm^3
Gallon (British)	1	= 4.54 *l*

Units of weight

Milligram-force	1 mgf	
Gram-force	1 gf	= 1000 mgf
Kilogram-force	1 kgf	= 1000 gf
Ton	1 t	= 1000 kgf
Ounce	1	= 28.35 gf
Pound	1 lbs	= 0.454 kgf
Long ton	1	= 1016 kgf
Short ton	1	= 907 kgf

Time

Second	1 s	
Minute	1 m	= 60 s
Hour	1 h	= 60 m

Angle

Second	1″	grade	1ᵍ
Minute	1′ = 60″	100ᵍ	= 90°
Degree	1° = 60′		
1°	$= \frac{\pi}{180}$ rad	1 rad	= 57.3°

Work

Kilogram-force	1 kgfm	= 9.80665 J
Metre	1 kgfm	= 9.80665 Ws
Joule	1 J	= 1 Nm = 1 Ws
Watt-second	1 Ws	= 0.102 kgfm
Kilowatt hour	1 kWh	= 3.6 × 10^6 J = 859.845 $kcal_{IT}$
I.T. Kilocalorie	1 $kcal_{IT}$	= 426.9 kgfm

Power

Kilogram-force metre/second

1 kgfm/s	= 9.80665 W
Kilowatt	1 kW = 1000 W = 1000 J/s = 102 kgfm/s (approx.)
Metric horse power	1 HP = 75 kgfm/s = 0.736 kW
I. T. Kilocalorie/hour	= 1 $kcal_{IT/h}$ = 1.163 W

Pressure

Pascal	1 Pa	= 1 N/m^2		1 atm	= 101325 Pa
Bar	1 bar	= 100000 Pa	Torr	1 torr	$= \frac{101325}{760}$ Pa
Atmosphere	1 atm	≈ 1 kgf/cm^2	1 kgf/cm^2 = 735.6 mm of mercury		

Mechanics—Dynamics and Kinetics

IS: 1890 Part I/III

Units of physical quantities, fundamental laws of mechanics, resolution of forces

SI Unit of force

1 newton, is that force which, when applied to a body having a mass of one kilogram, gives it an acceleration of one metre per second squared —(1 m/s²).

Conventional Unit of force

1 kilogram-force, is the force which, when applied to a body having a mass of 1 kg, gives it an acceleration of 9.80665 m/s²=standard acceleration of free fall.

Weight

The weight of a body is that force which, when applied to the body, would give it an acceleration equal to the local acceleration of force fall. Unit of weight is kgf. The quantity here defined has commonly been called the local "gravitational" force on the body. It is noteworthy that the "weight" arises not only from the gravitational forces existing at the place where the body is, but also from the centrifugal force. As the effect of atmospheric buoyancy is excluded, the weight defined is the weight in vacuum.

Mass

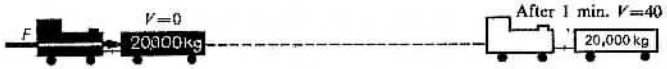
Unit of Mass is **kilogram**. 1 kilogram is the mass of the platinum-iridium cylinder deposited at the International Bureau of Weights and Measures and declared international proto-type of the kilogram by the First General Conference of Weights and Measures.

Fundamental law of Dynamics

Force=Mass × Acceleration $\boxed{F=m\times a}$

m=mass
a=acceleration (e.g. in m/s²)
F=force (e.g. in kgf)

Example: Mass of a train is 20,000 kilograms. How much tractive force a locomotive should exert to accelerate this train starting from rest so that it reaches a velocity of 40 km/h after one minute ?



The acceleration $a=\frac{V}{t}=\frac{40\text{ km/h}}{1\text{ min.}}=\frac{40{,}000\text{m}}{3600\text{5.60s}}=0.185\text{ m/s}^2$

The tractive force $F=m\times a=20{,}000\text{ kg}\times 0.185\text{ m/s}^2=3700\text{ kgf}$

Graphical representation of Forces

The effect of a force depends upon its magnitude, its direction and its point of application

Forces are represented graphically by pointed arrow

Point of Application of Force=Origin of Arrow

Magnitude of Force =Length of Arrow, e.g. 1 cm =5 kgf
4 cm =20 kgf

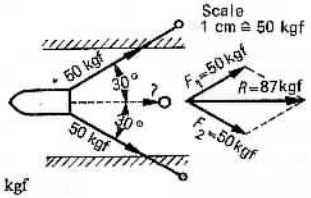
Direction of Force =Direction of Arrow

Law of Parallelogram of Forces

When two forces F_1 and F_2 act on a body at an angle, their resultant can be found out by representing the forces graphically and by completing the Parallelogram.

Example: Two ropes, one on each bank of a river, are holding a boat. Angle between the ropes is 60°. Force acting on each rope is 50 kgf. What force is needed to hold the boat if only one rope held in the middle of river is used ?

The diagonal R of the parallelogram shows the resultant force=87 kgf

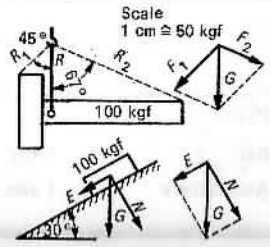


Resolving a Force

Using the parallelogram, one can resolve a force into two (or more) components.

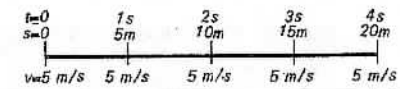
1. **Example**: A rope is to be replaced by two other ropes R_1 and R_2. What forces F_1 and F_2 are acting on the individual ropes ?

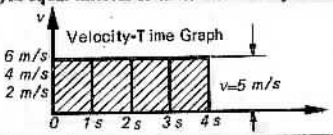
2. **Example**: A body weighing 100 kgf rests on an inclined plane having an angle of 30°. Find the component of the weight acting normal to the inclined plane. The force G caused by the weight of 100 kgf can be resolved into two components E and N by completing the parallelogram having its diagonal=$G\times N$=87 kgf. This problem can also be solved by applying trigonometry.



Uniform velocity

A body is said to have a uniform velocity if it moves equal distances (s) in equal interval of time. The velocity remains constant. Velocity (v) is the distance (s) travelled in unit time (t).





Linear and Curvilinear Motion

Velocity=$\frac{\text{Distance covered}}{\text{time}}$ $\boxed{v=\frac{s}{t}}$

Distance=Velocity × time $\boxed{s=v\times t}$

Time=$\frac{\text{Distance covered}}{\text{velocity}}$ $\boxed{t=\frac{s}{v}}$

Example: Planing

s=Length of workpiece=L=1800 mm=$\frac{1800}{1000}$ m

t=Time required per stroke=5.45=$\frac{5.4}{60}$ min

$v=\frac{L}{t}=\frac{\frac{1800}{1000}\text{ m}}{\frac{5.4}{60}\text{ min}}=\frac{1800\times 60}{1000\times 5.4}=20\text{ m/min}$

Rotary Motion

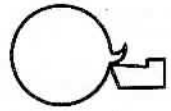
Peripheral speed=Circumference of workpiece multiplied by revolutions per unit time

$\boxed{v=\pi\times d\times n}$

Diameter=$\frac{\text{Peripheral speed}}{\pi\times\text{rpm}}$ $\boxed{d=\frac{v}{\pi\times n}}$

rpm=$\frac{\text{Peripheral speed}}{\text{Circumference of work}}$ $\boxed{n=\frac{v}{\pi\times d}}$

Example: Turning



d=60 mm=$\frac{60}{1000}$ m

n=100 rpm

$v=\pi\times d\times n=\frac{3.14\times 60\times 100}{1000}=18.89=18\text{ m/min}$

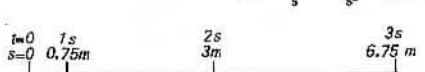
Uniformly accelerated motion

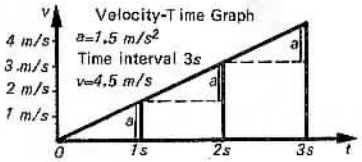
The velocity (v) gradually increases per second at a constant rate a. The acceleration (or retardation) is the gradual rate of increase (or decrease) of the velocity per unit of time.

Example: If the velocity increases per second by 1.5 m/s the acceleration

$a=\frac{1.5\text{m/s}}{\text{s}}=\frac{1.5\text{ m}}{\text{s}^2}=1.5\text{ ms}^{-2}$

Velocity—Time—Graph





Initial velocity=v=o

The velocity after t seconds $\boxed{v=a\times t}$ in m/s, Acceleration $\boxed{a=\frac{v}{t}}$ in m/s², Distance covered $s=\frac{v}{2}\times t=\frac{a\times t}{2}\times t=\frac{a}{2}\times t^2$ in m

Example: a=1 cm/s²; t=2h; v=?

$v=a\times t=1\text{ cm/s}^2\times 2h=\frac{1\text{ cm}\times 7200\text{s}}{\text{s}^2}=7200\text{ cm/s}=72\text{m/s}$

v=6m/s; t=3s; a=?; s=?

$a=\frac{v}{t}=\frac{6\text{m/s}}{3\text{s}}=2\text{m/s}^2$ or $a=2\text{ms}^{-2}$

$s=\frac{v}{2}\times t=\frac{3\text{m}}{\text{s}}\times 3\text{s}=9\text{m}.$

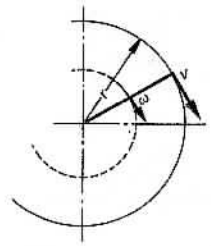
Angular velocity

Angular velocity of a rotating body is the distance covered per second by a point lying at a distance of 1 metre from the axis of rotation, along the periphery of the circle of 1 metre radius and having its centre on the axis of rotation.

Angular velocity ω (rad/s)=$\frac{3.14\text{ rpm}}{30}$ $\boxed{\omega=\frac{\pi\times n}{30}=0.105\times n}$ n=rpm, ω in rad/s

ω is approx.=1/10 × rpm

Angular velocity ω (rad/s)=$\frac{\text{Peripheral speed } v\text{ (m/s)}}{\text{Radius } r\text{ (m)}}$ $\boxed{\omega=\frac{v}{r}}$ v in m/s, r in m, ω in rad/s



Example: n=600 rpm; d=400 mm=0.4 m; ω=?; v=?

Solution: $\omega=\frac{\pi\times n}{30}=\frac{3.14\times 600}{30}=62.8\text{ rad/s}$; $v=\omega\times r=62.8\times 0.2=12.56\text{ m/s}$

Work

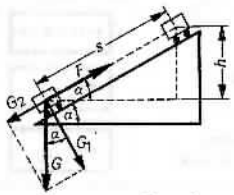
Work A=Force F×Distance covered s (s is measured in the direction of force).

$\boxed{A=F\times s}$ A, the work done is expressed in kgf m

$A=G\times h$

Example: When a body weighing 25 kgf is lifted to a height s=6 metres or alternatively a force of 25 kgf applied uniformly to a body displaces it by 6 metres, the work done

$A=F\times s$=25 kgf×6m=150 kgfm



The same amount of work is done while lifting a load along an inclined plane, as would have been done in lifting the same load vertically upwards.

Example: How much work is done in moving a vehicle weighing 5000 kgf, through a distance of 1 km, along an inclined plane having a gradient 1:100?

G=5000 kgf; $h=\frac{1000}{100}$=10 metres

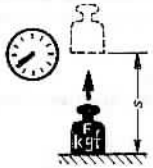
$A=G\times h$=5000 kgf×10 m=50,000 kgfm

F also can be found out by using the parallelogram of forces. G is resolved into two components G_1 and G_2. $F=G_2$, but acts in opposite direction.

$\therefore\ F=G_2=G\times\sin\alpha=5000\times\frac{1}{100}$=50 kgf, or $A=F\times s$=50 kgf×1000m=50,000 kgfm

Power

Power P=Work A per unit of time



$\boxed{P=\frac{A}{t}}$ If A is in kgfm; t in sec, then Power is expressed in kgf m/s

$\boxed{P=\frac{F\times s}{t}}$ F in kgf; s in m; t in sec then P is expressed in kgf m/s

$\boxed{P=F\times v}$ F in kgf, v in m/s P expressed in kgf m/s

Example: A load of 25 kgf lifted vertically upwards to a height h=6m in 5 seconds corresponds to a power P=30 kgf m/s

$P=\frac{A}{t}=\frac{F\times s}{t}=F\times v$

$=\frac{25\ \text{kgf}\times 6\ \text{m}}{5\ \text{sec}}$

=30 kgf m/s

Power P=Force F× Velocity v

F 5 kgf m/s=1 metric horse power=0.736 kW

$\boxed{\text{HP}=\frac{F\times v}{75}}$

Example: What should be the rating of a motor for lifting a load of 1000 kgf to a height of 12 metres in 1 min?

$A=F\times s$=1000 kgf. 12 m=12000 kgfm

102 kgf m/s=1 kW=1.36 metric HP.

$\boxed{\text{kW}=\frac{F\times v}{102}}$

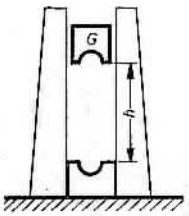
$P=\frac{A}{t}=\frac{12000\ \text{kgfm}}{60\ \text{s}}=200\ \text{kgfm/s}=\frac{200}{102}\ \text{kW}\approx 2\text{kW}$

Potential and kinetic energy

Potential energy E_p = the work of a body can do as a result of its position.
Kinetic energy E_k = the work a body can do as a result of its velocity.

Potential energy $\boxed{E_p=G\times h}=m\times g\times h$

m=Mass
g=Local acceleration of free fall
h=Height in m. Then
E_p=is expressed in kgfm



Kinetic energy $\boxed{E_k=\frac{m\times v^2}{2}}$ $m=\frac{G(\text{kgf})}{g(\text{m/s}^2)}$

$v^2=2\times g\times h$; E_k in kgfm

Example: Weight of drop forge hammer=G=250 kgf, Height of fall h=2m, then
Potential Energy=$E_p=G\times h$=250 kgf×2m=500 kgfm. The Kinetic energy attained by the hammer after falling down through the distance of 2m (i.e. at the time of striking the piece to be forged) must also be same as its original potential energy.

Example: $E_k=\frac{m\times v^2}{2}=\frac{250\ \text{kgf}\times 2\text{m}\times g\times 81\ \text{m/s}^2\times 2}{g\times 81\ \text{m/s}^2\times 2}$=500 kgfm

g=local acceleration of free fall, is taken as 9.81 m/s²

SPK

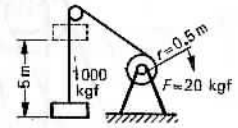
Efficiency

While transmitting power or changing energy from one form to another, part of the power or energy is lost due to friction, radiation, etc. Hence the output is always less than input.

Efficiency $\eta=\frac{\text{Output Work}}{\text{Input Work}}$ $\boxed{\eta=\frac{A\ (\text{output})}{A\ (\text{input})}}$ or $\eta=\frac{\text{Output Power}}{\text{Input Power}}$ $\%\eta=\frac{\text{Output}}{\text{Input}}\times 100$

The more η approaches unity, better the machine.

1. **Example**: 100 revolutions of the winch handle are required for raising a load of 1000 kgf up to a height of 5m. What is the efficiency?

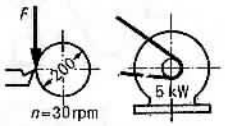


Output work=$G\times h$=1000 kgf×5m=5000 kgfm

Input work $=F\times s=F\times\pi\times d\times 100$=20kfg×π×1m×100=6280 kgfm

$=\frac{\text{Output work}}{\text{Input work}}=\frac{5000\ \text{kgfm}}{6280\ \text{kgfm}}=0.8=80\%$

2. **Example**: What cutting force is required while turning a work piece of 200 mm dia. at 30 rpm? Drive motor is of 5 kW and efficiency of drive =80% rpm.



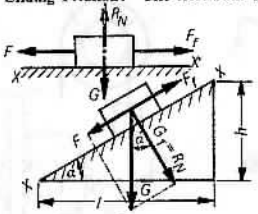
$\eta=\frac{\text{Output power}}{\text{Input power}}$ $P_{(\text{output})}=\eta\times P_{(\text{input})}=0.8\times 5$ kW=4 kW

$P_{(\text{output})}=F\times v$ $F=\frac{P_{(\text{input})}}{v}=\frac{4\text{kW}}{\pi dn}=\frac{4\times 102\ \text{kgf}\times\text{m/s}}{\pi\times 0.2\text{m}\times\frac{30}{60}\ s}$=1300 kgf

Efficiency	Steam Turbines	Water Turbines	Steam Engines	I. C. Engines	Electric Motors
	0.18...0.22	0.85...0.9	0.15...0.2	0.25...0.30	0.8...0.9

Friction

Sliding Friction: The frictional resistance F_f depends upon material of the body and sliding surface, its surface finish and the weight of the sliding body. For a body sliding uniformly on a horizontal plane, the driving force F equals the frictional resistance F_f. The ratio of the frictional force F_f to the normal reaction R_N (i.e. the reaction caused due to the weight of the body at right angles to the supporting plane xx). This ratio is known as coefficient of friction μ.



$\boxed{\mu=\frac{F_f}{R_N}}$

F_f=Force of friction kgf=F

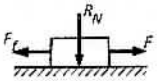
R_N=Normal reaction, acting at right angles to supporting plane, in kgf

$F=F_f=\mu\times R_N$ μ=Coefficient of friction

$\boxed{\mu=\tan\alpha=\frac{h}{l}}$

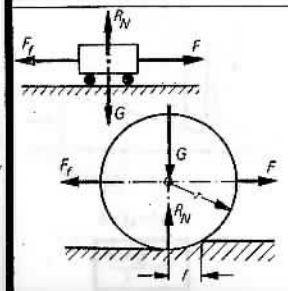
α=Angle of repose=The angle of inclination of the inclined plane when the body just starts sliding down.

1. **Example**: $R_N=G$=100 kgf, Find F, the Driving force, which will just make the body to slide



$F=F_f=\mu=R_N$=0.1×100 kgf
=10 kgf

Coefficient of friction for sliding	With static friction		With sliding friction	
	dry	greased	dry	greased
Steel on steel	0.15	0.1	0.1	0.01
Steel on grey cast iron, bronze	0.2	0.1	0.15	0.01
Wood on wood	0.65	0.2	0.3	0.1



Rolling friction: In rolling, the contact of roller with the surface on which the roller rolls is theoretically a point or line contact. However in practice, because of deformation of roller or surface or both the contact is not a line or point contact (see figure). The weight G will act through the centre of the roller at right angles to contact surface, and will have a lever arm=f, where f is the coefficient of rolling friction (f=0.05 cm for steel wheel upon steel rail).

r=radius of roller in cm F_f=Frictional resistance=F the driving force
$G=wt=R_N$=Normal reaction.

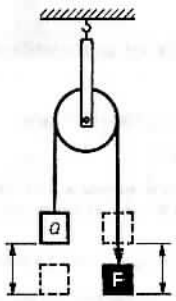
Then by law of moments $\boxed{F_f=F=\frac{f}{r}\times R_N=\frac{f}{r}\times G}$

The pulley

By using a machine we do not save work, we can only lift or move a bigger load by applying a smaller force. But if we save in force, this force has to be applied for a longer distance. Thus neglecting losses, the input work done by force=out put work done by load.

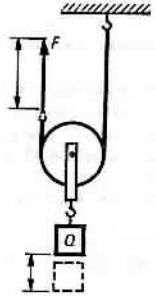
In the following diagrams F=force, and Q=load

Fixed pulley



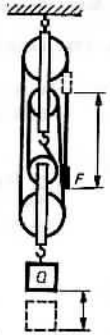
$F=Q$

Movable pulley (movable block)



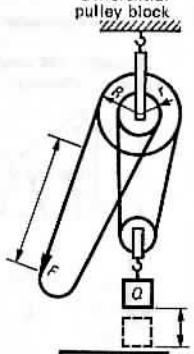
$F=\frac{1}{2}Q$

Set of pulleys



$F=\frac{Q}{n}$ n=number of blocks

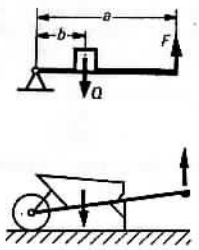
Differential pulley block



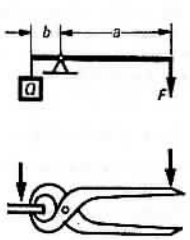
$F=Q\left(\frac{R-r}{2R}\right)$

The lever

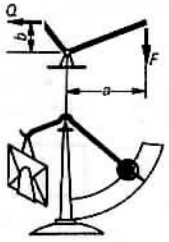
Second class lever



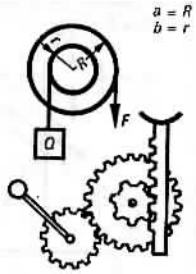
First class lever



Bellcrank lever



Wheel and axle
Winch



Winch

$F\times a=Q\times b$ (law of moments)

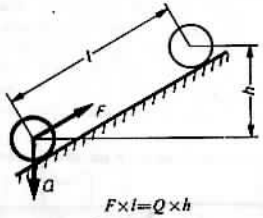
Force F × lever arm a = Load Q × load arm b

Moment of force = Moment of load

Sum of clockwise moments = Sum of anticlockwise moments

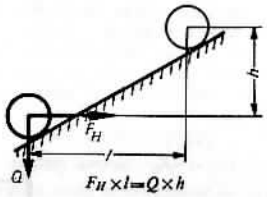
Lever arm is the perpendicular distance of the fulcrum from the line of action of the force acting on the lever.

The inclined plane, The wedge



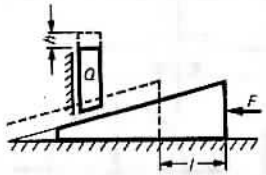
$F\times l=Q\times h$

$F=\frac{Q\times h}{l}$



$F_H\times l=Q\times h$

$F_H=\frac{Q\times h}{l}$

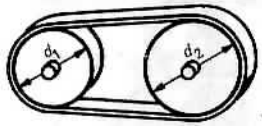


$F\times l=Q\times h$

$F=\frac{Q\times h}{l}$

Belt Drive

Simple drive (horizontal drive)
Driving pulley 1 — Driven pulley 2



n_1 rpm $\quad$ n_2 rpm

Diameter × rpm of the driving pulley = diameter × rpm of the driven pulley

$d_1\times n_1=d_2\times n_2$

$d_1=\frac{d_2\times n_2}{n_1}$ $\quad d_2=\frac{d_1\times n_1}{n_2}$ $\quad n_1=\frac{d_2\times n_2}{d_1}$ $\quad n_2=\frac{d_1\times n_1}{d_2}$

Transmission ratio, i

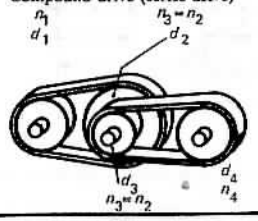
$=\frac{\text{rpm of the driving pulley}}{\text{rpm of the driven pulley}}$

$=\frac{\text{dia of the driven pulley}}{\text{dia of the driving pulley}}$

$i=\frac{n_1}{n_2}=\frac{d_2}{d_1}$

$n_2=\frac{n_1}{i}$

Compound drive (series drive)



Speed: $n_2=\frac{d_1\times n_1}{n_1}$ $\quad n_4=\frac{d_3\times n_3}{d_4}$

$n_4=n_1\times\frac{d_1}{d_2}\times\frac{d_3}{d_4}$

Total transmission ratio, $i=i_1\times i_2$

$i=i_1\times i_2$

$i_1=\frac{n_1}{n_2}$ $\quad i_2=\frac{n_3}{n_4}$

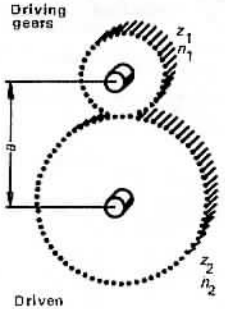
$i=\frac{n_1}{n_4}=\frac{d_2}{d_1}\times\frac{d_3}{d_2}$

$n_4=\frac{n_1}{i}$

Gear Drive

Spur Gear Drive

Single gear train



Number of teeth × rpm of the driving gear = Number of teeth × rpm of the driven gear

$z_1\times n_1=z_2\times n_2$

$z_1=\frac{z_2\times n_2}{n_1}$ $\quad z_2=\frac{z_1\times n_1}{n_2}$ $\quad n_1=\frac{z_2\times n_2}{z_1}$ $\quad n_2=\frac{z_1\times n_1}{z_2}$

Gear ratio = inverse ratio of rpm of gears

$\frac{z_1}{z_2}=\frac{n_2}{n_1}$

Gear ratio = diameter ratio

$\frac{z_1}{z_2}=\frac{d_1}{d_2}$

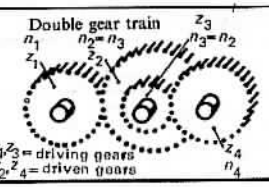
Gear ratio, i

$i=\frac{n_1}{n_2}=\frac{d_2}{d_1}=\frac{z_2}{z_1}$

Distance a of the axes (d=pitch circle dia)

$a=\frac{d_1+d_2}{2}$

Double gear train



Speeds $n_2=n_3=\frac{z_1\times n_1}{z_2}$ $\quad n_4=\frac{z_3\times n_3}{z_4}$

$n_4=n_1\times\frac{z_1}{z_2}\times\frac{z_3}{z_4}$

Total transmission, $i=i_1\times i_2$

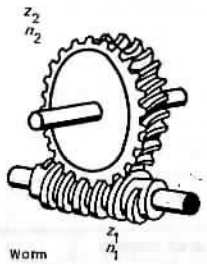
$i=i_1\times i_2$

Speed, $n_4=\frac{n_1}{i}$

$i=\frac{n_1}{n_4}=\frac{z_2}{z_1}\times\frac{z_4}{z_3}$

Worm Gear Drive

Worm wheel



rpm of the worm × thread number* of the worm = rpm of the worm wheel × number of teeth of the worm wheel

$n_1\times z_1=n_2\times z_2$

Transmission ratio, i

$i=\frac{n_1}{n_2}$

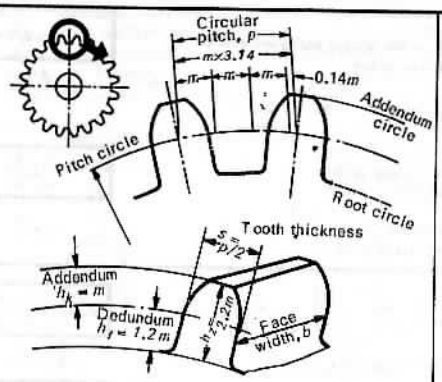
$=\frac{\text{speed of the worm}}{\text{speed of the worm wheel}}$

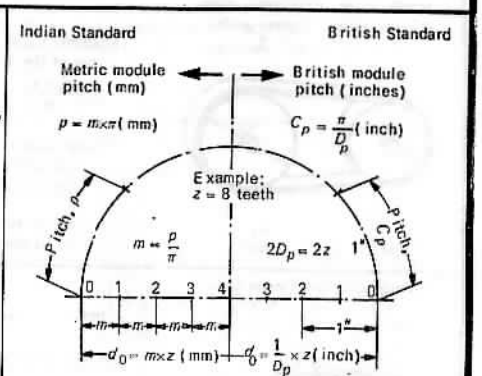
$=\frac{\text{number of teeth of warm wheel}}{\text{thread number of the worm}}$

$i=\frac{z_2}{z_1}$

*With single start thread worms $z_1=1$
With double start thread worms $z_1=2$

Dimensions for Gears



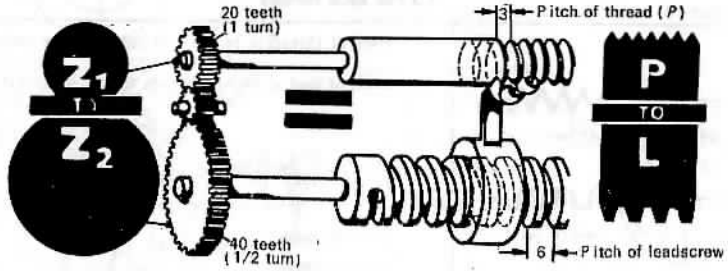


Recommended series of modules IS: 2535-1969	1	1.25	1.5	2	2.5	3	4	5	6
	8	10	12	16	20	25	32	40	50

Notation	Cylindrical gear: Spur gear	Cylindrical gear: Helical gear
Module m_n Real module (helical) m_s	$m=\frac{p}{\pi}=\frac{d_0}{z}=\frac{d_k}{z+2}=\frac{2a}{z_1+z_2}$	$m_n=m_s\times\cos\beta=\frac{p_n}{\pi}=\frac{p_s\times\cos\beta}{\pi}$
Pitch p_n Real pitch (helical) p_s	$p=m\times\pi=\frac{d_0\times\pi}{z}=\frac{d_k\times\pi}{z-2}$	$m_s=\frac{m_n}{\cos\beta}=\frac{p_s}{\pi}=\frac{p_n}{\pi\times\cos\beta}=\frac{d_0}{z}$ $p_n=m_n\times\pi=p_s\times\cos\beta=\frac{\pi\times d_0\times\cos\beta}{z}$
Pitch circle dia d_0 Pitch circle dia of the driving gear d_{01} Pitch circle dia of the driven gear d_{02}	$d_0=z\times m=d_k-2m=\frac{z\times d_k}{z+2}$ $d_{01}=\frac{2a}{1+i}$ i=gear ratio $d_{02}=\frac{2a\times i}{1+i}$	$p_s=m_s\times\pi=\frac{p_n}{\cos\beta}=\frac{m_n\times\pi}{\cos\beta}$ $d_0=z\times m_s=\frac{z\times m_n}{\cos\beta}=\frac{z\times d_k}{z+2\cos\beta}$ $d_{01}=\frac{2\times a}{1+i}$ $d_{02}=\frac{2a\times i}{1+i}$
Addendum circle dia d_k	$d_k=d_0+2m=m\ (z+2)$	$d_k=d_0+2m_n=m_n\left(\frac{z}{\cos\beta}+2\right)$
Number of teeth z	$z=\frac{d_0}{m}=\frac{d_0\times\pi}{p}=\frac{d_k-2m}{m}$	$z=\frac{d_0\times\pi}{p_s}=\frac{d_0}{m_s}=\frac{d_0\times\cos\beta}{m_n}$
Tooth depth h_z	$h_z=\frac{13}{6}\times m=2.2m$	$h_z=\frac{13}{6}\times m_n=2.2m_n$
Addendum h_k	$h_k=\frac{6}{6}\times m=m$	$h_k=\frac{6}{6}\times m_n=m_n$
Dedundum h_f	$h_f=\frac{7}{6}\times m=1.2m$	$h_f=\frac{7}{6}\times m_n=1.2m_n$
Tooth thickness s	$s=\frac{m\times\pi}{2}=\frac{p}{2}$	$s_n=\frac{m_n\times\pi}{2}=\frac{p_n}{2}$
Helix angle β	—	$\cos\beta=\frac{m_n}{m_s}=\frac{z\times m_n}{d_0}$ ($\beta\approx20°$ usually)
Face width b	$b=6m$ (with power wheels) $=10m$ (with normal wheels)	$b=10m_n$
Centre distance a	$a=\frac{d_{01}+d_{02}}{2}=\frac{m(z_1+z_2)}{2}$	$a=\frac{d_{01}+d_{02}}{2}=m_s\times\frac{z_1+z_2}{2}$

D_p=diametral pitch=ratio of the number of teeth to the number of inches in the pitch diameter
C_p=circular pitch=distance from centre of teeth measured on pitch circle

Change Gears



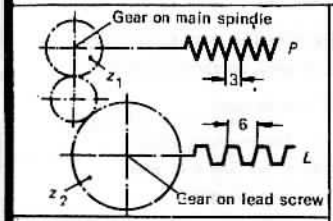
Number of teeth of the driving gear, z_1 = Pitch of thread, P
Number of teeth of the driven gear z_2 = Pitch of leadscrew, L

$$\boxed{\frac{z_1}{z_2}=\frac{P}{L}}$$

Change gear ratio, P/L must be transformed (by multiplying numerator and denominator by a suitable number) to obtain gears available in the change gear set. There are several change gear sets available. A typical set contains the following gears: 20 25 30 35 40 45 50 55 60 65 70 75 80 85 90 95 100 110 120 125 127

Lead screw with pitch in millimetre

To cut metric thread



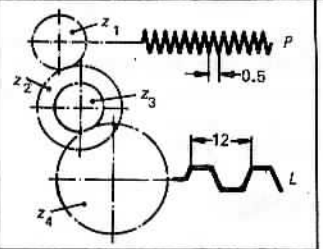
Single gear train

Pitch of thread to be cut=3 mm

Pitch of lead screw=6 mm

$$\frac{z_1}{z_2}=\frac{P}{L}=\frac{3}{6}=\frac{1}{2}=\frac{20}{40}$$

z_1=20 teeth, z_2=40 teeth } or 25 teeth, 50 teeth



Double gear train

Pitch of thread to be cut=0.5 mm

Pitch of lead screw=12 mm

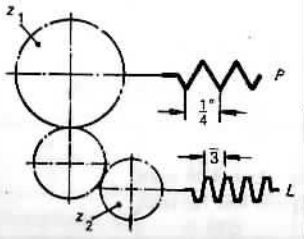
$$\frac{P}{L}=\frac{0.5}{12}=\frac{1}{24}$$

Since no convenient numbers of teeth can be found, the ratio must be split into two factors

$$\frac{z_1}{z_2}\times\frac{z_3}{z_4}=\frac{1}{24}=\frac{1}{4}\times\frac{1}{6}=\frac{25}{100}\times\frac{20}{120}$$

Driving gears z_1 and z_3=25 and 20 teeth
Driven gears z_2 and z_4=100 and 120 teeth

To cut inch thread



Pitch of thread to be cut=4 threads in one inch=$\frac{1"}{4}=\frac{25.4\text{ mm}}{4}$

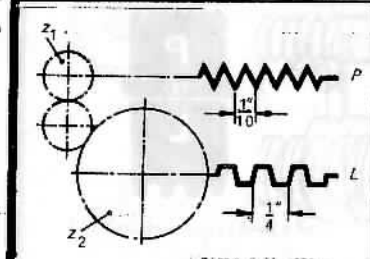
Pitch of lead screw=3 mm

$$\frac{z_1}{z_2}=\frac{P}{L}=\frac{1/4"}{3}=\frac{25.4}{4}\times\frac{1}{3}=\frac{12.7}{2}\times\frac{1}{3}=\frac{127}{60}$$

z_1=127 teeth
z_2= 60 teeth

Lead screw with pitch in inches

To cut inch thread



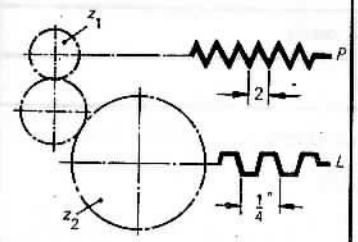
Pitch of thread to be cut=10 threads to an inch=1/10"

Pitch of lead screw=4 threads to an inch=1/4"

$$\frac{z_1}{z_2}=\frac{P}{L}=\frac{1/10}{1/4}=\frac{4}{10}=\frac{40}{100}$$

z_1= 40 teeth
z_2=100 teeth

To cut metric thread

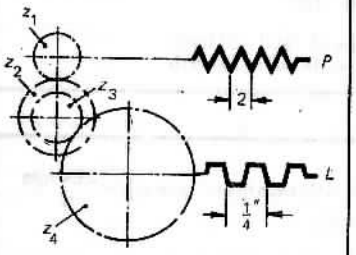


Pitch of thread to be cut=2 mm
Pitch of lead screw=4 threads to an inch=1/4"

$$\frac{z_1}{z_2}=\frac{P}{L}=\frac{2}{25.4/4}=\frac{2\times4}{25.4}=\frac{4}{12.7}=\frac{40}{127}$$

z_1= 40 teeth
z_2=127 teeth

In such cases, where gear wheel of teeth 127 is not available or cannot be used, an approximated value, e.g., $\frac{1600}{36}$ or $\frac{432}{17}$ is to be used instead of the value 25.4. The resulting error in pitch will amount to 0.16 mm and 0.43 mm respectively per 1000 mm thread length.



Let 25.4 be replaced by $\frac{1600}{63}$

$$\frac{z_1}{z_2}=\frac{P}{L}=\frac{2}{25.4/4}=\frac{8}{25.4}=\frac{8}{1600/63}=\frac{8\times63}{1600}$$

$$=\frac{63}{200}=\frac{7}{10}\times\frac{9}{20}=\frac{35}{50}\times\frac{45}{100}$$

Driving gears, z_1 and z_3=35 and 45 teeth
Driving gears, z_2 and z_4=45 and 100 teeth

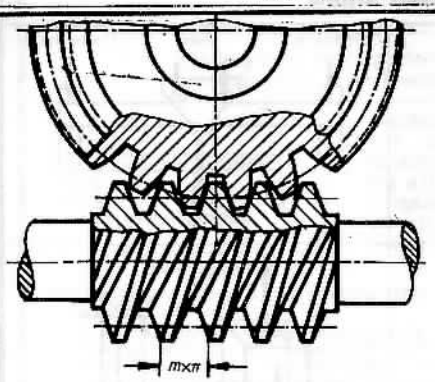
Let 25.4 be replaced by $\frac{432}{17}$

$$\frac{z_1}{z_2}=\frac{P}{L}=\frac{2}{25.4/4}=\frac{8}{25.4}=\frac{8}{432/17}=\frac{8\times17}{432}$$

$$=\frac{17}{54}=\frac{17}{9}\times\frac{1}{6}=\frac{85}{45}\times\frac{20}{120}$$

Driving gears, z_1 and z_3=85 and 20 teeth
Driving gears, z_2 and z_4=45 and 120 teeth

Cut thread based on module (worm)



Thread based on module is normally found in worm gears.

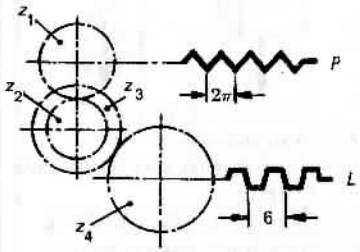
Module is a multiple of π=3.14

Module 1=$m\times\pi=1\times3.14=3.14$ mm
Module 2=$m\times\pi=2\times3.14=6.28$ mm
Module 3=$m\times\pi=3\times3.14=9.42$ mm

With multistart worms the module will be equal to $m\times\pi\times$number of starts.

Module m has the pitch $m\times\pi$

Lead screw with Pitch in millimetre



To cut worm of module 2 mm, single start pitch=$2\times\pi$ mm; pitch of lead screw=6 mm

$$\frac{z_1}{z_2}=\frac{P}{L}=\frac{2\times\pi}{6}$$

For π, the approximate value=$\frac{32\times27}{25\times11}$ may be used

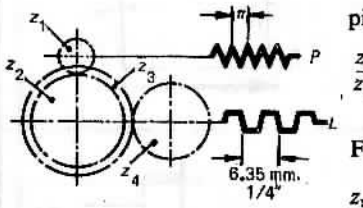
$$\frac{z_1}{z_2}=\frac{P}{L}=\frac{2\times\pi}{6}=\frac{2\times32\times27}{6\times25\times11}=\frac{32\times9}{25\times11}=\frac{32}{25}\times\frac{36}{44}$$

Driving gears, z_1 and z_3 are 32 and 36 teeth gears
Driving gears, z_2 and z_4 are 25 and 44 teeth gears

Approximation to π

$\pi\approx$	$\frac{8\times97}{13\times19}$	$\frac{25\times47}{22\times17}$	$\frac{19\times21}{127}$	$\frac{32\times27}{25\times11}$	$\frac{22}{7}$
Diviation in mm/m	0.034	0.038	0.044	0.072	0.402

Lead screw with pitch in inches



To cut worm of module 1 mm, single start pitch=$1\times\pi$ mm; pitch of lead screw=4 threads to an inch=1/4"=$\frac{25.4\text{ mm}}{4}$

$$\frac{z_1}{z_2}=\frac{P}{L}=\frac{1\times\pi}{\frac{25.4}{4}}=\frac{4\times\pi}{25.4\times1}$$

For $\frac{\pi}{25.4}=\frac{\pi}{1"}$ the approximation $\frac{5\times19}{32\times24}$ may be used

$$\frac{z_1}{z_2}=\frac{P}{L}=\frac{\pi}{25.4}\times\frac{4}{1}=\frac{5\times19}{32\times24}\times\frac{4}{1}=\frac{5\times19}{8\times24}=\frac{25}{80}\times\frac{95}{60}$$

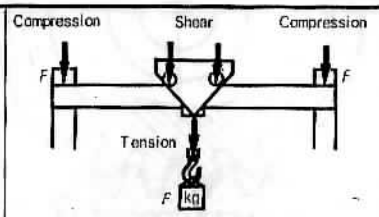
Driving gears, z_1 and z_3 are 25 and 95 teeth gears
Driving gears, z_2 and z_4 are 80 and 60 teeth gears

Approximation to $\pi/1"$

$\frac{\pi}{1"}\approx$	$\frac{47}{4\times95}$	$\frac{5\times19}{32\times24}$	$\frac{12}{97}$	$\frac{22\times5}{7\times127}$	$\frac{5\times9}{26\times14}$
Deviations in mm/m	0.005	0.016	0.214	0.402	0.472

Tensile and compressive strength

Strength : The strength of a metal is its ability to resist the application of force without rupture. In service a material may have to withstand tension, compression or shear forces.



A force F (kgf) acting on a rod, causes to create a counter force of tension or compression, σ (kgf/cm²) per square centimetre in the material which counteracts the external applied force F.

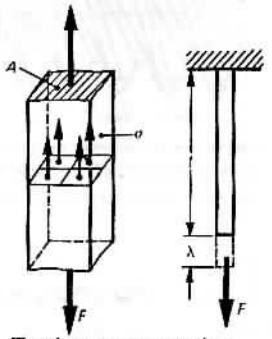
$\sigma = \frac{F}{A}$ [kgf/cm²]	$F = A \times \sigma$ [kgf]	$A = \frac{F}{\sigma}$ [cm²]

Tensile strength is the stress corresponding to maximum load in a tension test. Machine parts are being designed such that the design stresses are $^1/_{10}, \ldots, ^1/_5$ of the stress figure at failure.

Elongation, ϵ = Change in length per unit length

$$\epsilon = \frac{\lambda}{l} = \frac{\text{change in length}}{\text{initial length}}$$

This is usually expressed as a percentage of original gauge length.



σ = Tension or compression
F = Force applied
A = Cross-sectional, area (mm² or cm²)
ϵ = Elongation
λ = Change in length (mm, or cm)
l = Initial length (mm or cm)

Resistance against shearing

Shear force: A type of force which causes or tends to cause two contiguous parts of the body to slide relative to each other in a direction parallel to their plane of contact.

Shear strength: The stress required to produce fracture in the plane of cross-section, acted on by the shear force.

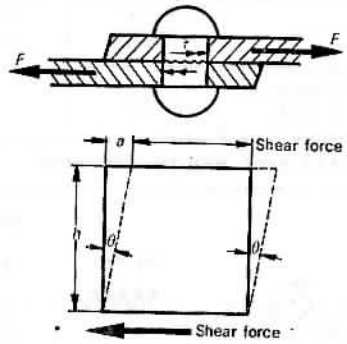
Shear strain: It is the tangent of the angle of rotation of lines originally drawn perpendicular to two parallel lines on the body under the action of a force.

Shear strain = $\tan \theta = a/h$

θ = shear angle

a = linear displacement under shear force

h = distance between the parallel lines drawn originally on the body

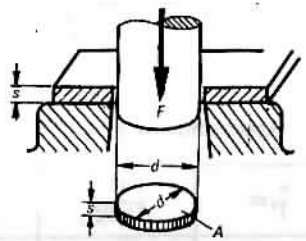


The shear strength of a number of materials is listed on page 121.

The shear strength is approximately 4/5 of the tensile strength.

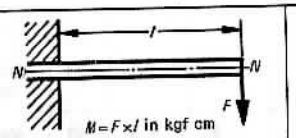
Example : A sheet metal 2 mm thick having a shear strength of 40 kgf/mm² is to receive a punch hole of 35 mm diameter. Find the force required to make the punching.

$F = A \times \tau = \pi \times d \times s \times \tau = 110 \text{ mm} \times 2 \text{ mm} \times 40 \text{ kgf/mm}^2$
$= 8800 \text{ kg}$



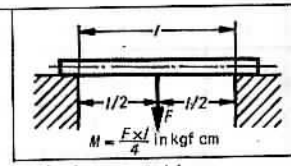
Bending strength

The larger the bending effort F (kgf) and the longer the arm of the lever, the larger is the bending moment.



Cantilever beam

Bending moment, M at any cross section of a member subjected to bending is the algebraic sum of couples or moments of all external forces or both, acting on one side.



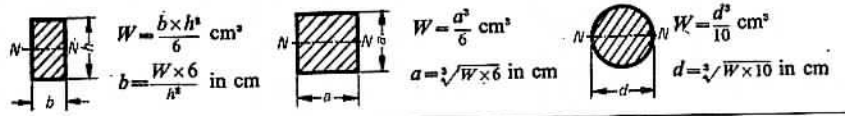
Simply supported beam

The bending moment is at its maximum either where a lever (bar, rod, etc.) is seated or mounted, or in the middle of said part.

The bending moment along the axis (axis line N-N) is completely neutralized by the moment of resistance of the beam. The statical conditions of equilibrium show that the moment of resistance and the bending moment are numerically equal.

The bending stresses are the larger, the more they are off the neutral or zero-axis and the less the resistance of the cross-section is against bending. Distances and bending resistances are thus expressed by the moment of resistance, (W).

The following cross-sections have the properties expressed in formulae:



The bending moment may be taken into account when the permissible tension, P_t is larger and the moment of resistance of particular material is larger.

$M = P_t \times W$ kgf/cm

$M = P_t \times W$ kgf/cm

Generally we know the bending moment, M (product of $F \times l$) and the permissible tension of a particular material. The moment of resistance thus can be computed and data derived to choose cross-sectional dimensions.

$$W = \frac{M}{P_t} \text{ cm}^3$$

Example: Find the diameter of a horizontal bar of 160 cm length. A load placed in the middle of said bar weighs 100 kg.

$P_t = 700$ kgf/cm² (design stress)

$$W = \frac{M}{Pt} = \frac{F \times l}{4 \times Pt.} = \frac{100 \times 160}{4 \times 700} = 5.7 \text{ cm}^3 \qquad d = \sqrt{W.10} = \sqrt{57} = 3.87 \text{ cm}$$

Allowable stresses for plates, sections and bars (in kgf/cm² maximum) [IS : 800-1962]

Nature of force	Material	Thickness or diameter	Allowable stress
Tension	St 42-S, St 42-0, St 42-W	upto and including 20 mm	1500
		over 20 mm	1420
	St 58-HT	upto and including 45 mm	2125
		over 45 mm	1810
	St 55-HTW	upto and including 32 mm	2125
		over 45 mm	1810
Compression	St 42-S	All	1250
	St 42-0, St 42-W, St 55-HTW, St 58-HT	upto and including 32 mm	1719
		upto and including 45 mm	1719
Shearing	St 42-S, St 42-0, St 42-W	All	1100
		upto and including 45 mm	1575
	St 58-HT	over 45 mm	1340
		upto and including 32 mm	1575
	St 55-HTW	over 32 mm	1340
Bending	St 42-S, St 42-0, St 42-W	upto and including 20 mm	1650
		over 20 mm	1575
		upto and including 45 mm	2285
	St 58-HT	over 45 mm and upto and including 60 mm	1970
		upto and including 32 mm	2285
	St 55-HTW	over 32 mm	1970

Preferred Numbers and Sizes

Preferred numbers IS: [illegible]–1963				Preferred nominal sizes in millimeters IS: 1136–1958							
Basic series				For wrought metal products							
R5	*R10*	*R20*	*R40*								
1.00	1.00	1.00	1.00	0.10	**1.0**	**10**	**100**				370
			1.06				105				375
		1.12	1.12		**1.1**	**11**	**110**			38	380
			1.18				115				390
	1.25	1.25	1.25	**0.12**	**1.2**	**12**	120	**0.4**	**4.0**	**40**	**400**
			1.32				**125**				410
		1.40	1.40			13	130			42	420
			1.50				135				430
1.60	1.60	1.60	1.60							44	440
			1.70		**1.4**	**14**	**140**		4.5	45	450
		1.80	1.80				145			46	460
			1.90		1.5	15	150				470
	2.00	2.00	2.00				155			48	480
			2.12	**0.16**	**1.6**	**16**	**160**	**0.5**	**5.0**	**50**	**500**
		2.24	2.24				165				
			2.36			17	170			52	520
2.50	2.50	2.50	2.50				175			53	530
			2.65		1.8	18	180		5.5	55	550
		2.80	2.80				185				
			3.00			19	190			56	560
	3.15	3.15	3.15				195			58	580
			3.35	**0.20**	**2.0**	**20**	**200**	**0.6**	**6.0**	60	600
		3.55	3.55			21	210			62	
			3.75		2.2	22	220			**63**	**630**
4.00	4.00	4.00	4.00			23	230				650
			4.25			24	240				
		4.50	4.50	**0.25**	**2.5**	**25**	**250**				670
			4.75			26	260			68	
	5.00	5.00	5.00				270				
			5.30		2.8	28	280		7.0	70	700
		5.60	5.60				290			71	710
			6.00	**0.30**	3.0	30	300			72	
6.30	6.30	6.30	6.30				310			75	750
			6.70				315			78	
		7.10	7.10		**3.2**	**32**	**320**	**0.8**	**8.0**	**80**	**800**
			7.50				330			82	
	8.00	8.00	8.00			34	340			85	850
			8.50							88	
		9.00	9.00		3.5	35	350		9.0	90	900
			9.50				355			92	
						36	360			95	950
10.00	10.00	10.00	10.00							98	

Preferred numbers may be applied with advantage either to the grading of complete articles by length, area, volume, weight, power, rating, etc. or to the individual dimensions of component parts in engineering design.

Series *R5* is to be preferred to series *R10* and *R10* series in turn is to be preferred to series *R20*.

Numbers above 10 or below 1 (table values) can be extended, in that proper factors are used to get the required multiples or sub-multiples (10, 100, etc., 0·1, 0·01, etc).

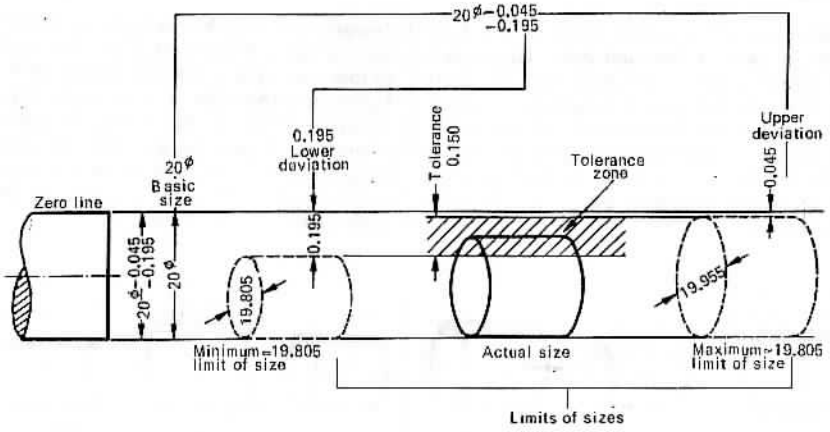
Numbers indicated in bold type are the preferred ones.

Selection of a single size: The selection of a single size for a product irrespective of any idea of grading should be from one of the preferred sizes. In the selection of a single size, the sizes derived from *R10* series of preferred numbers should be given preference over the sizes derived from the *R20* series. The sizes derived from the *R20* series should, in turn, be given preference over the remaining sizes.

Selection of a series of sizes: Where a product is available in more than one size, the series of sizes for the product should be derived from the series of preferred sizes. While doing so, a series of sizes with a coarser gradation, when applicable, is to be preferred to a finer gradation series. The series of sizes derived from the *R10* series of preferred numbers is, therefore, to be used whenever practicable.

Terminology for ISO System

Dimensioning



Basic size: For the sake of convenience, a basic size, which is usually a whole number (e.g., 20, 75, etc.) is given to represent the dimension of a part. This is the design size and the allowable maximum and minimum limits of the dimensions of the part are fixed in relation to the basic size. Basic size relates to the position of zero line.

Upper deviation: The algebraic difference between the maximum limit and the corresponding basic size.

Lower deviation: The algebraic difference between the minimum limit and the corresponding basic size.

Tolerance: Due to limitations of cost and inaccuracies of manufacturing methods, it is not feasible to make a part precisely to a given dimension and may only be made to lie between two limits, maximum and minimum. The difference between the maximum and minimum limits of sizes is the tolerance. It is also equal to the algebraic difference between the upper and lower deviations.

Actual size: The size of a part as may be found by measurement.

Zero line: In a graphical representation of limits and fits, the zero line represents the basic size. Upper and lower deviations are referred to zero line. When the zero line is drawn horizontally, positive deviations are shown above and negative below it.

Tolerance zone: In a graphical representation of tolerance, the zone bounded by the two limits of size of the part (shown by hatched line in the figure above).

Symbols for upper and lower deviations:

	Hole	Shaft
Upper deviation	ES	es
Lower deviation	EI	ei

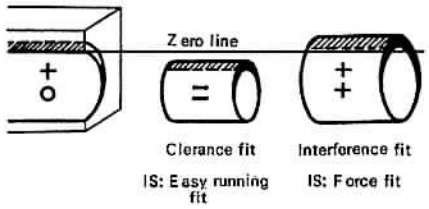
Limits and Fits

IS: 919–1963

Basic hole system

Taking the basic size of a common reference for both the systems, the following co-relation exists:

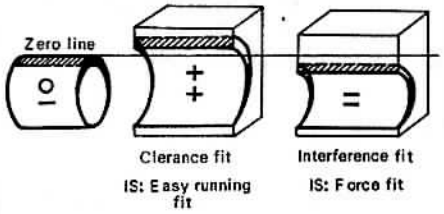
Design size of the hole is taken as the basic size for both, i.e. hole and shaft. The position of tolerance zone of the shaft may be above or below the zero line or in some cases (e.g. js) the zero line may lie within the tolerance zone of the shaft. Depending upon the position of tolerance zone of the shaft, the shaft may give a certain type of fit with a hole in the basic hole system.



For hole lower deviation is zero
Minimum limit = Basic size
Application: Machine and engine building, locomotive construction.

Basic shaft system

Design size of the shaft is taken as the basic size for both, i.e. hole and shaft. The position of tolerance zone of the hole may be above or below the zero line or in some cases (e.g. JS) the zero line may lie within the tolerance zone of the hole. Depending upon the position of tolerance zone of the hole, the hole may give a certain type of fit with a shaft in the basic shaft system.

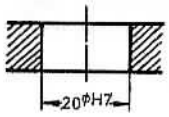


For shaft upper deviation is zero
∴ Maximum limit = Basic size
Application: Electric motors, ball bearings, power transmission, products made from bright drawn bars.

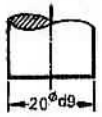
IS Symbols

While giving dimensions on drawings, the deviations are not shown directly added to the basic size, but the toleranced size is indicated by its basic size followed by a symbol consisting of a letter (sometimes two letters, e.g. JS) and a number.

For example, instead of writing $20\phi^{+0.021}_{+0}$, it is mentioned as $20\phi^{H7}$. The actual values of deviations corresponding to the size so mentioned can be read from tables (see pages 80–83).



Deviations according to tables of fits, in μ
($1\mu = 1/1000$ mm)
$= 20\phi^{+21}_{0}$



Deviations according to tables of fits in μ
($1\mu = 1/1000$mm)
$= 20\phi^{-65}_{-117}$

Holes: capital letters (e.g. H, G, M) are used to designate the position of tolerance zone.

Shafts: small letters (e.g. h, g, m) are used to designate the position of tolerance zone.

The numbers 0, 01, 1,..., 16 decide the amount of tolerance and indicate the grade of accuracy of hole (in all 18 grades).

(Tolerance shown is exaggerated)

Grade	0, 01, 1—4 finer gr.	5—11 fine to coarse gr.	12—16 coarse gr.
Application	Gauge	Machine building	Rolling, casting

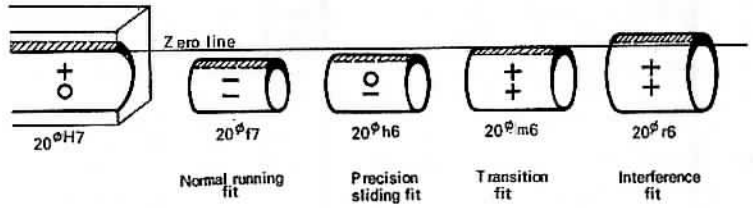
I.S. Symbols

The letters *a...js, j...z, za, zb, zc* (letters *i, l, o, q, w* are omitted) determine the position of tolerance zone with reference to zero line (in all 25 positions). Capital letters denote holes and small letters denote shafts. The position of tolerance zone determines the kind of fit which can be obtained.

Basic hole system

Holes gets the capital letter *H*

Shafts get different letters to decide the position of tolerance zone to obtain a desired fit:

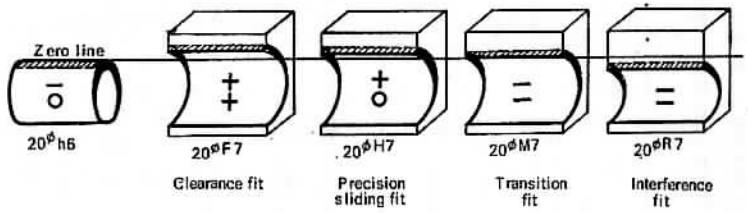


For hole lower deviation = zero, hence minimum limit = basic size

Basic shaft system

Shaft gets the small letter *h*

Holes get different letters to decide the position of tolerance zone to obtain a desired fit.



For shaft upper deviation = zero, maximum limit = basic size

Recommended holes for machine building industry		Recommended shafts	
Grade	Holes	Grade	Shafts
5	H	5	*g*, h*, j*, k*, m**
6	G, H*, J*, K*, M*, N*, P,	6	*f*, g*, h*, j*, k*, m*, n*, p*, r*, s*, t*
7	F*, G*, H*, K*, M*, N*, P, R, S	7	*e, f*, h, j, k, m, n, p r, s, t, u*
8	E* F* H*	8	*c, d, e*, f*, h**
9	B, C, D, E, H P*	9	*b, c, d*, e*, h*
10	D, F*	10	*h*
11	A*, B, C, D*, H*	11	*a*, b, c, d*, h**

* Shows first preference
Without asterisk mark: Second preference

IS FITS (Basic hole system)

IS: 919–1963 & IS: 2709–1964

Black numbers = Deviations at go side
Red numbers = Deviations at no go side

Deviations in μ (1μ = 0.001 mm)

Diameter steps in mm		Hole	Shafts						Hole	Shafts						
Over	To	H6	f6	g5	h5	j5	k5	m5	H7	f7	g6	h6	j6	k6	m6	n6
0	3	+ 6 0	− 6 − 12	− 2 − 6	0 − 4	+ 2 − 2	+ 4 0	+ 6 + 2	+10 0	− 6 − 16	− 2 − 8	0 − 6	+ 4 − 2	+ 6 0	+ 8 + 2	+10 + 4
3	6	+ 8 0	− 10 − 18	− 4 − 9	0 − 5	+ 3 − 2	+ 6 + 1	+ 9 + 4	+12 0	− 10 − 22	− 4 −12	0 − 8	+ 6 − 2	+ 9 + 1	+12 + 4	+16 + 8
6	10	+ 9 0	− 13 − 22	− 5 −11	0 − 6	+ 4 − 2	+ 7 + 1	+12 + 6	+15 0	− 13 − 28	− 5 −14	0 − 9	+ 7 − 2	+10 + 1	+15 + 6	+19 +10
10	14	+11	− 16	− 6	0	+ 5	+ 9	+15	+18	− 16	− 6	− 0	+ 8	+12	+18	+23
14	18	0	− 27	−14	− 8	− 3	+ 1	+ 7	0	− 34	−17	−11	− 3	+ 1	+ 7	+12
18	24	+13	− 20	− 7	0	+ 5	+11	+17	+21	− 20	− 7	0	+ 9	+15	+21	+28
24	30	0	− 33	−16	− 9	− 4	+ 2	+ 8	0	− 41	−20	−13	− 4	+ 2	+ 8	+15
30	40	+16	− 25	− 9	0	+ 6	+13	+20	+25	− 25	− 9	0	+11	+18	+25	+33
40	50	0	− 41	−20	−11	− 5	+ 2	+ 9	0	− 50	−25	−16	− 5	+ 2	+ 9	+17
50	65	+19	− 30	−10	0	+ 6	+15	+24	+30	− 30	−10	0	+12	+21	+30	+39
65	80	0	− 49	−23	−13	− 7	+ 2	+11	0	− 60	−29	−19	− 7	+ 2	+11	+20
80	100	+22	− 36	−12	0	+ 6	+18	+28	+35	− 36	−12	0	+1	+25	+35	+45
100	120	0	− 58	−27	−15	− 9	+ 3	+13	0	− 71	−34	−22	− 9	+ 3	+13	+23
120	140	+25	− 43	−14	0	+ 7	+21	+33	+40	− 43	−14	0	+14	+28	+40	+52
140	160	0	− 68	−32	−18	−11	+ 3	+15	0	− 83	−39	−25	−11	+ 3	+15	+27
160	180															
180	200	+29	− 50	−15	0	+ 7	+24	+37	+46	− 50	−15	0	+16	+33	+46	+60
200	225	0	− 79	−35	−20	−13	+ 4	+17	0	− 96	−44	−29	−13	+ 4	+17	+31
225	250															
250	280	+32	− 56	−17	0	+ 7	+27	+43	+52	− 56	−17	0	+16	+36	+52	+66
280	315	0	− 88	−40	−23	−16	+ 4	+20	0	−108	−49	−32	−16	+ 4	+20	+34
315	355	+36	− 62	−18	0	+ 7	+29	+46	+57	− 62	−18	0	+18	+40	+57	+73
355	400	0	− 98	−43	−25	−18	+ 4	+21	0	−119	−54	−36	−18	+ 4	+21	+37
400	450	+40	− 68	−20	0	+ 7	+32	+50	+63	− 68	−20	0	+20	+45	+63	+80
450	500	0	−108	−47	−27	−20	+ 5	+23	0	−131	−60	−40	−20	+ 5	+23	+40

Note: When *t5* to *t7* deviations are missing *u5* to *u7* deviations should be substituted for them.

IS FITS (Basic hole system)

SPK.

Deviations in μ (1μ = 0.001 mm)

			Hole	Shafts								Hole	Shafts	
p6	r6	s6	H8	c9	d8	e8	f8	h8	j7	t7	u7	H11	a11	h11
+ 12 + 6	+ 16 + 10	+ 20 + 14	+14 0	− 60 − 85	− 20 − 34	− 14 − 28	− 6 − 20	0 −14	+ 6 − 4	− −	+ 28 + 18	+ 60 0	− 270 − 330	0 − 60
+ 20 + 12	+ 23 + 15	+ 27 + 19	+18 0	− 70 −100	− 30 − 48	− 20 − 38	− 10 − 28	0 −18	+ 8 − 4	− −	+ 35 + 23	+ 75 0	− 270 − 345	0 − 75
+ 24 + 15	+ 28 + 19	+ 32 + 23	+22 0	− 80 −116	− 40 − 62	− 25 − 47	− 13 − 35	0 −22	+10 − 5	− −	+ 43 + 28	+ 90 0	− 280 − 370	0 − 90
+ 29 + 18	+ 34 + 23	+ 39 + 28	+27 0	− 95 −138	− 50 − 77	− 32 − 59	− 16 − 43	0 −27	+12 − 6	− −	+ 51 + 33	+110 0	− 290 − 400	0 −110
+ 35 + 22	+ 41 + 28	+ 48 + 35	+33 0	−110 −162	− 65 − 98	− 40 − 73	− 20 − 53	0 −33	+13 − 8	− + 62 + 41	+ 62 + 41 + 69 + 48	+130 0	− 300 − 430	0 −130
+ 42 + 26	+ 50 + 34	+ 59 + 43	+39 0	−120 −182 −130 −192	− 80 −119	− 50 − 89	− 25 − 64	0 −39	+15 −10	+ 73 + 48 + 79 + 54	+ 85 + 60 + 95 + 70	+160 0	− 310 − 470 − 320 − 480	0 −160
+ 51 + 32	+ 60 + 41 + 62 + 43	+ 72 + 53 + 78 + 59	+46 0	−140 −214 −150 −224	−100 −146	− 60 −106	− 30 − 76	0 −46	+18 −12	+ 96 + 66 +105 + 75	+117 + 87 +132 +102	+190 0	− 340 − 530 − 360 − 550	0 −190
+ 59 + 37	+ 73 + 51 + 76 + 54	+ 93 + 71 +101 + 79	+54 0	−170 −257 −180 −267	−120 −174	− 72 −126	− 36 − 90	0 −54	+20 −15	+126 + 91 +139 +104	+159 +124 +179 +144	+220 0	− 380 − 600 − 410 − 630	0 −220
+ 68 + 43	+ 88 + 63 + 90 + 65 + 93 + 68	+117 + 92 +125 +100 +133 +108	+63 0	−200 −300 −210 −310 −230 −330	−145 −208	− 85 −148	− 43 −106	0 −63	+22 −18	+162 +122 +174 +134 +186 +146	+210 +170 +230 +190 +250 +210	+250 0	− 460 − 710 − 520 − 770 − 580 − 830	0 −250
+ 79 + 50	+106 + 77 +109 + 80 +113 + 84	+151 +122 +159 +130 +169 +140	+72 0	−240 −355 −260 −375 −280 −395	−170 −242	−100 −172	− 50 −122	0 −72	+25 −21	+212 +166 +226 +180 +242 +196	+282 +236 +304 +258 +330 +284	+290 0	− 660 − 950 − 740 −1030 − 820 −1110	0 −290
+ 88 + 56	+126 + 94 +130 + 98	+190 +158 +202 +170	+81 0	−300 −430 −330 −460	−190 −271	−110 −191	− 56 −137	0 −81	+26 −26	+270 +218 +292 +240	+367 +315 +402 +350	+320 0	− 920 −1240 −1050 −1370	0 −320
+ 98 + 62	+144 +108 +150 +114	+226 +190 +244 +208	+89 0	−360 −500 −400 −540	−210 −299	−125 −214	− 62 −151	0 −89	+29 −28	+325 +268 +351 +294	+447 +390 +492 +435	+360 0	−1200 −1560 −1350 −1710	0 −360
+108 + 68	+166 +126 +172 +132	+272 +232 +292 +252	+97 0	−440 −595 −480 −635	−230 −327	−135 −232	− 68 −165	0 −97	+31 −32	+393 +330 +423 +360	+553 +490 +603 +540	+400 0	−1500 −1900 −1650 −2050	0 −400

Note; *a11* Shaft is not provided for diameters up to 1 mm.

IS FITS (Basic shaft system)

Ref: IS: 919–1963 & IS: 2709–1964

Black numbers = Deviations at go side
Red numbers = Deviations at no go side

Deviations in μ (1μ = 0.001 mm)

Diameter steps in mm		Shaft	Holes				Shaft	Holes							
Over	To	h5	G6	J6	K6	M6	h6	F6	G7	J7	K7	M7	N7	P7	R7	S7
0	3	0 −4	+8 +2	+2 −4	0 −6	−2 −8	0 −6	+12 +6	+12 +2	+4 −6	0 −10	−2 −12	−4 −14	−6 −16	−10 −20	−14 −24
3	6	0 −5	+12 +4	+5 −3	+2 −6	−1 −9	0 −8	+18 +10	+16 +4	+6 −6	+3 −9	0 −12	−4 −16	−8 −20	−11 −23	−15 −27
6	10	0 −6	+14 +5	+5 −4	+2 −7	−3 −12	0 −9	+22 +13	+20 +5	+8 −7	+5 −10	0 −15	−4 −19	−9 −24	−13 −28	−17 −32
10 14	14 18	0 −8	+17 +6	+6 −5	+2 −9	−4 −15	0 −11	+27 +16	+24 +6	+10 −8	+6 −12	0 −18	−5 −23	−11 −29	−16 −34	−21 −39
18 24	24 30	0 −9	+20 +7	+8 −5	+2 −11	−4 −17	0 −13	+33 +20	+28 +7	+12 −9	+6 −15	0 −21	−7 −28	−14 −35	−20 −41	−27 −48
30	40	0	+25	+10	+3	−4	0	+41	+34	+14	+7	0	−8	−17	−25	−34
40	50	−11	+9	−6	−13	−20	−16	+25	+9	−11	−18	−25	−33	−42	−50	−59
50	65	0	+29	+13	+4	−5	0	+49	+40	+18	+9	0	−9	−21	−30 −60	−42 −72
65	80	−13	+10	−6	−15	−24	−19	+30	+10	−12	−21	−30	−39	−51	−32 −62	−48 −78
80	100	0	+34	+16	+4	−6	0	+58	+47	+22	+10	0	−10	−24	−38 −73	−58 −93
100	120	−15	+12	−6	−18	−28	−22	+36	+12	−13	−25	−35	−45	−59	−41 −76	−66 −101
120	140	0	+39	+18	+4	−8	0	+68	+54	+26	+12	0	−12	−28	−48 −88	−77 −117
140	160	−18	+14	−7	−21	−33	−25	+43	+14	−14	−28	−40	−52	−68	−50 −90	−85 −125
160	180														−53 −93	−93 −133
180	200	0	+44	+22	+5	−8	0	+79	+61	+30	+13	0	−14	−33	−60 −106	−105 −151
200	225	−20	+15	−7	−24	−37	−29	+50	+15	−16	−33	−46	−60	−79	−63 −109	−113 −159
225	250														−67 −113	−123 −169
250	280	0	+49	+25	+5	−9	0	+88	+69	+36	+16	0	−14	−36	−74 −126	−138 −190
280	315	−23	+17	−7	−27	−41	−32	+56	+17	−16	−36	−52	−66	−88	−78 −130	−150 −202
315	355	0	+54	+29	+7	−10	0	+98	+75	+39	+17	0	−16	−41	−87 −144	−169 −226
355	400	−25	+18	−7	−29	−46	−36	+62	+18	−18	−40	−57	−73	−98	−93 −150	−187 −244
400	450	0	+60	+33	+8	−10	0	+108	+83	+43	+18	0	−17	−45	−103 −166	−209 −272
450	500	−27	+20	−7	−32	−50	−40	+68	+20	−20	−45	−63	−80	−108	−109 −172	−229 −292

IS FITS (Basic shaft system)

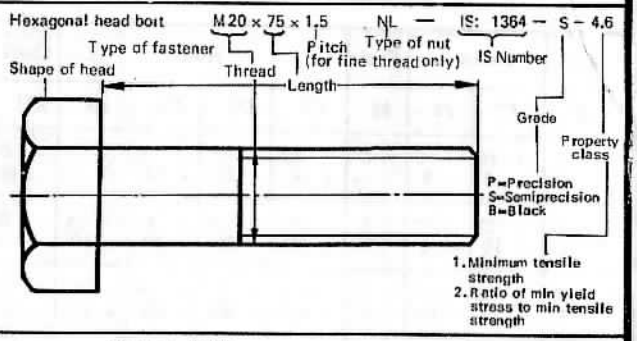
Deviations in μ (1μ = 0.001 mm)

Shaft	Holes		Shaft	Holes				Shaft	Holes		
h7	F7	J8	h8	C9	D9	E8	F8	h11	A11	C11	D11
0 −10	+16 +6	+6 −8	0 −14	+85 +60	+45 +20	+28 +14	+20 +6	0 −60	+330 +270	+120 +60	+80 +20
0 −12	+22 +10	+10 −8	0 −18	+100 +70	+60 +30	+38 +20	+28 +10	0 −75	+345 +270	+145 +70	+105 +30
0 −15	+28 +13	+12 −10	0 −22	+116 +80	+76 +40	+47 +25	+35 +13	0 −90	+370 +280	+170 +80	+130 +40
0 −18	+34 +16	+15 −12	0 −27	+138 +95	+93 +50	+59 +32	+43 +16	0 −110	+400 +290	+205 +95	+160 +50
0 −21	+41 +20	+20 −13	0 −33	+162 +110	+117 +65	+73 +40	+53 +20	0 −130	+430 +300	+240 +110	+195 +65
0	+50	+24	0	+182 +120	+142	+89	+64	0	+470 +310	+280 +120	+240
−25	+25	−15	−39	+192 +130	+80	+50	+25	−160	+480 +320	+290 +130	+80
0	+60	+28	0	+214 +140	+174	+106	+76	0	+530 +340	+330 +140	+290
−30	+30	−18	−46	+224 +150	+100	+60	+30	−190	+550 +360	+340 +150	+100
0	+71	+34	0	+257 +170	+207	+126	+90	0	+600 +380	+390 +170	+340
−35	+36	−20	−54	+267 +180	+120	+72	+36	−220	+630 +410	+400 +180	+120
0	+83	+41	0	+300 +200	+245	+148	+106	0	+710 +460	+450 +200	+395
−40	+43	−22	−63	+310 +210	+145	+85	+43	−250	+770 +520	+460 +210	+145
				+330 +230					+830 +580	+480 +230	
0	+96	+47	0	+355 +240	+285	+172	+122	0	+950 +660	+530 +240	+460
−46	+50	−25	−72	+375 +260	+170	+100	+50	−290	+1030 +740	+550 +260	+170
				+395 +280					+1110 +820	+570 +280	
0	+108	+55	0	+430 +300	+320	+191	+137	0	+1240 +920	+620 +300	+510
−52	+56	−26	−81	+460 +330	+190	+110	+56	−320	+1370 +1050	+650 +330	+190
0	+119	+60	0	+500 +360	+350	+214	+151	0	+1560 +1200	+720 +360	+570
−57	+62	−29	−89	+540 +400	+210	+125	+62	−360	+1710 +1350	+760 +400	+210
0	+131	+66	0	+595 +440	+385	+232	+165	0	+1900 +1500	+840 +440	+630
−63	+68	−31	−97	+635 +480	+230	+135	+68	−400	+2050 +1650	+880 +480	+230

Bolt

System of designation — IS:1367–1967

Example:
A hexagon head bolt, M 20, 75 mm long with fine thread pitch = 1.5 with nut and lock. Nut as per IS: 1364, of semi-precision grade made of steel with minimum tensile strength=40 kgf/mm² and having a ratio of minimum yield stress to minimum tensile strength =0.6



Surface finish

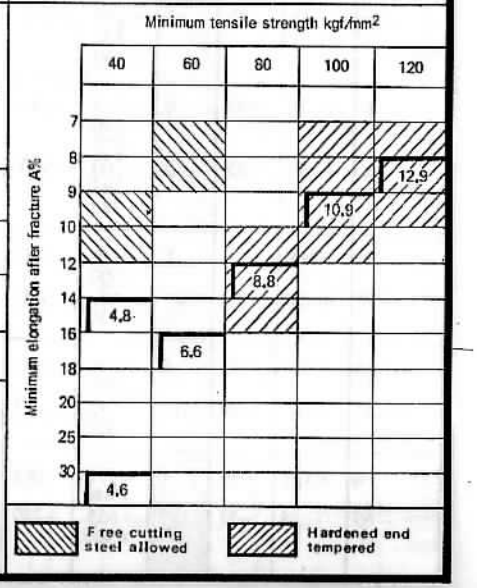
Fastener grade	Surfaces: Bolt	Surfaces: Nut	Surface finish symbol
P	All surfaces and threads except points	All surfaces and threads	▽▽
	Point of bolt	Minor dia of thread	~
S	Thread, shank and bearing face	Thread and bearing face	▽▽
	All other faces	Flats and minor dia of thread	~
B	Flank and minor dia of thread	Flank and major dia of thread	▽▽
	Major dia of thread	Minor dia of thread	~
	All other faces	All other faces	~

Property classes for bolt

Code number giving the minimum tensile strength	Minimum tensile strength kgf/mm²	Code number giving the ratio of min yield stress to min tensile strength	Ratio of min yield stress to min tensile strength in %
4	40	0.6	60
6	60	0.7	70
8	80	0.8	80
10	100		
12	120	0.9	90

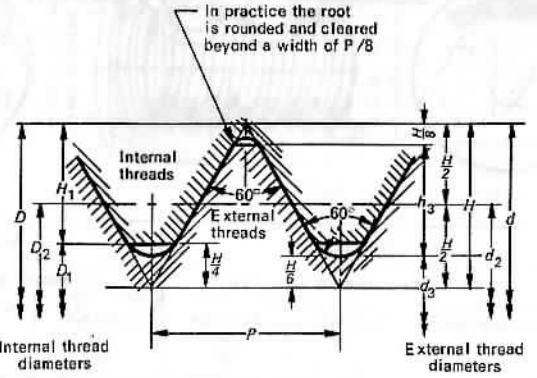
Typical examples showing the bolt designations :
Hex bolt M 16 × 70 NL–IS: 1363
Hex bolt M 20 × 1.5 × 75–IS: 1364–S–4.6
Hex bolt M 4 × 20–IS: 2389–4.6
Hex bolt M 42 × 100–IS: 3138–B–4D

Suitable steels for bolts



ISO Metric Screw Threads

Design profiles of internal and external threads



$D_2 = D - 2 \times \frac{3}{8} H$
$= D - 0.649\ 519\ 053\ P$
$D_1 = D - 2 \times \frac{5}{8} H$
$= D - 1.082\ 531\ 755\ P$
$d_2 = d - 2 \times \frac{3}{8} H$
$= d - 0.649\ 519\ 053\ P$
$d_3 = d - 2 \times \frac{17}{24} H$
$= d - 1.226\ 869\ 322\ P$

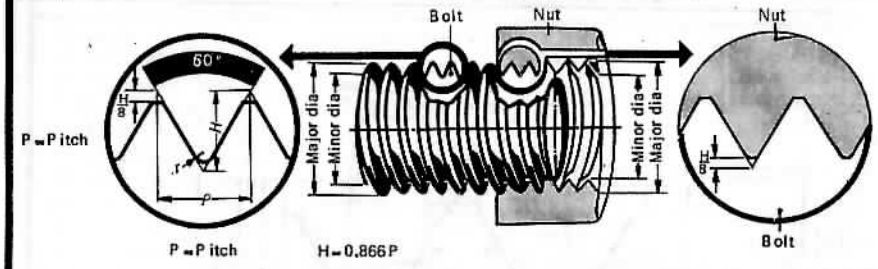
Tensile stress area $= \frac{\pi}{4}\left(\frac{d_2 + d_3}{2}\right)^2$

Thread size	BOLT				NUT			WASHER	
	Minor dia [1]	Tensile stress area	Pitch P	Thickness of head	Thickness of nut	Width across flats	Width across corners	Outside dia	Thickness of washer
d, D mm	d_3 mm	mm²	mm	mm	mm	mm	Min mm	mm	mm
M1.6	1.171	1.27	0.35	1.1	1.3	3.2	3.48	4	0.3
M2	1.509	2.07	0.4	1.4	1.6	4	4.38	5	0.3
M2.5	1.948	3.39	0.45	1.7	2	5	5.51	6.5	0.5
M3	2.387	5.03	0.5	2	2.4	5.5	6.08	7	0.5
M4	3.141	8.78	0.7	2.8	3.2	7	7.74	9	0.8
M5	4.018	14.2	0.8	3.5	4	8	8.87	10	1.0
M6	4.773	20.1	1	4.0	5	10	11.05	12.5	1.6
M8	6.466	36.6	1.25	5.5	6.5	13	14.38	17	1.6
M10	8.160	58.0	1.5	7.0	8	17	18.90	21	2
M12	9.853	84.3	1.75	8.0	10	19	21.10	24	2.5
M16	13.546	157	2	10	13	24	26.75	30	3
M20	16.933	245	2.5	13	16	30	33.53	37	3
M24	20.319	353	3	15	19	36	39.98	44	4
M30	25.706	561	3.5	19	24	46	51.28	56	4
M36	31.092	817	4	23	29	55	61.31	66	5

ISO Metric screw threads IS: 4218 (Part III)—1967
Hexagon bolts and screws IS: 2389—1968 and IS: 1364—1967
Plain washer for bolts and screws IS: 2016—1967

[1] For minor diameter of nut D_1 refer to page 86

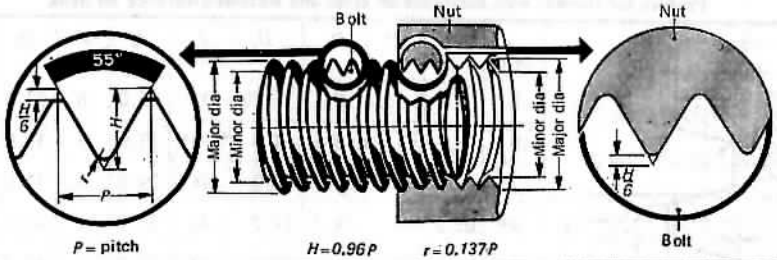
ISO Metric Screw Threads



Size	Pitch	Nut threads: Minor diameter D_1 in mm			Bolt threads: Minor diameter d_3 in mm		
		Tolerance class 5H	Tolerance class 6H	Tolerance class 7H	Tolerance class 4h	Tolerance class 6g	Tolerance class 8g
Coarse series							
M1	0.25	0.729... 0.785	—	—	0.641... 0.693	0.604... 0.675	—
M1.2	0.25	0.929... 0.985	—	—	0.841... 0.893	0.804... 0.875	—
M1.6	0.35	1.221... 1.301	1.221... 1.321	—	1.106... 1.171	1.063... 1.151	—
M2	0.4	1.567... 1.657	1.567... 1.679	—	1.438... 1.509	1.394... 1.490	—
M2.5	0.45	2.013... 2.113	2.013... 2.138	—	1.871... 1.948	1.825... 1.928	—
M3	0.5	2.459... 2.571	2.459... 2.599	2.459... 2.639	2.303... 2.387	2.256... 2.367	—
M4	0.7	3.242... 3.283	3.242... 3.422	3.242... 3.466	3.035... 3.141	2.979... 3.119	—
M5	0.8	4.134... 4.294	4.134... 4.334	4.134... 4.384	3.901... 4.019	3.842... 3.995	3.787... 3.995
M6	1	4.917... 5.107	4.917... 5.153	4.917... 5.217	4.630... 4.773	4.563... 4.747	4.495... 4.747
M8	1.25	6.647... 6.859	6.647... 6.912	6.647... 6.982	6.301... 6.466	6.230... 6.438	6.158... 6.438
M10	1.5	8.376... 8.612	8.376... 3.676	8.376... 8.751	7.967... 8.160	7.888... 8.128	7.808... 8.128
M12	1.75	10.106...10.371	10.106...10.441	10.106...10.531	9.632... 9.853	9.543... 9.819	9.457... 9.819
M16	2	13.835...14.135	13.835...14.210	13.835...14.310	13.302...13.546	13.204...13.508	13.114...13.508
M20	2.5	17.294...17.649	17.294...17.744	17.294...17.854	16.647...16.933	16.541...16.891	16.440...16.891
M24	3	20.752...21.152	20.752...21.252	20.752...21.382	19.978...20.319	19.855...20.271	19.740...20.271
M30	3.5	26.211...26.661	26.211...26.771	26.211...26.921	25.322...25.706	25.189...25.653	25.066...25.653
M36	4	31.670...32.145	31.670...32.270	31.670...35.420	30.665...31.093	30.521...31.033	30.390 ..31.033
Fine series							
M8×1	1	6.917... 7.107	6.917... 7.153	6.917... 7.217	6.630... 6.773	6.563... 6.747	6.495... 6.747
M10×1.25	1.25	8.647... 8.859	8.647... 8.912	8.647... 8.982	8.301... 8.466	8.230... 8.438	8.158... 8.438
M12×1.25	1.24	10.647...10.859	10.644...10.912	10.677...10.982	10.291...10.466	10.216...10.438	10.136...10.438
M16×1.5	1.5	14.376...14.612	14.376...14.676	14.376...14.751	13.962...14.160	13.880...14.128	13.796...14.128
M20×1.5	1.5	18.376...18.612	18.376...18.676	18.376...18.751	17.962...18.160	17.880...18.128	17.796...18.128
M24×2	2	21.835...22.135	21.835...22.210	21.835...22.310	21.296...21.546	21.194...21.508	21.099...21.508
M30×2	2	27.835...28.135	27.835...28.210	27.835...28.310	27.296...27.546	27.194...27.508	27.099...27.508
M36×3	3	32.752...33.152	32.752...33.252	32.752...33.382	31.978...32.319	31.855...32.271	31.740...32.271

Limits of sizes for commercial bolts and nuts: IS: 4218 (Part VI)—1967
(diameter range 1 to 39 mm)

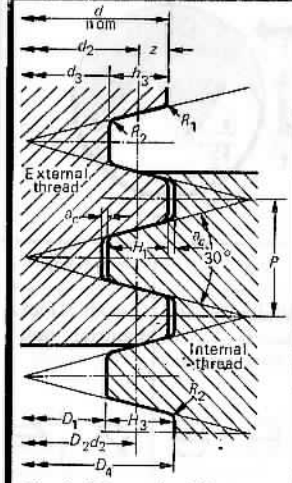
British standard whitworth threads—BSW



Thread size d_1 D inches	Bolt and nut: Major diameter d mm	Minor diameter d_3 mm	Tensile stress area mm²	Threads per inch z	Pitch P mm	Thickness of head k	Nut: Thickness (0.8 d) m	Width across flats SW	Width across corners e	Washer: Outside diameter d_3	Thickness s
1/4	6.35	4.72	0.175	20	1.27	5	5.5	11	12.7	14	1.5
5/16	7.94	6.13	0.295	18	1.41	6	6.5	14	16.2	18	2
3/8	9.53	7.49	0.441	16	1.59	7	8	17	19.6	22	2.5
1/2	12.70	9.99	0.784	12	2.12	9	11	22	25.4	28	3
5/8	15.88	12.92	1.311	11	2.31	11	13	27	31.2	34	3
3/4	19.05	15.80	1.960	10	2.54	13	16	32	36.9	40	4
7/8	22.23	18.61	2.720	9	2.82	16	18	36	41.6	45	4
1	25.40	21.34	3.575	8	3.18	18	20	41	47.3	52	5
1 1/8	28.58	23.93	4.497	7	3.63	20	22	46	53.1	58	5
1 1/4	31.75	27.10	5.770	7	3.63	22	25	50	57.7	62	5
1 3/8	34.93	29.51	6.837	6	4.23	24	28	55	63.5	68	6
1 1/2	38.10	32.68	8.388	6	4.23	27	30	60	69.3	75	6
1 5/8	41.28	34.77	9.495	5	5.08	30	32	65	75.0	80	7
1 3/4	44.45	37.95	11.310	5	5.08	32	35	70	80.8	85	7
2	50.80	43.57	14.912	4 1/2	5.65	36	40	80	92.4	98	8
2 1/4	57.15	49.02	18.873	4	6.35	40	45	85	98.2	105	9
2 1/2	63.50	55.37	24.079	4	6.35	45	50	95	109.7	120	9
2 3/4	69.85	60.56	28.804	3 1/2	7.26	50	55	105	121.3	130	10
3	76.20	66.91	35.161	3 1/2	7.26	54	60	110	127.1	135	10
3 1/4	82.55	72.54	41.333	3 1/4	7.82	58	65	120	138.6	150	12
3 1/2	88.90	78.89	48.885	3 1/4	7.82	62	70	130	150.2	160	12
3 3/4	95.25	84.41	55.959	3	8.47	67	75	135	155.9	165	12
4	101.60	90.76	64.697	3	8.47	70	80	145	167.5	180	14
4 1/4	107.95	96.64	73.349	2 7/8	8.84	75	85	155	179.0	190	14
4 1/2	114.30	102.99	83.307	2 7/8	8.84	80	90	165	190.5	205	14
4 3/4	120.66	108.83	93.014	2 3/4	9.24	84	95	175	202.1	215	16
5	127.01	115.18	104.185	2 3/4	9.24	90	100	180	207.9	220	16
5 1/4	133.36	120.96	114.922	2 5/8	9.68	94	105	190	219.5	230	16
5 1/2	139.71	127.31	127.304	2 5/8	9.68	98	110	200	231.0	245	18
5 3/4	146.06	133.04	139.022	2 1/2	10.16	102	115	210	242.6	255	18
6	152.41	139.39	152.608	2 1/2	10.16	106	120	220	254.1	270	18

ISO Metric Trapezoidal Screw Threads — IS: 7008–1973

Profiles for threads with clearance on crest and without clearance on flank



$H_1 = 0.5P$ $\quad h_3 = H_1 + a_c$

$d_3 = d - 2H_3$ $\quad d_2 = D_2 = d - 2z$

$z = 0.25P = \frac{H_1}{2}$

R_1 Max $= 0.5a_c$ $\quad R_2$ Max $= a_c$

d Nom	d_2	P	D_4	D_1	d Nom	d_3	P	D_4	D_1
8	6.2	1.5	8.3	6.50	32	25.0	6	33.0	26.0
10	7.5	2	10.5	8.0	34*	27.0	6	35.0	28.0
12	8.5	3	12.5	9.0	36	29.0	6	37.0	30.0
14*	10.5	3	14.5	11.0	40	32.0	7	41.0	33.0
16	11.5	4	16.5	12.0	44	36.0	7	45.0	37.0
18*	13.5	4	18.5	14.0	48	39.0	8	49.0	40.0
20	15.5	4	20.5	16.0	52	43.0	8	53.0	44.0
22*	16.5	5	22.5	17.0	60	50.0	9	61.0	51.0
24	18.5	5	24.5	19.0	70	59.0	10	71.0	60.0
26*	20.5	5	26.5	21.0	80	69.0	10	81.0	70.0
28	22.5	5	28.5	23.0	90	77.0	12	91.0	78.0
30*	23.0	6	31.0	24.0	100	87.0	12	101.0	88.0

*Non-preferred sizes

Designation: Examples: Tr 40×7
A single start ISO metric trapezoidal screw thread of Nom. dia = 40mm and pitch = 7mm.

Tr 40×14 (*P*7) LH
A multiple start left hand ISO metric trapezoidal screw thread of Nom. dia = 40mm and lead = 14mm and pitch = 7mm

Square Threads — IS: 4694–1968

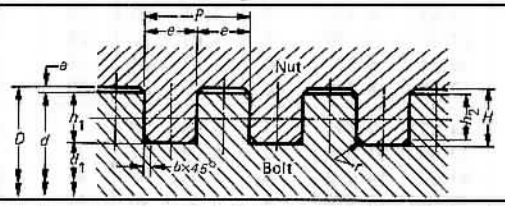
Basic profile of square threads

$h_1 = 0.5P$

$H = 0.5P + a$

$h_3 = 0.5P - b$

Area of core $= \frac{\pi}{4} d_1^2$



$e = 0.5P$

$D = d + 2a$

$d_1 = d - 2h_1$

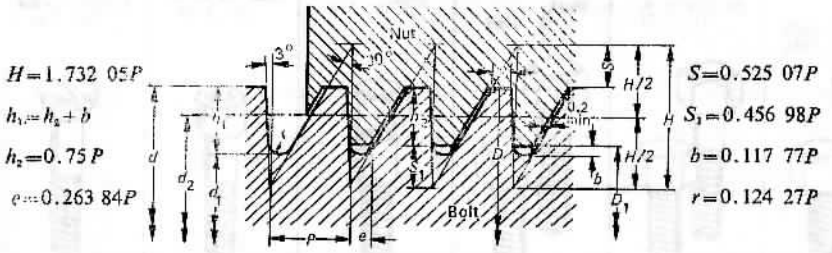
Basic dimensions for square threads—normal series in millimetres

Nominal diameter	Major diameter		Minor diameter, d_1	Pitch, P	Area of core mm²
	Bolt, d	Nut, D			
22	22	22.5	17	5	227
26	26	26.5	21	5	346
30	30	30.5	24	6	452
36	36	36.5	30	6	707
40	40	40.5	33	7	855
44	44	44.5	37	7	1075
48	48	48.5	40	8	1257
52	52	52.5	44	8	1521
60	60	60.5	51	9	2043

Designation: Example: SQ 30×6–IS: 4694–1968
A square thread of nominal dia. 30mm and pitch 6mm

Saw Tooth Threads — IS: 4696–1968

Basic profile of saw tooth threads



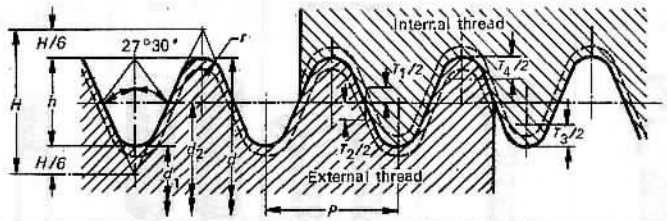
Basic dimensions for saw tooth threads—normal series (in mm)

Nominal diameter	Bolt			Pitch diameter d_2	Pitch P	Nut	
	Major diameter d	Minor diameter d_1	Area of core mm²			Major diameter D	Minor diameter D_1
22	22	13.322	139	18.590	5	22	14.5
24	24	15.322	184	20.590	5	24	16.5
26	26	17.322	236	22.590	5	26	18.5
28	28	19.322	293	24.590	5	28	20.5
30	30	19.586	301	25.909	6	30	21
32	32	21.586	370	27.909	6	32	23
36	36	25.586	514	31.909	6	36	27
40	40	27.852	609	35.227	7	40	29.5
48	48	34.116	914	42.545	8	48	36
50	50	36.116	1024	44.545	8	50	38

Designation: Example— ST 48×3–IS: 4696
A saw tooth thread of nominal diameter 48 mm and pitch 3 mm

Pipe Threads — IS: 2643–1964

Dimensions for pipe threads for fastening purposes — in mm



$H = 0.960\ 491P$

$h = 0.640\ 327P$

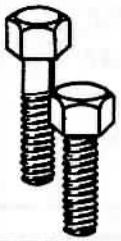
$r = 0.137\ 329P$

Size designation	Pitch P	Basic diameter: Internal and external threads: Major d	Basic diameter: Internal and external threads: Pitch d_2	Basic diameter: Internal and external threads: Minor d_1	Pitch diameter: Internal thread: $d_2 + T_1$ max	Pitch diameter: Internal thread: d_2 min	Pitch diameter: Class A and Class B d_2 max	Pitch diameter: External thread: Class A $(d_2 - T_2)$ min	Pitch diameter: External thread: Class B $(d_2 - 2T_2)$ min	Minor diameter for internal thread: $d_2 + T_3$ max	Minor diameter for internal thread: d_1 min	Major diameter for external thread: d max	Major diameter for external thread: $d - T_4$ min
PF1/8	0.907	9.728	9.147	8.566	9.254	9.147	9.147	9.040	8.933	8.848	8.566	9.728	9.514
PF1/4	1.337	13.157	12.301	11.445	12.426	12.301	12.301	12.176	12.051	11.890	11.445	13.157	12.907
PF3/8	1.337	16.662	15.806	14.940	15.931	15.806	15.806	15.681	15.556	15.395	14.940	16.662	16.412
PF1/2	1.814	20.955	19.793	18.631	19.935	19.793	19.793	19.651	19.509	19.172	18.631	20.955	20.671
PF3/4	1.814	26.441	25.279	24.117	25.421	25.279	25.279	25.137	24.995	24.658	26.117	24.441	26.157
PF1	2.309	33.249	31.770	30.291	31.950	31.770	31.770	31.590	31.410	30.931	30.291	33.249	32.889
PF1 1/4	2.309	41.910	40.431	38.952	44.611	40.431	40.431	40.258	40.071	39.592	38.952	41.910	41.550
PF1 1/4	2.309	47.803	46.324	44.845	46.504	46.324	46.324	46.144	45.964	45.485	44.485	47.803	47.443
PF2	2.309	59.614	58.135	56.656	58.315	58.135	58.135	57.955	57.775	57.296	56.656	59.614	59.254

Designations: (1) Ext. FP 2B—External pipe thread for fastening purposes of size 2 with class B tolerance.
(2) Int. FP 2—Internal pipe thread for fastening purposes of size 2.

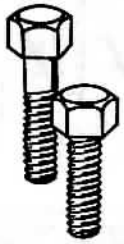
Standard bolts, screws and nuts

Hexagon bolts and screws Precision IS : 2389 IS : 1364



Hexagon bolts and screws Semiprecision IS : 1364

Hexagon bolts and screws Black IS : 1363

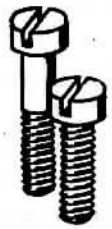


Hexagon fit bolts IS : 3640

Hexagon socket head cap screws IS : 2269

Countersunk head screws with hexagon socket IS : 6761

Slotted cheese head screws IS : 1366



Slotted roundhead wood screw IS : 6739

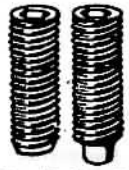
Slotted countersunk head screws IS : 1365

Slotted raised countersunk head screws IS : 1365

Hexagon socket grub screws Type A: Cup point Type C: Conical end IS : 6094

Type A Type C

Hexagon socket grub screws Type E: Flat end Type G: Cylindrical dog point IS : 6094



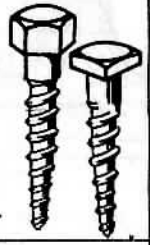
Type E Type G

Slotted raised countersunk head wood screw IS : 6736

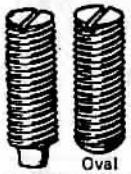
Slotted pan head screws IS : 6101

Slotted countersunk head wood screw IS : 6760

Hexagon and square head coach screws IS : 1120



Slotted grub screws IS 2388



Tapered dog point Oval point

Studs Precision and black IS : 1862

Hexagon nut Precision, IS : 2389 Semiprecision, IS : 1364 Black, IS : 2585

Pipe nut IS : 3468

Castle nut IS : 2232 and IS : 5368

Slotted round nut IS : 6731

Slotted nut IS : 2232

Domed cap nut IS : 2687

Cap nut IS : 2687

Thumbscrews IS : 3726

Knurled thumbscrew

Slotted thumbscrew

Knurled nuts IS : 3460

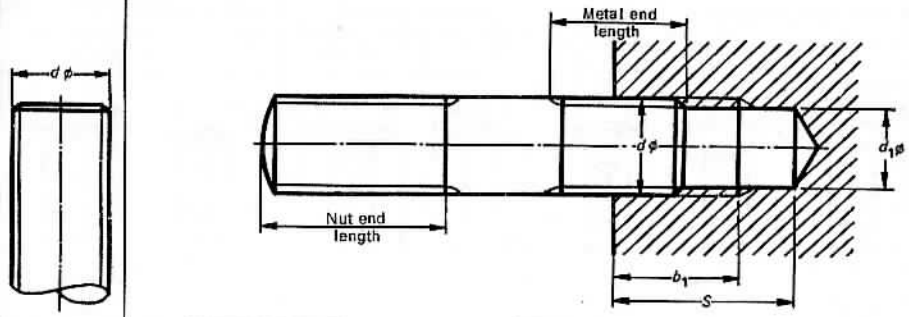
Type A

Type B

Wing screw IS : 3727

Wing nut IS : 2636

Threaded length, threaded end of stud and depth of threaded hole



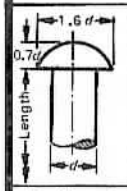
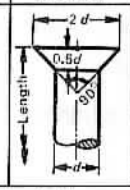
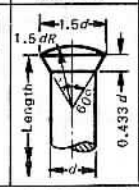
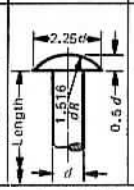
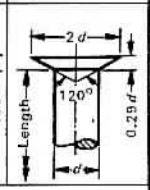
Nominal size $d\phi$ in mm	Tap drill d_1	STUD IS: 1862–1967 Threaded length Nut end length	Metal end length Type A	Metal end length Type B	HOLE IS: 4499–1968 Threaded end b_1 of the thread Stud type A	Threaded end b_1 of the thread Stud type B	Depth S of the threaded hole Stud type A	Depth S of the threaded hole Stud type B
M3	2.5	12	3.00– 4.00	4.50– 5.70	3	4.5	6	8
M4	3.3	14	4.00– 5.20	6.00– 7.20	4	6	8	10
M5	4.2	16	5.00– 6.20	7.50– 9.00	5	7.5	9	12
M6	5	18	6.00– 7.20	9.00–10.50	6	9	11	14
M8	6.75	22	8.00– 9.50	12.00–13.80	8	12	13	17
M10	8.5	26	10.00–11.50	15.00–16.80	10	15	16	21
M12	10.25	30	12.00–13.80	18.00–19.80	12	18	18	24
M16	14	38	16.00–17.80	24.00–26.10	16	24	23	31
M20	17.5	46	20.00–22.10	30.00–32.10	20	30	28	38
M24	21	54	24.00–26.10	36.00–38.50	24	36	33	45
M30	26.5	66	30.00–32.10	45.00–47.50	30	45	40	55
M36	32	78	36.00–38.50	54.00–67.00	36	54	47	65

Designation of screws—Examples

Hex screw M8×30—IS: 1364–P 8.8	Hexagon screw, precision grade Property class 8.8
Slotted countersunk head screw BM10×30—IS: 1365–4.8	B=Type B (partially threaded) Property class 4.8
Slotted cheese head screw AM10×30—IS: 1366–4.8	A=Type A (fully threaded) Property class 4.8
Hexagon socket head cap screw M10×60—IS: 2269–P–8.8	P=Precision grade Property class 8.8
Pan head screw AM5×50—IS: 6101–4.8	A=Type A (fully threaded) Property class 4.8
Screw M10×60—IS: 6761–8.8	Countersunk head screws with hexagon socket,

Rivets

Rivets for general purposes (below 12 mm diameter) IS: 2155–1962

Snap head	Flat countersunk head 90°	Round countersunk head 60°	Mushroom head	Countersunk head 120°
				

Preferred shank diameter: 1.6, 2, 2.5, 3, 4, 5, 6, 8, 10 mm

Material:

(1) Hot rolled steel rivet bars for structural purposes (IS: 1148–1973)

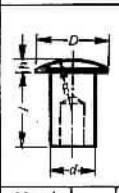
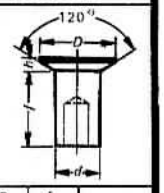
$c = 0.23$ max $\quad S = 0.055$ max $\quad P = 0.055$ max

(2) High tensile rivet bars for structural purposes (IS: 1149–1964)

$c = 0.22$ max $\quad S = 0.05$ max $\quad P = 0.05$ max

Designation: Example—Pan head rivet 6 × 30 IS: 2155
A pan head rivet of 6 mm dia and length 30 mm

Tubular and semi-tubular rivets IS: 4040—1967

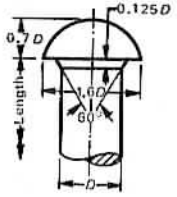
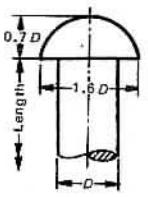
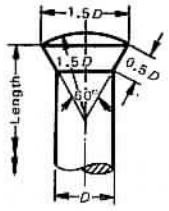
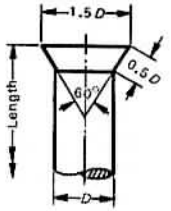
Oval head	Countersunk head 120°
	

Nominal size mm d	D mm	R Nom mm	d_1 Nom mm	h mm
2	4	4	1.0	0.6
2.5	5	5	1.3	0.7
3	6	5.5	1.6	0.85
4	8	8	2.7	1.2
5	9.5	8.5	3.3	1.3
6	10	8.5	3.5	1.3

Material: Steel C 15 grade

Designation : Example—Oval Hd tubular rivet 4 × 12 IS: 4040 steel. Oval head tubular rivet of size 4 mm, length 12 mm.

Rivets for general purposes and boiler rivets (12 to 48 mm diameter) IS: 1929–1961 IS: 1928–1961

Snap head rivet for boilers	Snap head rivet for general purpose	Rounded countersunk head for general purpose and boilers	Flat countersunk head for general purpose and boilers
			

Basic size of rivets, D, mm: 12 14 16 18 20 22 24 27 30 33 36 39 42 48

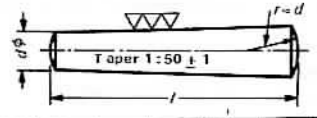
Preferred length and diameter combinations for rivets for general purposes and boiler rivets

Length mm	Diameter, mm													
	12	14	16	18	20	22	24	27	30	33	36	39	42	48
28	×	–	–	–	–	–	–	–	–	–	–	–	–	–
35.5	×	×	×	–	–	–	–	–	–	–	–	–	–	–
40	×	×	×	×	–	–	–	–	–	–	–	–	–	–
50	×	×	×	×	×	×	–	–	–	–	–	–	–	–
63	×	×	×	×	×	×	×	×	–	–	–	–	–	–
80	×	×	×	×	×	×	×	×	×	–	–	–	–	–
100	–	–	×	×	×	×	×	×	×	×	×	–	–	–
150	–	–	–	–	–	–	×	×	×	×	×	×	×	×
200	–	–	–	–	–	–	–	–	×	×	×	×	×	×
250	–	–	–	–	–	–	–	–	–	–	–	–	×	×

Note: Preferred combinations are indicated by "×"

Taper pins

IS: 6688–1972



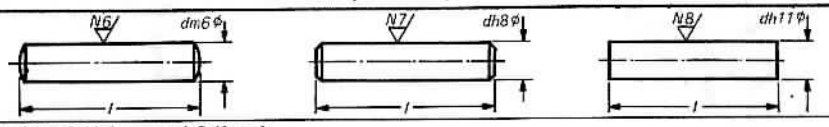
Material: Cold drawn steel C 40 grade. Nominal size is the diameter of the thin end of the taper pin.
Surface finish: N6 for type A and N7 for type B
Designation: Example: Solid taper pin A 10 × 50 IS: 6688. A solid taper pin of nominal diameter 10mm of type A and nominal length 50mm

d_φ	1.5	2	2.5	3	4	5	6	8	10	12	16	20	25	30	40	50
l	8... 25	10... 35	10... 35	12... 45	14... 55	20... 60	25... 90	25... 130	30... 160	35... 180	40... 180	45... 200	50... 200	55... 200	60... 200	65... 200

Standard sizes: 8, 10, 12, 14, 16, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 90, 100, 110, 120, 130, 140, 150, 160, 170, 180, 190, 200,

Cylindrical pins

IS: 2393–1972



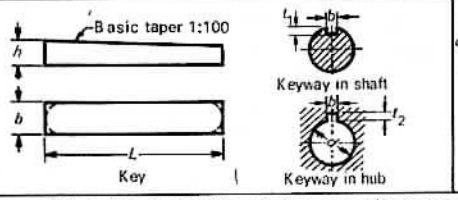
Material: Cold drawn steel C 40 grade
Designation: Example: Cylindrical pin 10*h*8 × 20 IS: 2393
A cylindrical pin of nominal diameter 10mm, tolerance *h*8 and nominal length 20mm

d_φ	1.5	2	2.5	3	4	5	6	8	10	12	16	20	25	30	40	50
l	4... 16	6... 25	6... 25	8... 30	8... 45	10... 50	12... 60	14... 80	20... 100	25... 150	30... 180	40... 200	50... 200	60... 200	80... 200	100... 200

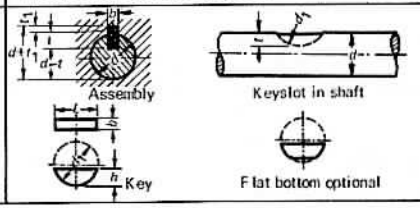
Standard sizes: 4, 5, 6, 8, 10, 12, 14, 16, 20, 25, 30 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 90, 100, 110, 120, 130, 140, 150, 160, 170, 180, 190, 200

Keys and keyways

Taper keys and keyways IS: 2292–1963



Woodruff keys and keyslots IS: 2294–1963



Material: Steel with minimum tensile strength 58 kgf/mm²

Designation: Example: Taper key 10 × 8 × 80 IS: 2292; a taper key of width 10mm, height 8mm and length 80mm

Designation: Example: Woodruff key 4 × 5 IS: 2294; a woodruff key of width 4mm and height 5mm

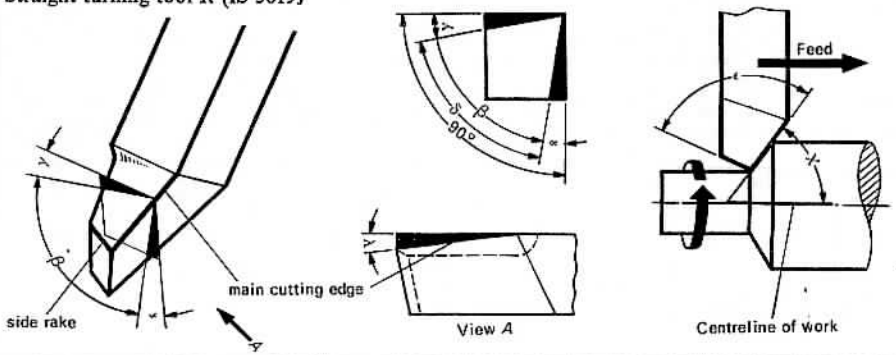
Shaft diameter ∅ d	Key width × height $b \times h$	Keyway depth in shaft t_1	Keyway depth in hub t_2	Shaft diameter ∅ d	Woodruff key: Width b	Height h	Length L	Dia of key d_1	Keyslot depth in shaft t	Keyslot depth in hub t_1
6... 8	3 × 2	1.2	0.5	Over 3... 4	1	1.4	3.82	4.0	1.0	0.6
Over 8...10	3 × 3	1.8	0.9	Over 4... 6	1.5	2.6	6.76	7.0	2.0	0.8
Over 10...12	4 × 4	2.5	1.2	Over 6... 8	2	2.6	6.76	7.0	1.8	1.0
Over 12...17	5 × 5	3.0	1.7			3.7	9.66	10.0	2.9	1.0
Over 17...22	6 × 6	3.5	2.1	Over 8...10	2.5	3.7	9.66	10.0	2.9	1.0
Over 22...30	8 × 7	4.0	2.5		3	3.7	9.66	10.0	2.5	1.4
Over 30...38	10 × 8	5.0	2.5		3	5	12.65	13.0	3.8	1.4
Over 38...44	12 × 8	5.0	2.5	Over 10...12	4	5	12.65	13.0	3.5	1.7
Over 44...50	14 × 9	5.5	2.9			6.5	15.72	16.0	5.0	1.7
Over 50...58	16 × 10	6.0	3.4							
Over 58...65	18 × 11	7.0	3.3	Over 12...17	5	6.5	15.72	16.0	4.5	2.2
Over 65...75	20 × 12	7.5	3.8			7.5	18.57	19.0	5.5	2.2

Metal Cutting Operations

Turning

Angles on turning tools

Straight turning tool *R* (IS 3019)



Angles on turning tools

α=Side clearance

β=Lip angle α+β+γ=90°

γ=Side rake angle

δ=Cutting angle δ=α+β

ϵ=Nose angle

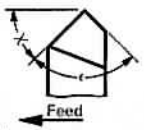
χ=Setting angle

λ=Front clearance angle

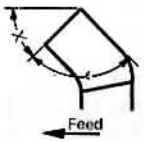
Standard values for angles on turning tools

Tools from high speed steel and tool steel			Material to be turned	Carbide tipped tools		
						Front clearance
α	β	γ		α	γ	λ
6°	84°	0°	Cast iron, brass, bronze (harder qualities)	4°– 6°	0°	5°– 8°
8°	74°	8°	Steel and cast steel above 70 kgf/mm² tensile strength; grey cast iron, gun metal, brass, bronce	4°– 6°	10°	3°– 5°
8°	68°	14°	Steel and cast steel with 50–70 kgf/mm² tensile strength; grey cast iron, brass (softer qualities)	4°– 6°	12°	3°– 5°
8°	62°	20°	Steel and grey cast iron with 34–50 kgf/mm² tensile strength	4°– 6°	15°	3°– 5°
8°	55°	27°	Mild steel, bronze (softer qualities)	4°– 6°	20°	3°– 5°
10°	40°	40°	Aluminium and other soft materials	8°–10°	30°	5°–10°

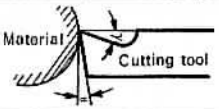
Setting angle and nose angle for roughing tools



Setting angle χ	Nose angle ϵ
≈45°	≈100°–110°
≈65°	≈ 90°
≈85°	≈ 80°



Standard values for cutting speeds—angles—specific cutting force



Values given apply to dry cutting, using high-speed steel tools at a cutting speed V_{60} (tool life 60 min); tungston carbide tools at a cutting speed V_{40} (tool life 40 min). Setting angle λ=45°, nose angle ϵ=90°, front clearance λ=5...8°. Front clearance λ=10...25° for soft materials.

Values of specific cutting force (pressure), tangential to the work surface are applicable for depth of cut four to eight times of the feed.

Material	Strength in kgf/mm²	Cutting tool	Side clearance α°	Side rake γ°	Feed *s* in mm per revolution: 0.1	0.2	0.4	0.8	1.6	3.2	0.1	0.2	0.4	0.8
					Cuttting speed *V* m/min						Specific cutting force K_s kgf/mm² chip section			
Plain carbon structural steel St 34, St 37, St 42	up to 45	HSS TCT[1]	8 5	14 10	280	60 236	45 200	34 170	25 67	19 56	360	260	190	136
St 50, St 63	50... 70	HSS TCT	8 5	14 10	240	44 205	32 175	24 145	18 50	14 42	410	225	215	154
St 78	70... 85	HSS TCT	8 5	14 10	200	32 170	24 132	18 106	13 34	10 27	440	315	230	164
Cast steel	50... 75	HSS TCT	8 5	10 6	118	34 100	25 85	19 71	14 24	11 20	360	260	190	136
Alloy steels	85...100	HSS TCT	8 5	10 6	150	24 118	17 95	12 75	8.5 24	(6) 20	500	360	260	185
Mn-steel, Cr-Ni-steel, Cr-Mo steel	100...140	HSS TCT	8 5	6 6	95	16 75	11 60	8 50	(5-6) 16	13	530	380	275	200
Other steel alloys	140...180	HSS TCT	8 5	6 6	60	9.5 48	6 38	32	10	8	570	410	300	215
Tool steel	150...180	HSS TCT	8 5	6 6	50	40	32	27	8.5	6.7	570	410	300	215
Austenitic-steel Manganese steel		HSS TCT	5	6	40	32	25	20	6.7	5.3	660	480	350	252
Cast iron Grade 15	BHN ...200	HSS TCT	8 5	0 0	140	48 118	27 95	18 80	14 67	9.5	190	136	100	72
Cast iron Grade 25	BHN 200...250	HSS TCT	8 5	0 0	106	32 90	18 75	13 63	9.5 53	6.3	290	208	150	108
Malleable Cast iron		HSS TCT	8 5	10 10	106	43 90	28 75	20 63	13 53	9	240	175	125	92
Copper alloys Brass	BHN 80...120	HSS TCT	8 5	0 6	600	125 530	85 450	56 400	36 355		160	115	85	60
Tin bronze		HSS TCT	8 5	0 6	500	85 450	63 375	48 335	34 300	24	140	100	70	52
Cast bronze		HSS TCT	8 5	0 6	355	63 280	53 236	43 200	36 180	28	340	245	180	125
Light alloys Pure aluminium		HSS TCT	12 12	30 30	400 1320	300 1120	200 950	118 850	75 710		105	76	55	40
Al-alloys (11...13% S:)		HSS TCT	12 12	18 18	100 224	67 190	45 160	30 140	118		140	100	70	52
Piston alloys		HSS TCT	12 12	14 14	25	22	20	18	17		125	90	65	48
Magnesium-alloys		HSS TCT	8 5	6 6	1000 1800	900 1500	800 1250	750 1060	710 900		58	42	30	22
Synthetic and moulded materials Ebonite (Hard rubber)		HSS TCT	12 12	10 10	300	280	250	224	200		48	35	25	18
Moulded plastic, Bakelite		HSS TCT	12 12	14 14	280	212	170	132	100		48	35	25	18

[1]TCT: Tungston carbide tipped tools
Cutting speeds given apply to the hard TCT grades

Material removal rate—machine power

Chip section $A = a \times s$ (mm²)
Material removal rate $V = a \times s \times u$ (cm³/min)
Tangential cutting force $F = a \times s \times Ks$ (kgf/mm²)
Machine power $P = \frac{a \times s \times Ks \times v}{60 \times 120 \times \eta}$ (kW)

a=depth of cut
s=feed in mm/rev
v=cutting speed in m/min
Ks=specific cutting force kgf/mm²
η=efficiency

Example: Material St 37; a=4 mm; s=0.4 mm/rev.; Ks=190 kgf/mm²; u=45 m/min; η=0.75
What amount of machine power is required?

Solution: $P = \frac{4 \times 0.4 \times 190 \times 45}{60 \times 120 \times 0.75} = 2.98 \approx 3$ kW

Chip cross-section at machine power P=1 kW

Cutting speed v m/min	Specific cutting force Ks in kgf/mm²																	
	660	570	500	480	410	360	300	260	215	200	190	160	140	125	100	70	50	35
	Chip section A in mm² P=kW (η=0.75)																	
5	1.39	1.61	1.83	1.91	2.23	2.55	3.06	3.53	4.27	4.59	4.83	5.73	6.55	7.34	9.18	13.11	18.36	26.22
6	1.15	1.34	1.53	1.59	1.86	2.12	2.53	2.94	3.55	3.82	4.02	4.78	5.46	6.12	7.65	10.92	15.30	21.85
8	0.87	1.00	1.14	1.19	1.39	1.59	1.91	2.20	2.66	2.86	3.02	3.56	4.09	4.59	5.73	8.19	11.47	16.39
10	0.69	0.80	0.91	0.95	1.11	1.27	1.53	1.76	2.13	2.29	2.41	2.86	3.27	3.67	4.59	6.55	9.18	13.11
13	0.53	0.61	0.70	0.73	0.83	0.98	1.17	1.53	1.64	1.76	1.85	2.20	2 52	2.82	3.53	5.04	7.06	10.08
14	0.49	0.57	0.65	0.68	0.79	0.91	1.09	1.26	1.52	1.64	1.72	2.04	2.36	2.62	3.27	4.68	6.55	9.36
15	0.46	0.53	0.61	0.63	0.74	0.85	1.02	1.17	1.42	1.53	1.61	1.91	2.18	2.44	3.06	4.37	6.12	8.74
18	0.38	0.44	0.51	0.53	0.62	0.70	0.85	0.98	1.18	1.26	1.34	1.59	1.82	2.04	2.55	3.64	5.10	7.28
20	0.34	0.40	0.45	0.47	0.55	0.63	0.76	0 88	1.06	1.14	1.20	1.43	1.63	1.83	2.29	3.27	4.59	6.55
24	0.29	0.33	0.38	0.39	0.46	0.53	0.63	0.73	0.88	0.95	1.00	1.19	1.36	1.53	1.91	2.18	3.82	5.46
25	0.27	0.32	0.36	0.38	0.44	0.51	0.61	0.70	0.85	0.91	0.98	1.14	1.31	1.46	1.83	2.62	3.67	5.24
27	0.25	0.29	0.34	0.35	0.41	0.47	0 57	0.65	0.79	0.85	0.89	1.06	1.21	1.36	1.70	2.42	3.40	4.85
32	0.21	0.25	0.28	0.29	0.35	0.39	0.47	0.55	0.66	0.71	0.75	0.89	1.02	1.14	1.43	2.04	2.86	4.09
34	0.20	0.23	0.26	0.28	0.33	0.37	0.45	0.52	0.62	0.67	0.71	0.84	0.96	1.08	1.35	1.92	2.70	3.85
40	0.17	0.20	0.22	0.23	0.27	0.31	0.38	0.44	0.53	0.57	0.60	0.71	0.81	0.91	1.14	1.63	2.29	3.27
45	0.15	0.18	0.20	0.21	0.25	0.28	0.34	0.38	0.47	0.51	0.53	0.63	0.72	0.81	1.02	1.48	2.04	2.91
50	0.13	0.16	0.18	0.19	0.22	0.25	0.30	0.35	0.42	0.46	0.49	0.57	0.65	0.73	0.91	1.31	1.83	2.62
53		0.15	0.17	0.18	0.21	0.24	0.28	0.33	0.40	0 43	0.46	0.54	0.61	0.69	0.86	1.23	1.73	2.47
60		0.13	0.15	0.16	0.18	0.21	0.25	0.29	0.35	0.38	0.40	0.47	0.54	0.61	0.76	1.09	1.53	2.18
63		0.12	0.14	0.15	0.17	0.20	0.24	0.28	0.33	0.36	0.38	0.45	0.52	0.58	0.73	1.04	1.45	2.08
67		0.11	0.13	0.14	0.16	0.19	0.22	0.26	0.31	0.34	0.36	0.42	0.49	0.54	0.68	0.97	1.37	1.95
75		0.10	0.12	0.12	0.14	0.17	0.20	0.23	0.28	0.30	0.32	0.38	0.42	0.48	0.60	0.87	1.22	1.74
85			0.10	0.11	0.13	0.15	0.18	0.20	0.25	0.27	0.28	0.33	0.38	0.43	0.54	0.77	1.08	1.54
95	Example:		0.09	0.10	0.11	0.13	0.16	0.18	0.22	0.24	0.25	0.30	0.34	0.38	0.48	0.69	0.96	1.37
100	P=? kW		00.9	0.09	0.11	0.13	0.15	0.17	0.21	0.23	0.24	0.28	0.32	0.36	0.45	0.65	0.91	1.31
106	St 63		00.8	0.09	0.10	0.12	0.14	0.16	0.20	0.21	0.22	0.27	0.30	0.34	0.43	0.61	0.86	1.23
118			00.7	0.08	0.09	0.10	0.13	0.15	0.18	0.19	0.20	0.24	0.28	0.31	0.38	0.55	0.78	1.11
150	s=4 min/rev			0.06	0.07	0.08	0.10	0.11	0.14	0.15	0.16	0.19	0.21	0.24	0.30	0.43	0.61	0.87
170	a=4 mm			0.05	0.06	0.07	0.09	0.10	0.12	0.13	0.14	0.16	0.19	0.21	0.27	0.38	0.54	0.77
200	v=32 m/min For P=1 kW is			0.04	0.05	0.06	0.07	0.08	0.10	0.11	0.12	0.14	0.16	0.18	0.22	0.32	0.45	0.65
224	A=0.66 mm²				0.05	0.05	0.06	0.08	0.09	0.10	0.11	0.13	0.14	0.16	0.20	0.29	0.40	0.58
236	For A=1.6 mm² is				0.04	0.05	0.06	0.07	0.08	0.09	0.10	0.12	0.13	0.15	0.19	0.27	0.38	0.55
250					0.04	0.04	0.05	0.07	0.08	0.08	0.09	0.11	0.13	0.14	0.18	0.26	0.36	0.52
280	$P = \frac{1 \times 1.6}{0.66} = 2.4$ kW				0.03	0.04	0.05	0.06	0.07	0.08	0.08	0.10	0.11	0.13	0.16	0.23	0.32	0.46
300						0.03	0.05	0.05	0.07	0.07	0.08	0.09	0.10	0.12	0.15	0.21	0.30	0.43
400						0.03	0.04	0.04	0.05	0.05	0.06	0.07	0.08	0.09	0.11	0.16	0.22	0.32
500							0.03	0.03	0.04	0.04	0.04	0.05	0.06	0.07	0.09	0.13	0.18	0.26
600							0.02	0.03	0.03	0.03	0.04	0.04	0.05	0.06	0.07	0.10	0.15	0.21
300									0.02	0.02	0.03	0.03	0.04	0.04	0.05	0.08	6.11	0.16
1000																0.06	0.09	0.13
1500																0.04	0.06	0.09

Taper Turning

1. Swivelling the compound slide

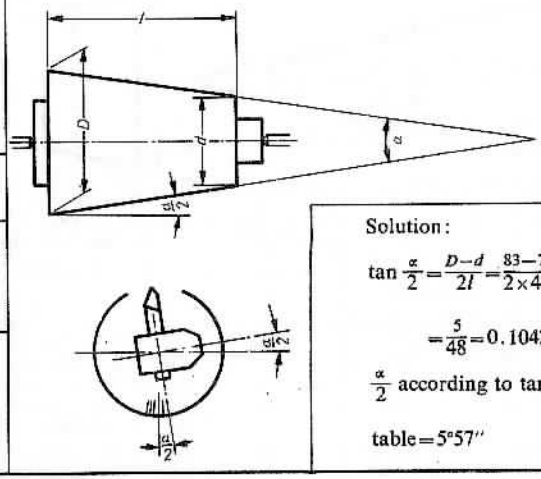
Set to degrees

The compound slide is shifted by $\alpha/2$ degrees (half taper angle); setting angle $\alpha/2$ is to be computed using the formula:

$\tan \frac{\alpha}{2} = \frac{D-d}{2} : l = \frac{D-d}{2 \times l}$

The result of the computation can now be looked up in the tangent table on page 59.

Example:
D=83 mm
d=73 mm
l=48 mm



Solution:

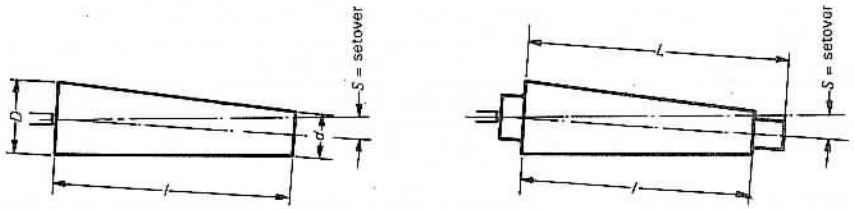
$\tan \frac{\alpha}{2} = \frac{D-d}{2l} = \frac{83-73}{2 \times 48}$

$= \frac{5}{48} = 0.1042$

$\frac{\alpha}{2}$ according to tangent table = 5°57″

2. Setting over the tail stock

Shift the axis of rotation of the workpiece, at an angle to the lathe axis, and feed the tool parallel to the lathe axis. The angle at which the axis of rotation of workpiece is shifted is equal to half the angle of taper.



When length of taper = length of work

$S = \frac{D-d}{2}$

Note: Applicable for slender tapers only

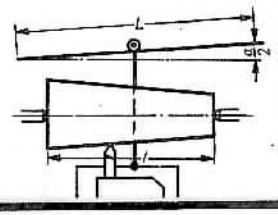
$S \leqslant \frac{L}{50}$; $\frac{\alpha}{2} \leqslant 3°$

Length of the taper is less than length of workpiece

$S = \frac{D-d}{2 \times l} \times L$

3. Taper turning by a taper attachment

For tapers upto a setting angle of 10°, and a length of max. 500 mm, depending on the length of the guide bar.

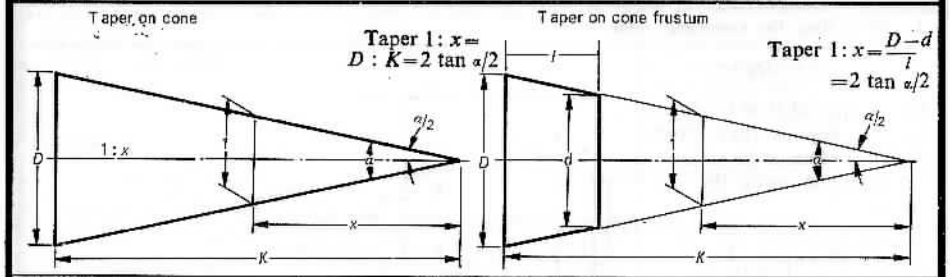


The guide bar is to be displaced by the setting angle $\alpha/2$ (half cone angle). The angle can be computed as above

$\tan \frac{\alpha}{2} = \frac{D-d}{2 \times l}$

Details of tapers

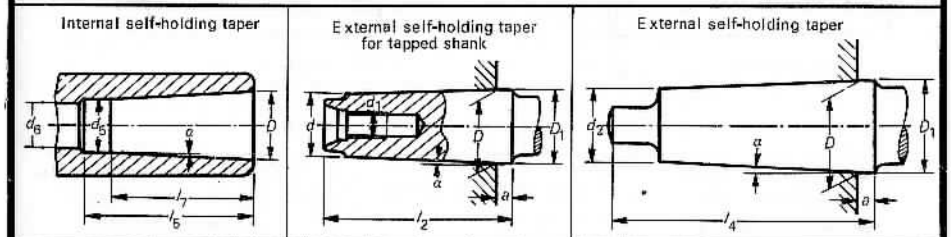
IS: 3458–1966



Nominal taper 1 : x (Ratio of dia D to its axial length K)	Cone angle α	Setting angle for machine taper-angle = α/2	Applications M = Machine construction W = Tool design
1 : 0.289	120°	60°	(M) countersink for centre bore
1 : 0.500	90°	45°	(M) valve cone, shoulders on piston rod, tip of lathe centre, countersunk screws, locknut for pipeline
1 : 0.866	60°	30°	(M) veeslots, centres bores, packing cone (W) tips of lathe centre(dead or live centre)
1 : 1.50	30°52′12″	18°26′6″	(M) conical seals for heavy pipe screw units
1 : 3.429	16°35′40″	8°17′50″	(W) steep angle taper, milling spindle/head, and milling cutter
1 : 5	11°25′16″	5°42′38″	(M) pivot journal, friction clutch, bores in pulleys, locking device for abrasive disc, cones for stop valve
1 : 6	9°31′38″	4°45′49″	(M) conical seal for cocks, cross head pins for locomotive, die sinking mill
1 : 10	5°43′30″	2°51′45″	(M) taper shaft ends, adj. bearing bushes, die sinking mills, machine parts stressed across or along the axis, injection syringes
1 : 15	3°49′6″	1°54′33″	(M) piston rod for locomotive, propeller hub for ships
1 : 20	2°51′52″	1°25′56″	(W) metric tapers, tool tapers, tool shanks and taper nose spindle for machine tools, metric taper fine threads for soldering gear
1 : 30	1°54′34″	0°57′17″	(W) bores in shell reamers and shell drills
1 : 50	1°8′46″	0°34′23″	(M) taper pins, bores in spool for spooling machine, conical pipe thread

Self-holding tapers

IS: 1715–1963

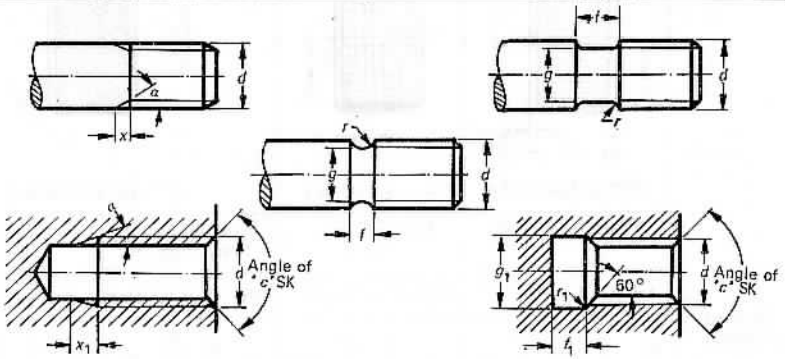


Designation of taper	α	a	D	D1	d	d1	d2	d5	d6	l2	l4	l5	l7
Metric 4	1°25′26″	2.0	4.0	4.1	2.9	—	—	3.0	—	25	—	25	20
Metric 6	1°25′56″	3.0	6.0	6.2	4.4	—	—	4.6	—	35	—	34	28
Morse 0	1°29′27″	3.0	9.045	9.2	6.4	—	6.1	6.7	—	53	59.5	52	45
Morse 1	1°25′43″	3.5	12.065	12.2	9.4	M6	9.0	9.7	7	57	65.5	56	47
Morse 2	1°25′50″	5.0	17.78	18.0	14.6	M10	14.0	14.9	11.5	69	80.0	67	58
Morse 3	1°26′16″	5.0	23.825	24.1	19.8	M12	19.1	20.2	14.0	86	99.0	84	72
Morse 4	1°29′15″	6.5	31.267	31.6	25.9	M16	25.2	26.5	18.0	109	124.0	107	92
Morse 5	1°30′26″	6.5	44.399	44.7	37.6	M20	36.5	38.2	23.0	136	156.0	135	118
Morse 6	1°29′36″	8.0	63.348	63.8	53.9	M24	52.4	54.8	27.0	190	218.0	188	164
Metric 80	1°25′56″	8.0	80.00	80.4	70.2	M30	69.0	71.5	33.0	204	228.0	202	170

Thread run-out and undercut for ISO threads

IS: 1369–1961

Runout of threads	Thread undercut	
For bright and semi-bright screws	For $d \leqslant M5/h \leqslant 0.8$	For $d > M5$ $(h > 0.8)$



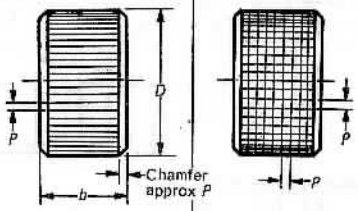
Thread diameter d	For external threads: Thread Runout X (max) for throat angle α			Width undercut (f) for throat angle α			Radius	Nominal undercut diameter	For internal threads: Thread runout X_1 (max) for lead angle α		Clearance a (min) for lead angle α		Width of undercut (f_1) for lead angle α		Radius	Normal undercut diameter
Coarse	⩾ 15°	⩾ 22.5°	⩾ 30°	⩾ 15°	⩾ 22.5°	⩾ 30°	r	g	⩾ 20°	⩾ 25°	> 20°	> 25°	⩾ 20°	⩾ 25°	r_1	g_1
series	Large	Normal	Small	Normal	Normal	Small			Normal	Small	Normal	Small	Normal	Small		
M 2	1.0	0.6	0.4	—	1	—	0.5	—	0.8	0.6	1.0	0.8	—	—	—	—
M 3	1.2	0.8	0.6	—	1.2	—	0.6	—	1.0	0.8	1.8	1.2	—	—	—	—
M 4	1.6	1.0	0.8	—	1.6	—	0.8	—	1.4	1.0	2.0	1.4	—	—	—	—
M 5	2.0	1.2	0.8	—	1.6	—	0.8	—	1.6	1.2	2.0	1.4	—	—	—	—
M6, M7	2.5	1.6	1.0	2	—	1.6	0.8	—	2.0	1.5	2.5	1.6	—	—	—	—
M8, M9	3.0	2.0	1.4	2.5	—	2.0	1.0	d—1.7	2.5	1.8	2.5	1.6	—	—	—	—
M10, M11	3.5	2.5	1.6	3.0	—	2.0	1.0	d—2.1	3.0	2.0	2.5	1.6	5.0	3.0	1.0	d+0.5
M12	4.0	2.5	2.0	4.0	—	2.5	1.2	d—2.4	3.5	2.5	2.5	1.6	5.0	3.5	1.0	d+0.5
M14, M16	5.0	3.0	2.5	4.0	—	2.5	1.2	d—2.7	4.0	3.0	2.5	1.6	6.0	4.0	1.2	d+0.5
M18, M20	6.0	4.0	3.0	5.0	—	3.0	1.6	d—3.3	5.0	3.5	2.5	1.6	8.0	5.0	1.6	d+0.5
M24, M27	7.0	4.5	3.5	6.0	—	4.0	2.0	d—4.0	6.0	4.0	2.5	1.6	10.0	6.0	2.0	d+0.5
M30, M33	8.0	5.0	4.0	8.0	—	5.0	2.0	d—4.6	7.0	5.0	3.0	2.0	10.0	6.0	2.0	d+0.5
M36, M39	10.0	6.0	4.5	8.0	—	5.0	2.0	d—5.2	8.0	6.0	3.0	2.0	12.0	8.0	2.0	d+0.5
M42, M45	11.0	7.0	5.0	10.0	—	6.0	2.5	d—5.9	9.0	6.0	3.0	2.0	14.0	8.0	2.5	d+0.5
M48, M52	12.0	8.0	6.0	10.0	—	6.0	2.5	d—6.5	10.0	7.0	3.0	2.0	14.0	8.0	2.5	d+0.5

For Whitworth Threads

Thread ϕ d	Thread runout max for α = 22½°		Width undercut for α = 22½°		Undercut		Radius	
	X	X_1	f_1	f_1	g	g_1	r	r_1
1/4″	2.0	3.0	2.5	4.0	4.4	6.5	1.0	1.0
3/8″	2.5	4.0	3.0	5.0	7.0	10.0	1.0	1.0
1/2″	3.0	5.0	4.0	6.0	9.5	13.0	1.2	1.2
5/8″	4.0	6.0	5.0	8.0	12.5	16.5	1.6	1.6
3/4″	4.0	6.0	5.0	8.0	15.0	19.5	1.6	1.6
7/8″	4.0	7.0	6.0	8.0	18.0	23.0	2.0	1.6
1″	5.0	8.0	6.0	10.0	20.0	26.0	2.0	2.0
1⅛″	6.0	9.0	8.0	10.0	26.0	32.5	2.0	2.0
1¼″	7.0	10.0	8.0	12.0	32.0	39.0	2.0	2.0
1½″	8.0	12.0	10.0	14.0	37.0	45.0	2.5	2.5
2″	9.0	14.0	12.0	16.0	42.0	52.0	3.0	3.0
2½″	10.0	14.0	12.0	16.0	54.0	64.0	3.0	3.0
3″	11.0	15.0	16.0	18.0	66.0	77.0	4.0	4.0

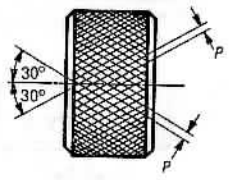
Knurling — IS: 3403–1966

Pitches for straight, cross and diamond knurling



Straight knurling

Cross knurling



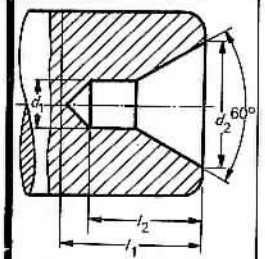
Diamond knurling

Turned diameter D in mm	For all materials: Pitch P for width b					For vulcanite, etc.: Pitch P for width b				For light alloys brass fibre, etc.: Pitch P for width b				For steel: Pitch P for width b			
	Up to 2	Over 4 to 6	Over 6 to 16	Over 16 to 32	Over 32	Up to 6	Over 6 to 16	Over 16 to 32	Over 32	Up to 6	Over 6 to 16	Over 16 to 32	Over 32	Up to 6	Over 6 to 16	Over 16 to 32	Over 32
Up to 8	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
Over 8 to 16	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.8	0.8	0.8	0.8
Over 16 to 32	0.5	0.5	0.8	0.8	0.8	0.5	0.8	0.8	0.8	0.5	0.8	0.8	0.8	0.8	1.0	1.0	1.0
Over 32 to 63	0.5	0.5	0.8	1.0	1.0	0.5	0.8	1.0	1.0	0.5	0.8	1.0	1.0	0.8	1.0	1.2	1.2
Over 63 to 100	0.8	0.8	0.8	1.0	1.2	0.8	0.8	1.0	1.2	0.8	0.8	1.0	1.2	0.8	1.0	1.2	1.6
Over 100	0.8	1.0	1.0	1.0	1.2	0.8	1.0	1.2	1.6	0.8	1.0	1.2	1.6	1.0	1.2	1.6	2.0

Centre Holes — IS: 2473–1963

Without chamfer (Type A) | With chamfer (Type B)

Type A



Dimensions of centre holes in mm

Type A				Type B				
d_1	l_1*	d_2	l_2 min	l_1* approx	b	d_3	d_4	l_3 min
1	3	2.5	2.2	3.5	0.4	2.5	4	2.6
1.6	5	4	3.5	5.5	0.5	4	5.6	4
2.5	7	6.3	5.5	8.3	0.8	6.3	9	6.3
4	11	10	9	12.7	1.2	10	14	10.2
6.3	18	16	14	20	1.8	16	22.4	15.8
10	28	25	22	—	—	—	—	—
16	70	63	60	—	—	—	—	—

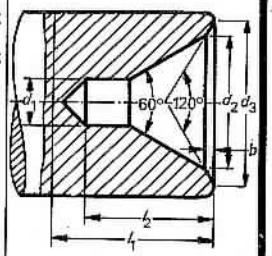
*Parting off dimension, if the centre hole is to be removed from the finished workpiece.

Designation:

A centre hole of Type A and diameter, d_1=4 mm is designated as

Centre Hole A×4 IS: 2473

Type B



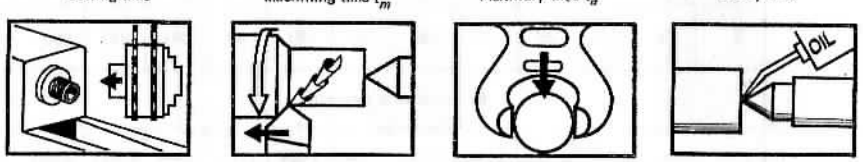
n = Revolutions per min — Cutting speed v — Diameter d

ϕ in mm d	Cutting speed v in m/min: 8	10	15	20	25	30	35	40	50	80	100	150
	Revolutions per min											
5	510	636	955	1272	1590	1912	2230	2548	3180	5095	6360	9550
6	425	531	797	1060	1325	1593	1856	2124	2650	4240	5300	8000
7	364	455	683	910	1136	1365	1593	1820	2275	3630	4550	6800
8	318	400	597	796	996	1194	1393	1592	1990	3180	3980	6000
9	283	354	530	708	886	1060	1240	1415	1770	2830	3540	5200
10	255	318	478	637	796	956	1125	1274	1590	2550	3180	4800
11	231	289	434	580	724	868	1013	1157	1445	2310	2890	4350
12	212	265	398	531	663	796	928	1060	1325	2130	2660	4000
14	182	228	341	455	568	682	796	910	1136	1820	2280	3410
16	159	199	298	398	497	597	695	796	995	1590	1990	2980
18	142	177	265	354	443	530	620	708	885	1420	1770	2660
20	128	159	239	319	398	478	558	637	795	1270	1590	2390
22	116	145	217	290	362	434	506	579	723	1150	1450	2170
25	102	128	192	255	319	383	446	510	638	1020	1280	1910
28	91	114	171	227	284	341	398	455	568	910	1140	1710
32	80	100	149	199	249	298	348	398	498	800	1000	1490
36	71	89	133	177	221	265	310	354	442	710	890	1330
40	64	80	119	159	199	239	278	318	393	640	800	1200
45	57	71	106	142	177	214	248	283	354	570	710	1060
50	51	64	96	127	159	191	223	255	318	510	640	950
55	46	58	87	116	145	174	203	231	298	460	580	870
60	43	53	80	106	133	159	186	212	265	420	530	800
70	36	46	68	91	144	136	169	182	227	360	450	680
80	32	40	60	80	100	119	139	159	199	320	400	600
90	28	35	53	71	89	106	124	142	177	285	355	530
100	26	32	48	64	80	96	111	127	159	255	320	480
110	23	29	43	58	73	87	101	116	145	232	290	435
125	20	26	38	51	64	76	89	102	127	200	255	380
140	18	23	34	46	57	68	80	91	114	180	228	340
160	16	20	30	40	50	60	70	80	100	160	200	300
180	14	17	27	35	44	53	53	71	88	140	175	265
200	12	16	24	32	40	48	62	64	80	125	160	240
220	11.6	14	22	29	36	43	50	57	71	114	143	210
250	10.2	12.7	19	25	32	38	44	51	64	100	125	190
275	9.2	11.6	17	23	29	35	40	47	58	93	115	175
300	8.5	10.6	16	21	26	32	37	43	53	85	105	160
350	7.2	9.1	14	18	22	28	32	36	45	73	91	135
400	6.3	7.9	12	16	20	24	28	32	40	64	80	120
450	5.6	7.1	10.6	14	18	21	24	28	36	57	71	105
500	5	6.4	9.5	13	16	19	22	26	32	51	64	95

$v = \pi \times d \times n$ $\quad n = \frac{v}{\pi \times d}$ $\quad d = \frac{v}{\pi \times n}$ put d in metre in the formula

Calculating the machining time

To enable proper estimation of the time required for operating machine tools, the following distinctions are made



Setting up the machine: Getting tools, study of drawings

Actual time in which the tool is cutting

Clamping job, setting the tool, measuring, checking

Lubricant machine trouble shooting or repair work, short breaks

Longitudinal turning

rpm known

l = length to be turned

s_r = feed mm/rev

n = rpm

Feed per minute:

$$s = s_r \times n$$

$$t_m = \frac{l}{s_r \times n} \text{ (min)}$$

Example:

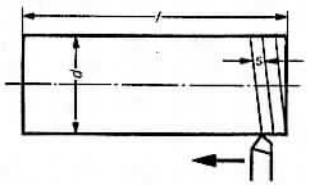
l = 600 mm

s_r = 0.5 mm/rev

n = 50 rpm

$$t_m = \frac{600 \text{ mm}}{0.5 \text{ mm/rev} \times 50 \text{ rpm}}$$

= 24 min



Machining time = length to be turned / feed per minute

Note: The rpm calculated will be different from the rpm available with a particular machine.

rpm unknown

d = diameter (m)

v = cutting speed m/min

s_r = feed mm/rev

l = length to be turned

$$n = \frac{v}{\pi \times d}$$

$$t_m = \frac{l \times \pi \times d}{s_r \times v} \text{ (min)}$$

Example:

d = 0.125 m

v = 20 m/min

s_r = 0.5 mm/rev

l = 600 mm

$$t_m = \frac{600 \text{ mm} \times 3.14 \times 0.125 \text{ m}}{0.5 \text{ mm/rev} \times 20 \text{ m/min}}$$

= 23.5 min

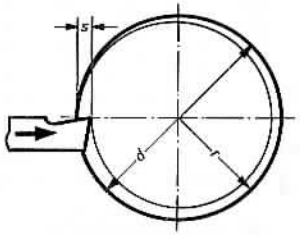
Facing

$$n = \frac{v}{\pi \times d}$$

$$t_m = \frac{r}{s_r \times n}$$

In order to obtain a uniform cutting speed, the rpm should be varied. For a constant rpm an average cutting speed should be considered.

For facing the radius r can be considered, as the length to be turned



Example:

d = 0.250 m

v = 20 m/min

s_r = 0.5 mm/rev

$$n = \frac{v}{\pi \times d} = \frac{20 \text{ m/min}}{3.14 \times 0.25 \text{ m}}$$

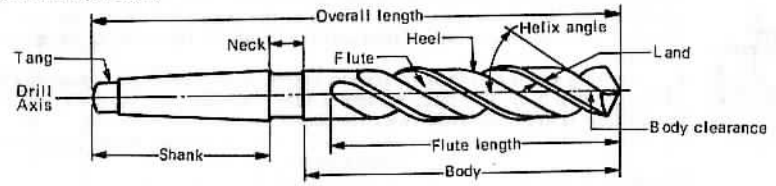
= 25 rpm

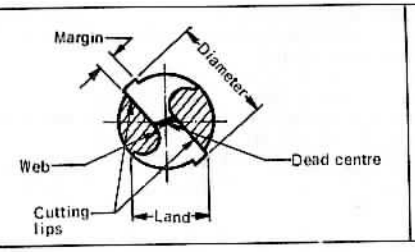
$$t_m = \frac{r}{s_r \times n} = \frac{125 \text{ mm}}{0.5 \text{ mm} \times 25 \text{ rpm}}$$

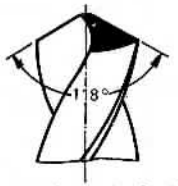
= 10 min

Drills

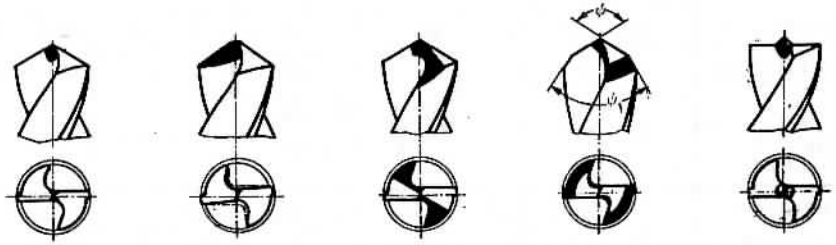
Twist drill with taper shank







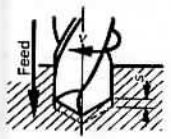
Normal cutting angle of a drill



Recommended drills

Material to be drilled	Point angle	Helix angle d=3.2...5	5–10	10	Material to be drilled	Point angle	Helix angle d 3.5...5 mm	5 mm
Steel and cast steel up to 70 kgf/mm² strength Gray cast iron Malleable cast iron Brass German silver, nickel	118°	22°	25°	30°	Copper (up to 30 mm drill diameter) Al-alloys, forming curly chips; Celluloid	140°	35°	40°
Brass, CuZn 40	118°	12°	13°	13°	Austenitic steels Magnesium-alloys	140°	12°	13°
Steel and cast steel 70...120 kgf/mm²	118°	22°	25°	30°	Moulded plastics (with thickness $s \leqslant d$)	80°	35°	40°
Stainless steels; Copper (drill diameter more than 30 mm) Al-alloy, forming short broken chips	140°	22°	25°	30°	Moulded plastics, with thickness $s \leqslant d$, Laminated plastics, hard rubber (ebonite) marble, slate, coal	80°	12°	13°
					Zinc-alloys	118°	35°	40°

Cutting speed *v*—Feed *s*—Coolents



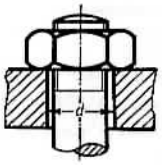
v = Cutting speed: Peripheral speed in m/min. [The speed depends on the material which is to be processed as well as the type of drill to be used; it further depends on the rate of feed and the depth of the hole to be drilled.]
s = Feed.

Cutting speed and feed values are taken on the basis of a tool life for drilling a depth of 2000 mm. assuming that the depth to be drilled in one single hole is approximately twice the diameter.

Material	Cutting speed using tool steel drills	Feed *s* (mm/rev) / Cutting speed *v* in m/min with low-alloy high speed steel / Diameter of drill: 5φ	10φ	15φ	20φ	25φ	30φ	35φ	Cooling and lubricating agents
Steel up to 40 kgf/mm²	...20	0.1 / 15	0.18 / 18	0.25 / 22	0.28 / 26	0.31 / 29	0.34 / 32	0.36 / 35	Soluble oil mineral oil
Up to 60 kgf/mm²	...14	13	16	20	23	26	28	29	
Up to 80 kgf/mm²	...10	0.07 / 12	0.13 / 14	0.16 / 16	0.19 / 18	0.21 / 21	0.23 / 23	0.25 / 24	Sulphurized and chlorinated oil
Up to 100 kgf/mm²	—	8	10	13	15	17	18	19	
Beyond 100 kgf/mm²	—	0.015...0.17 mm/rev / 6...12 m/min							
Grey cast iron Up to 18 kgf/mm²	...14	0.15 / 24	0.24 / 28	0.3 / 32	0.32 / 34	0.35 / 37	0.38 / 39	0.4 / 40	Dry or plenty soluble oil
Up to 22 kgf/mm²	...10	16	18	21	24	26	27	28	
Grey cast iron Up to 30 kgf/mm²	...8	0.1 / 12	0.16 / 14	0.2 / 16	0.24 / 18	0.28 / 20	0.3 / 21	0.3 / 22	
Brass Up to 40 kgf/mm²	...40	0.1 / 60...70 m/min	0.15	0.22	0.27	0.3	0.32	0.36	2/3 lard oil 1/3 kerosene
Up to 60 kgf/mm²	...25	0.07 / 40...60 m/min.	0.12	0.18	0.24	0.25	0.28	0.32	
Bronze Up to 30 kgf/mm²	...15	0.1 / 30...40m/min	0.15	0.22	0.27	0.3	0.32	0.36	
Up to 70 kgf/mm²	...12	0.05 / 25...35 m/min.	0.08	0.12	0.18	0.2	0.22	0.26	
Aluminium (pure)	...50	0.05 / 80...120 m/min	0.12	0.2	0.3	0.35	0.4	0.46	2/3 lard oil 1/3 kerosene
Aluminium alloys	...40	0.12 / 100...150 m/min	0.2	0.3	0.4	0.46	0.5	0.6	
Magnesium alloys	...80	0.15 / 200...250 m/min.	0.2	0.3	0.38	0.4	0.45	0.5	Dry or special oil
Moulded plastics Pressed materials	...15	0.04 / 35...45 m/min	0.05	0.07	0.1	0.12	0.15	0.17	Compressed air

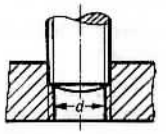
SPK

Drill diameter for



Through holes

Threaded holes (core diameter *d*)



For metric threads

Thread size	Through hole: Fine	Through hole: Medium	Threaded hole: Steel cast steel malleable iron	Threaded hole: Gray cast iron brass bronze
M1	1.1	1.3	0.75	0.7
M1.2	1.3	1.5	0.95	0.91
M1.6	1.7	1.8	1.3	1.2
M2	2.2	2.4	1.6	1.5
M2.5	2.7	3.0	2.1	2.0
M3	3.2	3.6	2.5	2.4
M4	4.3	4.8	3.3	3.2
M5	5.3	5.8	4.2	4.1
M6	6.4	7	5	4.8
M8	8.4	9.5	6.7	6.5
M10	10.5	11.5	8.4	8.2
M12	13	14	10	9.9
M16	17	18	13.75	13.5
M20	21	23	17.25	17.5
M24	25	27	20.75	20.5
M30	31	33	26	25.75
M36	37	39	31.5	31
M42	43	45	37	36.5
M45	46	48	40	39.5
M48	50	52	42.5	42
M52	54	56	46.5	46
M56	58	62		
M60	62	65		
M64	66	70		
M68	70	74		

For Whitworth threads

Thread size	Through hole: Fine	Through hole: Medium	Threaded hole: Steel cast steel malleable iron	Threaded hole: Gray cast iron brass bronze
1/4"	6.7	7.4	5.0	4.8
5/16"	8.4	9.5	6.4	6.2
3/8"	10	11.5	7.7	7.5
7/16"	12	13	9.25	9.0
1/2"	13.5	15	10.25	10.0
5/8"	17	18	13.25	13.0
3/4"	20	22	16.25	16.0
7/8"	23	25	19	18.75
1"	26	28	22	21.5
1 1/8"	30	32	24.75	24.25
1 1/4"	33	35	27.75	27.5
1 3/8"	36	38	30.5	30
1 1/2"	40	42	33.5	33
1 5/8"	43	45	35.5	35
1 3/4"	46	48	39	38
1 7/8"	53	52	41.5	41
2"	54	55	44.5	44
2 1/4"	60	62		
2 1/2"	66	68		
2 3/4"	72	74		
3"	78	82		
3 1/4"	85	88		
3 1/2"	92	95		
3 3/4"	98	102		
4"	105	108		

Conversion of drill sizes inches—mm

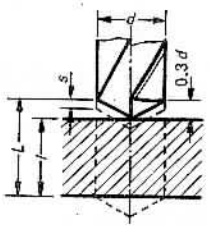
Fractions of inch	Inches	mm	Fractions of inch	Inches	mm	Fractions of inch	Inches	mm
1/32	0.031	0.8	3/16	0.188	4.8	17/32	0.531	13.5
3/64	0.042	1.2	7/32	0.219	5.6	9/16	0.562	14.3
1/16	0.063	1.6	1/4	0.25	6.4	19/32	0.594	15.1
5/64	0.078	2.0	9/32	0.281	7.1	5/8	0.625	15.9
3/32	0.094	2.4	5/16	0.313	7.9	11/16	0.678	17.5
7/64	0.109	2.8	11/32	0.344	8.7	3/4	0.75	19.1
1/8	0.125	3.2	3/8	0.375	9.5	13/16	0.813	20.6
9/64	0.141	3.6	13/32	0.406	10.3	7/8	0.875	22.2
5/32	0.156	4.0	7/16	0.437	11.3	15/16	1.938	23 8
11/64	0.172	4.4	1/2	0.5	12.7	1	1.000	25.4

Calculating machining time for drilling operations

The machining (drilling) time is the period of time in which the machine performs the actual drilling operation

rpm—known

L = length of drill travel in mm
$L = l + 0.3\ d$
d = diameter of the drill in mm
n = revolutions per minute
s_r = feed in mm/revolution
s = feed per minute



rpm—unknown

$L = l + 0.3\ d$
d = diameter of the drill in mm
s_r = feed in mm/revolution
v = cutting speed in m/min

$$v = \frac{\pi \times d \times n}{1000}$$

$$n = \frac{v \times 1000}{\pi \times d} \text{ min}$$

t_m = machining time

t_m = machining time

$$t_m = \frac{L}{s_r \times n} \text{ min}$$

Machining time = length of tool travel / feed per min

$$t_m = \frac{L \times \pi \times d}{s_r \times v \times 1000} \text{ min}$$

Example: l=35 mm d=30 mm
s_r=0.2 mm/rev, n=300 rpm
$L = l + 0.3d = 35 + 0.3 \times 30$
$= 35 + 9 = 44$ mm

$$t_m = \frac{L}{s_r \times n} = \frac{44}{0.2 \times 300} = 0.73 \text{ min}$$

Example: l = 35 mm d=30 mm
s_r = 0.2 mm/rev v=28 m/min
$L = l + 0.3\ d$ =35+9=44 mm

$$t_m = \frac{L \times \pi \times d}{s_r \times v \times 1000} = \frac{44 \times \pi \times 30}{0.2 \times 28 \times 1000}$$

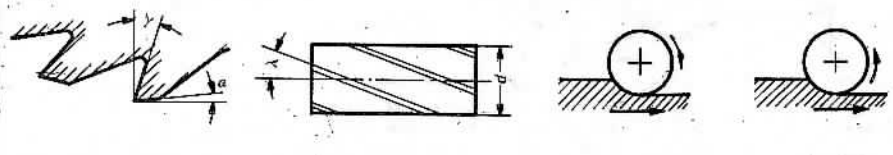
=0.73 min

Machining time in minutes per 10 mm length of tool travel

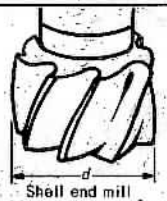
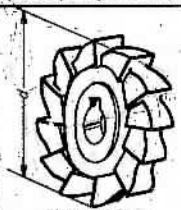
rpm n	Feed s in mm/rev								
	0.1	0.12	0.16	0.2	0.25	0.32	0.4	0.5	0.65
	Machining time t_m in min/10 mm								
11.2	8.93	7.44	5.85	4.46	3.57	2.79	2.23	1.78	1.38
14	7.15	5.95	4.46	3.57	2.85	2.23	1.78	1.43	1.09
18	5.56	4.36	3.47	2.77	2.22	1.73	1.39	1.11	0.85
22.4	4.47	3.71	2.79	2.23	1.78	1.39	1.11	0.89	0.68
28	3.57	2.97	2.23	1.78	1.42	1.11	0.88	0.71	0.55
35.5	2.82	2.34	1.76	1.40	1.12	0.86	0.70	0.56	0.43
45	2.22	1.85	1.39	1.11	0.88	0.69	0.55	0.44	0.34
56	1.79	1.31	1.12	0.89	0.71	0.55	0.44	0.36	0.27
71	1.41	1.17	0.88	0.70	0.53	0.44	0.35	0.28	0.21
90	1.11	0.92	0.69	0.55	0.44	0.34	0.28	0.22	0.17
112	0.89	0.74	0.58	0.44	0.36	0.28	0.22	0.18	0.14
140	0.71	0.59	0.44	0.36	0.28	0.22	0.18	0.14	0.109
180	0.55	0.43	0.35	0.28	0.22	0.17	0.14	0.111	0.085
224	0.45	0.37	0.28	0.22	0.18	0.13	0.111	0.081	0.068
280	0.36	0.29	0.22	0.18	0.14	0.111	0.088	0.071	0.055
355	0.28	0.23	0.17	0.14	0.112	0.086	0.070	0.056	0.043
450	0.22	0.18	0.14	0.111	0.088	0.069	0.055	0.044	0.034
560	0.18	0.13	0.112	0.089	0.071	0.055	0.044	0.036	0.027
710	0.14	0.117	0.088	0.070	0.053	0.044	0.035	0.028	0.021
900	0.111	0.092	0.069	0.055	0.044	0.034	0.028	0.022	0.017
1120	0.089	0.074	0.058	0.044	0.036	0.028	0.022	0.018	0.014
1400	0.071	0.059	0.044	0.036	0.028	0.022	0.018	0.014	0.011
1800	0.055	0.043	0.035	0.028	0.022	0.017	0.014	0.011	0.008
2240	0.045	0.037	0.028	0.022	0.018	0.013	0.011	0.009	0.007
2800	0.035	0.029	0.022	0.018	0.014	0.011	0.009	0.007	0.006

Milling

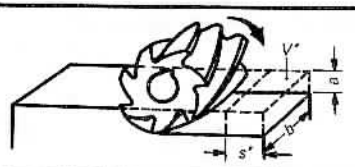
Suggested number of teeth and angles of high-speed steel cutters



α=clearance angle
γ=radial rake angle
λ=helix ange

Type of cutter	Plain carbon steels up to 75 kgf/mm² strength			Tough materials up to 100 kgf/mm² strength			Light alloys		
	ϕ	Number of teeth	Angles $\measuredangle$	ϕ	Number of teeth	Angles $\measuredangle$	ϕ	Number of teeth	Angles $\measuredangle$
	d	z	α γ λ	d	z_1	α γ λ	d	z	α γ λ
Cylindrical cutter or slab milling cutter	40	6	up-milling	40	10	up-milling	40	4	up-milling
	50	6	7° 10° 38°	50	10	4° 5° 35°	50	4	8° 25° 45°
	63	6		63	10		63	5	
	80	8		80	12		80	5	
	100	8		100	16		100	6	
	125	10	down-milling	125	16	down-milling	125	6	down-milling
	160	10	12° 16° 35°	160	20	8° 12° 30°	160	8	14° 30° 45°
 Shell end mill (face milling cutter)	40	8		40	12		40	4	
	50	10		50	14		50	5	
	63	10	up-milling	63	14	up-milling	63	6	up-milling
	80	12	7° 10° 20°	80	18	4° 5° 20°	80	6	8° 25° 35°
	100	12		100	20		100	7	
	125	14		125	22		125	8	
	160	16		160	24		160	10	
 Side and face milling cutter	50	10	up-milling	50	16	up-milling	50	4	up-milling
	63	10	α γ λ	63	16	α γ λ	63	6	α γ λ
	80	12	7° 12° 15°	80	18	5° 6° 10°	80	6	8° 25° 30°
	100	14		100	20		100	8	
	125	16	down-milling	125	22	down-milling	125	8	down-milling
	160	18	α γ λ	160	24	α γ λ	160	10	α γ λ
	180	18	12° 18° 15°	180	26	8° 14° 12°	180	12	14° 30° 30°
	200	20		200	30		200	12	
End milling cutter	10	4		10	6		10	3	
	12	4		12	6		12	3	
	16	5	up-milling	16	8	up-milling	16	3	up-milling
	20	6	7° 8° 15°	20	8	4° 6° 15°	20	4	8° 20° 25°
	25	6		25	8		25	4	
	32	6		32	10		32	4	
	40	6		40	10		40	5	

Maximum chip removal rate



The max. chip removal V' in cm³ per kW/min

Method of milling	Carbon steel 35...60 kgf/mm² strength	Alloy steel (annealed) 60...80 kgf/mm² strength	Alloy steel (heat treated) up to a strength of 100 kgf/mm²	Grey cast (moderately hard)	Brass and gun metal	Light alloys Light metal
Slab milling	12	10	8	22	30	60
	cm³/kW min					
Face milling	15	12	10	28	40	75

Calculating the rate of feed

The maximum chip removal V can be found by multiplying the cutting capacity V' with the rated power P of the machine

Max. chip removal/min = cutting capacity × rated power of the machine

$$V = V' \times P$$

The chip removal can be found by multiplying the depth of cut a by the width of cut b by the rate of feed s

$$\text{Metal removal rate (volume of chips produced per minute)} = \frac{\text{Depth of cut} \times \text{width of cut} \times \text{rate of feed}}{1000}$$

$$V = \frac{a \times b \times s}{1000}$$

The maximum permissable rate of feed s thus becomes

$$s = \frac{V \times 1000}{a \times b} \text{ mm/min}$$

Example: A workpiece made of St 50 is to be milled. The driving power of the milling machine is 2.5 kW. Depth of cut $a = 5$ mm, width of cut $b = 100$ mm.

Find the max. permissable rate of feed s

Maximum chip removal per minute $= V = V' \times P$ ($V' = 12$ cm³/kW min—value taken from table)

$V = 12 \times 2.5$

$V = 30$ cm³/min

$$\text{The rate of feed} = \frac{V \times 1000}{a \times b} = \frac{30 \times 100}{5 \times 100} = 60 \text{ mm/m}$$

Feed rate s based on the permissible quantity V' of chips produced

Permissible chip removal V' cm³/kW min	Depth of cut a mm	s with a machine drive power $P = 1$ kW; Width b of cut in mm: 40	50	60	80	100	120	140	160	180
8	3	66	53	44	33	26	22	19	16	15
	5	40	32	27	20	16	13	11	10	9
	8	25	20	16	12	10	8	7	6	5.5
10	3	83	66	55	41	33	27	23	20	18
	5	50	40	33	25	20	16	14	12	11
	8	31	25	21	15	12.5	10	9	8	7
12	3	100	80	67	50	40	33	29	25	22
	5	60	48	40	30	24	20	17	15	13
	8	37	30	25	19	15	12	10	9	8
15	3	125	100	84	62	50	42	36	31	28
	5	75	60	50	37	30	25	21	19	16
	8	47	37	31	21	19	15	13	11	10
22	3	184	146	121	92	73	61	52	46	41
	5	110	88	73	55	44	37	31	27	24
	8	69	55	46	34	27	23	19	17	15
28	3	230	185	155	116	94	78	67	58	52
	5	140	110	93	70	56	47	40	35	31
	8	87	70	58	44	35	29	25	22	19
60	3	500	400	335	250	200	165	142	125	110
	5	300	240	200	150	120	100	86	75	67
	8	185	150	125	94	75	62	53	47	42
75	3	625	500	415	310	250	205	178	156	140
	5	375	300	250	185	150	125	105	94	83
	8	235	185	155	115	94	78	67	58	52

* Calculated values shown in the table are to be multiplied by the factors 2.5 or 5 in case the machine drive power P is 2.5 or 5 kW respectively.

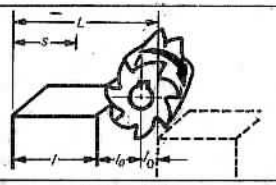
Estimation of machining time

$$\text{Machining time} = \frac{\text{Total length of travel}}{\text{Rate of feed}}$$

$$t_m = \frac{L}{s}$$

The total length of travel depends on the length of the workpiece, the size of cutter used and the method of milling employed.

Travel L

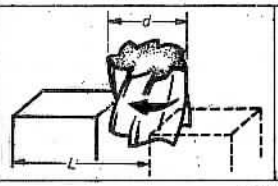


Slab milling

Roughing and finishing cutters

L = length of workpiece + approach + over travel

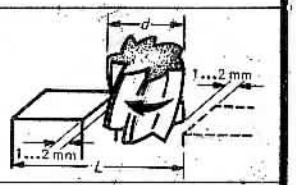
$L = l + l_a + l_o$



Face milling

Roughing cut

$L = l + \frac{d}{2} + 2$

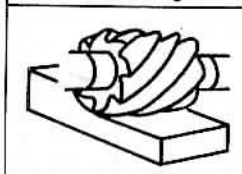
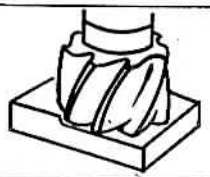
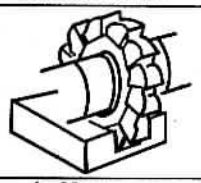


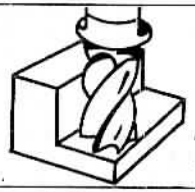
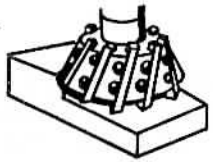
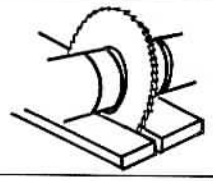
Face milling

Finish cut

$L = l + d + 4$

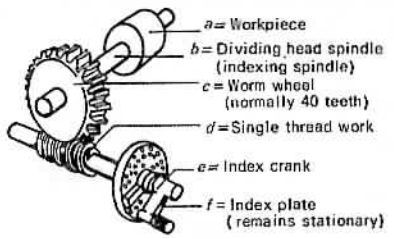
Suggested cutting speed and feed*

	Slab milling				Shell-end mill				Site and face-mill			
												
Width of cut b	b=100 mm				b=70 mm				b=20 mm			
	Roughing ▽		Finishing ▽▽		Roughing ▽		Finishing ▽▽		Roughing ▽		Finishing ▽▽	
Depth of cut a	a=5 mm		a=0.5 mm		a=5 mm		a=0.5 mm		a=10 mm			
	Cutting speed v m/min	Feed s' mm/min	Cutting speed v m/min	Feed s' mm/min	Cutting speed v m/min	Feed s' mm/min	Cutting speed v m/min	Feed s' mm/min	Cutting speed v m/min	Feed s' mm/min	Cutting speed v m/min	Feed s' mm/min
Carbon steel up to 63 kgf/mm²	17	100	22	60	17	100	22	70	18	100	22	40
Alloy steel annealed up to 78 kgf/mm²	14	80	18	50	14	90	18	55	14	80	18	30
Alloy steel heat treated up to 100 kgf/mm²	10	50	14	36	10	55	14	42	12	50	14	25
Gray cast iron up to HB 180	12	120	18	60	12	140	18	70	14	120	18	40
Brass (Cu Zn 40)	35	70	35	50	36	190	55	150	36	150	55	75
Light alloy	200	200	250	100	200	250	250	110	200	200	250	100

	End milling cutter				Inserted tooth face milling cutter				Circular saw			
												
Width of cut b	b=25 mm				b=180 mm				b=2.5 mm			
	Roughing ▽		Finishing ▽▽		Roughing ▽		Finishing ▽▽		Roughing ▽			
Depth of cut a	a=5mm		a=0.5 mm		a=5 mm		a=0.5 mm		a=10 mm			
	Cutting speed v m/min	Feed s' mm/min	Cutting speed v m/min	Feed s' mm/min	Cutting speed v m/min	Feed s' mm/min	Cutting speed v m/min	Feed s' mm/min	Cutting speed v m/min	Feed s' mm/min		
Carbon steel up to 63 kgf/mm²	17	50	22	120	20	65	30	50	45	50		
Alloy steel annealed up to 78 kgf/mm²	15	40	19	100	16	36	23	40	35	40		
Alloy steel heat treated up to 100 kgf/mm²	13	20	17	65	14	20	18	30	25	30		
Grey cast iron up to HB 180	15	80	19	120	16	100	24	90	35	50		
Brass (Cu Zn 40)	35	80	55	120	50	200	60	120	350	200		
Light alloys	160	90	180	120	250	250	300	90	320	180		

* Values to be adjusted against available machine power (refer page 109).

Simple indexing



A typical set of plates has the following number of holes

I	15	16	17	18	19	20
II	21	23	27	19	31	33
III	37	39	41	43	47	49

To turn a workpiece once, turn the index crank 40 times.

For N equal divisions, make $\frac{40}{N}$ turns with the indexing spindle to get a given division.

$$\text{Number of crank turns for a particular indexing} = \frac{\text{Number of teeth of worm wheel}}{\text{Number of equally spaced divisions}}$$

$$\text{Index crank} \quad n_c = \frac{40}{N}$$

Example: $N=16$; $Z_w=40$

$n_c = \frac{Z}{N} = \frac{40}{16} = 2\ \frac{8}{16}$ number of holes / turns / hole circle

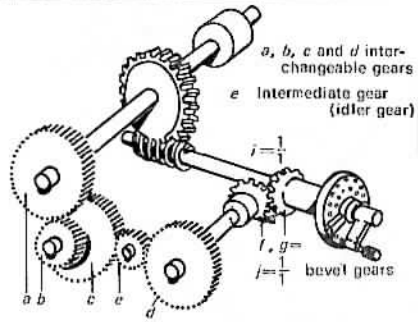
2 complete cranks and 8 more holes on the 16-hole circle

Example: $N=45$; $Z_w=40$

$n_c = \frac{Z}{N} = \frac{40}{45} = \frac{8}{9}$, extended to an available hole circle $\frac{8}{9} = \frac{8}{9} \cdot \frac{2}{2} = \frac{16}{18}$

16 holes on the hole circle 18

Differential indexing



Differential indexing is made possible by connecting the index plate to the head stock spindle by means of a gear train. The index plate can be made to move either in the same direction (positive) or in the opposite direction (negative) to the index crank. This causes the movement of the index plate to be either faster or slower, travelling either more or less than the movement of the index crank

N = Number of divisions required to be indexed per one complete revolution of the work piece

n = Replacement for the number N. [n must be chosen such that it can be used in the simple indexing method]

$$\text{Number of crank turns required} = \frac{\text{Number of teeth 40}}{\text{Selected number of equal divisions}}$$

$$n_c = \frac{40}{N}$$

$$\text{Number of teeth (gear teeth)} \times \text{gear ratio} = \text{Number of crank turns (number of equal divisions required minus selected number of equal divisions)}$$

Selected change gear sets					
24	24	28	32	40	44
48	56	64	72	86	100

Change gears:

$$Z = \frac{40}{n}(n-N)$$

If the number chosen for n is greater than the number of divisions required (N) the index crank and the index plate are said to move in the same direction (positive motion). If the number chosen for n is less than the number N, the opposite direction results (negative motion).

Example:

$N=53$; $Z_w=40$

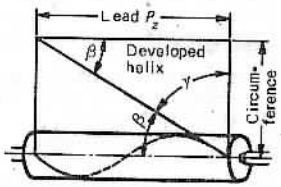
$n=56$ crank turns $n_c = \frac{40}{56} = \frac{15}{21}$, 15 holes on the hole circle 21

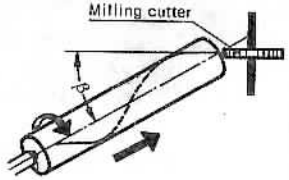
Change gears $= \frac{40}{n}(n-N) = \frac{40}{56}(56-53) = \frac{45}{21} = \frac{9}{3} \times \frac{5}{7} = \frac{72}{24} \times \frac{40}{56}$

Change gears; $Z_a=72$; $Z_b=24$; $Z_c=40$; $Z_d=56$

Milling helical slots and keyways

The workpiece is moved along its longitudinal axis while simultaneously radial motion is also given resulting in the cutter generating helical grove. Both motions are given by the table feed screw. It causes workpiece to move along its longitudinal axis while at the same time a set of change gears is allowed to operate the index spindle. The index plate is to be loosened such that it can rotate. The table must be set to the angle β of adjustment.





Nomenclature:
- P_z lead of helix on workpiece
- p_z lead of the table feed spindle
- d diameter of workpiece
- β helix angle (table angle)
- γ lead angle
- n_c number of crank turns required to turn the workpiece through one complete turn

$(i = n_c : 1 = 40:1)$

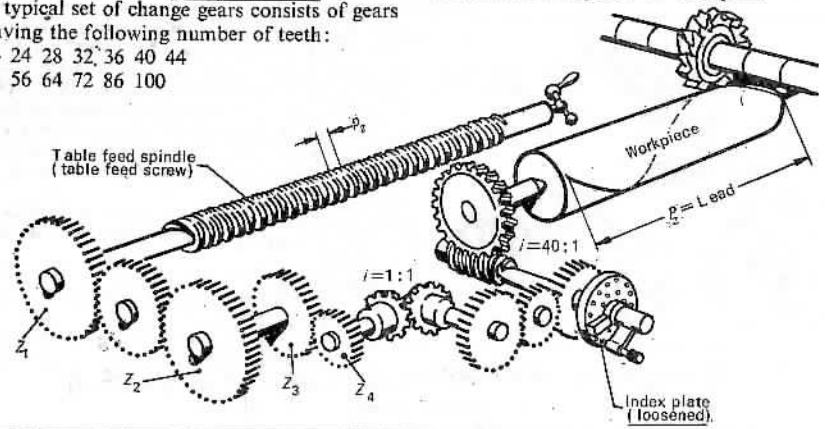
Lead $P_z = \tan\gamma \times \pi d$

Helix angle $\boxed{\beta = 90° - \gamma°}$ or $\boxed{\tan\beta = \frac{\pi \times d}{P_z}}$

Lead angle $\tan\gamma = \frac{P_z}{\pi \times d}$

Gear ratio $\boxed{i = \frac{Z_1}{Z_2}\cdot\frac{Z_3}{Z_4} = \frac{p_z n_c}{P_z}} = \frac{\text{Lead of table feed spindle} \times \text{number of crank turns (40)}}{\text{Lead of helix required on workpiece}}$

A typical set of change gears consists of gears having the following number of teeth:
24 24 28 32 36 40 44
48 56 64 72 86 100



Example: $d = 40$ mm; $P_z = 450$; $p_z = 6$ mm; $n_c = 40$

Solution: Helix angle: $\tan\beta = \frac{\pi d}{P_z} = \frac{3.14 \times 40}{450} = 0.2791$; $\beta = 15°36'$

Change gears $= \frac{Z_1}{Z_2}\times\frac{Z_3}{Z_4} = \frac{p_z n_c}{P_z} = \frac{6\times40}{450} = \frac{2\times4}{5\times3} = \frac{40\times32}{100\times24}$

Check:

$P_z = \frac{Z_3 \times Z_4 \times p_z \times n_c}{Z_1 \times Z_3} = \frac{100\times24\times6\times40}{40\times22}$

$= 450$ mm

Example: $d = 42$ mm; $P_z = 26\frac{1}{4}" \approx 667$ mm; $p_z = 4$ threads per inch $= \frac{1}{4}"$

Solution: Helix angle $\tan\beta = \frac{\pi \times d}{P_z} = \frac{3.14\times42}{667} = 0.1977$; $\beta = 11°11'$

Change gears: $\frac{Z_1}{Z_2}\times\frac{Z_3}{Z_4} = \frac{p_z n_c}{P_z} = \frac{\frac{1}{4}"\times70}{105/4} = \frac{1\times40\times4}{4\times105} = \frac{8}{21} = \frac{2\times4}{3\times7}$

$= \frac{24\times32}{36\times56}$

Check:

$P_z = \frac{Z_2 \times Z_4 \times p_z \times n_c}{Z_1 \times Z_3}$

$= \frac{36\times54}{24\times32}\times\frac{1}{4}\times40$

$= \frac{210}{8} = 26\frac{1}{2}"$

Shaping and planing

Cutting speed and feed

	Type of tool	Steel	Steel	Cast steel	Gray casting	Gun metal	Light alloy
		Cutting speed v in m/min					
Roughing ▽	Tool steel	10...15	8...12	9...12	8...12	15...20	—
	High speed steel	15...20	12...16	12...16	12...16	20...25	30...35
Finishing ▽▽	Tool steel	15...20	12...16	12...16	14...18	20...25	—
	High speed steel	20...25	16...20	16...20	18...22	30...40	50...60
		Feed s in mm per stroke					
	Tool steel	0.2...6					—
	High speed steel	0.6...12					0.1...1

Calculating the required machining time

L = Length of stroke = Length of workpiece l plus l_a and l_o (approach and overtravel)
s = Feed per stroke in mm
V_C = Cutting stroke speed in m/min
V_R = Return stroke speed in m/min

Time taken by cutting stroke

$t_C = \frac{\text{Length of stroke}}{\text{Cutting stroke speed}}$ $\quad t_C = \frac{L}{V_C \times 1000}$ min

Time taken by return storke

$t_R = \frac{\text{Length of stroke}}{\text{Return stroke speed}}$ $\quad t_R = \frac{L}{V_R \times 1000}$ min

Time taken by one complete stroke

t = Time for cutting-stroke plus time for return stroke $\quad t = \frac{L}{V_C \times 1000} + \frac{L}{V_R \times 1000}$

Number of complete stroke

$Z = \frac{\text{Width of work}}{\text{Feed}}$ $\quad Z = \frac{b}{s}$

Required machining time t_m = Number of complete strokes × time taken by one complete stroke

$\boxed{t_m = \frac{b}{s} \times \left(\frac{L}{V_C \times 1000} + \frac{L}{V_R \times 1000}\right)}$ in min or $\boxed{t_m = \frac{b}{s} \times \frac{2L}{V_m \times 1000}}$ in min

Average speed $V_A = \frac{2V_C \times V_R}{V_C + V_R}$ m/min

When n, the number of complete strokes per min is known, the average speed V_A can be calculated using the formula

$V_A = \frac{2L \times n}{1000}$ m/min

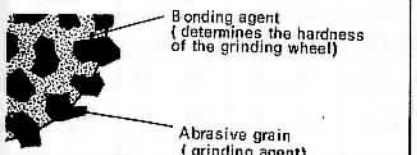
Example: Width of workpiece $b = 200$, Length of workpiece plus over travel $L = 400$ mm
$V_C = 10$ m/min $\quad V_R = 20$ m/min $\quad s = 5$ mm

$t_m = \frac{b}{s}\left(\frac{L}{V_C \times 1000} + \frac{L}{V_R \times 1000}\right) = \frac{200}{5}\left(\frac{400}{10{,}000} + \frac{400}{20{,}000}\right) = \frac{200}{5} \times \frac{1200}{2000} = 2.4$ min

Grinding

Hardness and granulation of the grinding wheel

Hardness of the Grinding wheel is the hardness of the bonding agent.



Granulation is the grain size distribution.

Sieves are used to sort out grains by their size.

Sieve mesh number= abrasives grain size number

Grade of the grinding wheel	Grade code	Grain size (grit)	Code grit
Soft	A B C D E F G H	Coarse	10, 12, 14, 16, 20, 24
Medium	I J K L M N O P	Medium	30, 36, 46, 54, 60
Hard	Q R S T U V W X Y Z	Fine	80, 100, 120, 150, 180
		Very fine	220, 240, 280, 320, 400, 500, 600

Properties of material to be ground and kind of abrasive to be used

Hard materials above 35 kgf/mm², e.g. steel, malleable iron cast steel	Aluminium oxide
Soft and brittle materials up to 35 kgf/mm², e.g. gray cast iron, brass, bronze, copper, aluminium, plastic	Silicon carbide

Grinding and types of bonds

General grinding of various materials	Vitrified bond
Face grinding with large area of wheel in contact. Grinding of items sensitive to heat, such as cutters and precision tools	Silicate bond
Super finish grinding of chill castings, cast iron rollers, hardened steel cams, Aluminion pistons	Rubber bond

Structure of grinding wheel

Finish grinding and polishing Hard and brittle materials	Dense structure (Low porosity)
Rough grinding	Coarse structure (High porosity)

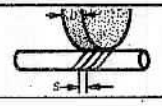
Material and hardness of the grinding wheel	Material and grit size
Soft materials—hard wheels	Soft material—coarse grains
Hard materials—soft wheels	Hard material—fine grains

Peripheral speeds of grinding wheels

Type of grinding	Peripheral speed	
Cylindrical grinding	25 ... 30 m/s	Use higher values to grind workpieces from steel
Internal grinding	15 ... 20 m/s	
Surface grinding	20 ... 25 m/s	
Tool grinding	18 ... 20 m/s	
Parting-off grinding	... 80 m/s	Use lower values to grind workpieces from grey cast iron

Longitudinal feed *s*

Rate of feed *s* per revolution of the workpiece in fraction of the breadth (width) *b* of the grinding wheel

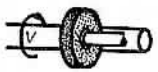


	Cylindrical grinding		Internal grinding	
	Steel	Gray cast iron	Steel	Gray cast iron
Rough grinding	$\frac{2}{3}$ ... $\frac{3}{4}$	$\frac{3}{4}$ ... $\frac{5}{6}$	$\frac{1}{2}$ ... $\frac{3}{4}$	$\frac{2}{3}$ ... $\frac{4}{5}$
Finish grinding	$\frac{1}{4}$ ... $\frac{1}{3}$	$\frac{1}{3}$ ... $\frac{1}{2}$	$\frac{1}{4}$ ... $\frac{1}{3}$	$\frac{1}{4}$ ... $\frac{1}{3}$

Depth of cut

Material	Rough grinding	Finish grinding
Steel	0.01 mm ... 0.06 mm	0.005 mm ... 0.01 mm

Peripheral speed of the workpiece; Hardness and grit of the grinding wheel

Material	Type of grinding	Cylindrical grinding		Internal grinding		Face grinding
		Peripheral speed *v* m/min	 Grain/Hardness	Peripheral speed *v* m/min	Grain/Hardness	Grain/Hardness
Annealed Steel	rough gr. finish gr.	12...15 9...12	46L...M	16...21 —	45...50 J...O	30...60J
Hardened Steel	rough gr. finish gr.	14...16 9...12	46K	— 18...23	46K...60H	30...60H...K
Gray cast iron	rough gr. finish gr.	12...15 9...12	46K	— 18...23	40...46K...M	16...30 J...K
Brass	rough gr. finish gr.	18...20 14...16	36K...46J	— 25...30	36K...46J	—
Aluminium	rough gr. finish gr.	40...50 28...30	30K...40J	— 32...35	30H	—

Number of revolutions for grinding wheels n = rmp

Grinding wheel Ø mm	Peripheral speed in m/s: 15m	20m	25m	30m	35m	Grinding wheel Ø mm	Peripheral speed in m/s: 15m	20m	25m	30m	35m
	rmp of the wheel						rmp of the wheel				
10	28600	38200	47700	57300	68600	130	2200	2950	3670	4400	5150
15	19100	25500	31800	38200	44600	150	1900	2550	3200	3800	4450
20	14300	19100	23900	23900	33400	175	1635	2200	2730	3270	3800
25	11500	15300	19100	23000	26750	200	1440	1910	2390	2875	3350
30	9500	12700	15900	19100	22200	225	1275	1700	2100	2550	2975
35	8100	10900	13600	16300	19100	250	1150	1525	1900	2300	2675
40	7160	9550	11940	14320	16700	275	1030	1400	1700	2060	2400
45	6300	8490	10600	12740	14860	300	950	1275	1590	1900	2230
50	5730	7650	9550	11450	13400	350	820	1090	1370	1640	1900
60	4750	6350	7950	9950	11100	400	725	960	1200	1450	1675
65	4400	5900	7350	8800	10300	450	635	850	1060	1275	1485
70	4050	5450	6800	8150	9550	500	575	770	960	1150	1340
75	3825	5100	6380	7650	9000	550	515	700	850	1030	1200
80	3580	4775	5970	7160	8350	600	475	640	800	950	1110
90	3185	4245	5300	6370	7430	650	440	590	730	875	1030
100	2865	3825	4775	5730	6700	700	405	540	675	810	950
115	2490	3320	4150	4980	5815	750	380	510	635	765	890
125	2300	3015	3800	4600	5300	800	360	475	600	715	835

Number of revolutions for workpiece n = rmp

Workpiece d Ø mm	Peripheral speed of the workpiece in m/min: 6m	8m	10m	12m	15m	18m	20m	24m	28m	32m
	rpm of workpiece									
5	382	510	636	764	956	1148	1280	1528	1784	2038
8	238	318	396	477	597	716	797	955	1114	1273
10	191	255	318	382	478	574	640	764	892	1019
12	159	212	265	318	398	477	631	637	743	849
14	136	182	227	273	341	409	455	546	637	728
16	119	159	199	239	298	358	398	477	557	637
18	106	141	177	212	265	318	354	424	495	566
20	95	128	159	191	239	287	319	382	446	509
22	87	115	145	174	217	260	289	347	405	459
25	76	102	127	153	190	229	225	306	357	408
28	68	99	114	136	171	205	228	273	318	364
32	59	79	99	119	149	179	199	239	279	318
36	53	71	88	106	132	159	177	212	247	283
40	47	63	79	95	119	143	159	191	223	254
45	42	56	70	85	106	127	141	170	198	226
50	38	51	63	76	95	115	127	153	178	204
56	34	45	57	68	85	102	114	136	159	182
63	30	40	51	61	76	99	101	121	141	162
70	27	36	45	55	68	82	91	109	127	145
80	23	31	39	47	59	71	79	95	111	125
90	21	28	35	42	53	63	71	85	99	112
100	19	25	31	38	47	57	63	76	89	102
110	17	23	29	35	43	52	58	69	81	93
125	15	20	25	30	38	45	51	61	71	81
140	13	18	23	27	34	41	45	55	64	73
160	12	16	19	24	29	36	39	48	56	64

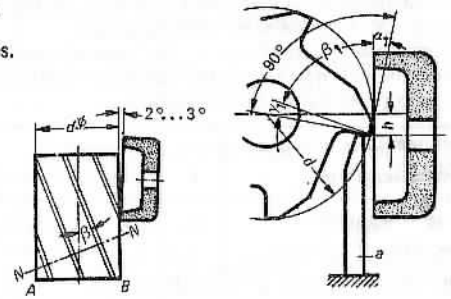
Reference offset h at sharpening of helical cutters

Arbor and finger a are to be set by the offset h so that the clearance angle α can be obtained.

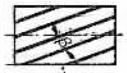
The cup wheel is tilted through 2...3 degrees.

d = Diameter of cutter
β = Helix angle (angle of helix to the longitudinal axis of cutter)
α = Effective clearance angle measured perpendicular to N–N

α_1 = Clearance angle actually ground by setting h

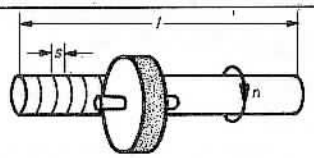


To grind tool steel and high speed steel use the following wheels: Aluminium oxide wheels, Normal grinding 46...60 J...L, Finishing: 60, K...M. To grind carbide tools use silicon-carbide wheels, Finishing: 80...100 G...H

Helix angle β	Clearance angle α	Clearance angle α_1 (actually ground)	Diameter d of cutter in mm: 10	20	30	40	50	60	75	90	110	130	150	160
			Reference offset h in mm											
 20°	3°	2°49′	0.25	0.49	0.74	0.98	1.23	1.47	1.84	2.21	2.70	3.19	3.68	3.94
	4°	3°46′	0.33	0.66	0.99	1.31	1.64	1.97	2.46	2.96	3.61	4.27	4.93	5.25
	5°	4°42′	0.41	0.82	1.23	1.64	2.05	2.46	3.07	3.69	4.51	5.33	6.14	6.55
	6°	5°38′	0.49	0.98	1.47	1.96	2.45	2.94	3.68	4.42	5.40	6.38	7.36	7.85
	7°	6°35′	0.57	1.15	1.72	2.29	2.87	3.44	4.30	5.16	6.30	7.45	8.60	9.17
	8°	7°31′	0.65	1.31	1.96	2.62	3.27	3.92	4.90	5.88	7.19	8.50	9.81	10.47
45°	3°	2° 7′	0.18	0.37	0.55	0.74	0.92	1.11	1.38	1.66	2.03	2.40	2.77	2.95
	4°	2°50′	0.25	0.49	0.74	0.99	1.24	1.48	1.85	2.22	2.72	3.21	3.71	3.95
	5°	3°32′	0.31	0.62	0.92	1.23	1.54	1.85	2.31	2.77	3.39	4.00	4.61	4.92
	6°	4°15′	0.37	0.74	1.11	1.48	1.85	2.22	2.78	3.33	4.08	4.82	5.56	5.93
	7°	4°58′	0.43	0.87	1.30	1.73	2.16	2.60	3.24	3.89	4.76	5.63	6.49	6.93
	8°	5°40′	0.49	0.99	1.48	1.97	2.47	2.96	3.70	4.44	5.43	6.42	7.40	7.90
60°	3°	1°30′	0.13	0.26	0.39	0.52	0.65	0.78	0.98	1.18	1.44	1.70	1.96	2.10
	4°	2°	0.17	0.35	0.52	0.70	0.87	1.05	1.31	1.57	1.92	2.27	2.62	2.79
	5°	2°30′	0.22	0.44	0.65	0.87	1.09	1.31	1.64	1.96	2.40	2.83	3.27	3.49
	6°	3°	0.26	0.52	0.79	1.05	1.31	1.57	1.96	2.36	2.88	3.40	3.93	4.19
	7°	3°31′	0.31	0.61	0.92	1.23	1.53	1.84	2.30	3.76	3.37	3.99	4.60	4.91
	8°	4° 1′	0.35	0.70	1.05	1.40	1.75	2.10	2.63	3.15	3.85	4.55	5.25	5.60

Machining time in grinding

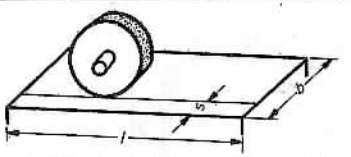
Machining time for cylindrical and internal grinding



l = Length of workpiece to be ground
s = Feed in mm/rev of workpiece
n = Revolutions per min of workpiece
x = Number of cuts

$$t_m = \frac{l \times x}{n \times s}$$

Machining time for surface grinding

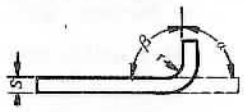


l = Length of workpiece to be ground
b = Width of workpiece to be ground
v = Velocity of table in m/min
s = Feed in mm/stroke

$$t_m = \frac{l \times b \times x}{v \times 1000 \times s}$$

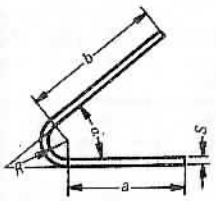
Folding, edging and cold bending

Smallest allowable radius of bend r bending angles $\alpha \leqq 120°$
(for $\alpha > 120°$ take the next following greater computed value from the tables; the same rule stands if the folding and/or bending is to take place along the axis the material had been rolled during the production).



Steel quantities having a tensile strength of	Up to thickness s of														
	1	1.5	2.5	3	4	5	6	7	8	10	12	14	16	18	20
Up to 40 kg/mm²	1	1.6	2.5	3	5	6	8	10	12	16	20	25	28	36	40
from 40 to 50 kg/mm²	1.2	2	3	4	5	8	10	12	16	20	25	28	36	40	45
from 50 to 65 kg/mm²	1.6	2.5	4	5	6	8	10	12	16	20	25	32	36	45	60

The flat bank length can be calculated as follows:



L=Flat blank length

a, b=Length of legs

R=Bend radius

q=Correction factor

s=Material thickness

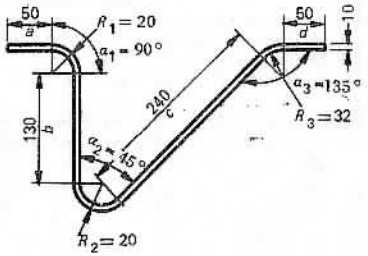
α=Bend angle

$$L = a + (R + q \times s/2)\,\frac{\pi\alpha}{180} + b + \ldots$$

Correction factor q

Ratio R:s	5.0	3.0	2.0	1.2	0.8	0.5
Correction factor	1.0	0.9	0.8	0.7	0.6	0.5

Examples for computing the length before bending



Length of legs

$a=50,\quad b=130,\quad c=240,$

$d=50,$

Bend radius

$R_1=20,\quad R_2=20$

$R_3=32,\quad s=10$

Bend angles $\alpha_1=90°,\ \alpha_2=45°,\ \alpha_3=135°$
Correction factors $q_1=0.8,\ q_2=0.8,\ q_3=0.96$

$$L = a + (R_1 + q_1 \times s/2)\,\frac{\pi\alpha_1}{180} + b + (R_2 + q_2 \times s/2)\,\frac{\pi\alpha_2}{180} + c + (R_3 + q_3 \times s/2)\,\frac{\pi\alpha_3}{180} + d$$

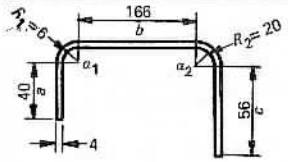
$$= 50 + \left(20 + 0.8 \times \frac{10}{2}\right)\frac{22}{7} \times \frac{90}{180} + 130 + \left(20 + 0.8 \times \frac{10}{2}\right)\frac{22}{7} \times \frac{45}{180} + 240 + \left(32 + 0.96 \times \frac{10}{2}\right)\frac{22}{7} \times \frac{135}{180} + 50$$

$$= 50 + (24)\,\frac{11}{7} + 130 + (24)\,\frac{11}{14} + 240 + (36.8)\,\frac{13}{14} + 50$$

$$= 50 + 24 \times 1.57 + 130 + 12 \times 1.57 + 240 + 36.8 \times 2.35 + 50$$

$$= 50 + 37.68 + 130 + 18.84 + 240 + 86.48 + 50$$

$$= 613.00 \text{ mm straight unbent length}$$



Length of legs: $a=40,\ b=166,\ c=56$

Bend radius: $R_1=6,\ R_2=20$

Material thickness: $s=4.0$

Bend angle: $\alpha_1=\alpha_2=90°$

Correction factor: $q_1=0.7\ q_2=1$

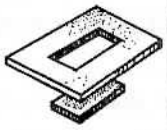
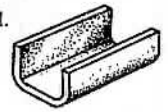
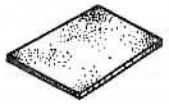
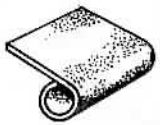
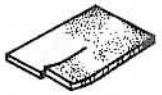
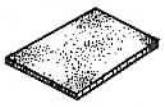
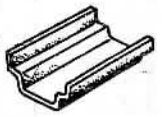
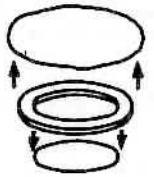
$$L = a + (R_1 + q_1 \times s/2)\,\frac{\pi\alpha_1}{180} + b + (R_2 + q_2 \times s/2)\,\frac{\pi\alpha_2}{180} + c$$

$$= 40 + 11.618 + 166.0 + 34 - 54 + 56.0$$

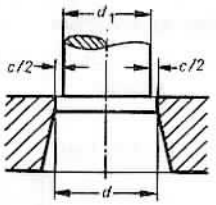
$$= 308.15 \text{ mm straight unbent length}$$

Presstool operations

SPK

Punching operations	Forming operations
Piercing	**Bending**
Cutting a required shape in a strip of blank. Punchings are considered as waste. 	Forming with a bending tool. 
Blanking	**Curling**
Cutting required profile of a component. Punchings are the products. 	Bending operation done by beading dies or sliding presses. 
Slitting	**Deep-drawing**
Partial shearing of the sheet 	Forming between enclosed punch and die.
Cropping	**Press-braking**
Cutting off a part from the stock strip without cutting the sides of the blank. 	Bending of longer profiles enabling as series of bending operations. 
Trimming	**Collar-drawing**
Cutting off excess stock from forged or pressed component, getting the exact profile.	Pushing a collar through a pre-pierced hole or profile.
Shaving	**Calibrating**
Post cutting in opposite direction to the initial cut in order to smoothen the sides of a profile and to achieve closer tolerances. 	Repressing operation on formed products in order to achieve exact sizes and forms.
Lancing	**Embossing**
Partial shearing and bending.	Changing the thickness of the stock without any deformation on the opposite side.

Clearance between Die and Punch



Clearance=The measured space between the mating members of a die set=c

∴ Clearance per side=$c/2$

$$c=d-d_1$$

Clearance for blanking and piercing.

Blanking: For a blank of definite diameter clearance should be on the punch.

Piercing: For a hole of definite diameter, clearance should be on the die.

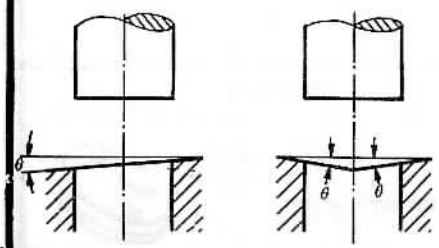
Clearance per side $(c/2)=0.01\times s\times\sqrt{T_B}$, where s=material thickness, T_B=shear's stress in kg/mm²

Sheet thickness in mm.	Clearance between punch and die					
	Mild steel	Steel moderately hard	Steel hard	Brass soft	Brass hard	Aluminium
0.25	0.01	0.015	0.02	0.01	0.025	0.02
0.5	0.025	0.03	0.035	0.025	0.03	0.05
0.75	0.04	0.45	0.05	0.03	0.04	0.07
1.0	0.05	0.06	0.07	0.04	0.06	0.10
1.25	0.06	0.075	0.09	0.05	0.07	0.12
1.50	0.075	0.09	0.10	0.06	0.08	0.15
1.75	0.09	0.1	0.12	0.075	0.09	0.17
2.0	0.10	0.12	0.14	0.08	0.10	0.20
2.25	0.11	0.14	0.16	0.09	0.11	0.22
2.5	0.13	0.15	0.18	0.10	0.13	0.25
2.75	0.14	0.17	0.20	0.12	0.14	0.28
3.0	0.15	0.18	0.21	0.13	0.16	0.30
3.3	0.17	0.20	0.23	0.15	0.18	0.33
3.5	0.18	0.21	0.25	0.16	0.19	0.35
3.8	0.19	0.23	0.27	0.19	0.22	0.38
4.0	0.20	0.24	0.28	0.21	0.24	0.40
4.3	0.22	0.26	0.30	0.23	0.27	0.43
4.5	0.23	0.27	0.32	0.26	0.30	0.45
4.8	0.24	0.29	9.34	0.29	0.33	0.48
5.0	0.25	0.30	0.36	0.33	0.36	0.50

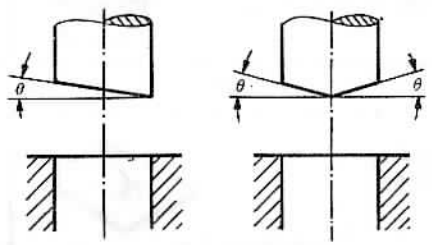
Angle of shear for blanking and piercing

In order to reduce the load on the press, the punch or die is given an 'angle of shear' to cause a gradual shearing.

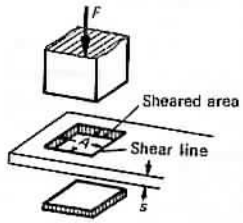
Blanking: Angle given on die



Piercing: Angle given on punch



Cutting force and stripping force



The cutting force F is the force required to act on the punch to cut a blank or slug. It depends on the shearing stress and cutting area A

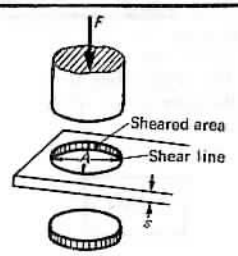
s = material thickness

l = total cutting length

$A=l\times s$=sheared area

τ_B = shear strength

$$F=A\times\tau_B$$ in kg



For diameter smaller than stock strip

$$F=\frac{\pi\times d\times s\times\sigma_B}{\sqrt[4]{d/s}}$$

d = dia. of punch

σ_B = tensile strength

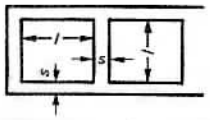
Stripping force for blanking and piercing:

$$P=\frac{l\times s\times\tau_B}{10}$$

Shear strength and tensile strength of various materials

Material	Shear strength τ_B kgf/mm²		Tensile strength σ_B kgf/mm²	Material	Shear strength τ_B kgf/mm²		Tensile strength σ_B kgf/mm²
	Annealed	Not annealed			Annealed	Not annealed	
C 10	24	32	40...50	Rolled bronze	32...40	40...60	
C 20	30	40	50...60	Rolled copper	18...22	25...30	
C 30	36	48	60...70	Rolled brass	22...30	35...40	
C 40	45	56	70...80	Zinc	12	20	
C 50	50	65	80...90	Tin	3	4	
C 60	55	72	90...100	Lead	2	3	
C 80	70	90	105...120	Aluminium	7...9	13...16	
C 98	80	105	120...140	Al-G-Mg	22	38	40
St 32	30		32...40	Al-Mg 5	19	24	21
St 37	35		37...45	Laminated plastic (Paper b.)	4...6		5.0...8.4
St 50	40		50...60	(Fabric b.)	5...6		6.2...7.0
St 63	50		63...75	PVC sheet	2.5...3		3.8...4.5
St 78	65		78...90	Leather	0.6...0.8		1

Web and stem



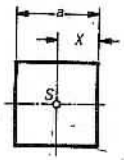
The width of web and stem depend on the thickness of the material to be cut and on the length of the shear line

Web length (l) in) mm	Thickness of sheet in mm											
	0.5	0.75	1	1.25	1.5	1.75	2	2.25	2.5	3	3.5	4
10	1.5	1.2	1	1.4	1.5	1.8	2	2	2	2	2.5	2.5
50	2	1.7	1.75	1.9	2	2.2	2.5	2.8	3	3.5	3.7	4
100	3	2.4	2	2.4	2.5	2.7	3	3.2	3.5	4	4.2	4.5
150	3.5	2.9	2.5	2.9	3	3.2	3.5	3.7	4	4.5	4.7	5
250	4	3·4	3	3.4	3.5	3.7	4	4.2	4.5	5	5.2	5.5
350	4.5	3.5	3.5	3.9	4	4.2	4.5	4.7	5	5.5	5.7	6

Location of centre of gravity S of certain lines

Square

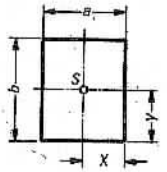
The point S lies at the intersection of the centre lines



$$X=\frac{a}{2}$$

Rectangle

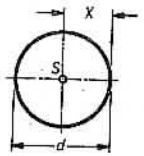
S lies at the intersection of the centre lines of the sides a and b



$$X=\frac{\sqrt{a^2+b^2}}{2}$$

Circle

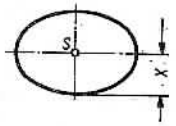
S lies at the intersection of two diameters



$$X=\frac{d}{2}$$

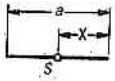
Ellipse

S lies at the intersection of the major and minor axis



Straight line

S lies at the centre of the line



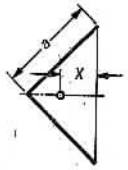
$$X=\frac{a}{2}$$

Triangle

S lies at the centre of the inscribed circle

Right angle with equal sides

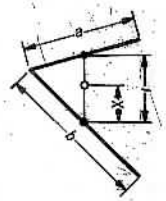
S lies on the angle bisector



$$X=0.707\times\frac{a}{2}$$

Angle with unequal lengths

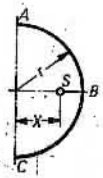
S lies on the straight line joining the centres of the sides a and b



$$X=\frac{a\times l}{a+b}$$

Semicircle

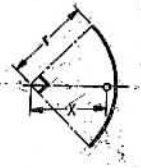
S lies on the centre line of arc ABC, at a distance $0.637\times r$ from the centre of arc



$$X=0.637\times r$$

Quadrant

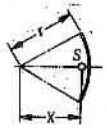
S lies on the centre line of arc at a distance $0.9\times r$ from the centre of arc



$$X=0.9\times r$$

Sextant

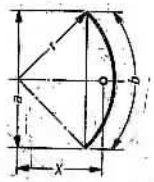
S lies on the centre line of arc, at a distance $0.955\times r$ from the centre of arc



$$X=0.955\times r$$

Segment

S lies on the centre line of arc and distance X from centre of arc



$$X=\frac{a\times r}{b}$$

Calculation of the plug point

Finding the plug point for a single punch with symmetrical cross section

The shank of a presstool must be located at the centre of gravity of the shearing lines which is called plugpoint.

Arithmetical method (according to law of moment–see page 68)

Fulcrum becomes axis of rotation D–D

Force becomes length of shearing line l

Lever arm becomes distance of the individual centre of gravity from the axis of rotation d

Load becomes sum of all shearing lines, $L=l_1+l_2+\dots$

Load arm becomes distance of the total centre of gravity S, x

Force × Lever arm = load × load arm

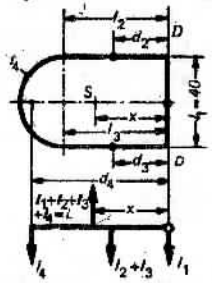
Or sum of moment of all forces = movement of the resultant force

$$\Sigma\, l\times d=L\times x$$

$$x=\frac{\Sigma\, l\times d}{L}$$

Example 1:

Symmetrical cross section



Solution:

$l_1=40$ mm $\quad d_1=0$

$l_2=50$ mm $\quad d_2=25$ mm

$l_3=50$ mm $\quad d_3=25$ mm

$l_4=628$ mm $\quad d_4=0.637\times20+50=62.7$ mm

$L=202.8$ mm

$$x=\frac{\Sigma\, l\times d}{L}=\frac{40\times0+50\times25+50\times25+62.8\times62.7}{202.8}$$

$x=31.7$ mm

Finding the plug point for a tool with two punches

Fulcrum becomes centre of gravity of punch No. 1

Force becomes length of shearing line of punch No. 2 l_2

Lever arm becomes distance between centres of gravity d

Load becomes length of shearing lines of both punches L

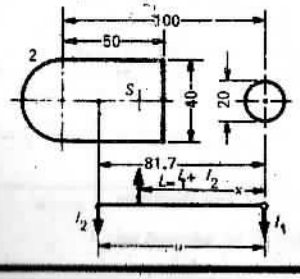
Load arm becomes distance of the total centre of gravity S (plug point) from the fulcrum x

$$l\times d=L\times x$$

$$x=\frac{l\times d}{L}$$

Example 2:

Symmetrical cross section



Solution:

The centre of gravity for punch No. 2 was found in Example 1

Length of shearing line of punch No. 1 $l_1=62.8$ mm

Length of shearing line of punch No. 2 $l_2=202.8$ mm

Distance between centres of gravity $d=81.7$ mm

$$x=\frac{l_2\times d}{L}=\frac{202.8\times81.7}{265.6}=62.4\text{ mm}$$

Machining properties of synthetic materials

Types of machining processes	Plastics (not laminated)			Laminated plastics
Filing	▽ Rough filing: rasp, bastrad file ▽▽ Fine filing : smooth file, dead smooth file			
Sawing	Circular saw	Thickness of material	0.5...4 mm	5...8 mm	10 mm and more
		Thickness of saw disc	3 mm	4 mm	5 mm
		Tooth pitch≈10 mm teeth upset or hollow disc			
		Cutting speed v=50 m/min Feed by hand			

Drilling				
Point angle: 90°...116° drilling	Drilling of hand materials	Point angle	Rake angle
Thin walled workpieces: 50°	parallel to the fabric	70°	0...20°
Flute angle 60°	Perpendicular to the fabric	100...110°	10...12°
Cutting speed v=70 m/min	Cutting speed v=60...90 m/min		
Feed s=0.2...0.3 mm/rev	Feed s=0.5...0.2 mm/rev		
Sharp cutting edges must set be carefully on the material to prevent damaging the periphery of the hole. Lift drills for cooling. For deep holes use compressed air to cool the drill. Provide pieces of wood or hard fabric where the drill point is first to leave the material. Dip drill into mineral oil to get smooth hole surface. Chips and yapour must be immediately removed.			

Planing	
Use carbide tipped tools, well sharpened and lapped	
Cutting speed v=10...20 m/min Feed s=0.2...0.8 mm/stroke	

Turning	Clearance angle α=8° Rake angle γ=15°					
	▽ roughing	▽▽ fiinishing		▽ roughing	▽▽ finishing
v m/min	80...200	200...300	v m/min	100...200	200...300
s mm/rev	0.3...0.5	0.1...0.3	s mm/rev	0.2...0.5	0.1...0.3
Depth of cut	3...5 mm	2 mm	Depth of cut	3...5 mm	2 mm
Set the tool exactly at centre Round off point of cutting edge. Radius of curvature 1.5...2 mm Remove chips by suction or exhaustion					

Milling		
Carbide tipped tools. Clearances angle α=20°. Rake angle γ=20°...25° Number of teeth=$\frac{1}{2}$...$\frac{3}{4}$ of the teeth used in metal milling.		
Cutting speed v=300 m/min Feed s=0.5...0.8 mm/rev	Cutting speed v=120...250 m/min Feed s=0.5...0.8 mm/rev	

Thread cutting	
Use tap with wide flutes. Use wax and grease to lubricate the screw-tap	
High speed steel	High speed steel v=20...40 m/min Thread milling gives smoother surface than turning

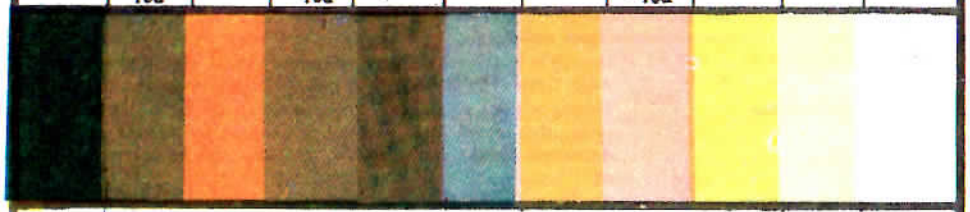
Grinding	
Belt grinding machines v≈6 m/min Grinding machine using sand paper v≈25 m/min Surface grinding: use water as coolent. Finish grinding: use silicon carbide wheels, grit 60 Rough grinding: use grit 20	

Shearing		
		Can be done up to 3 mm thickness Thicker materials to be warmed up

Heat treatment	Combination of heating and cooling operation in a prescribed manner (with respect to time, temperature and rate of heating and cooling) to induce desired properties in metals and alloys in the solid state. The conventional heating for hot working does not come within the scope of heat-treatment.
Rate of cooling	Temperature decrease per unit time.
Quenching	Rapid cooling.
Full annealing	Heating to and holding at some temperature above the transformation range, followed by cooling slowly through the transformation range.
Isothermal annealing	Heating to and holding at some temperature above the transformation range, then cooling to and holding at a suitable temperature until the austenite-to perlite transformation is complete, and finally cooling in air.
Sub-critical annealing	Heating to and holding at some temperature below the transformation range, followed by cooling at a suitable rate.
Bright annealing	Annealing under controlled atmosphere (by introducing neutral or reducing gas) to prevent oxidation of the surface and to preserve its brightness.
Close annealing	Annealing in closed container to reduce oxidation to minimum.
Open annealing (black annealing)	Annealing without any protective arrangement to exclude air.
Pack annealing	Close annealing of packs of sheets
Martempering	An isothermal treatment in which the object is quenched from a suitable austenising temperature to a temperature slightly above the Ms point and held at that temperature for a suitable length of time, for equalizing the temperature and then cooled in air. The martensite thus formed is then tempered as desired.
Case hardening	Hardening the surface by changing its composition followed by, if necessary, suitable heat treatment.
Carburizing (case carburizing)	A process of introducing carbon into the surface of a solid piece of steel by heating and holding above the transformation temperature in contact with a suitable carbonaceous medium.
Flame hardening	Rapid heating of the surface by means of an oxy-gas flame to a temperature above the transformation range, followed by immediate quenching.
Hardening	Any process which increases the hardness, for example, quenching from or above the transformation range, or cold working.
Induction hardening	Process of hardening by induction heating to the appropriate temperature and quenching in a suitable medium.
Nitriding	A process of surface hardening by introducing Nitrogen into the surface in a suitable steel by heating and holding it at approriate temperature in contact with cracked ammonia or other suitable nitrogenous medium.
Normalizing	A process of heat treatment for improving mechanical properties brought about by grain refinement and uniformity in structure. The process involves heating to and often holding for a specified time at a suitable temperature above the transformation range, followed by cooling freely in air.
Patenting (air patenting)	Heating to an appropriate temperature above the transformation range and then cooling at a controlled rate, in the desired medium (for example, air, lead, or a salt bath) to produce structure desirable for subsequent cold working. Usually this is applied to wire drawing.
Stress relieving (stabilizing)	Heating to and if necessary, holding at a sufficiently high temperature below the transformation range followed by slow cooling to remove internal stresses only. Also called stabilizing treatment.
Sub-zero treatment	Treatment given to certain hardened steels by cooling to temperatures below zero (0°C) to transform any retained austenite to martensite.
Tempering	Heating to elevated temperature but below transformation zone of hardened steels and holding for specified time at temperatures followed by slow cooling to develop desired mechanical properties in these steels.
Spheroidizing	A process of heat treatment to produce globular form of carbide in steel and consists of prolonged heating of steel to a suitable temperature, within or near the transformation range. This treatment improves machinability and facilitates finishing by cold work. It is also useful where such structure is desirable for further heat treatment in processing certain materials.

Colour scale for steel–hardening temperature range

Temperature °C	520 ... 580	... 650	... 750	... 780	... 800	... 830	... 880	... 1050	... 1150	...1250...1320
	Dark brown	Brownish red	Dark red	Dark cherry red	Cherry coloured	Light cherry	Orange	Yellowish red	Dark yellow	Light yellow	White



Colour scale for steel—tempering temperature range

Temperature °C	220	230	240	250	260	270	280	290	310	320	330
	Light yellow	Dark yellow	Yellowish orange	Yellowish brown	Brownish red	Dark red	Violet	Dark blue	Light blue	Grey	Greyish green

Forging temperatures for steel

Designation of steel	Hot working temperature °C	Tempering temperature °C	Normalising temperature °C
C 30	1200 ... 850	550 ... 660	860 ... 890
C 40	1200 ... 850	550 ... 660	830 ... 860
C 50	1200 ... 850	550 ... 660	810 ... 840
C 55 Mn 75	1200 ... 850	550 ... 660	810 ... 840
40 S 18	1200 ... 850	550 ... 660	830 ... 860
40 Mn 2 S 12	1200 ... 850	550 ... 660	840 ... 870
35 Mn 2 Mo 45	1200 ... 900	550 ... 660	—
55 Cr 70	1200 ... 850	500 ... 700	800 ... 850
40 Cr 1 Mo 28	1200 ... 850	550 ... 720	850 ... 880
25 Cr 3 Mo 55	1200 ... 850	550 ... 700	—
40 Cr 3 Mo 1 V 20	1200 ... 850	570 ... 650	—
35 Ni 1 Cr 60	1200 ... 850	550 ... 660	—
30 Ni 4 Cr 1	1200 ... 850	> 250	—
40 Ni 3 Cr 65 Mo 55	1200 ... 850	up to 660	830 ... 850

Hardening temperature for carbon steel

Carbon content in %								
0.35	0.4	0.5	0.6	0.7	0.8	0.9	1	Above 1
840°	820°	810°	800°	790°	780	770°	760°	760°

Low alloy steel: 750°...950°C High alloy steel: 1150°...1350°

Hardness Comparison Tables For Steel

Vickers Hardness Number, HV (30 kgf)	Brinell Hardness Number, HB (3000 kgf, 10 mm steel ball)	Brinell Hardness Number, HB (3000 kgf, 10 mm carbide ball)	Rockwell Hardness Number, HR (140 kgf / 90 kgf additional load, 10 kgf preload, diamond penet. dry)	Rockwell Hardness Number, HR (1.5875 mm steel ball)	Tensile strength	Hardness of various qualities of steel
30 kgf Load	3000 kgf Load [$F/D^2 = 30$]	3000 kgf Load	C Scale 150 kgf Load	B Scale 100 kgf Load	kgf/mm²	(untreated)
940	—	—	68.0	—		Glass Hardness
900	—	—	67.0	—		
860	—	—	65.9	—		
840	—	745	65.3	—		
820	—	733	64.7	—		
780	—	710	63.3	—		
740	—	684	61.8	—		
700	—	656	60.1	—		
660	—	620	58.3	—		
620	—	582	56.3	—		
580	—	545	54.1	—		
550	505	517	52.3	—		
500	465	471	49.1	—		
450	425	425	45.3	—	150	
420	397	397	42.7	—	142	
390	369	369	39.8	—	132	
370	350	350	37.7	—	126	
340	322	322	34.4	—	117	
320	303	303	32.2	—	110	
300	284	284	29.8	—	103	
285	270	270	27.8	—	98	
265	252	252	24.8	—	90	
250	238	238	22.2	—	86	
235	223	—	—	99.9	81	
225	214	—	—	98.3	78	
210	200	—	—	95.7	72	
200	190	—	—	93.8	69	
190	181	—	—	91.6	65	
175	166	—	—	87.9	60	
160	152	—	—	83.4	55	
145	138	—	—	78.1	50	
135	128	—	—	73.9	47	
120	114	—	—	66.3	42	
105	100	—	—	56.4	37	

Application: Can be applied on all materials. Suited specially for testing very hard and very thin materials and also case-hardened and nitrided steels.

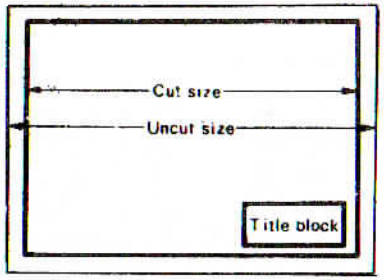
Application: Shall be limited to materials having a maximum Brinell Hardness of 400 with steel ball and of 700 with carbide ball.

Application: Used for testing materials having hardness values of HRc between 20 to 67.

Application: Used for testing materials having hardness values of HRb between 60 to 99.

Engineering Drawings

Standard sheet sizes and scales — IS: 696–1960

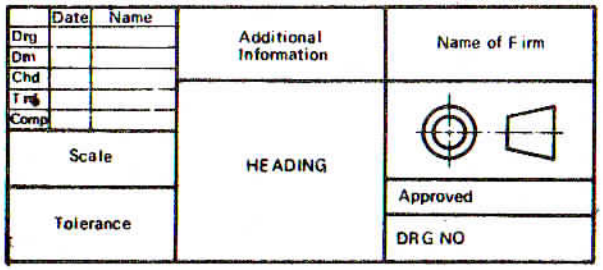


	Trimmed size	Untrimmed size
A0	841 × 1189	880 × 1230
A1	594 × 841	625 × 880
A2	420 × 594	450 × 625
A3	297 × 420	330 × 450
A4	210 × 297	240 × 330
A5	148 × 210	165 × 240

The sheets may be used vertically or horizontally. With the smaller sizes the vertical usage of sheets is preferred.

Scales: Full scale—1:1
Reduced scale— 1:2.5, 1:5, 1:10, 1:20, 1 50, 1:100, 1:200
Enlarged scale—10:1, 5:1, 2:1

The scale of the drawing should be indicated in the appropriate place in the title block. If more than one detail drawn to different scales occurs on a sheet, the corresponding scale should be shown under each relevant detail.



The title block is an important feature in a drawing because it facilitates obtaining uniformity and prevents details like title of drawing, name of organisation or firm, drawing number, scale, date of drawing, etc. in a definite manner. The title block should be placed at the bottom right hand corner of the sheet where it is readily seen when the prints are folded.
Size for A0, A1, A2, A3 IS 150 × 50 mm.
For A4 it may occupy the full width of the sheet.
For A5 horizontal 185 × 42 mm and vertical 123 × 50 mm.

It is sometimes found convenient to group a number of associated parts on a separate sheet distinct from the detailed drawing or drawings. A typical form of separate material or parts list is shown here.

Part No.	Description	Drawing No.or Ref. No.	Material	No. Off	Additional Information

R.No.	Date	Brief Record	Dated Initials

	Date	Name
Dm		
Chd		
Trd		
Comp		
Appd		

Name of Firm
HEADING
Part List No.
Supplement
Drg. No.

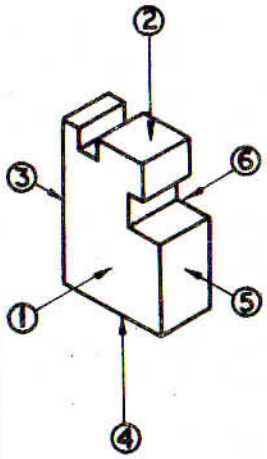
SPK.

Lines — IS: 696—1960

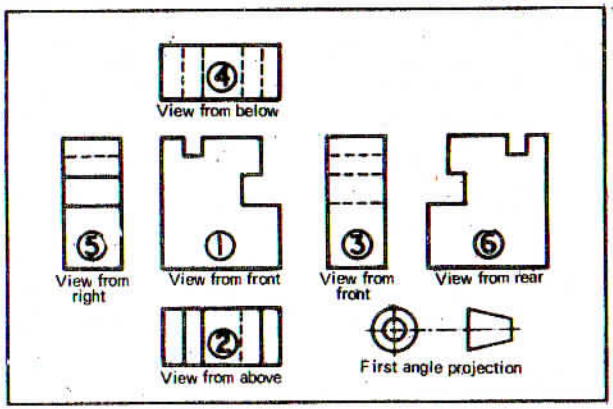
	Illustration thickness in mm			Description
Thick	1.00	0.80	0.60	**Outline of parts** The outline to be outstanding
Thin	0.30	0.25	0.20	**Dimension, extension and matching lines** For sectioning lines to be spaced evenly to make a shaded effect
Medium	0.50	0.45	0.40	**Hidden lines** Short dashes closely and evenly spaced
Thick	1.00	0.80	0.60	**Cutting plane lines** One long and two short dashes alternately and evenly spaced
Thin	0.30	0.25	0.20	**Centre and locus lines** Alternate long and short dashes in a proportion ranging form 6 : 1 to 4 : 1 closely and evenly spaced. In one drawing ratio selected must be maintained
Medium	1.00	0.80	0.60	**Brake lines for short brakes** Free hand lines

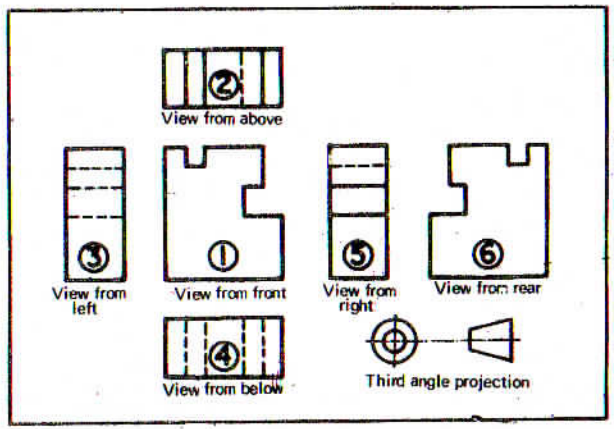
The outline should be outstanding in appearance on the drawing. Lines specified as thick should be three times as thick and lines specified as medium twice as thick than those specified as thin. The ratio between the thick, medium and thin types should, as far as possible, be maintained in any one drawing, the thick lines may be 1.00, 0.84, or 0.60 mm, the medium and thin lines having corresponding thickness as indicated in the table.

Methods of projection and arrangements of views



The standard method of projection recommended in IS is the third angle projection.



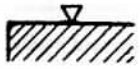
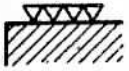


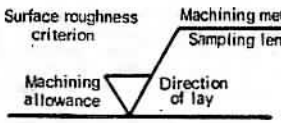
Assessment of surface roughness

IS: 3073–1967

Symbols on drawings

In order to simplify the method of denoting surface roughness on drawings, consideration should be given to requirements of general engineering industries. In most of the cases a general classification is enough. Hence, the most commonly known used and accepted system of indicating surface roughness is by triangles and the same is suggested here. They are classified into four groups.

	8 to 25 R_a μm		0.025 to 1.6 R_a μm
	1.6 to 8 R_a μm		< 0.025 R_a μm



Specifications in symbol:
To fully define the surface it is necessary to indicate the different characteristics of surface roughness along with the symbol and hence it is very essential to indicate a certain place for them in the symbol.

The R_a values are preferably determined as mean results from the measurement of several sampling lengths taken consecutively along the profile. These may be determined graphically or by direct reading instruments.

Surface roughness expected from manufacturing processes

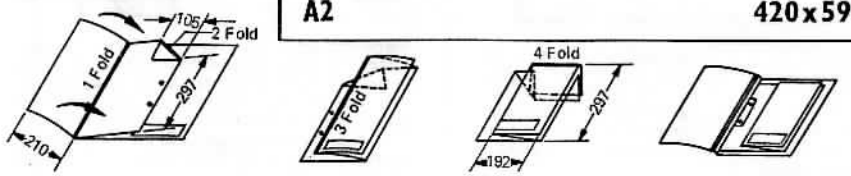
IS: 3073–1967

Manufacturing process	0.012	0.025	0.050	0.10	0.20	0.40	0.80	1.6	3.2	6.3	12.5	25	50	100	200
Flaming, sawing and chipping									6.3					100	
Hot rolling								2.5					50		
Planing								1.6					50		
Sand casting									5				50		
Turning and milling					0.32							25			
Filing					0.25							25			
Disc grinding								1.6				25			
Hand grinding									6.3			25			
Drilling								1.6				20			
Boring						0.4				6.3					
Radial cut-off sawing							1			6.3					
Permanent mold casting							0.8			6.3					
Surface and cylindrical grinding			0.063								5				
Extrusion				0.16							5				
Reaming, broaching and hobbing						0.4				3.2					
Die casting							0.8			3.2					
High pressure casting					0.32					2					
Burnishing			0.04							0.8					
Honing		0.025								0.4					
Super finishing	0.016									0.32					
Lapping	0.012									0.16					
Polishing			0.04							0.16					

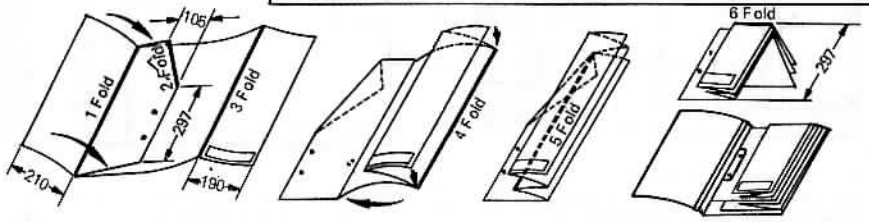
Folding of prints

A3 — 297 x 420

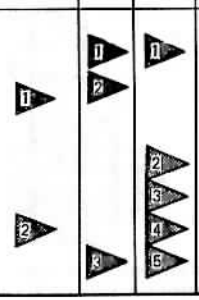
A2 — 420 x 594



A1 — 594 x 841



Folding of A3 | A2 | A1



The title block must be on top and be placed properly so that it is visible.
Folding sequence: Observe the numbered arrows at left.

Place folding guide (210×297 mm) at left side of print and fold towards right (Fold 1), make an additional fold at a distance of 297 mm from the bottom such that a triangular piece is folded in reverse direction (Fold 2).

Turn the print to the other side and make one additional fold (Fold 3) such that it is brought to coincide with fold number 1.

Make a fold at a distance of 190 mm from left side of print.

Make a reverse fold at a distance of 190 mm from right side of print (Fold 3).

Make another fold at a distance of 190 mm towards the front (Fold 4)

Fold the remaining part such that the title block comes on top and such that the overall width of the top sheet results in 210 mm (A3=Fold 2, A1=Fold 5).

Fold to a height of 297 mm (A2=Fold 4, A1=Fold 6).

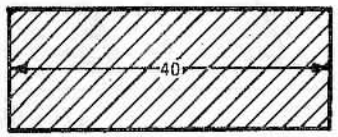
Lettering

Angle of inclination=75°

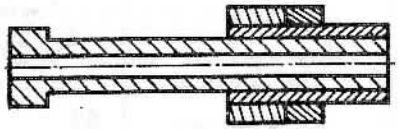
Sizes of letters height=2 2.5 3 4 5 6 8 10 12 16 20 25 mm

Let h be the height of upper-case letters:
the height of lower-case letters=$^6/_7\,h$,
the width (line width)=$^1/_7\,h$,
the line to line distance=$^{11}/_7\,h$.

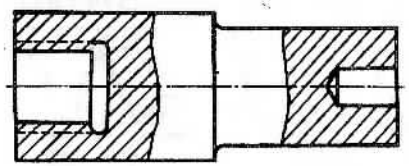
Sections and sectional views



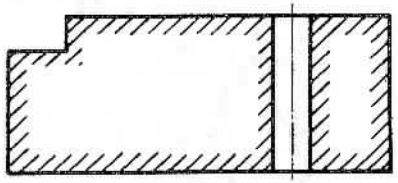
Hatching is used to make sections evident. It is done by thin lines at a well defined angle, preferably 45° to the axis or the main outline of the section. Spacing between the hatching lines should be in proportion to the hatched section. For figuring or lettering hatching may be interrupted.



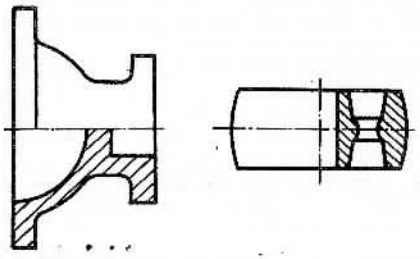
Separated areas of a section adjacent to each other are shown with different hatching. Two adjacent parts should be shown with lines at 45° and in opposite directions. The third part adjacent to the first two should have lines at 30° or 60°. For additional adjacent parts any suitable angle may be used to make each part stand out separately and clearly.



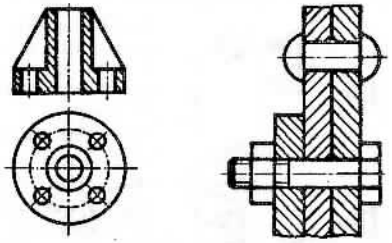
A local section may be drawn if a full or a half section is not necessary. In this case, the local break is shown by a thick, irregular, free-hand break line.



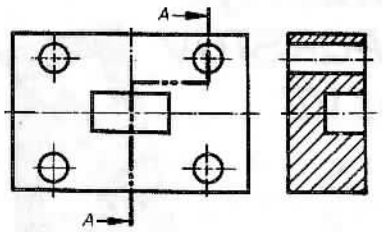
In the case of large areas, the hatching may be limited to a zone following the contour of the hatched area.



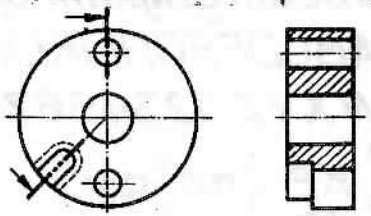
Symmetrical parts may be drawn half in full and half in section.



Section plane through pulley arms, shafts, bolts, nuts, rods, rivets, keys, pins and other similar parts should be shown by normal unsectioned views. When the cutting plane passes longitudinally through the centre or a rib or web cross-hatching should be eliminated.

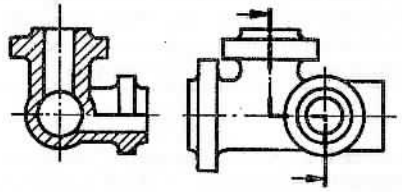


Cutting planes are indicated by a thick long dash and two short dashes alternately and evenly spaced. They are designated by capital letters, the direction of viewing being shown by arrows resting on the cutting line.

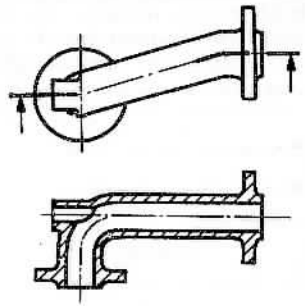


Section in two intersecting planes, one shown revolved into the projection.

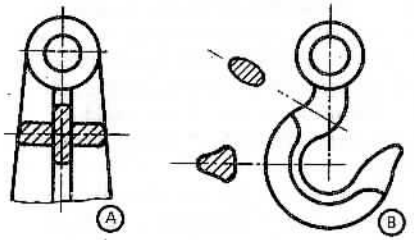
Sectioning and sectional views



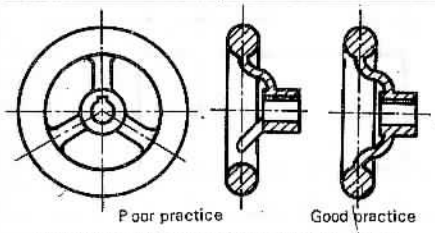
Section in two parallel planes



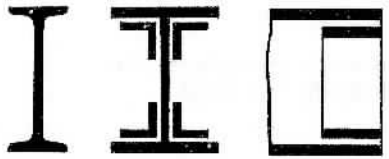
Section in three planes



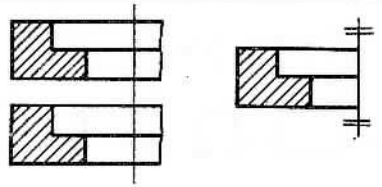
Cross sections may be revolved in place (A) or removed (B). In the first case, the outline is shown in continuous thick lines and further indentification is not necessary.



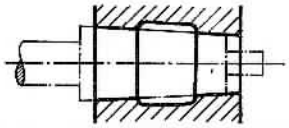
Any part having odd numbers of spokes, ribs or holes will give an unsymmetrical and misleading section if the principles of true projection are strictly adhered to. In such cases, those inclined elements or unsymmetrically placed elements should be rotated into the plane of paper.



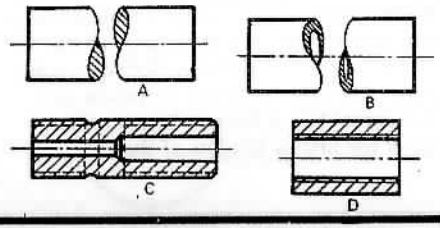
Thin sections may be shown entirely black. A thin space may be left between adjacent sections of this type.



The symmetrical objects may be drawn as a fraction of the whole. A line of symmetry is identified by two short parallel strokes drawn at each end and perpendicular to it.



The representation of parts adjacent to an object are drawn by a thin line made up of alternate long dash and dot. The adjacent part should not hide the principal part but may itself be hidden by the latter. These parts should be hatched only to avoid confusion, and should be only along the outlines.



A. Interrupted view of solid bar.

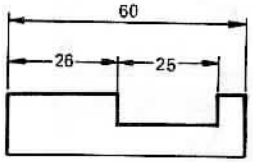
B. Interrupted view of tube.

C. External threads in section.

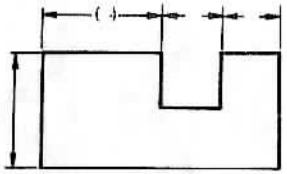
D. Internal threads in section.

Dimensioning

All the dimensions, tolerances, etc. necessary to provide for the correct functioning of the part should be expressed directly on the drawing together with any other information necessary to define the part completely in its finished form. It should not be necessary to deduce a functional dimension from other dimensions or to scale the drawing.

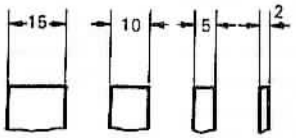


Dimension line is a light full line used to indicate the measurement the amount of which is denoted by figures in a space left in the dimension line or above the dimension line.

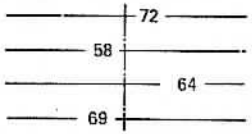


Extension lines (projection lines) are light full lines extending perpendicular to outline and are used when placing dimensions outside the object. There should be a gap of about 2 mm outside the outline and the line should extend about 3 mm beyond the dimensioning line. They could be oblique where necessary but preferably parallel to each other.

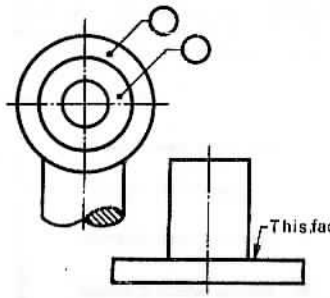
Arrow-heads are used to terminate dimension lines. Its length is about three times the depth and the space should be filled in. Adjacent arrow-heads may be replaced by clearly marked dots where space is limited.



For dimensioning in limited space, the arrow-heads should be reversed.

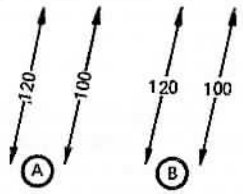


When there are several parallel dimension lines the figures should be staggered to avoid confusion.



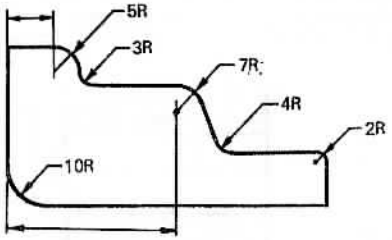
Leaders (pointer lines) are lines drawn from notes and figures to show where these apply. These are thin straight lines terminated by arrow-heads or dots. Arrow-heads should always terminate on a line; dots should be within the outline of the object. The leader should be curved or made free hand. The other end may terminate in a short horizontal bar at the mid-height of the lettering of the first or last line of the note.

Dimensioning



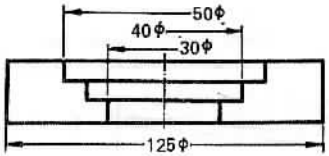
The two recommended systems of placing the dimensions are:

(a) Aligned system
(b) Unidirectional system

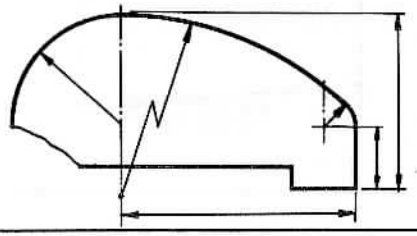


Wherever possible, the dimensions of small radii should be shown outside the outline of the object and should be followed by the letter *R*

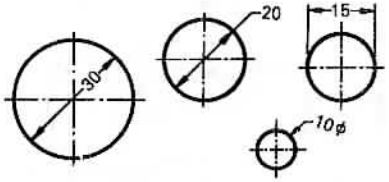
Where the centre point of a radius is dimensioned, a dot should be placed at the intersection of the lines which locate the centre of the radius.



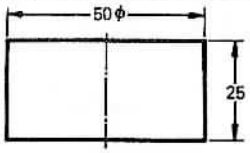
When space is restricted the method shown here can be used to place the dimensions of different diameters.



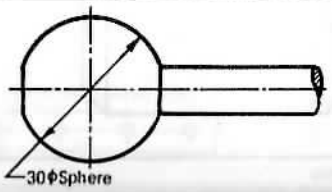
Where it is not possible to show the centre of an arc conveniently in its correct position, the method illustrated here should be used. The portion of the dimension line which touches the arc should be in line with the true centre.



The circles should be dimensioned by one of the methods illustrated here.

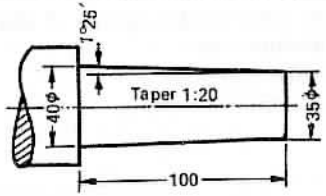


Where it is not obvious that the dimension represents a diameter or where the dimension is remote from the circle, the use of the symbol ϕ is very essential.

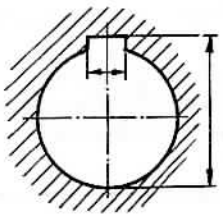


The diameter of a spherical surface should be dimensioned as shown here.

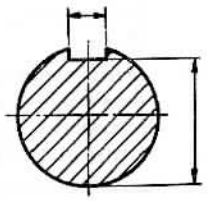
Taper, Keyways and Chamfers



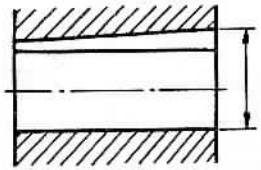
The following dimensions may be used in suitable combinations to define the size and form of tapered features:
(a) Diameter (or width) at each end. (b) Length. (c) Diameter (or width) at a selected cross-sectional plane which may be within or outside. (d) Dimension locating a cross-sectional plane at which a diameter or width is specified. (e) The rate of taper or the included angle.



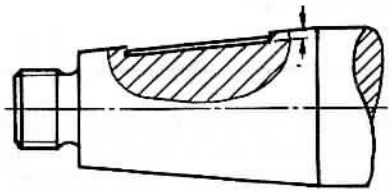
Parallel hub



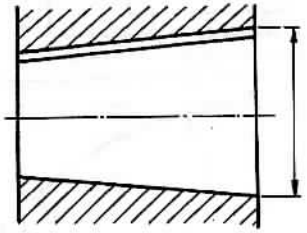
Parallel shaft



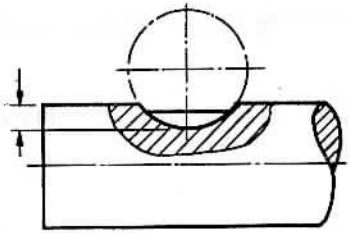
Tapered keyway in Parallel hub



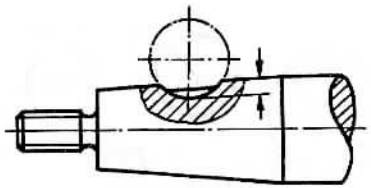
Parallel keyway in tapered shaft



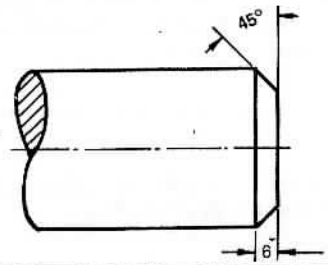
Parallel keyway in tapered hub



Woodruff keyway in parallel shaft

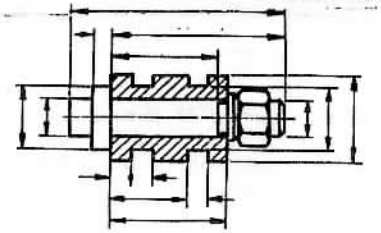


Woodruff keyway in tapered shaft

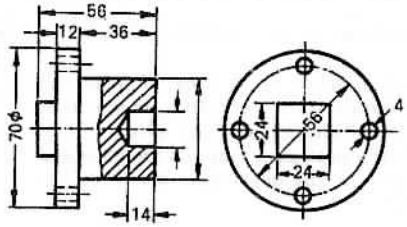


Dimensioning chamfers

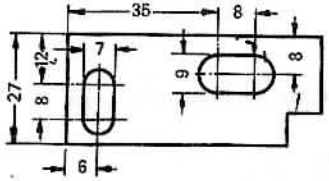
Dimensioning



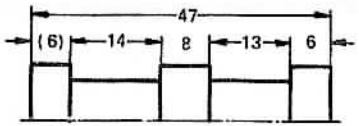
Dimensioning assembled parts:
When several parts are drawn in assembly, the groups of dimensions related to each part shall be kept separate as much as possible.



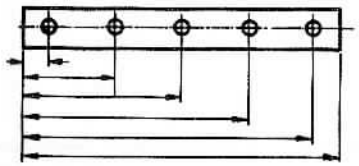
The dimensions should be quoted only once and should be placed on the view which shows the relevant features most clearly. In general, dimensions should be placed outside of views where possible and between views.



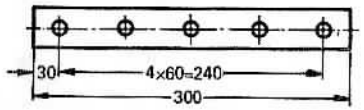
Dimensions should be taken from visible outlines rather than from hidden lines. They should be given from a base line, a centre line, an important hole or a finished surface which may be readily established according to the design requirements.



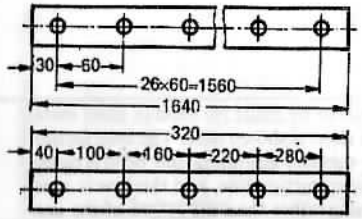
More dimensions than necessary should be avoided. When an overall dimension is used, dimensioning of one intermediate distance may be left out.



Where a number of dimensions has a common datum feature, method shown here should normally be used.



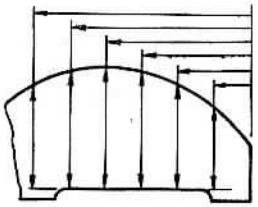
Where equidistant or regularly arranged elements appear on a drawing, the method shown here may be used for simplicity.



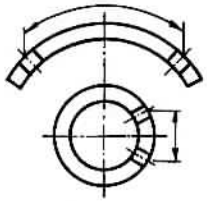
If there is any possibility of confusion between the pitch and the number of pitches, one pitch should be dimensioned.

Where there is no risk of confusion the method shown here may be used.

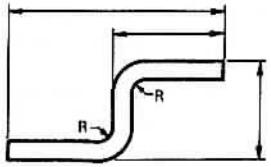
Dimensioning of curved surfaces, chords, arcs, bent parts, countersinks and counterbores



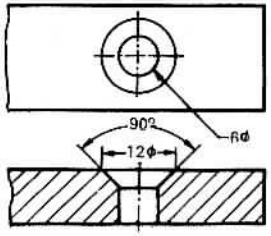
A curved line composed of circular arcs should preferably be dimensioned by radii, and co-ordinates should only be used if the preferred is impracticable.



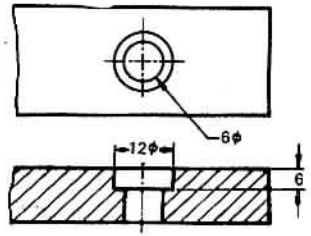
In dimensioning the spacing of holes and other features on curved parts, care should be taken to indicate clearly on which surface the dimensioned points are to be measured and whether the dimensions are cordal or circular.



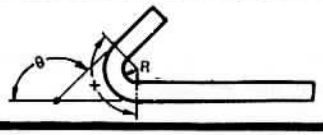
The method of dimensioning bent-up parts.



The method of dimensioning a counter sunk hole.

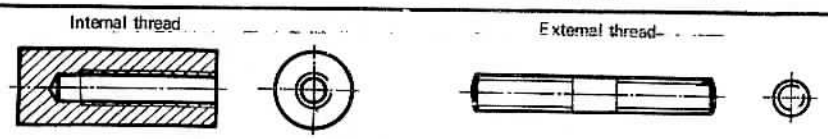


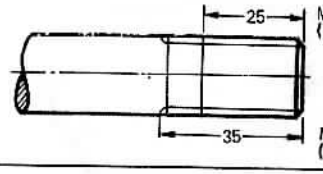
The method of dimensioning a counter bored hole.



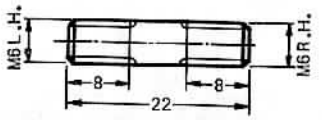
The allowance to be made for bend in sheet metal work or thin sheet is shown here. The intersection of the plane surfaces adjoining the bend is called a mould or construction line, and this line is used in dimensioning rather than the centre of the arc.

Representation of threads, bolts and joints

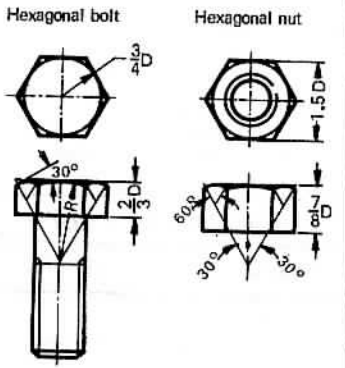


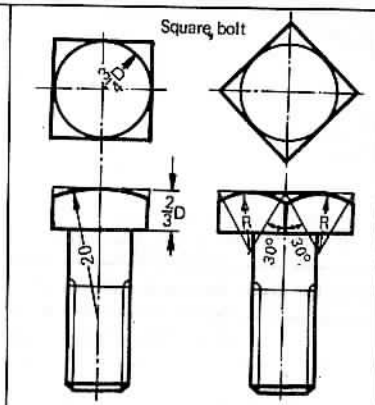


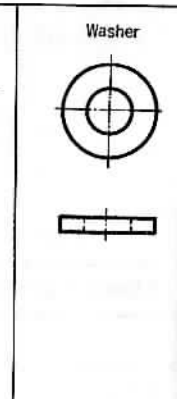
The length of full thread or the distance to the end of full thread should be specified where necessary, as shown here.

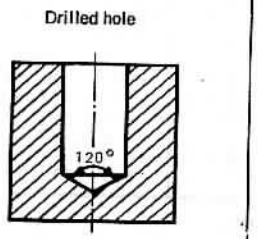


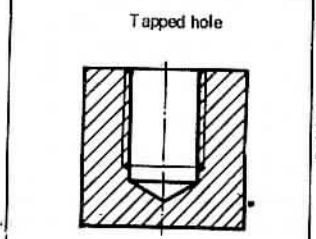
All threads on bolts, nuts and screws are assumed to be right-hand, unless otherwise stated. It is therefore necessary to state this on drawing where left-hand threads are required; it should be clearly indicated by the abbreviation 'LH'.

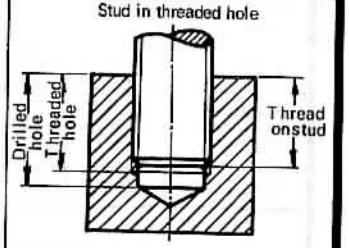


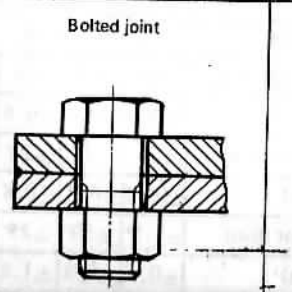


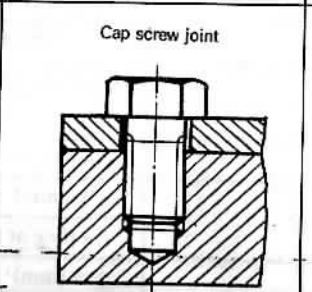


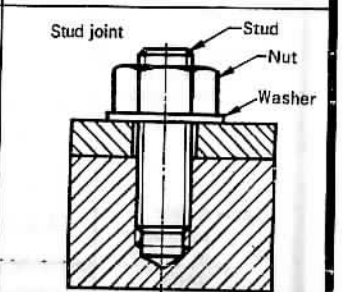




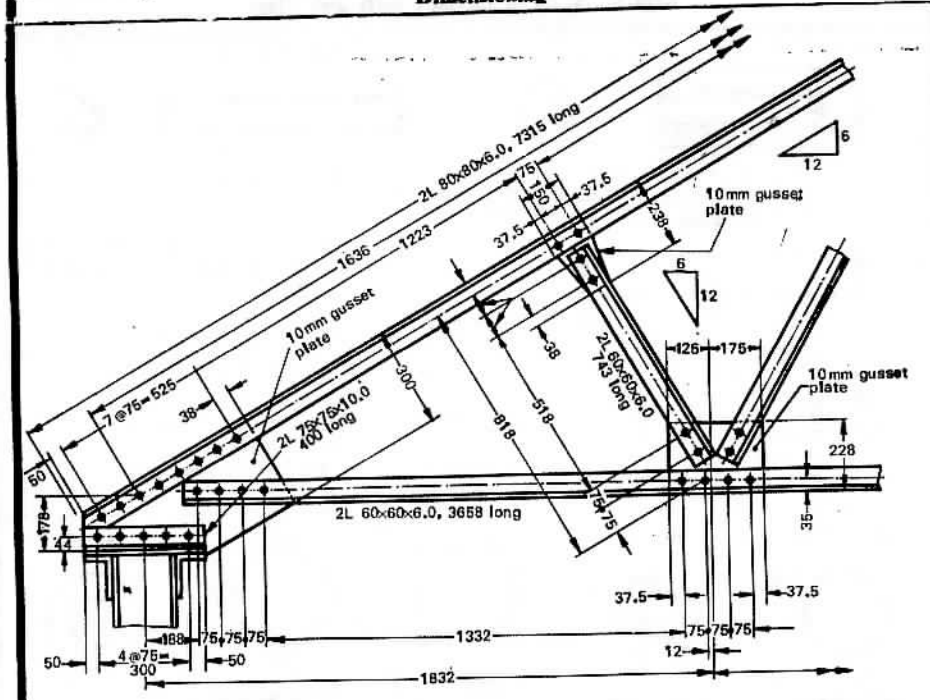








Dimensioning



Example of dimensioning for structural steel work (all dimensions in mm)

Allowable deviations for dimensions without specified tolerances IS: 2102–1969

Deviations for linear dimensions

Class of deviation	Above	0.5	3	6	30	120	315	1000	2000	4000	8000	12000	16000
	Upto and including	3	6	30	120	315	1000	2000	4000	8000	12000	16000	20000
Fine		±0.05	±0.05	±0.1	±0.15	±0.2	±0.3	±0.5	±0.8				
Medium		±0.1	±0.1	±0.2	±0.3	±0.5	±0.8	±1.2	±2	±3	±4	±5	±6
Coarse		–	±0.2	±0.5	±0.8	±1.2	±2	±3	±4	±5	±6	±7	±8
Extra coarse		–	±0.5	±1	±1.5	±2	±3	±4	±6	±8	±10	±12	±12

Deviations for angular dimensions

$\Delta\alpha$ = Angle tolerance in angular units

Δm = Angle tolerance in linear units

Class of deviations		Length (L) of shorter side of angle in mm			
	Above	—	10	50	120
	Up to and including	10	50	120	—
Fine and medium	Δm (mm)	±0.1	±0.2	±0.6	±0.8
	$\Delta\alpha$ (deg or min)	±1°	±30′	±20′	±10′
Coarse	Δm (mm)	±0.15	±0.3	±0.8	±1.2
	$\Delta\alpha$ (deg or min)	±1.5°	±50′	±25′	±15′
Extra coarse	Δm (mm)	±0.3	±0.8	±1.8	±2.4
	$\Delta\alpha$ (deg or min)	±3°	±2°	±1°	±30

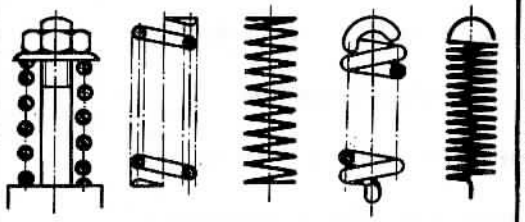
Designation of bars, stripes, tubes, etc., as per IS

Designation	Explanation
ISRO 20	Round bar, hot rolled steel, 20 mm
ISSQ 32	Square bar, hot rolled steel, 32 mm side width
10 ISF 3	Steel flat, hot rolled, 10 mm width, 3 mm thickness
ISST 100×2.5	Steel strip, 100 mm width 2.5 mm thickness
ISA 20 20	Equal angle, rolled steel, 20 mm size
ISA 30 20	Unequal angle, rolled steel, 30×20 mm depth
ISJB 150	Junior beam, rolled steel 150 mm depth
ISLB 75	Light weight beam, rolled steel, 75 mm depth
ISMB 100	Medium weight beam, rolled steel, 100 mm depth
ISWB 150	Wide flange beam, rolled steel, 150 mm depth
ISHB 200	Column section–H beam, rolled steel, 200 mm depth
ISJC 100	Junior channel, rolled steel, 100 mm depth
ISLC 75	Light channel, rolled steel, 75 mm depth
ISMC 125	Medium weight channel, rolled steel, 125 mm depth
ISNT 20	Normal tee bar, rolled steel, 20 mm size
ISDT 100	Deep legged tee bar, rolled steel, 100 mm size
ISLT 200	Slit light weight tee bar, rolled steel, 200 mm size
ISHT 75	Slit tee bar–H section, rolled steel, 75 mm size
ISALT 25×25–0.4	Aluminium T–section, 25 mm size, 0.4 kg weight per metre
ISALA 20×10×1.5	Aluminium, equal angle section, 20×10 mm size, 1.5 mm thickness
ISALB 40×20–0.4	Aluminium, beam section, 40×20 mm size, 0.4 kg weight per metre
ISALC 40×20×2	Aluminium, channel section, 40×20 mm size, 2 mm thickness

Symbols

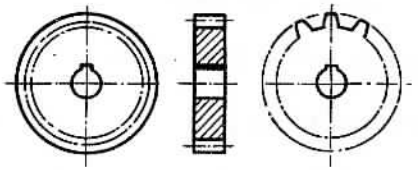
IS: 696—1960

	Shop snap headed rivets	Site CSK (near side) rivets	
	Shop CSK (near side) rivets	Site CSK (far side) rivets	
	Shop CSK (Far side) rivets	Site CSK (both side) rivets	
	Shop CSK (both side) rivets	Open hole	
	Site snap headed rivets		

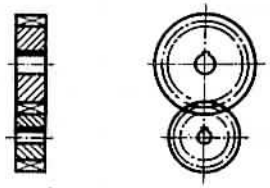


Springs

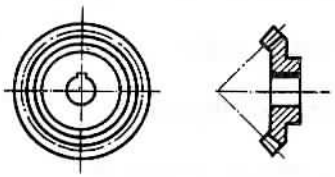
1. Compression spring used with bolt and nut
2. Compression spring with helical coils
3. Compression spring, schematic notation
4. Tension spring
5. Tension spring, schematic notation



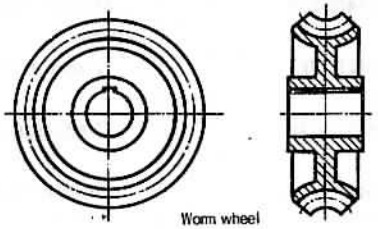
Spur gear — Alternative



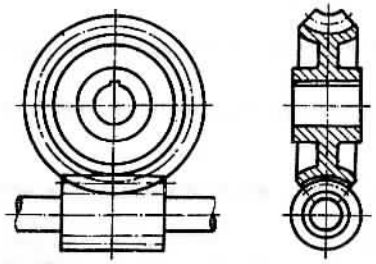
Spur gears (assembly)



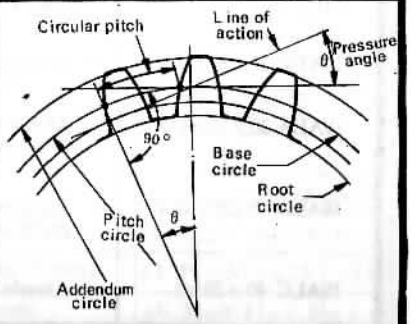
Bevel gear



Worm wheel



Worm wheel (assembly)



SPK.

Welding

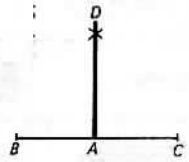
IS: 696—1960

Type size and position of welds in fusion-welded joints using the gas or arc welding process

Form of weld	Sectional representation	Appropriate symbol
Fillet		
Square butt		
Single V butt		
Double V butt		
Single U butt		
Double U butt		
Single bevel butt		
Double bevel butt		
Single J butt		
Double J butt		
Stud		
Bead edge or seal		
Sealing run		
Spot		
Seam		
Stitch		

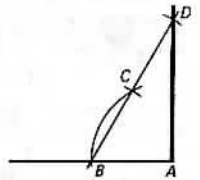
Form of weld	Sectional representation	Appropriate symbol
Plug or slot		
Backing strip		
Mashed seam	Before — After	
Mashed stitch	Before — After	
Projection	Before — After	
Flash	Pod or bar — Tube	
Butt resistance or pressure (upset)	Pod or bar — Tube	
Location of weld	The position of the arrow is chosen for purposes of clarity. Normally it would be placed immediately adjacent to the joint.	
Fillet weld one side of tee		
Fillet weld each side of tee		
Single V butt weld		
Double bevel butt		
Size of fillet weld	S — S — S	
Regular intermittent fillet welds commencing with a weld	50-100 — 50(100)	
Regular intermittent fillet welds commencing with an unwelded length	100-50 — (100)50(100)	
Staggered intermittent fillet welds	50-100 — 50-100 — (75)50(100) / 50(100)	
Size of unequal welds	10x6	

Geometrical Constructions



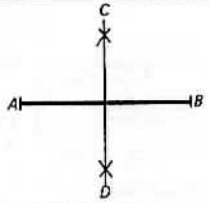
To erect a perpendicular to a straight line at point *A*

Draw arcs on both sides of point *A* having the same radius. Draw intersecting arcs from *B* and *C*. Connect point of intersection *D* with point *A*. *DA* is the perpendicular of *BC*.



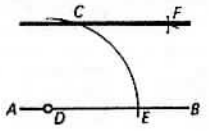
To erect a perpendicular to a straight line at its terminal point *A*.

With *A* as centre draw an arc cutting the base line at one end at *B*. With *B* as centre draw an arc to cut the former at *C*. With *C* as centre and with the same radius draw an arc in the same direction. Draw a straight line through *B* and *C* to cut the last arc at *D*. The line *AD* is perpendicular to *AB* at point *A*.



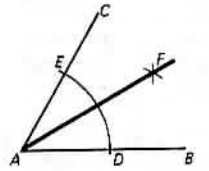
To bisect a line

With any radius greater than one-half *AB* draw two arcs with *A* and *B* as centres. The line *CD*, joining the points of intersection of the arcs, is the perpendicular bisector of the given line.



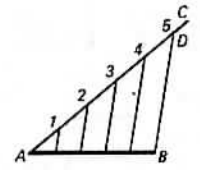
To draw a parallel line to *AB* through point *C*

With *D* as centre, at any point on *AB*, draw an arc through point *C*, one end cutting the line *AB* at *E*. With *C* and *E* as centres and with same radius draw arcs to intersect at *F*. A straight line drawn through points *C* and *F* is parallel to *AB*.



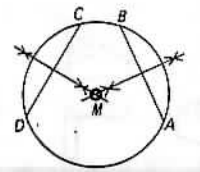
To bisect an angle

With *A* as centre draw an arc intersecting *AB* and *AC* at *D* and *E* respectively. With *D* and *E* as centres, draw arcs of equal radii intersecting at *F*. *AF* is the required bisector.



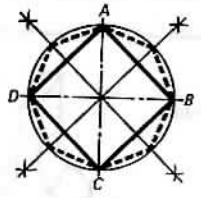
To divide a straight line *AB* into a given number (*n*) of equal parts

Draw the line *AC* at any convenient angle with *AB*, and mark off, with divider or scale *n* equal lengths from *A* to *D*. Join *D* to *B* and draw parallels to *BD* through the other points in *AD*. The intersections of these parallels with *AB* give the required equal parts.



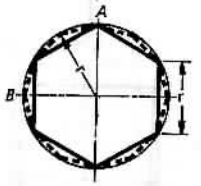
To locate centres of circle and arcs

Draw any two chords *AB* and *CD* in the circle. The intersecting point *M* of the perpendicular bisectors of the two chords is the centre required.



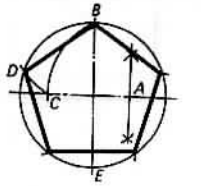
Square—Octagon

Draw two lines *AC*, *BD* at right angles and a circle whose centre is the point of intersection of these two lines. Join the points *ABCD* where the circle cuts the lines, to get the square. By drawing the two diagonals which bisect the lines *AB*, *BC*, *CD* and *DA*, one gets the other four corners required to complete the regular octagon.



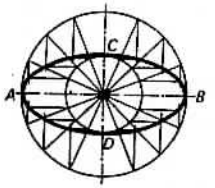
Hexagon—Dodecagon

Mark off by arcs, six points around the circle starting from *A*, equidistant as the radius of the circle. These points are the vertices of the required hexagon. If one is to start marking from point *B* with same radius, the additional points of intersection required to complete the regular dodecagon will be found.



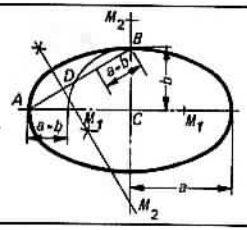
Pentagon—Decagon

Draw two diameters intersecting at right angles at the centre of the circle. Bisect one radius to get point *A* and from *A* with radius *AB*, cut the horizontal diameter at *C*. Length *BC* is side length of pentagon which can be marked off along the periphery. Starting at point *E* results in another set of points of intersection, required to complete the regular decagon.



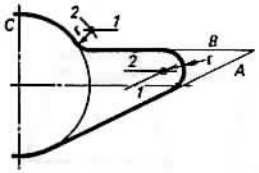
Ellipse by means of two circles

Draw two circles, with two diameters *AB* and *CD* to have major and minor axes respectively. Draw few more suitable number of diameters. Draw now vertical lines starting from the points of intersection of the bigger circle. Then draw equal number of horizontal lines starting from the points of intersection of the smaller circle to meet the vertical lines. A line drawn to join these intersecting points will complete ellipse.



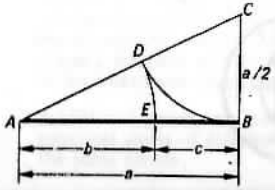
Ellipse

Join *A* and *B* with *C* as centre. Draw an arc having a radius $r = b$ (minor axis) cutting *AC*. From *B* mark the length $a - b$ to cut *AB* at *D*. The perpendicular bisector of *AD* intersects major and minor axes at M_1 and M_2 respectively as points required to complete the ellipse.



Curvature

Draw lines 1 and 2 parallel to *A* and *B* as well as *B* and *C* at a distance $= r$. The points of intersection are the centres of the required radii.

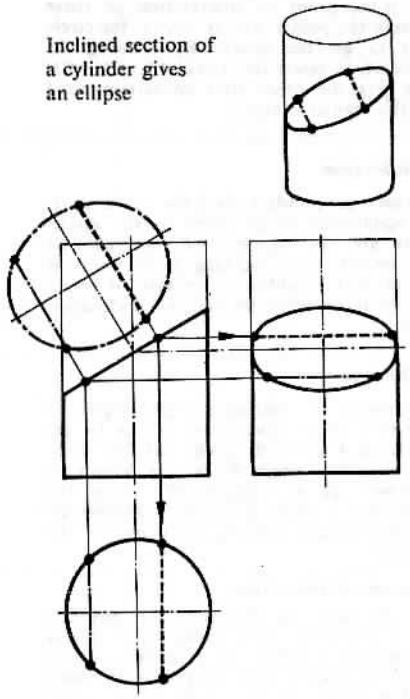


Golden section (division of a line *AB* so that segment *c* is to *b* what *b* is to *a*)

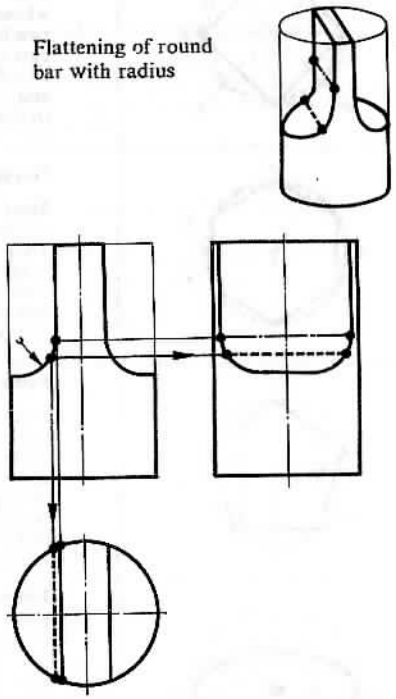
Draw a perpendicular $BC = a/2$, to *AB*. Join *A* and *C*. With *C* as centre and *CB* as radius draw an arc cutting *AC* at *D*. With *A* as centre and *AD* as radius draw an arc to cut *AB* at *E* to get the Golden Section of the line *AB*.

Section of solids

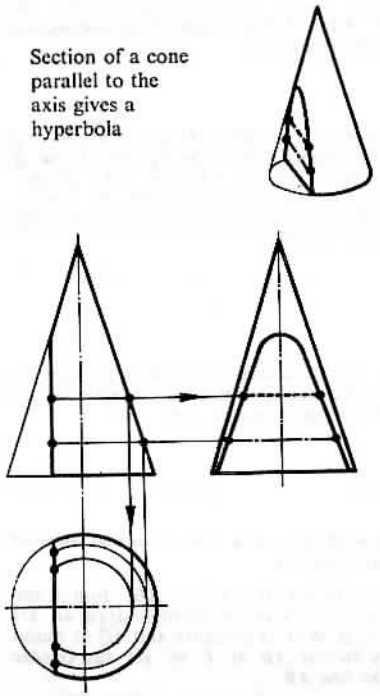
Inclined section of a cylinder gives an ellipse



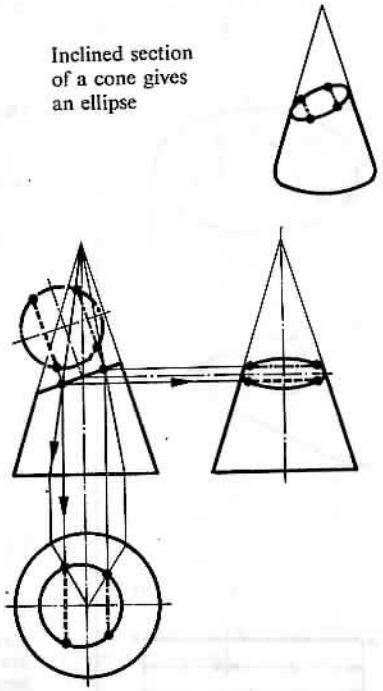
Flattening of round bar with radius



Section of a cone parallel to the axis gives a hyperbola

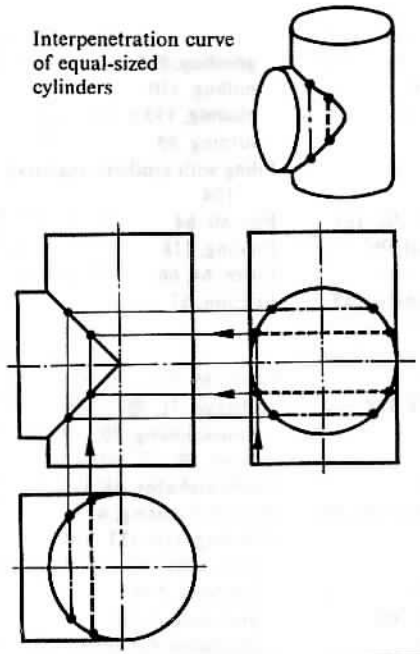


Inclined section of a cone gives an ellipse

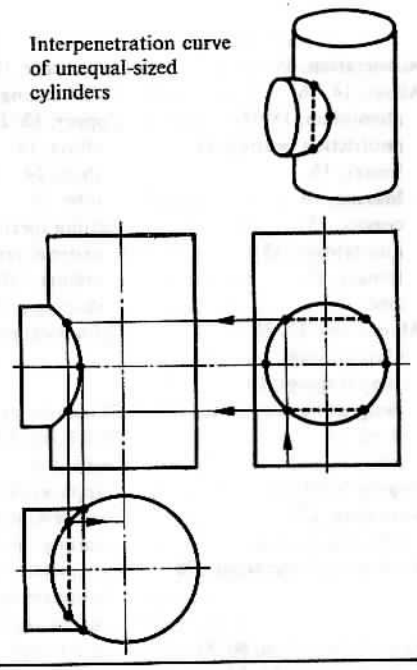


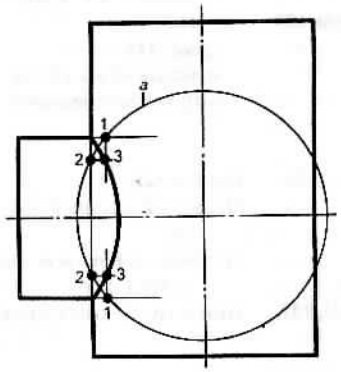
Interpenetration curves of solids

Interpenetration curve of equal-sized cylinders



Interpenetration curve of unequal-sized cylinders



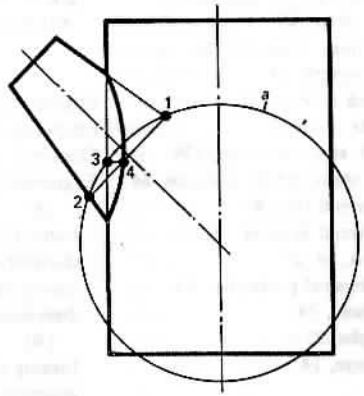


Interpenetration curve of unequal-sized cylinders—spherical shell construction method.

Finding a point of the interpenetration curve:

In the given projection the spherical shell α intersects the smaller cylinder at point 1. The larger cylinder intersects the shell at point 2. Horizontal and vertical lines through 1 and 2 intersect in 3 which is a point of the interpenetration curve.

Additional points may be found with spherical shells of different radii with the same centre.



Interpenetration curve of cylinder with an inclined cone–spherical shell construction method.

Finding a point of the interpenetration curve:

In the given projection the spherical shell α intersects the cone at points 1 and 2, and the cylinder at point 3. The horizontal line through 3 intersects with the connection of 1 and 2 in 4, which is a point of the interpenetration curve.

Additional points may be found with spherical shells of different radii with the same centre.

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SPK

Measures and Weights

Conversion of English and US Measures and Weights and Vice Versa

Linear measures	inch	foot	yard	cm	m
1 inch	1	0.0833	0.0278	2.540	0.0254
1 foot	12	1	0.3333	30.48	0.3048
1 yard	36	3	1	91.44	0.9144
1 cm (centimeter)	0.3937	0.0382	—	1	0.01
1 m (meter)	39.37	3.281	1.094	100	1

Square measures	sq. inch	sq. foot	sq. yard	cm^2	m^2
1 square inch	1	—	—	6.452	—
1 square foot	144	1	0.1111	929	0.0929
1 square yard	1296	9	1	8361	0.8361
1 cm^2	0.155	—	—	1	0.0001
1 m^2	1550	10.76	1.196	10000	1

Cubic measures	cu. inch	cu. foot	US gallon	cm^3	dm^3
1 cubic inch	1	—	—	16.39	0.0164
1 cubic foot	1728	1	7.481	—	28.32
1 US gallon	231	0.1337	1	3785	3.785
1 cm^3	0.061	—	—	1	0.001
1 dm^3	61.02	0.035	0.2642	1000	1

Weights	lb	Brit. cwt	Brit. t	kg
1 pound	1	0.009	—	0.454
1 long hundredweight	112	1	0.05	50.8
1 long ton (Brit.)	2240	20	1	1016
1 kg (kilogram)	2.205	0.020	0.00098	1

ALL ABOUT MACHINE TOOLS

REVISED TO INDIAN STANDARDS

Heinrich Gerling

This book deals with everything connected with machine tools, that is, everything important during the manufacture of workpieces by machining. It also deals with the machine tool itself and the tool required for it, about the machining operation and its physical principles. It is intended for all those who are directly concerned with machine tools, and particularly for the apprentices of all metal-working trades who are supposed to have some working knowledge about machine tools, their design and performance.

Contents: *Turning Operations*. Manufacture of Cylindrical Bolts. Manufacture of Eccentric Shaft. Profile Turning. Machining of Housings and Castings. Mass Production of Turned Parts. *Drilling and Boring Operations*. Drilling of Simple Holes. Countersinking and Counterboring Operations. Drilling of Smooth and Accurate Holes. Boring of Cross Holes on the Horizontal Boring Machine. Manufacture of Bushes. *Manufacture of Tapered Parts*. Lathe Centre. Holes for Taper Pins. *Milling Operations*. Plane Surfaces. Keyways. Slides. Hexagons. *Shaping and Planing Operations*. Shaping of V-blocks. Planing of Guide Gibs. *Manufacture of Parts on the Slotting Machine*. Slotting of Keyways. *Broaching Operations*. Broaching of Multiple Spline Bores. *Grinding Operations*. Shafts. Bores. Parallels. Fine-finishing Operations. *Threading Operations*. Threading Operations on the Turning Lathe with Taps and Dies. Cutting of Male Threads. Cutting of Internal Threads. *Manufacture of Gears*. Milling of Spur Gears by the Indexing Method. Index.

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